

Corso di Scienza delle Costruzioni

**IL PROBLEMA DI SAINT VENANT
ANALOGIA IDRODINAMICA
TORSIONE IN PARETE SOTTILE
TEORIA DI BREDT**

ANALOGIA IDRODINAMICA

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⇒ DETERMINARE "INTUITIVAMENTE" L'ANDAMENTO QUALITATIVO DI ζ

ANALOGIA IDRODINAMICA

⇒ DETERMINARE "INTUITIVAMENTE" L'ANDAMENTO QUALITATIVO DI ζ

↳ LINEA DI FLUSSO: LINEA TANGENTE A ζ

↳ ANALOGIA FORMALE tra $\left\{ \begin{array}{l} \cdot \text{ TENSIONI TANGENZIALI} \\ \cdot \text{ CAMPO DI VELOCITÀ DEL FLUIDO} \end{array} \right.$

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TORSIONE **UNIFORME**: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

$$\operatorname{div} \tau_z = 0 \quad \text{su } \Omega$$

Equilibrio

$$(\operatorname{rot} \tau_z) \cdot \mathbf{n} = \text{cost} \quad \text{su } \Omega$$

Congruenza

$$\tau_z \cdot \mathbf{n} = 0 \quad \text{su } \partial\Omega$$

Condizione di bordo

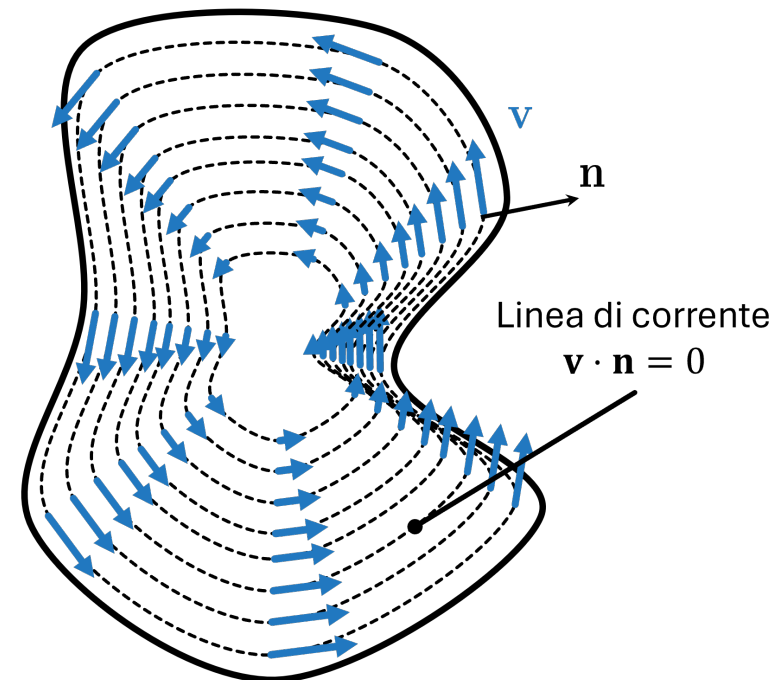
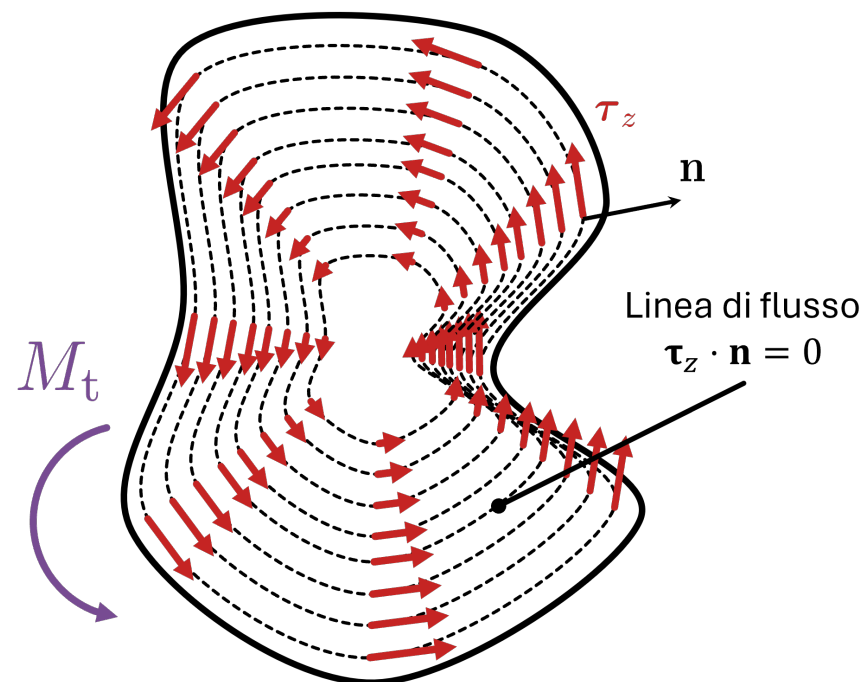
$$\operatorname{div} \mathbf{v} = 0 \quad \text{su } \Omega$$

Conservazione della massa
(fluido incompressibile)
Flusso inviscido

$$(\operatorname{rot} \mathbf{v}) \cdot \mathbf{n} = \text{cost} \quad \text{su } \Omega$$

$$\mathbf{v} \cdot \mathbf{n} = 0 \quad \text{su } \partial\Omega$$

Condizione di bordo
(pareti impenetrabili)



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TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME η_t

SEZIONI APERTE

$\zeta \parallel$ LINEA MEDIA

ζ VARIANO LINEARMENTE
SULLO SPESSORE

$\zeta_{\max, \min}$ SUI BORDI

$\neq$

SEZIONI CHIUSE

$\zeta \parallel$ LINEA MEDIA

ζ DISTRIBUITO UNIFORMEM.
SULLO SPESSORE

$\zeta = \text{cost.}$

In prima approssimazione!

ANALOGIA IDRODINAMICA

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SEZIONI APERTE

$z \parallel$ LINEA MEDIA

z VARIANO LINEARMENTE
SULLO SPESSORE

$z_{MAX, MIN}$ SUI BORDI

$\neq$

SEZIONI CHIUSE

$z \parallel$ LINEA MEDIA

z DISTRIBUITO UNIFORMEM.
SULLO SPESSORE

$z = \text{cost.}$

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

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$z \parallel$ LINEA MEDIA

z VARIANO LINEARMENTE
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$z_{MAX, MIN}$ SUI BORDI

$\neq$

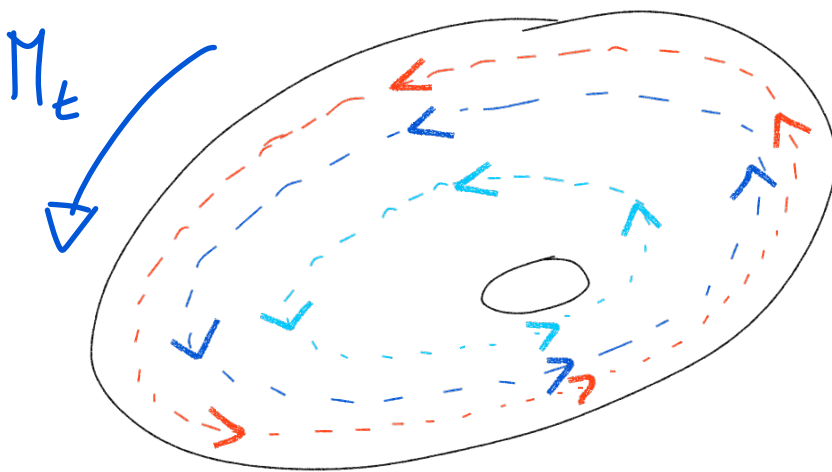
SEZIONI CHIUSE

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SULLO SPESSORE

$z = \text{cost.}$

SEZIONE COMPATTA



ANALOGIA IDRODINAMICA

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SEZIONI APERTE

$z \parallel$ LINEA MEDIA

z VARIANO LINEARMENTE
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$z_{MAX, MIN}$ SUI BORDI



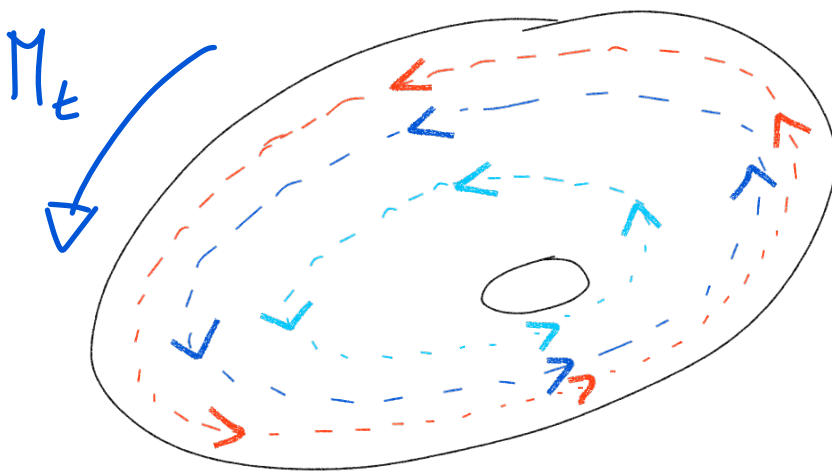
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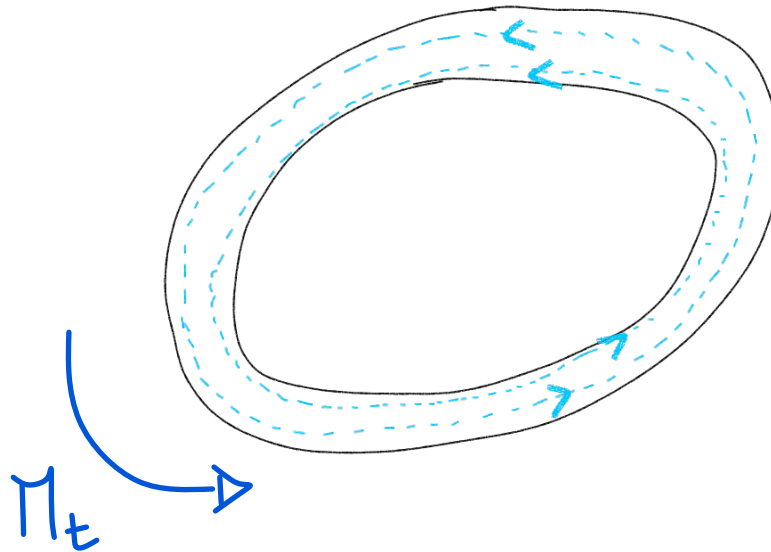
z DISTRIBUITO UNIFORMEM.
SULLO SPESSORE

$z = \text{cost.}$

SEZIONE COMPATTA



S. SOTTILE CHIUSA



ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

SEZIONI APERTE

$z \parallel$ LINEA MEDIA

z VARIANO LINEARMENTE
SULLO SPESSORE

$z_{MAX, MIN}$ SUI BORDI



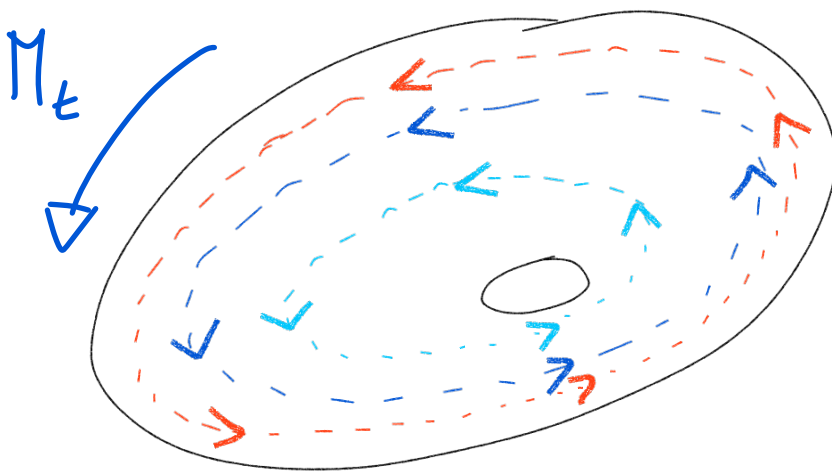
SEZIONI CHIUSE

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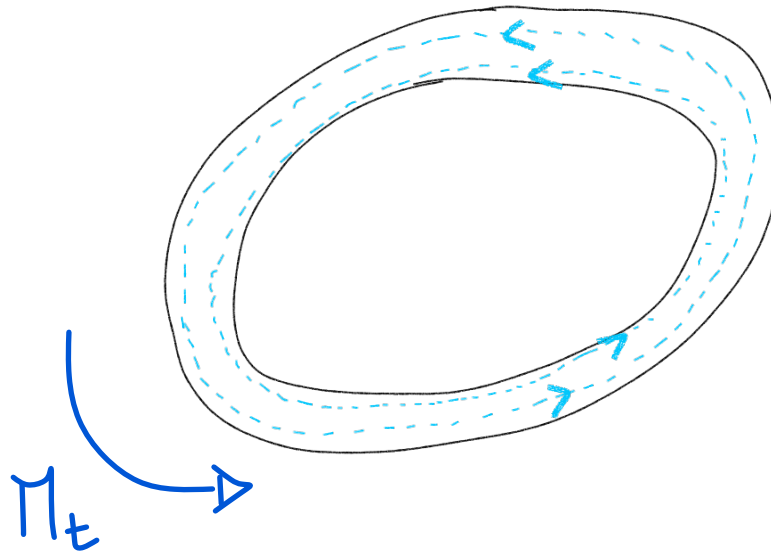
z DISTRIBUITO UNIFORMEM.
SULLO SPESSORE

$z = \text{cost.}$

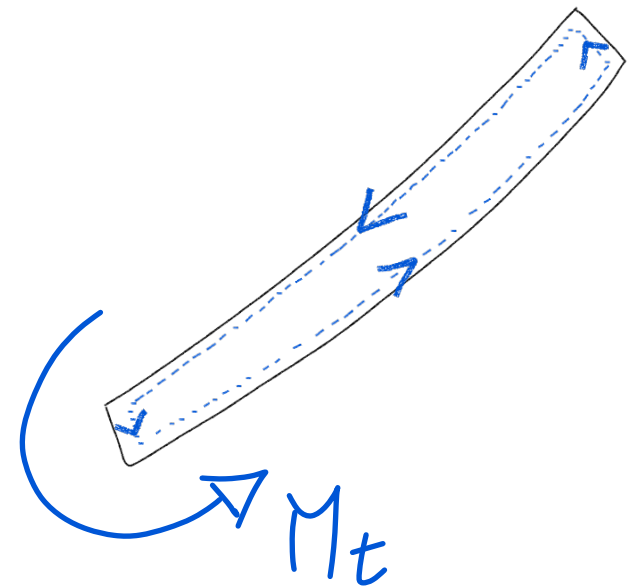
SEZIONE COMPATTA



S. SOTTILE CHIUSA



S. SOTTILE APERTA



ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME η_t

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME Π_t

DIM. CAMPO SOLENOIDALE

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME $\mathbf{n}_t$

DIM. CAMPO SOLENOIDALE

DETTO $\underline{\tau} = [\tau_{zx}, \tau_{zy}]$ (PIANO)

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME $\mathbf{n}_t$

DIM. CAMPO SOLENOIDALE

DETTO $\underline{\tau} = [\tau_{zx}, \tau_{zy}]$ (PIANO)

$$\operatorname{div}(\underline{\tau}) = \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} =$$

ANALOGIA IDRODINAMICA

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NEUMANN

$$\begin{cases} \tau_{zx} = \frac{M_t}{I_t} \left(\frac{\partial \psi(x,y)}{\partial x} - y \right) \\ \tau_{zy} = \frac{M_t}{I_t} \left(\frac{\partial \psi(x,y)}{\partial y} + x \right) \end{cases}$$

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TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

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← NEUMANN

$$\begin{cases} \tau_{zx} = \frac{M_t}{I_t} \left(\frac{\partial \psi(x,y)}{\partial x} - y \right) \\ \tau_{zy} = \frac{M_t}{I_t} \left(\frac{\partial \psi(x,y)}{\partial y} + x \right) \end{cases}$$

$$\Leftrightarrow \boxed{\frac{\partial^2 \psi(x,y)}{\partial x^2} + \frac{\partial^2 \psi(x,y)}{\partial y^2} = 0}$$

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$$\frac{\partial^2 \psi(x,y)}{\partial x^2} + \frac{\partial^2 \psi(x,y)}{\partial y^2} = 0$$

ANALOGO ANCHE
CON DERIVATA DI:

$$\begin{cases} \tau_{zx} = - \frac{M_t}{I_p} y \\ \tau_{zy} = + \frac{M_t}{I_p} x \end{cases}$$

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME Π_t

DIM. CAMPO ROTAZIONALE

ANALOGIA IDRODINAMICA

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DIM. CAMPO ROTAZIONALE

DETTO $\underline{\tau} = [\tau_{zx}, \tau_{zy}]$ (PIANO)

$$\text{rot}(\underline{\tau}) = \det \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \tau_{zx} & \tau_{zy} & 0 \end{bmatrix}$$

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$$= \frac{M_t}{I_t} \left[\frac{\partial^2 \gamma(x,y)}{\partial x \partial y} - \frac{\partial^2 \gamma(x,y)}{\partial x \partial y} + 2 \right] = 2 \frac{M_t}{I_t} \hat{k} =$$

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TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

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$$= \frac{M_t}{I_t} \left[\cancel{\frac{\partial^2 \gamma(x,y)}{\partial x \partial y}} - \cancel{\frac{\partial^2 \gamma(x,y)}{\partial x \partial y}} + 2 \right] = 2 \frac{M_t}{I_t} \hat{k} =$$

$\frac{M_t}{I_t} = G\Theta$ $\xrightarrow{\text{INERZ. TOR.}}$

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

DIM. CAMPO ROTAZIONALE

DETTO $\underline{\tau} = [\tau_{zx}, \tau_{zy}]$ (PIANO)

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NEUMANN $\begin{cases} \tau_{zx} = \frac{M_t}{I_t} \left(\frac{\partial \gamma(x,y)}{\partial x} - \gamma \right) \\ \tau_{zy} = \frac{M_t}{I_t} \left(\frac{\partial \gamma(x,y)}{\partial y} + x \right) \end{cases}$

$$= \frac{M_t}{I_t} \left[\cancel{\frac{\partial^2 \gamma(x,y)}{\partial x \partial y}} - \cancel{\frac{\partial^2 \gamma(x,y)}{\partial x \partial y}} + 2 \right] = 2 \frac{M_t}{I_t} \hat{k} = 2 G \Theta \hat{k}$$

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$\frac{M_t}{I_t} = G \Theta$ $\xrightarrow{\text{INERZ. TOR.}}$

ORIENTATO COME $\hat{k}$

c.v.d.

ANALOGIA IDRODINAMICA

TORSIONE UNIFORME: LINEE DI FLUSSO CHIUSE (ADDENSAMENTO SULLE STRIZIONI) ORIENTATE COME M_t

DIM. CAMPO ROTAZIONALE

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NEUMANN $\begin{cases} \tau_{zx} = \frac{M_t}{I_t} \left(\frac{\partial \gamma(x,y)}{\partial x} - y \right) \\ \tau_{zy} = \frac{M_t}{I_t} \left(\frac{\partial \gamma(x,y)}{\partial y} + x \right) \end{cases}$

$$= \frac{M_t}{I_t} \left[\cancel{\frac{\partial^2 \gamma(x,y)}{\partial x \partial y}} - \cancel{\frac{\partial^2 \gamma(x,y)}{\partial x \partial y}} + 2 \right] = 2 \frac{M_t}{I_t} \hat{k} = 2 G \Theta \hat{k} \Rightarrow \underline{\text{rot}(\underline{\tau}) = \text{cost}}$$

$\frac{M_t}{I_t} = G \Theta$ $\xrightarrow{\text{INERZ. TOR.}}$

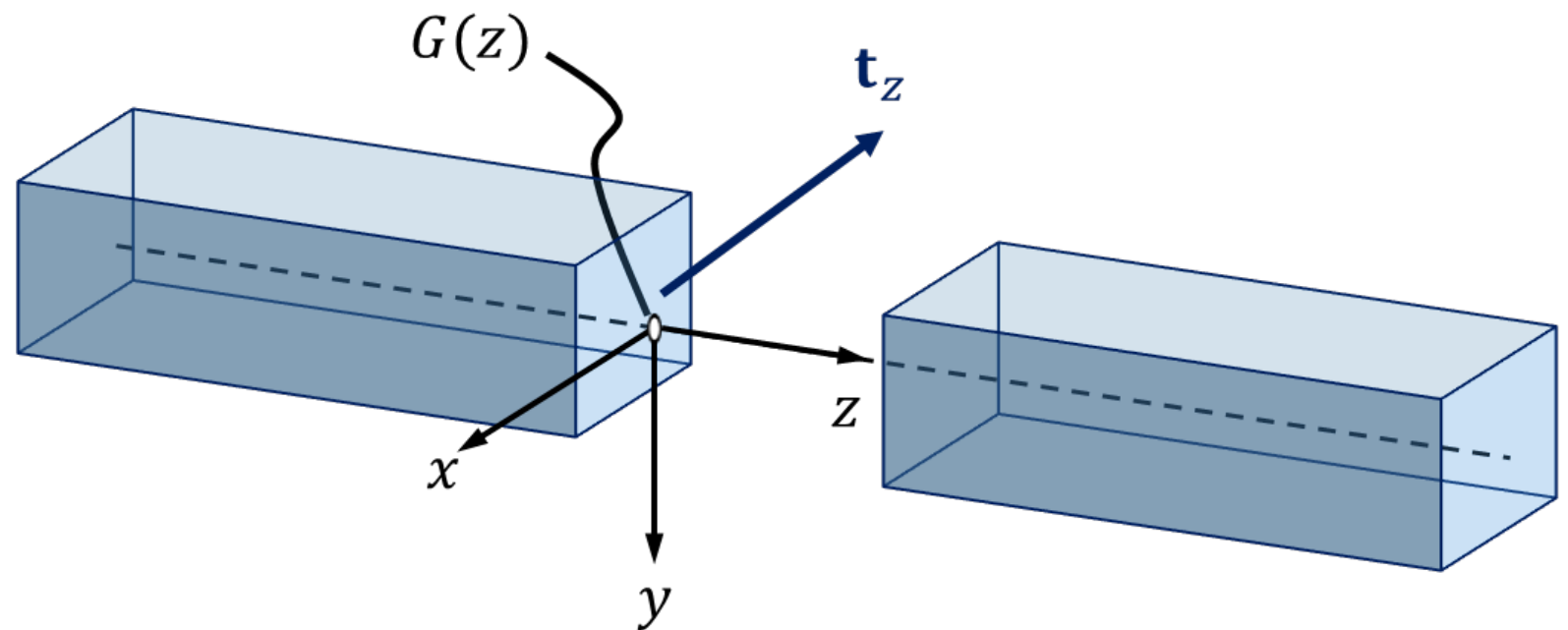
ORIENTATO COME $\hat{k}$

c.v.d.

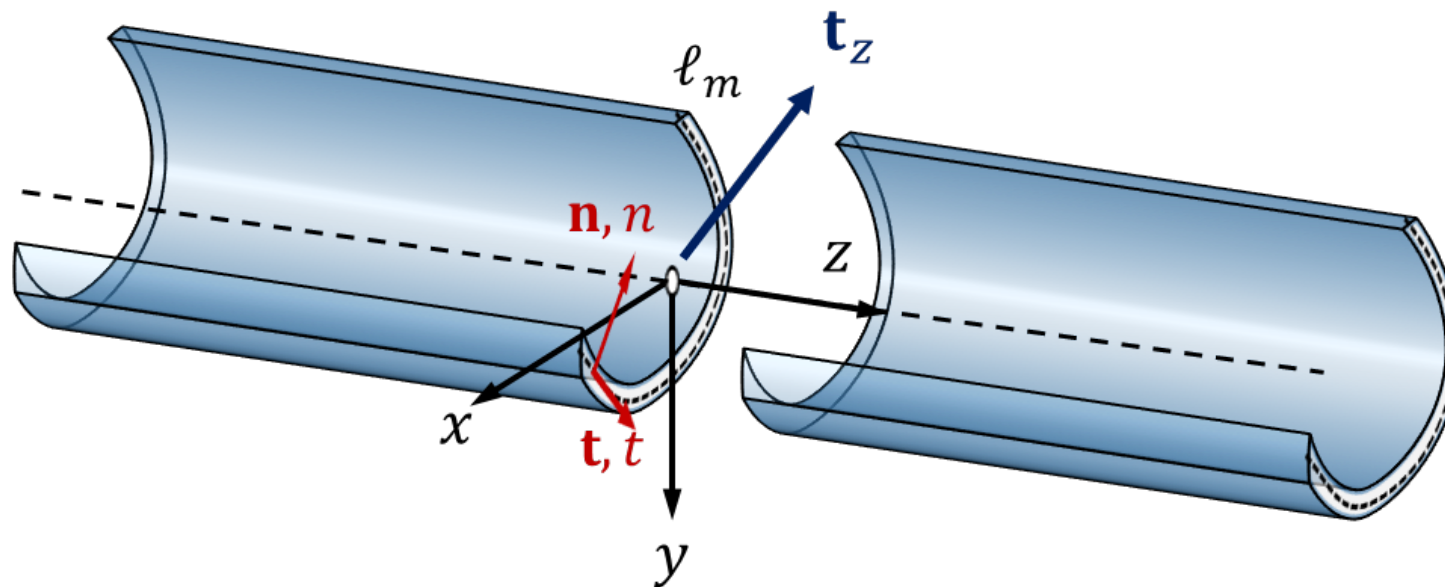
SEZIONI SOTTILI

Trave a sezione compatta

$$\begin{aligned}\tau_{zx} &:= \mathbf{t}_z \cdot \mathbf{e}_x \\ \tau_{zy} &:= \mathbf{t}_z \cdot \mathbf{e}_y \\ \sigma_z &:= \mathbf{t}_z \cdot \mathbf{e}_z\end{aligned}$$



$$\mathbf{t} \times \mathbf{n} = \mathbf{e}_z$$



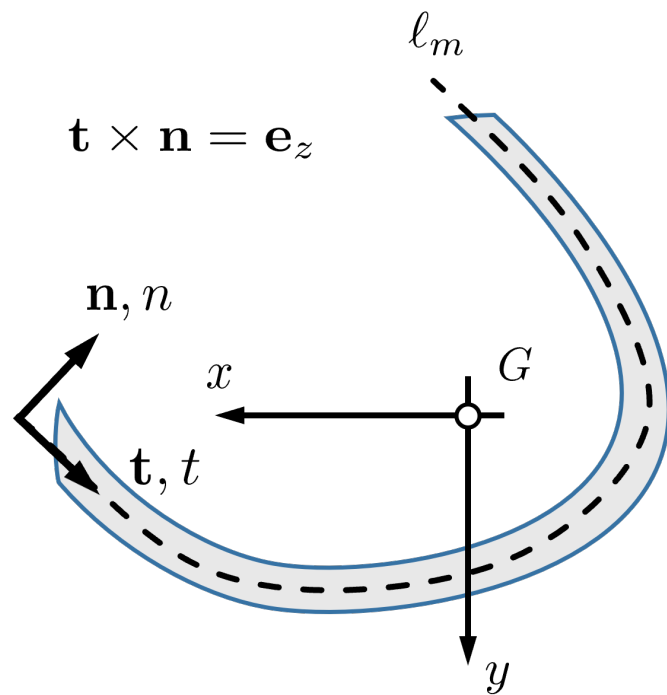
Trave a sezione sottile

Ci si riferisce alla base locale

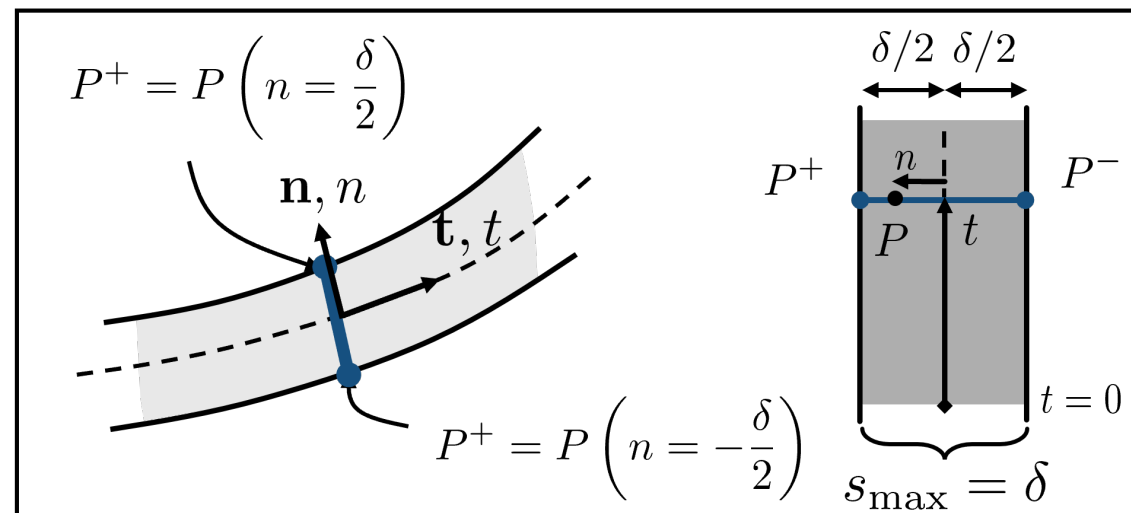
$$\begin{aligned}\tau_{zt} &:= \mathbf{t}_z \cdot \mathbf{t} \\ \tau_{zn} &:= \mathbf{t}_z \cdot \mathbf{n} \\ \sigma_z &:= \mathbf{t}_z \cdot \mathbf{e}_z\end{aligned}$$

SEZIONI SOTTILI

- Geometria completamente individuata dalla linea media ℓ_m , parametrizzata con un'ascissa curvilinea t , e una funzione spessore $s(t)$.
- I segmenti ortogonali punto per punto alla linea media P^-P^+ si chiamano **corde di spessore**
- Spessore: $s_{\max}(t) = \delta(t) = \|P^-P^+\|$
- Sistema di riferimento $\{\mathbf{t}, \mathbf{n}\}$ locale rispetto al quale si scrivono tutte le grandezze $(\mathbf{T}, \mathbf{E}, \mathbf{u})$



Nella base locale $\{\mathbf{t}, \mathbf{n}\}$:
 Coordinata lungo $\mathbf{t}$: t
 Coordinata lungo $\mathbf{n}$: n

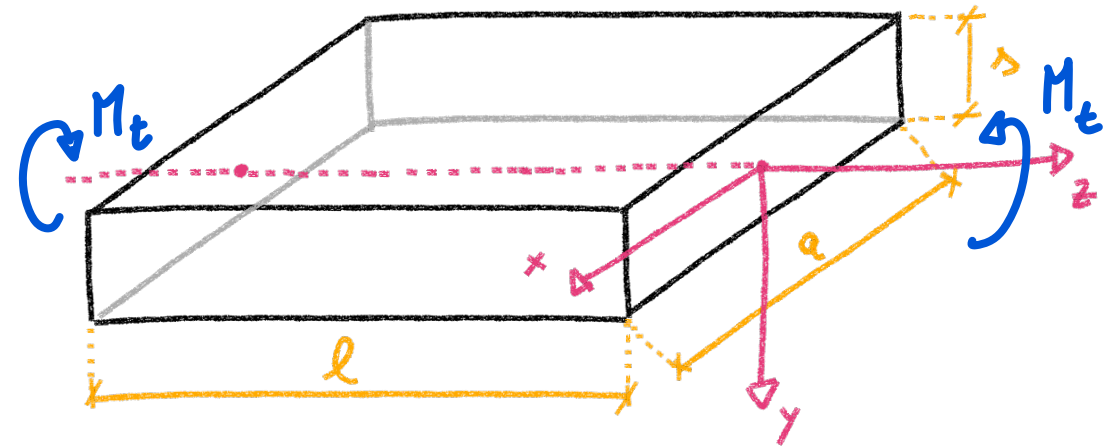


Problema di dSV per sezioni sottili

- σ_z dalla formula di Navier
- $\boldsymbol{\tau}_z = \tau_{zt}\mathbf{t} + \tau_{zn}\mathbf{n}$

SEZIONE RETTANGOLARE SOTTILE

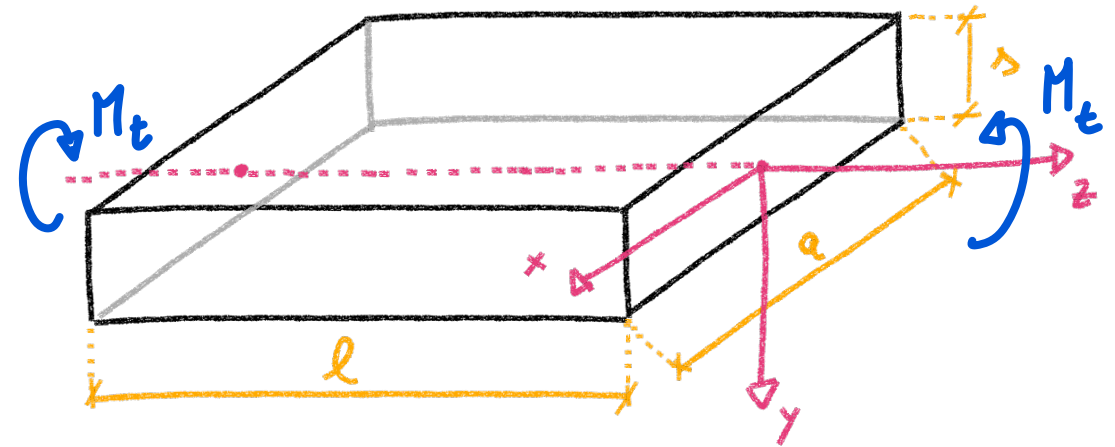
$$s \ll a, l$$



SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

$$s \ll a, l$$

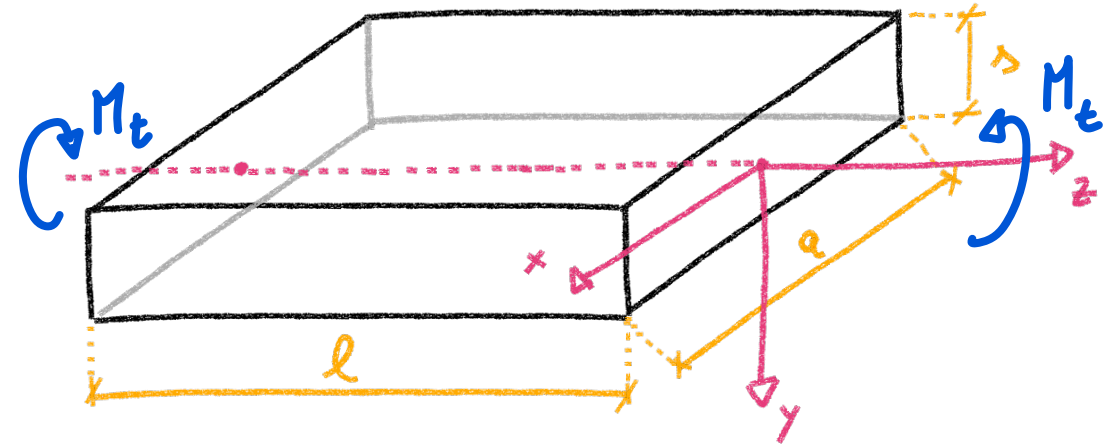


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI

$$s \ll a, l$$

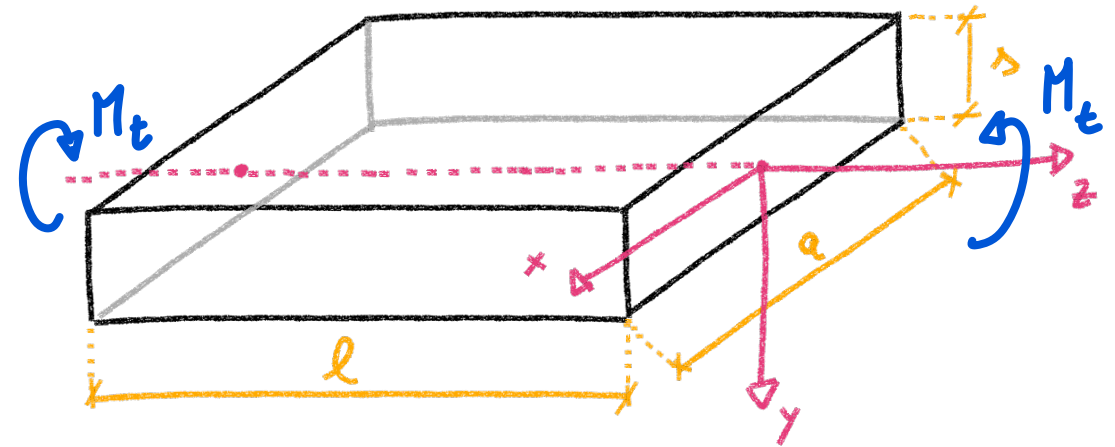


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- τ // LATI LUNGI
- τ LINEARI SU LINEA MEDIA $\begin{cases} z_{\max} \\ \text{ESTREMI} \end{cases}$

$$s \ll a, l$$

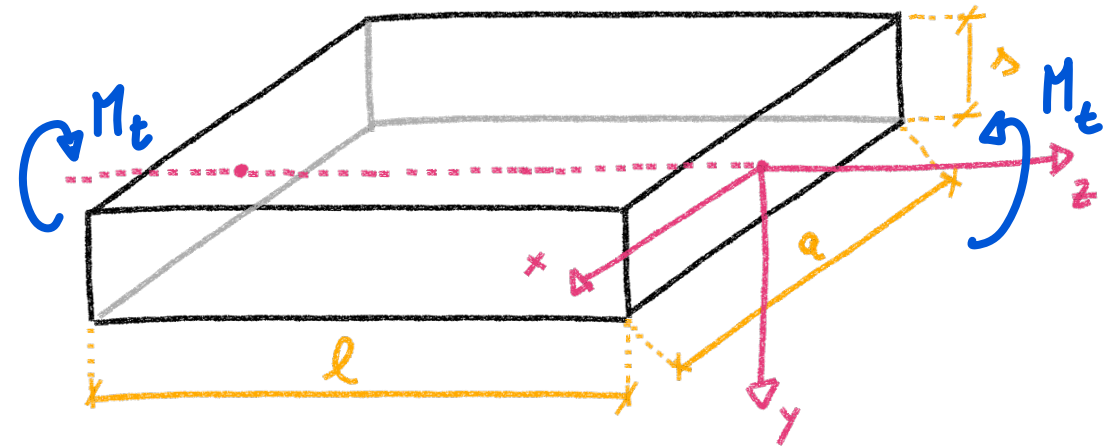


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $\tau \parallel$ LATI LUNGI
- τ LINEARI SU LINEA MEDIA $\left\{ \begin{array}{l} z_{\max} \\ \text{ESTREMI} \end{array} \right.$
- $\tau \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$

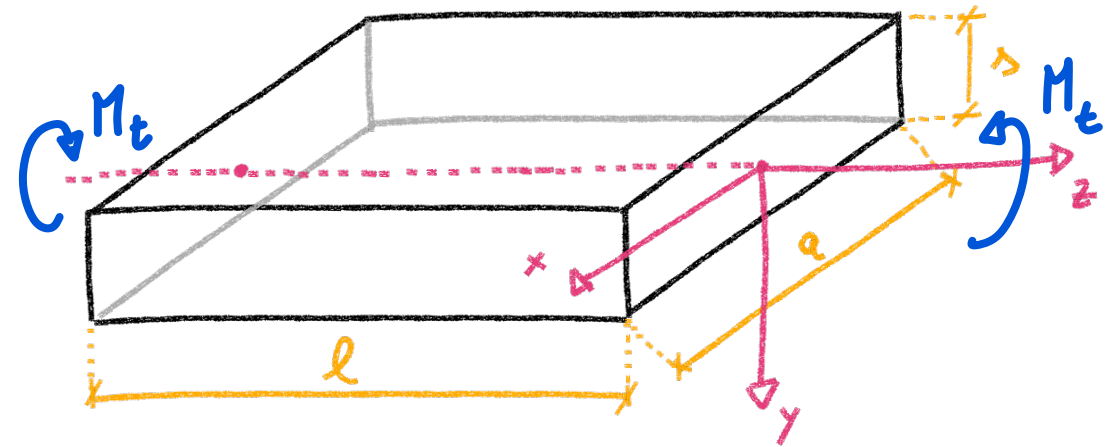


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $\tau \parallel$ LATI LUNGI
- τ LINEARI SU LINEA MEDIA $\left\{ \begin{array}{l} \tau_{\text{MAX}} \\ \text{ESTREMI} \end{array} \right.$
- $\tau \parallel$ LATI MINORI "TRASCURABILI"

$$\delta \ll a, l$$



SOLUZIONE ESATTA (NEUMANN)

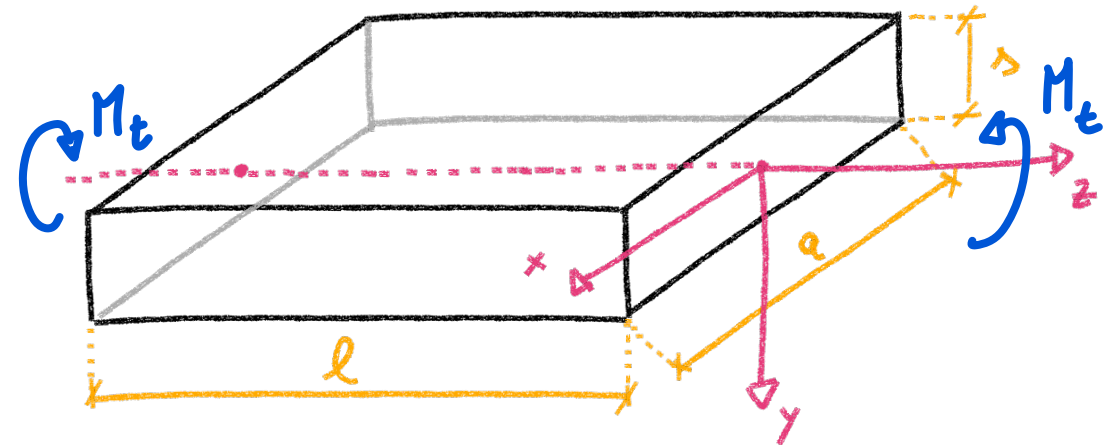
$$\left. \begin{array}{l} \tau_{\text{MAX}} = \frac{M_t}{k_1 a \delta^2} \\ I_t = k_2 a \delta^3 \end{array} \right\}$$

SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $\tau \parallel$ LATI LUNGI
- τ LINEARI SU LINEA MEDIA $\left\{ \begin{array}{l} \tau_{\text{MAX}} \\ \text{ESTREMI} \end{array} \right.$
- $\tau \parallel$ LATI MINORI "TRASCURABILI"

$$\boxed{\tau \ll a, l}$$



SOLUZIONE ESATTA (NEUMANN)

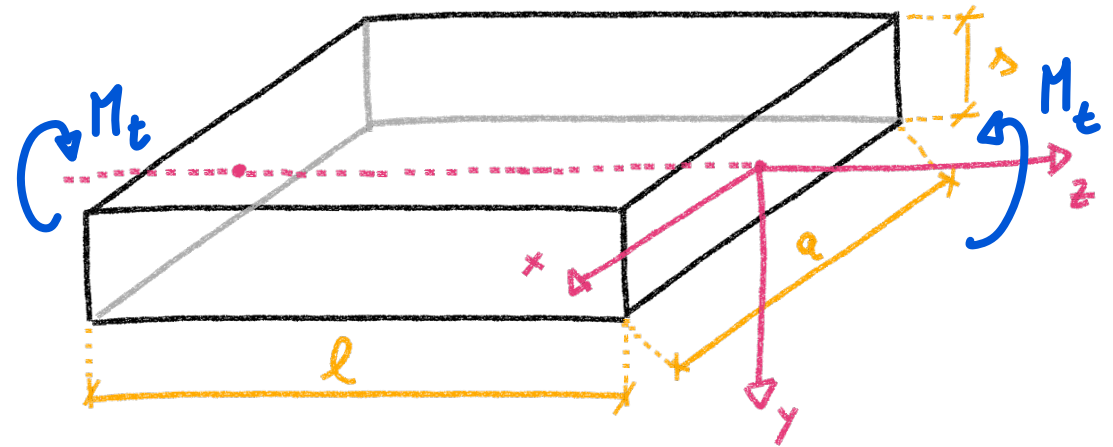
$$\left. \begin{array}{l} \tau_{\text{MAX}} = \frac{M_t}{k_1 a \tau^2} \\ I_t = k_2 a \tau^3 \end{array} \right\} \xrightarrow{\text{APPROX}} \frac{a}{\tau} \rightarrow \infty \Rightarrow k_1 = k_2 = \frac{1}{3}$$

SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\left\{ \begin{array}{l} z_{\text{MAX}} \\ \text{ESTREMI} \end{array} \right.$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$\boxed{h \ll a, l}$$



SOLUZIONE ESATTA (NEUMANN)

$$\left. \begin{array}{l} z_{\text{MAX}} = \frac{M_t}{k_1 a h^2} \\ I_t = k_2 a h^3 \end{array} \right\} \xrightarrow{\text{APPROX}} \frac{a}{h} \rightarrow \infty \Rightarrow k_1 = k_2 = \frac{1}{3}$$

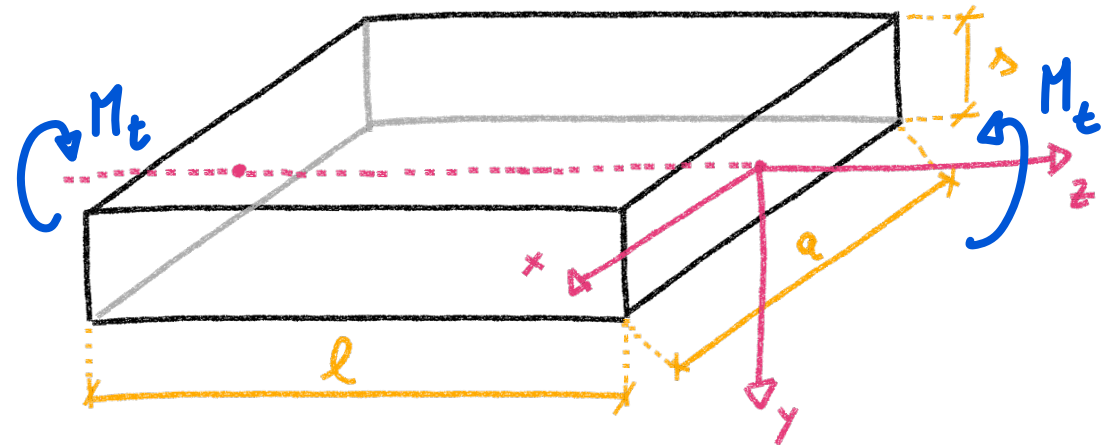
$$\begin{aligned} z_{zx} &= -2 \frac{M_t}{I_t} y \\ I_t &= \frac{1}{3} a h^3 \end{aligned}$$

SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$\boxed{b \ll a, l}$$



SOLUZIONE ESATTA (NEUMANN)

$$\left. \begin{aligned} z_{\text{MAX}} &= \frac{M_t}{k_1 a b^2} \\ I_t &= k_2 a b^3 \end{aligned} \right\} \xrightarrow{\text{APPROX}} \frac{a}{b} \rightarrow \infty \Rightarrow k_1 = k_2 = \frac{1}{3}$$

$y_{\text{MAX}} = \pm b/2$

$$\begin{aligned} z_{zx} &= -2 \frac{M_t}{I_t} y \\ I_t &= \frac{1}{3} a b^3 \end{aligned}$$

$$\rightarrow z_{\text{MAX}} = \frac{M_t}{I_t} b$$

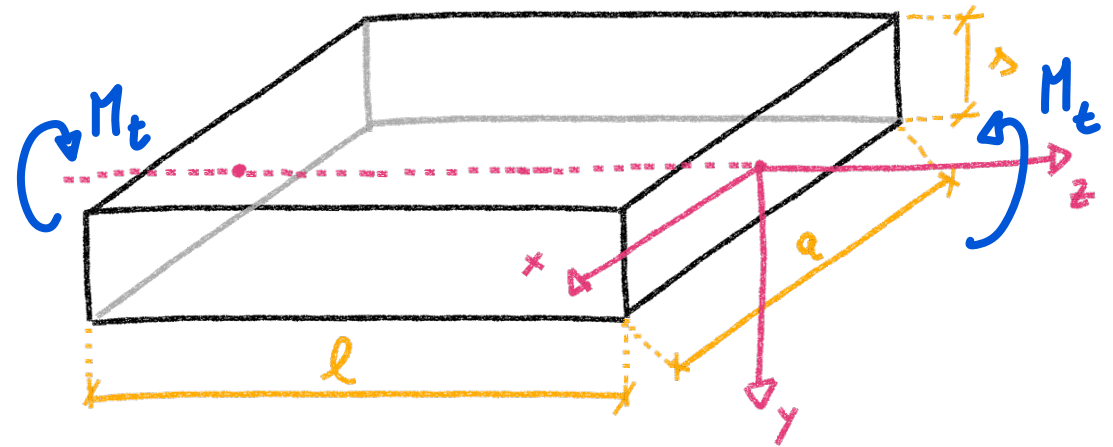
$$\Theta = \frac{M_t}{G I_t}$$

SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI" ★

$$\Delta \ll a, l$$



SOLUZIONE ESATTA (NEUMANN)

$$\left. \begin{aligned} z_{\text{MAX}} &= \frac{M_t}{k_1 a \Delta^2} \\ I_t &= k_2 a \Delta^3 \end{aligned} \right\} \xrightarrow{\text{APPROX}} \frac{a}{\Delta} \rightarrow \infty \Rightarrow k_1 = k_2 = \frac{1}{3}$$

$y_{\text{MAX}} = \pm \Delta/2$

$$\begin{aligned} z_{zx} &= -2 \frac{M_t}{I_t} y \\ I_t &= \frac{1}{3} a \Delta^3 \end{aligned}$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} \Delta$$

$$\Theta = \frac{M_t}{G I_t}$$

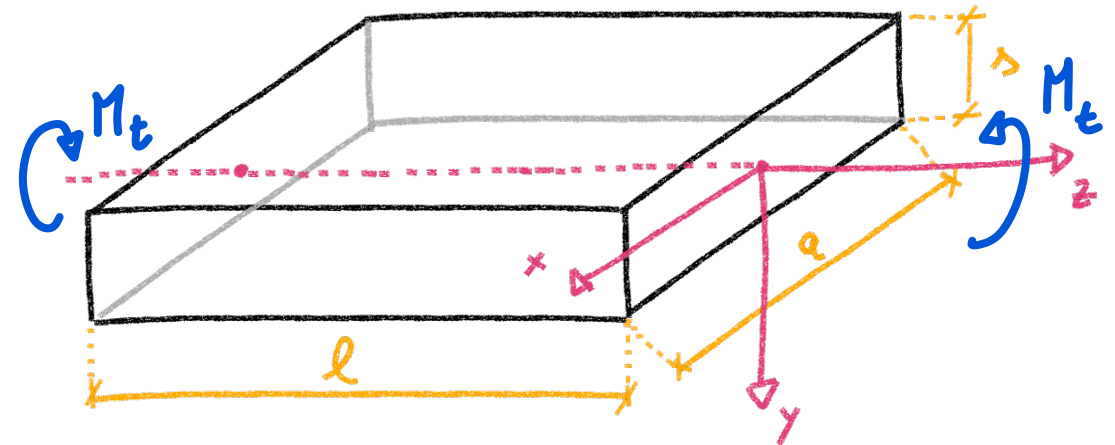
$$\star z_{zy} = 2 \frac{M_t}{I_t} x$$

SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

$$\boxed{b \ll a, l}$$

- $\tau \parallel$ LATI LUNGI
- τ LINEARI SU LINEA MEDIA $\begin{cases} \tau_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $\tau \parallel$ LATI MINORI "TRASCURABILI" ★



SOLUZIONE ESATTA (NEUMANN)

$$\left. \begin{aligned} \tau_{\text{MAX}} &= \frac{M_t}{k_1 a b^2} \\ I_t &= k_2 a b^3 \end{aligned} \right\} \xrightarrow{\text{APPROX}} \frac{a}{b} \rightarrow \infty \Rightarrow k_1 = k_2 = \frac{1}{3}$$

$y_{\text{MAX}} = \pm b/2$

$$\begin{aligned} \tau_{zx} &= -2 \frac{M_t}{I_t} y \\ I_t &= \frac{1}{3} a b^3 \end{aligned}$$

$$0 < y < \frac{b}{2}$$

$$\tau_{\text{MAX}} = \frac{M_t}{I_t} b$$

$$\Theta = \frac{M_t}{G I_t}$$

$$\star \tau_{zy} = 2 \frac{M_t}{I_t} x$$

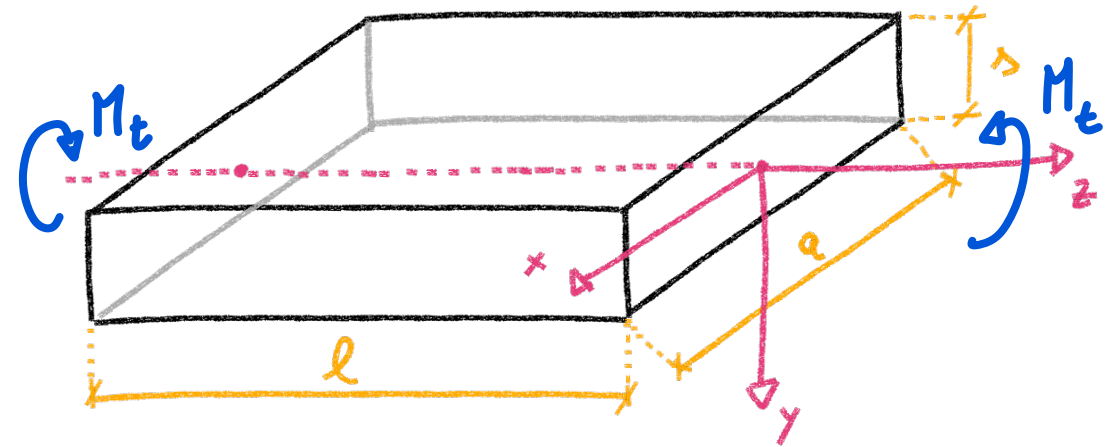
→ DIAGRAM.

SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$\lambda \ll a, l$$



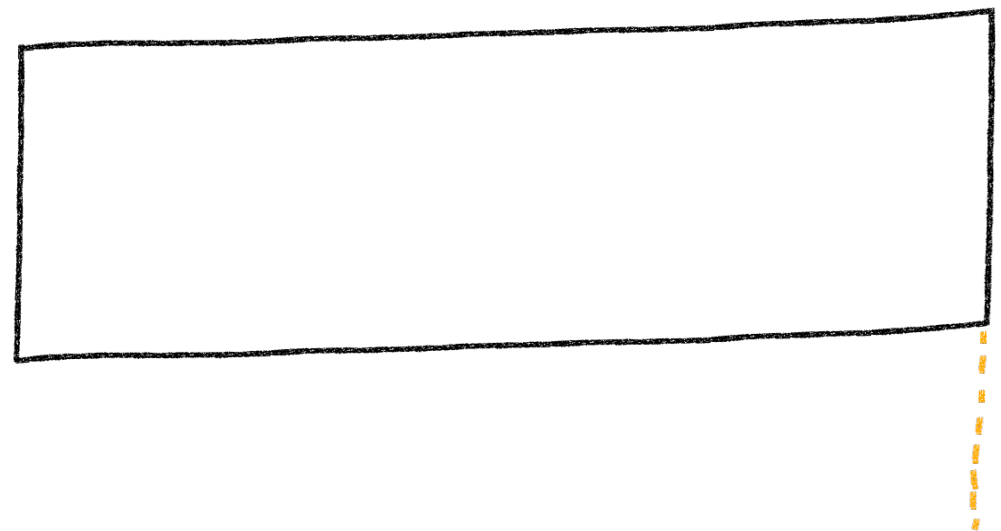
$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a \lambda^3$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} \lambda$$

$$y_{\text{MAX}} = \pm \lambda/2$$

zy

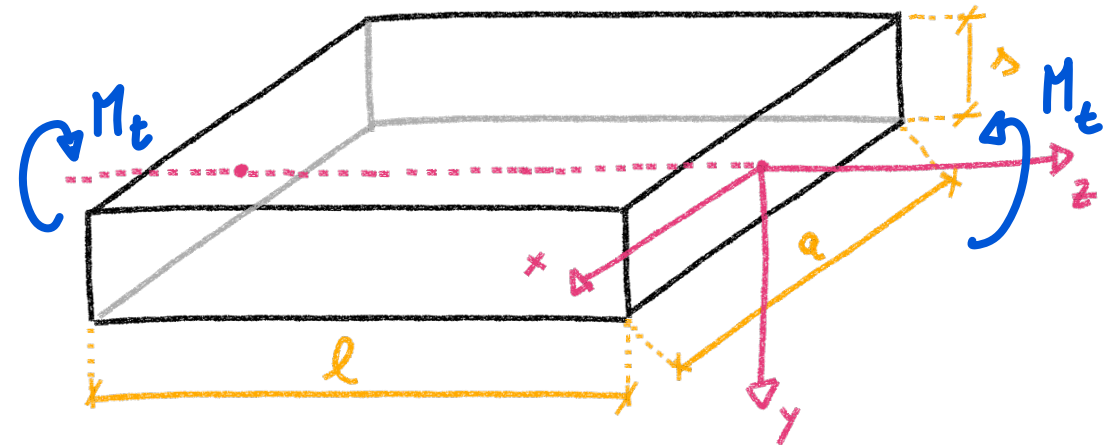


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$

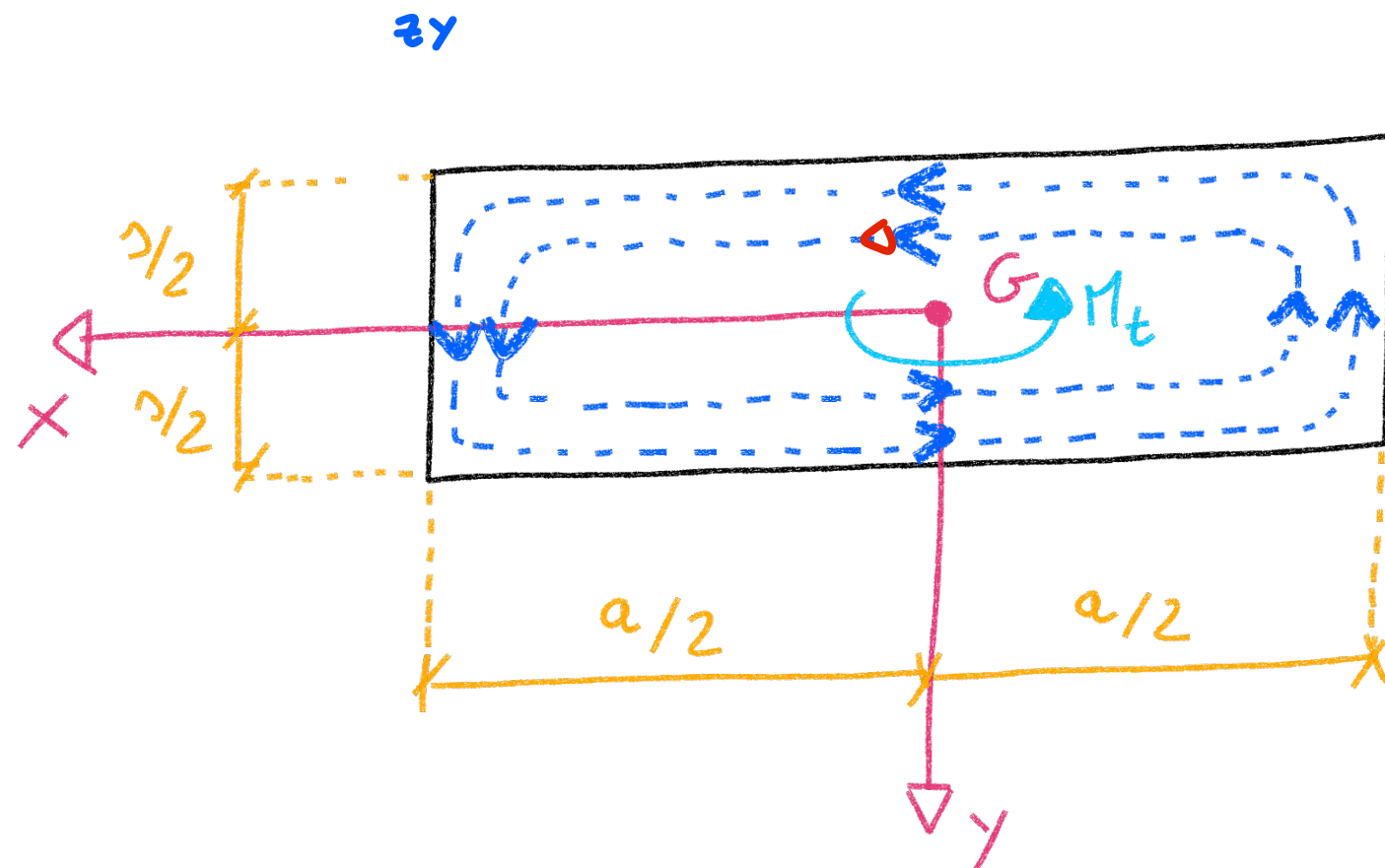


$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} s$$

$$y_{\text{MAX}} = \pm s/2$$

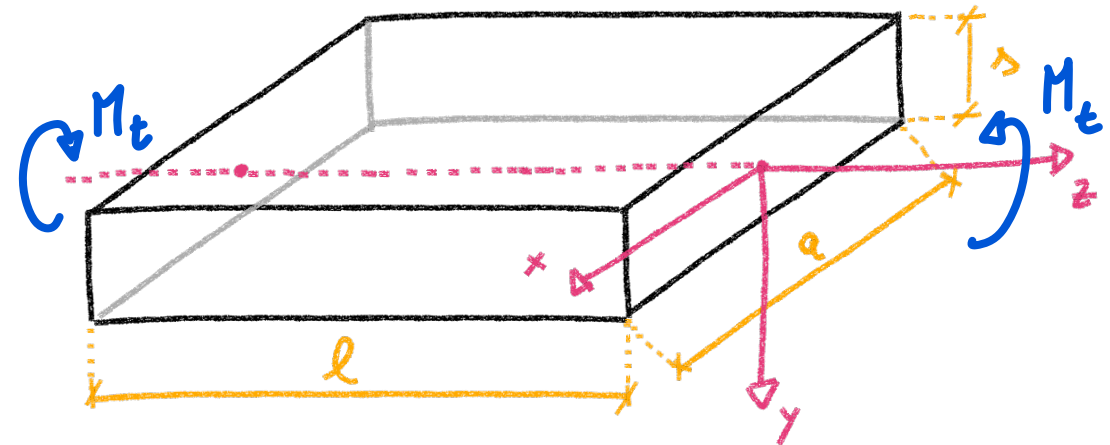


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$



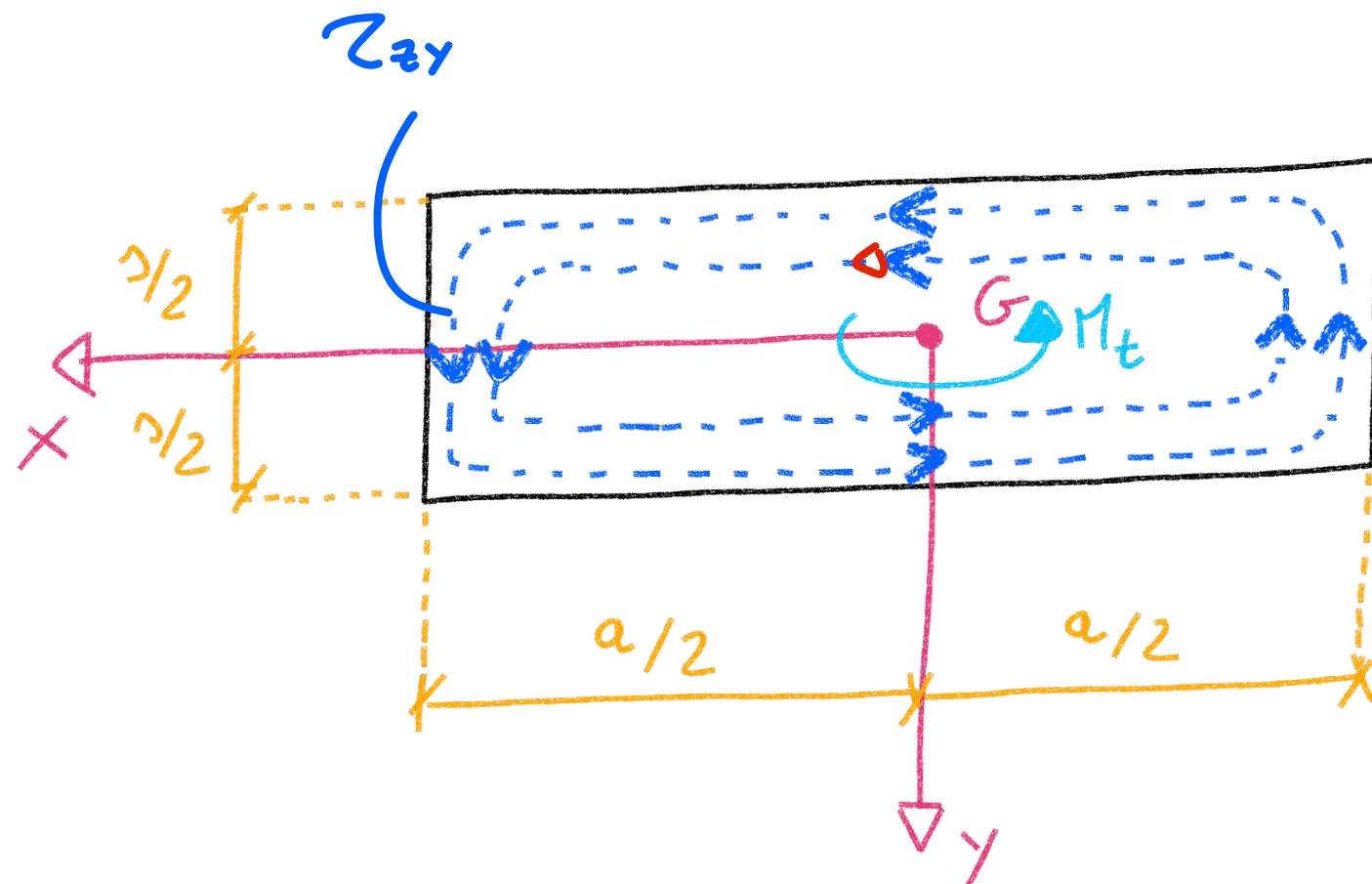
$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} s$$

$$y_{\text{MAX}} = \pm s/2$$

$$\textcircled{*} z_{zy} = 2 \frac{M_t}{I_t} x$$

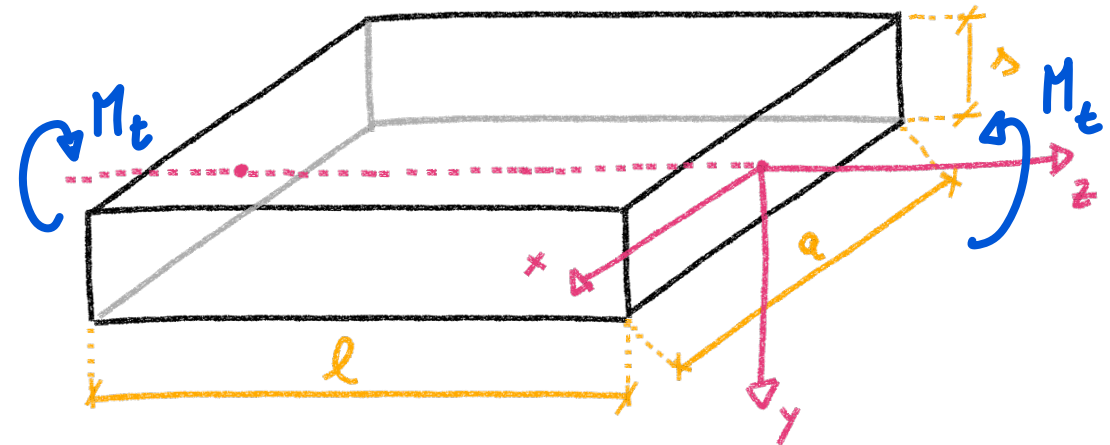


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$



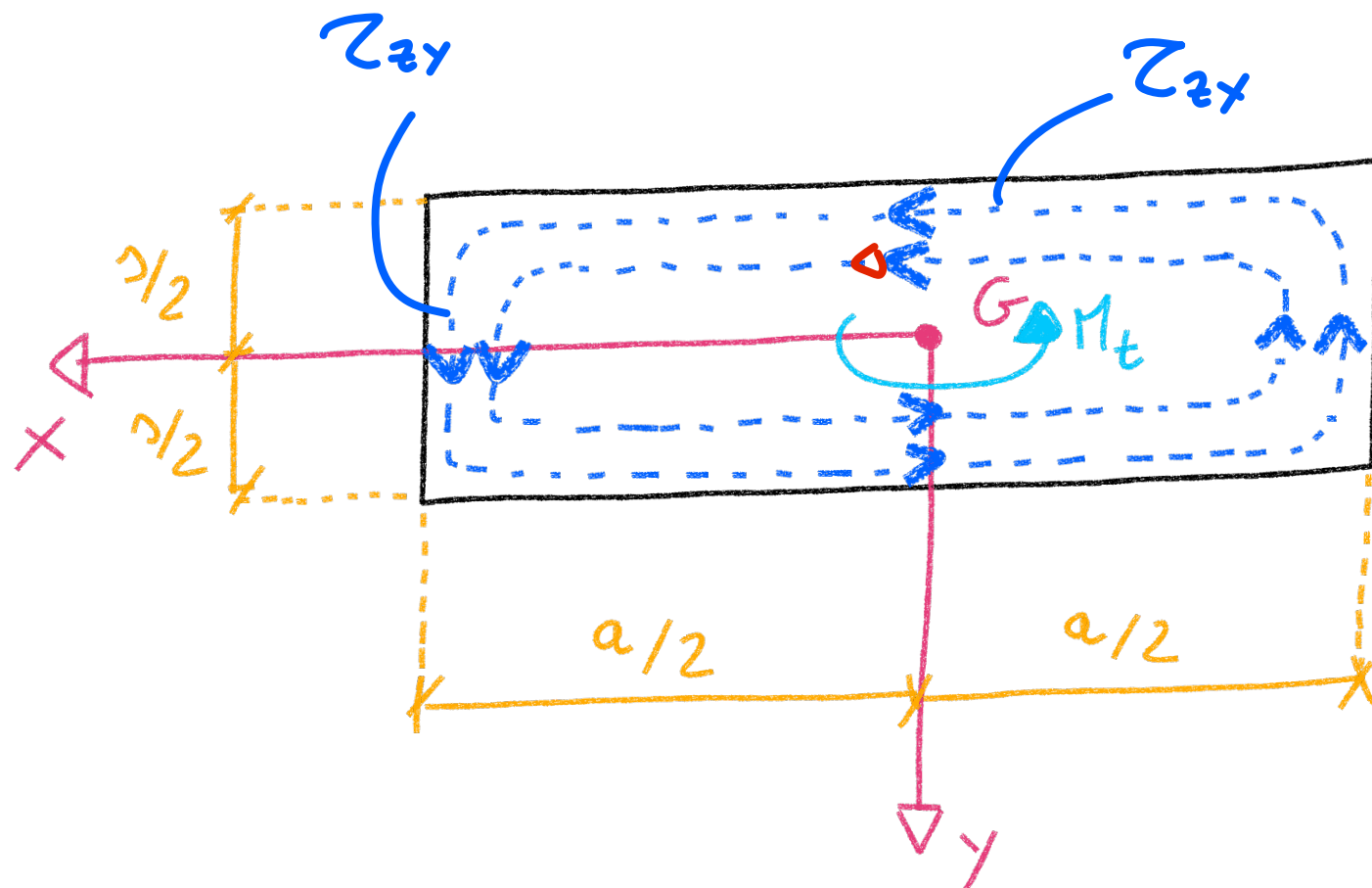
$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} s$$

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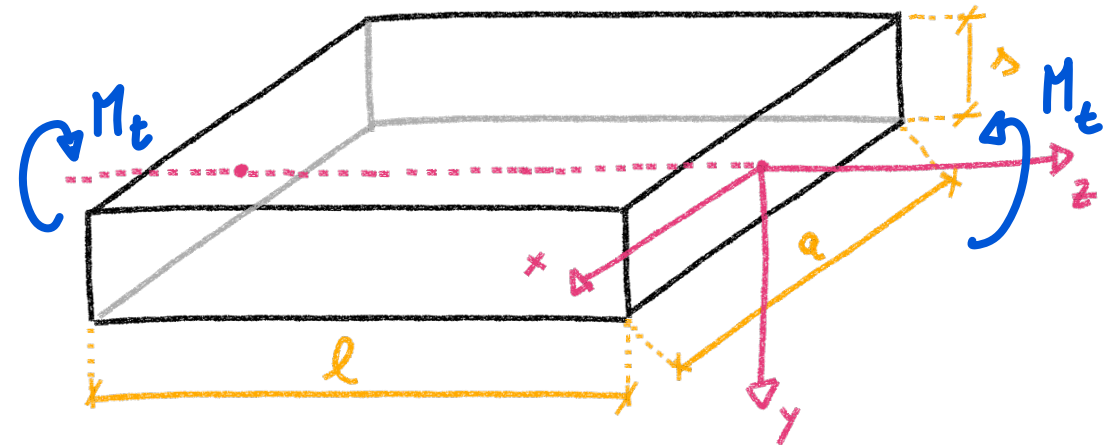


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$



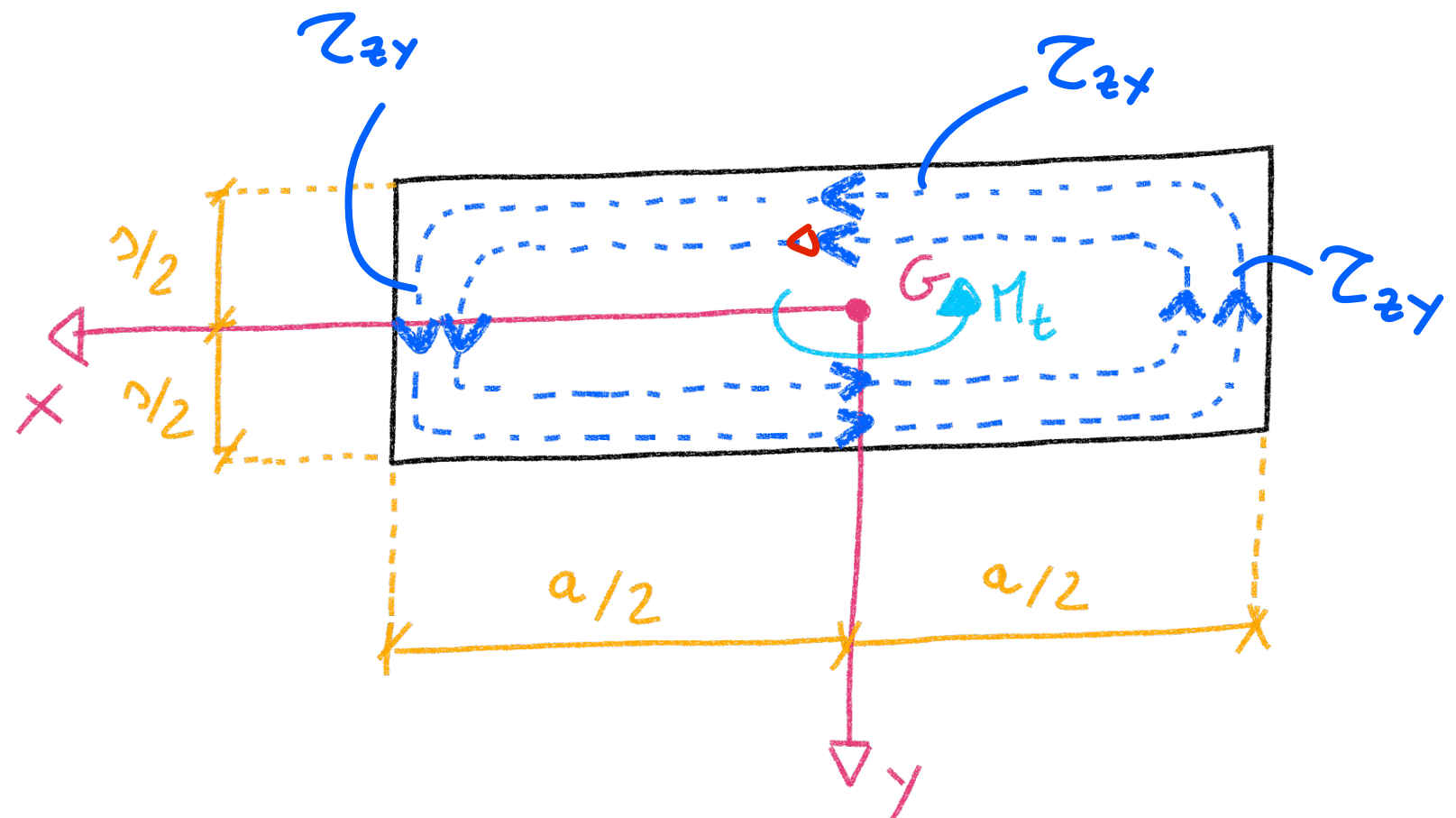
$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} s$$

$$y_{\text{MAX}} = \pm s/2$$

$$\textcircled{*} z_{zy} = 2 \frac{M_t}{I_t} x$$

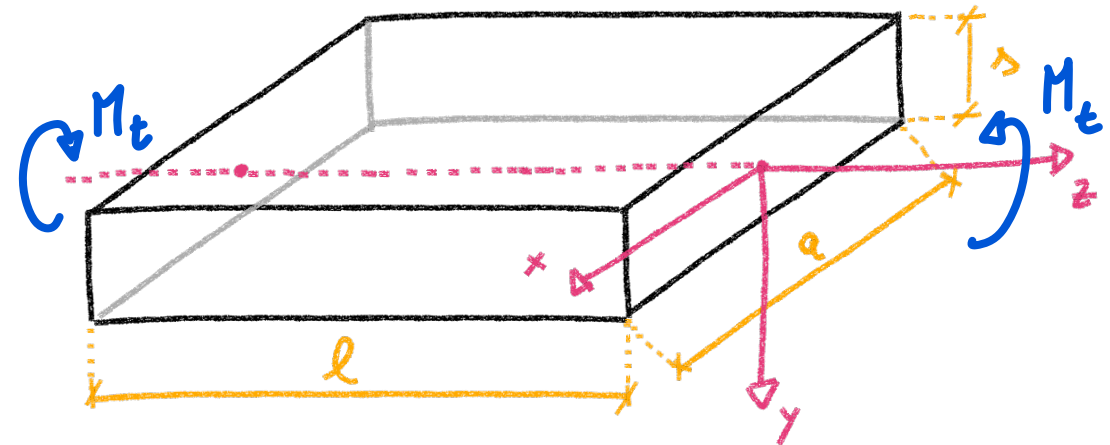


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$



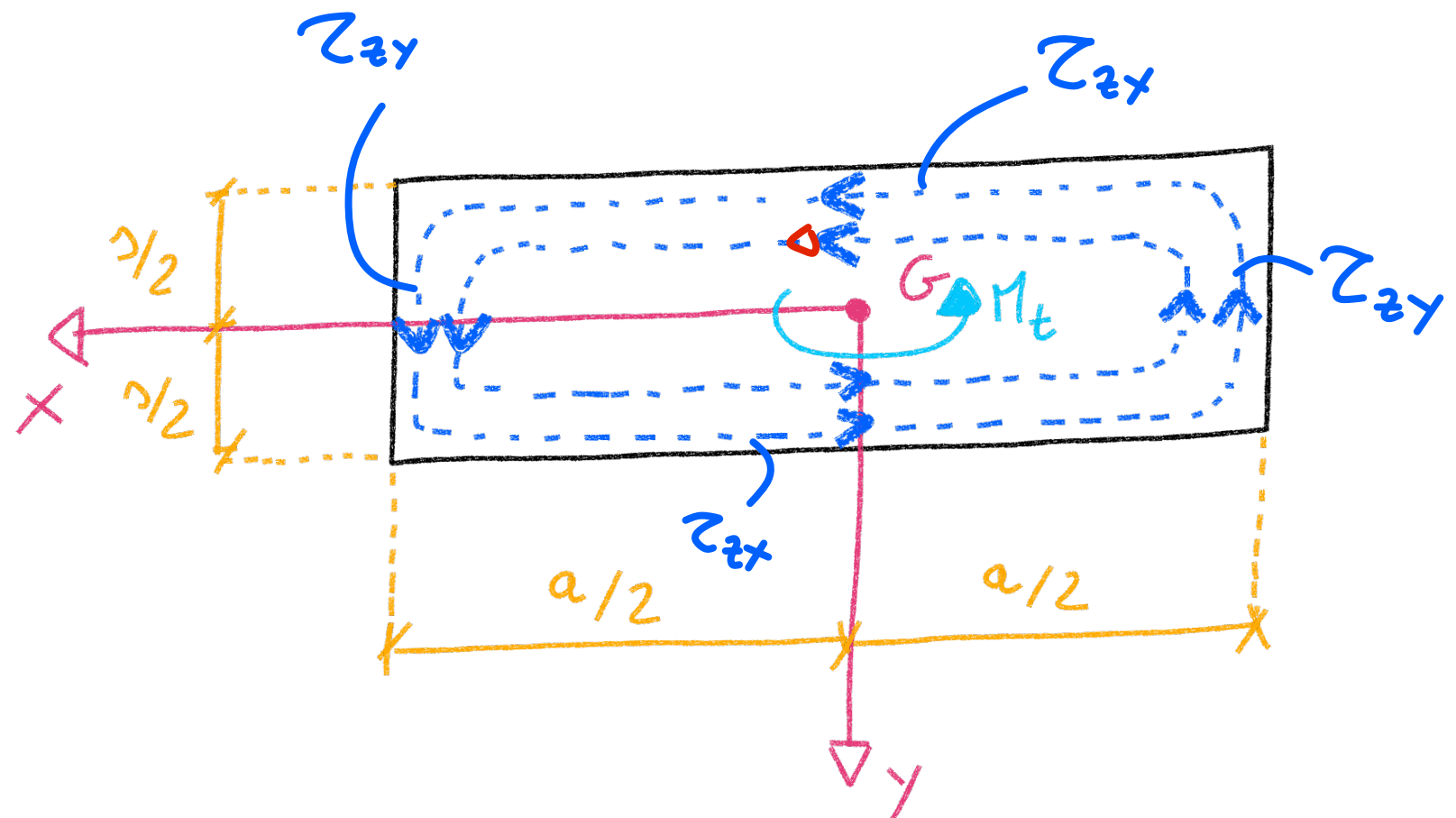
$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

$$z_{\text{MAX}} = \frac{M_t}{I_t} s$$

$$y_{\text{MAX}} = \pm s/2$$

$$\textcircled{*} z_{zy} = 2 \frac{M_t}{I_t} x$$

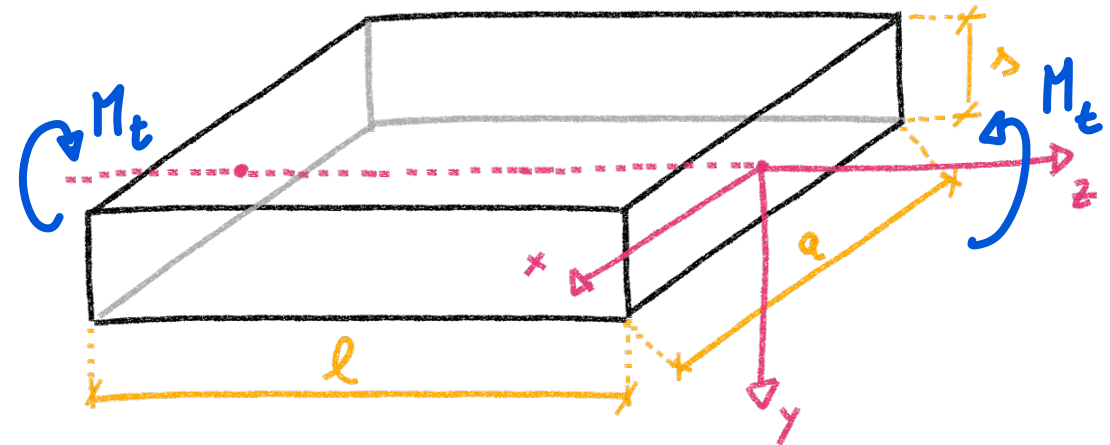


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

- $z \parallel$ LATI LUNGI
- z LINEARI SU LINEA MEDIA $\begin{cases} z_{MAX} \\ ESTREMI \end{cases}$
- $z \parallel$ LATI MINORI "TRASCURABILI"

$$s \ll a, l$$



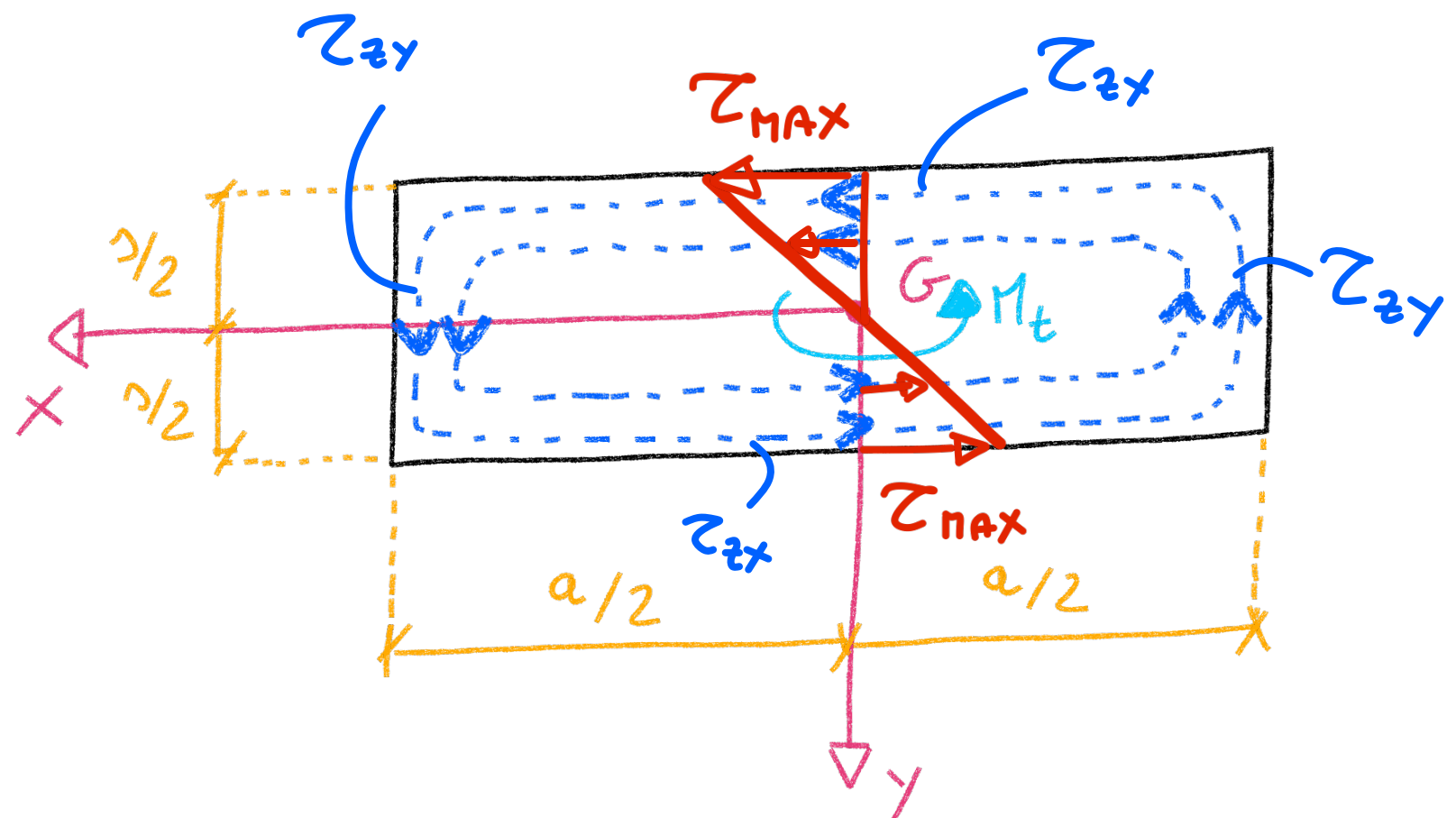
$$z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

$$z_{MAX} = \frac{M_t}{I_t} s$$

$$y_{MAX} = \pm s/2$$

$$\textcircled{*} z_{zy} = 2 \frac{M_t}{I_t} x$$

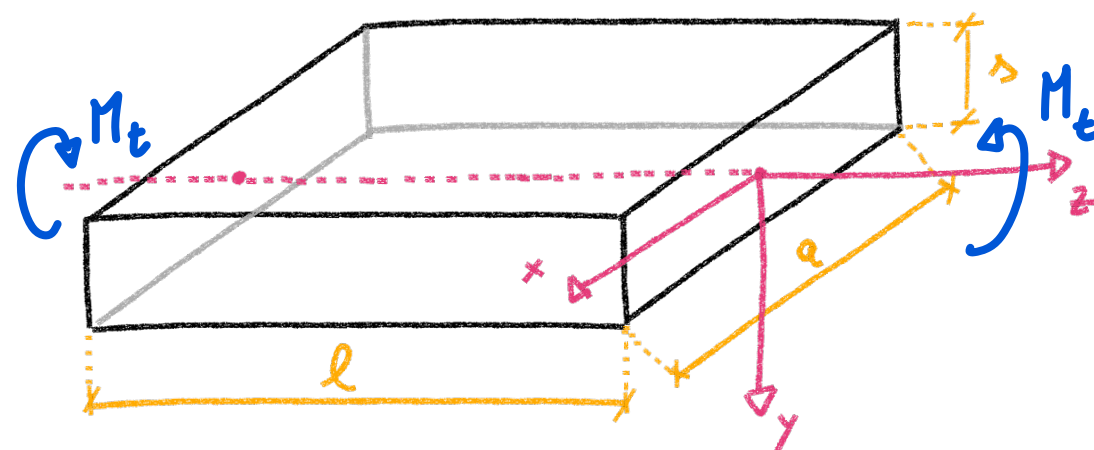


SEZIONE RETTANGOLARE SOTTILE

ANALOGIA IDRODINAMICA:

$$s \ll a, l$$

- $\tau \parallel$ LATI LUNGI
- τ LINEARI SU LINEA MEDIA $\begin{cases} \tau_{\text{MAX}} \\ \text{ESTREMI} \end{cases}$
- $\tau \parallel$ LATI MINORI "TRASCURABILI" ★



→ ???
ooo

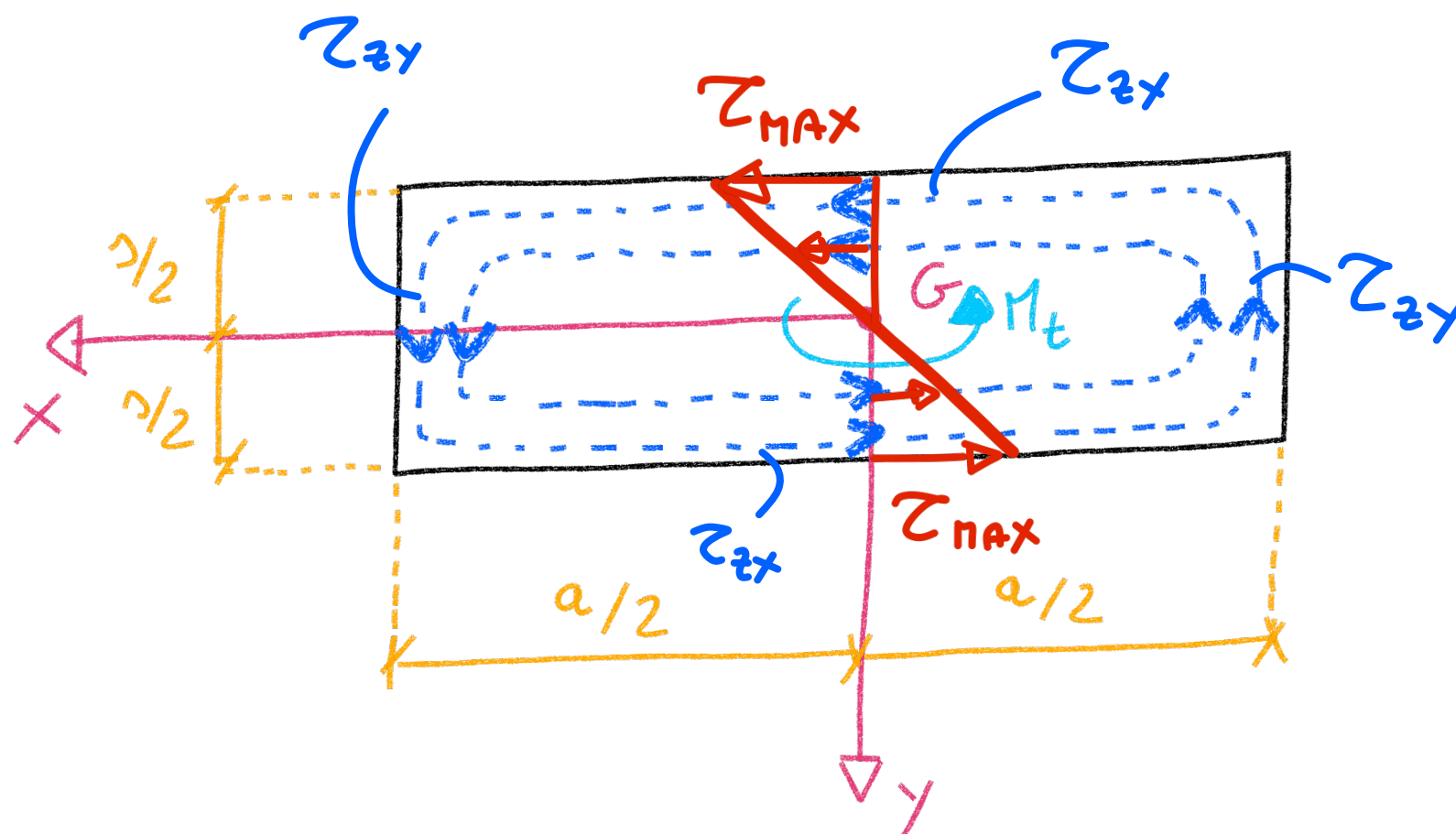
$$\tau_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a s^3$$

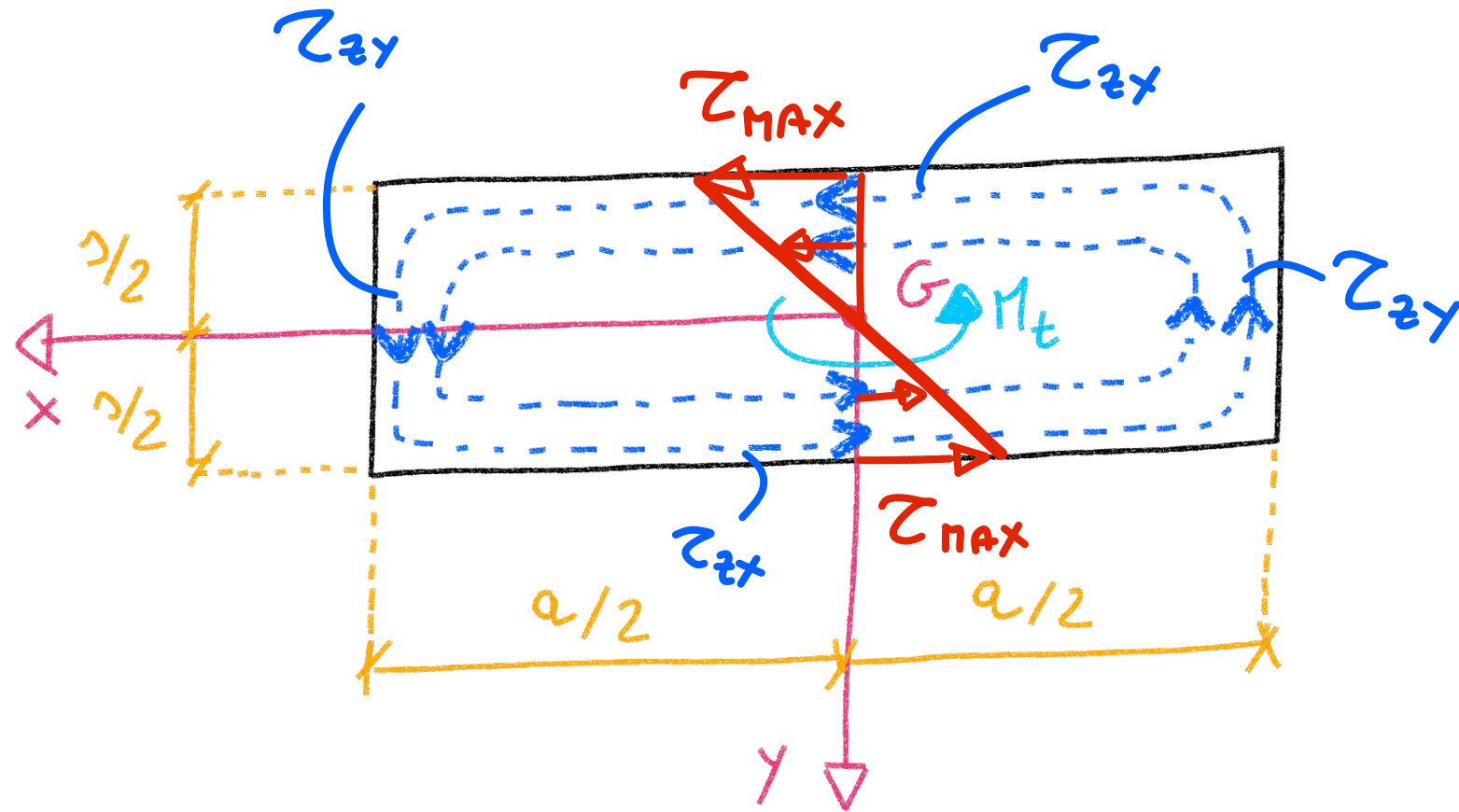
$$\tau_{\text{MAX}} = \frac{M_t}{I_t} s$$

$$y_{\text{MAX}} = \pm s/2$$

$$\textcircled{*} \tau_{zy} = 2 \frac{M_t}{I_t} x \quad \star$$

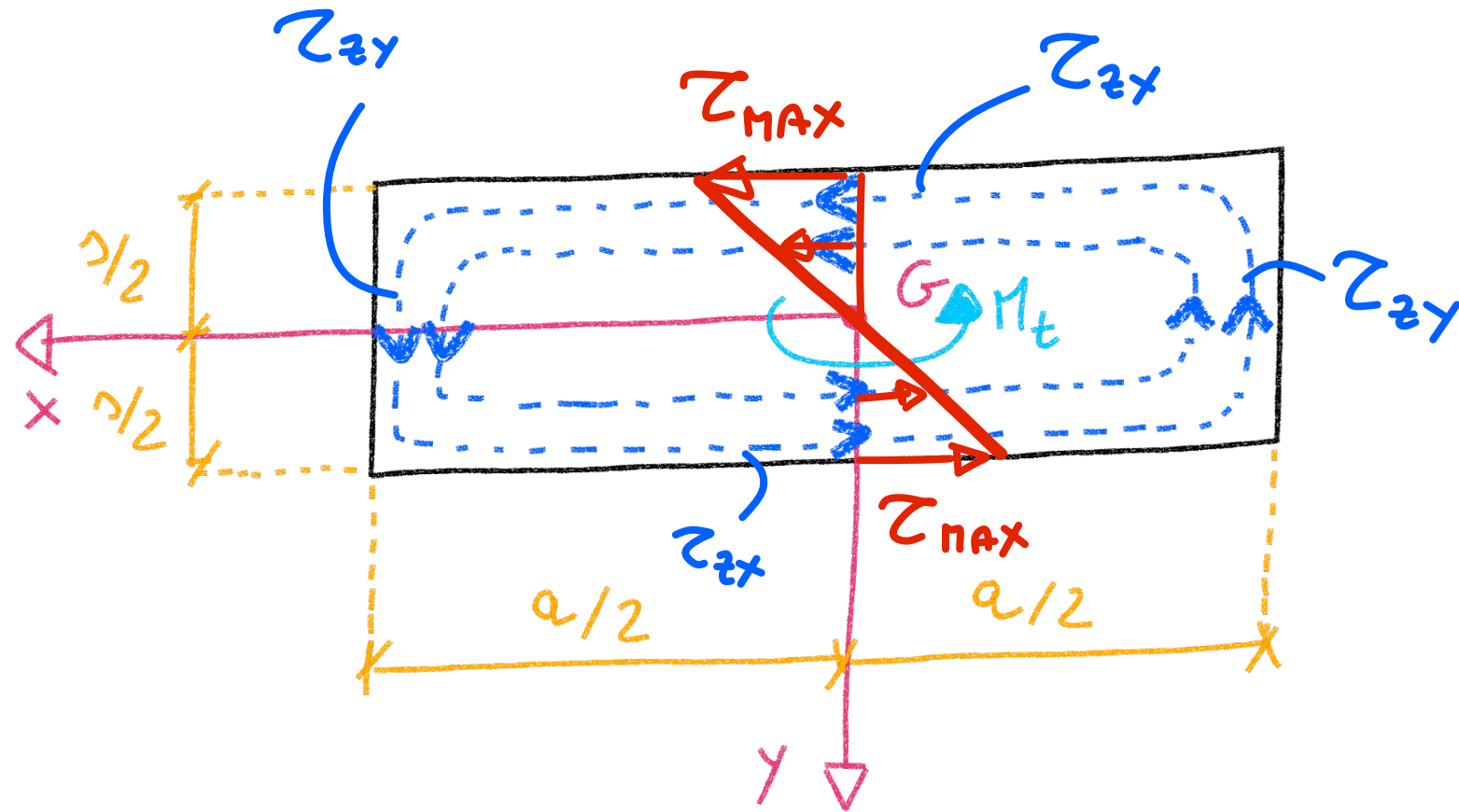


SEZIONE RETTANGOLARE SOTTILE



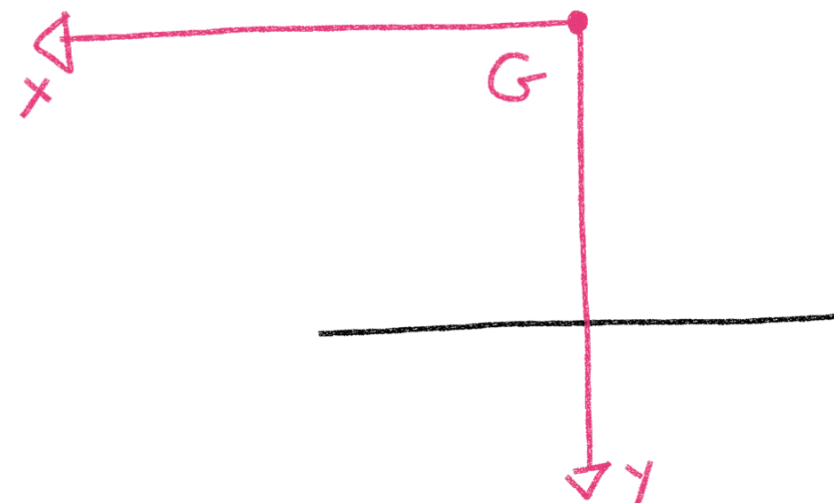
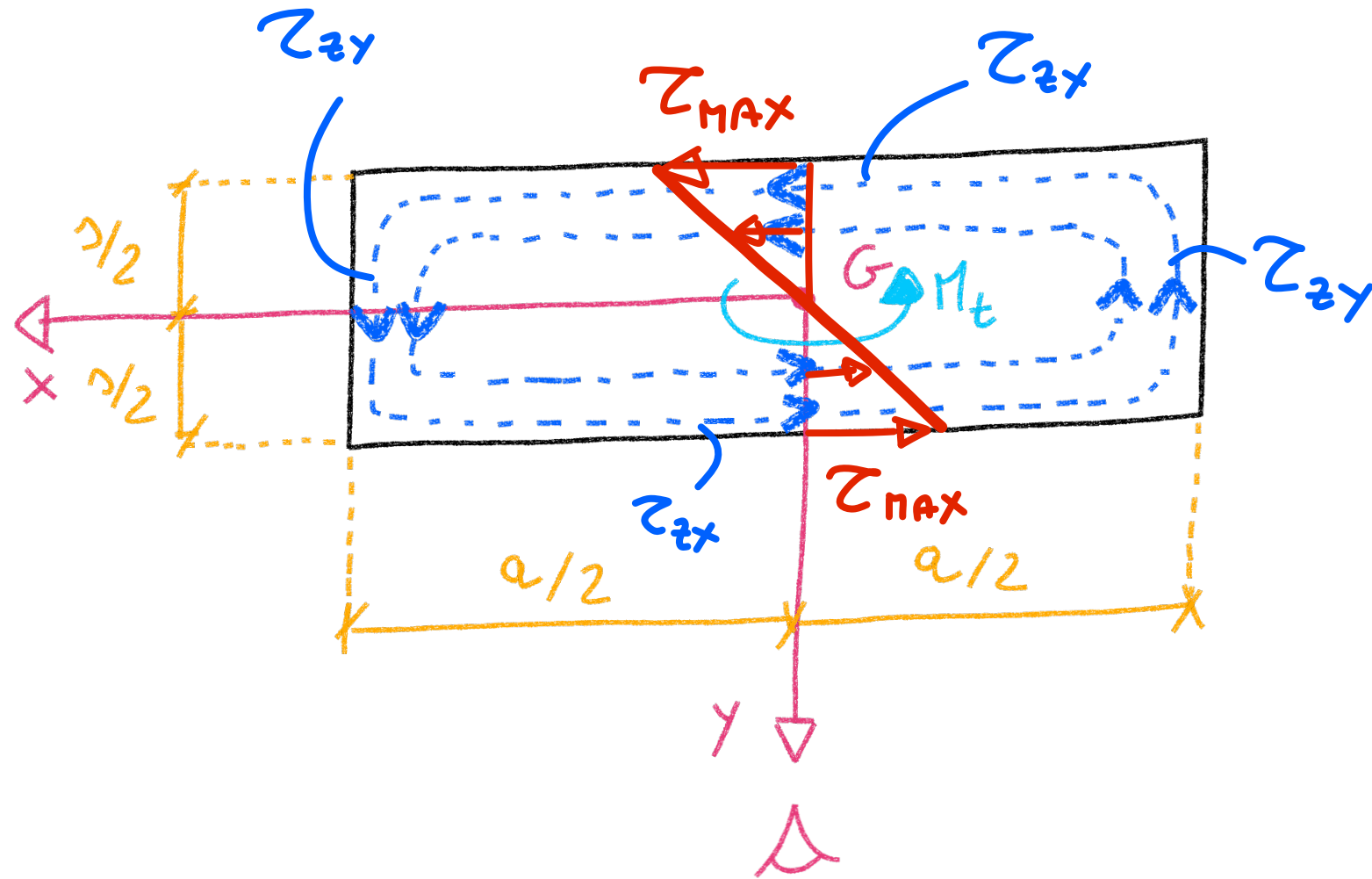
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}



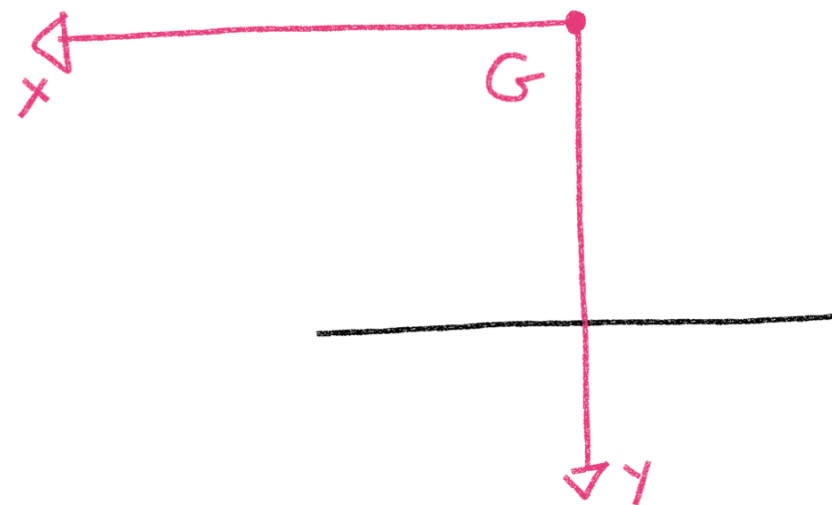
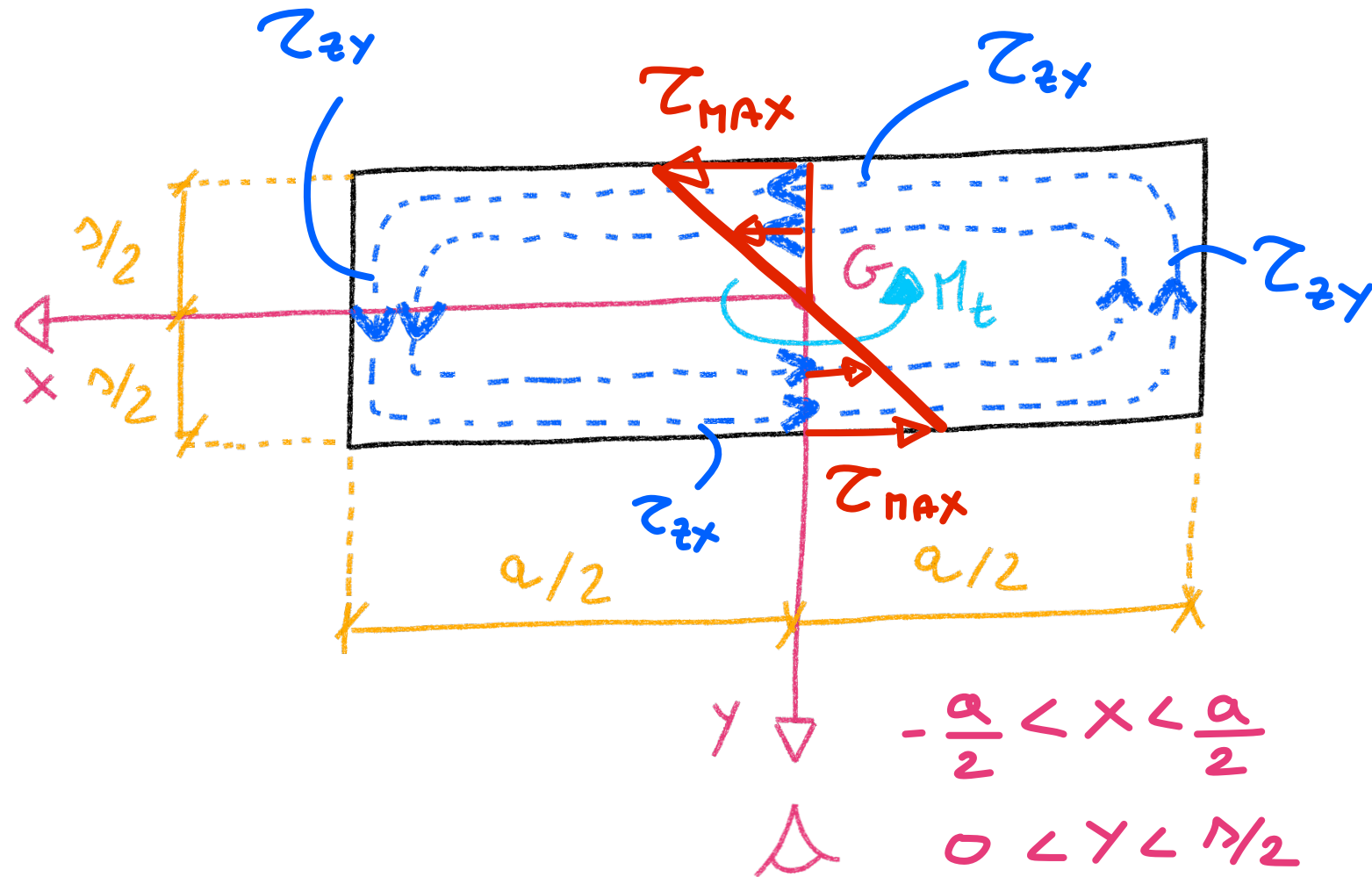
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}



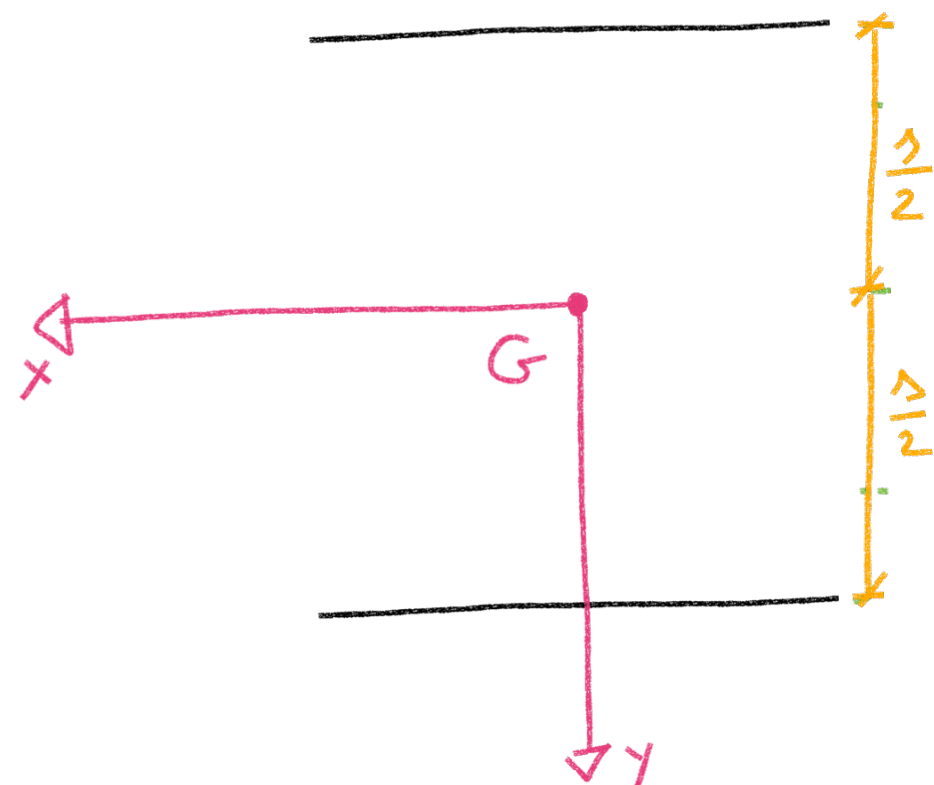
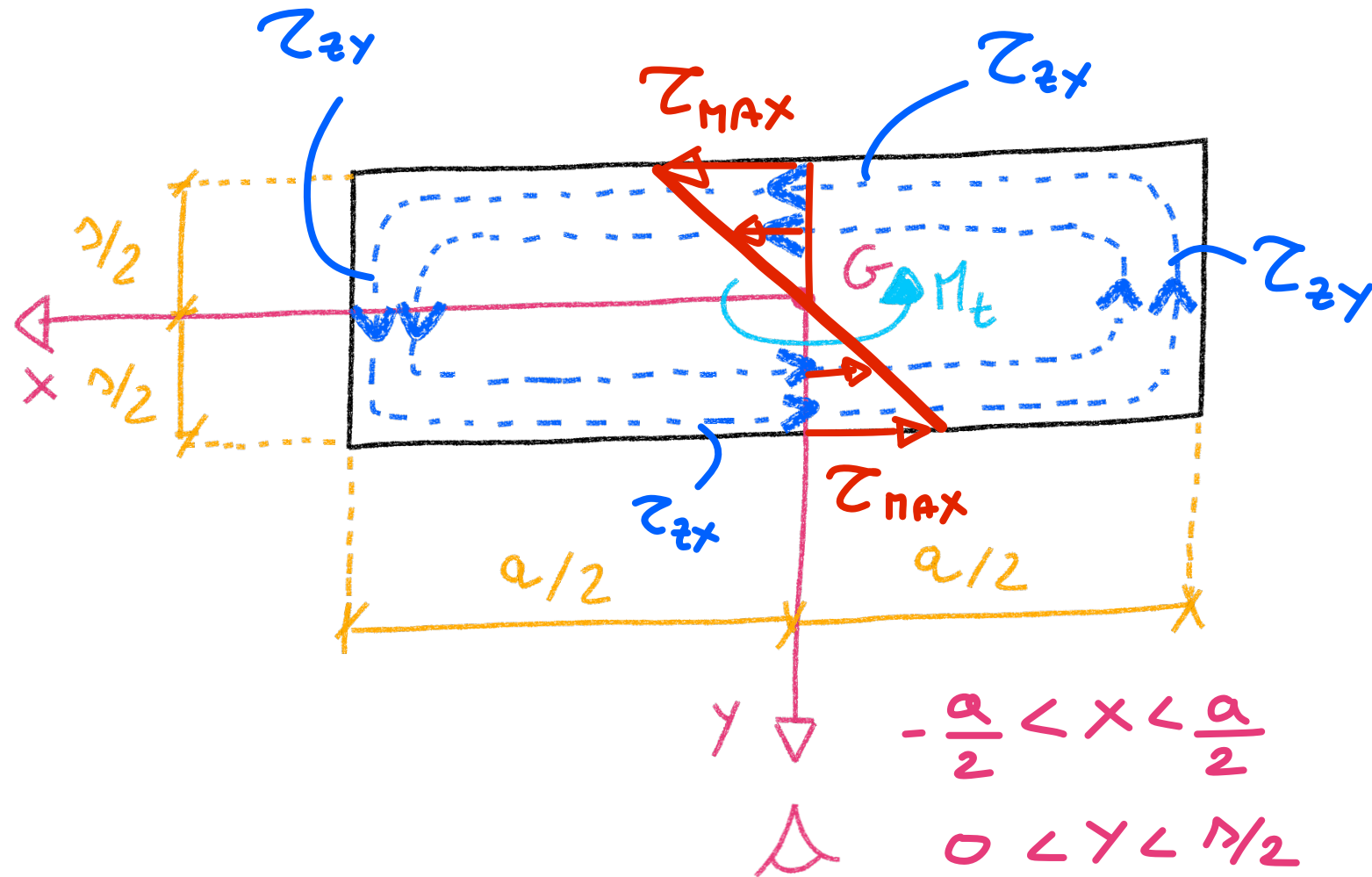
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}



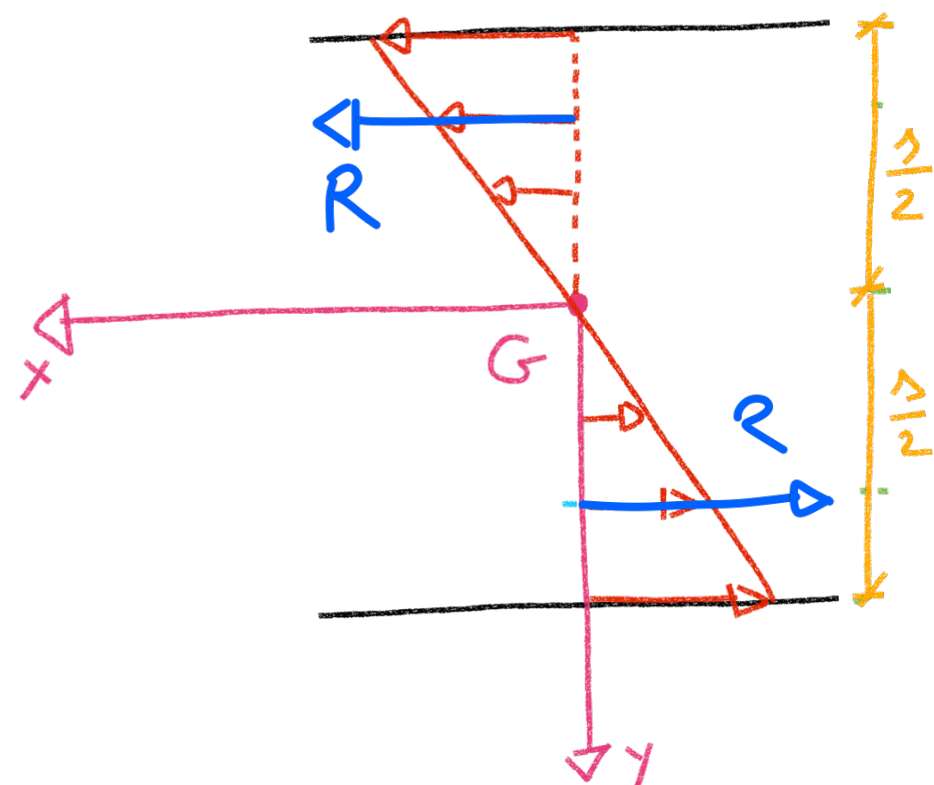
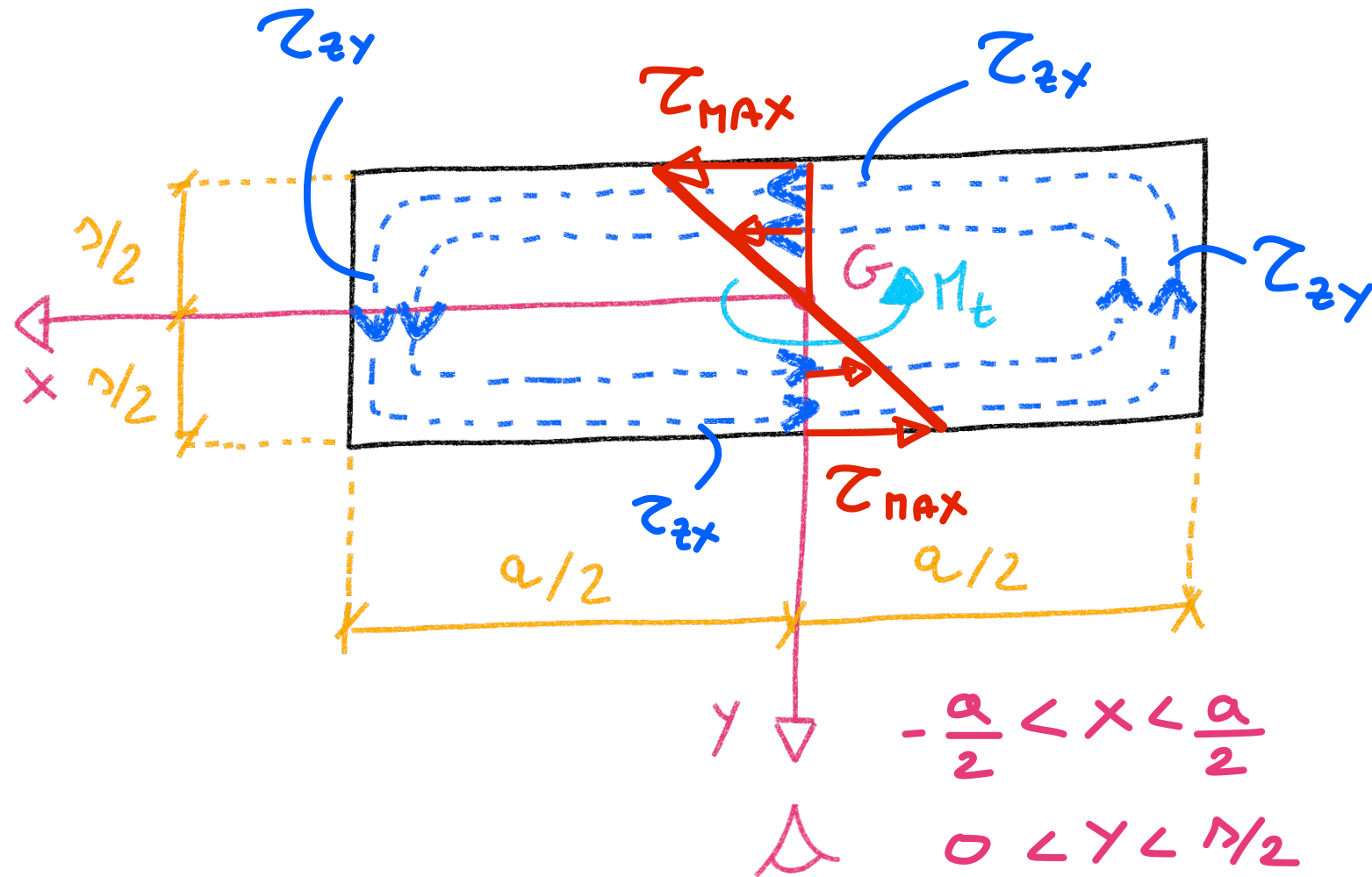
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}



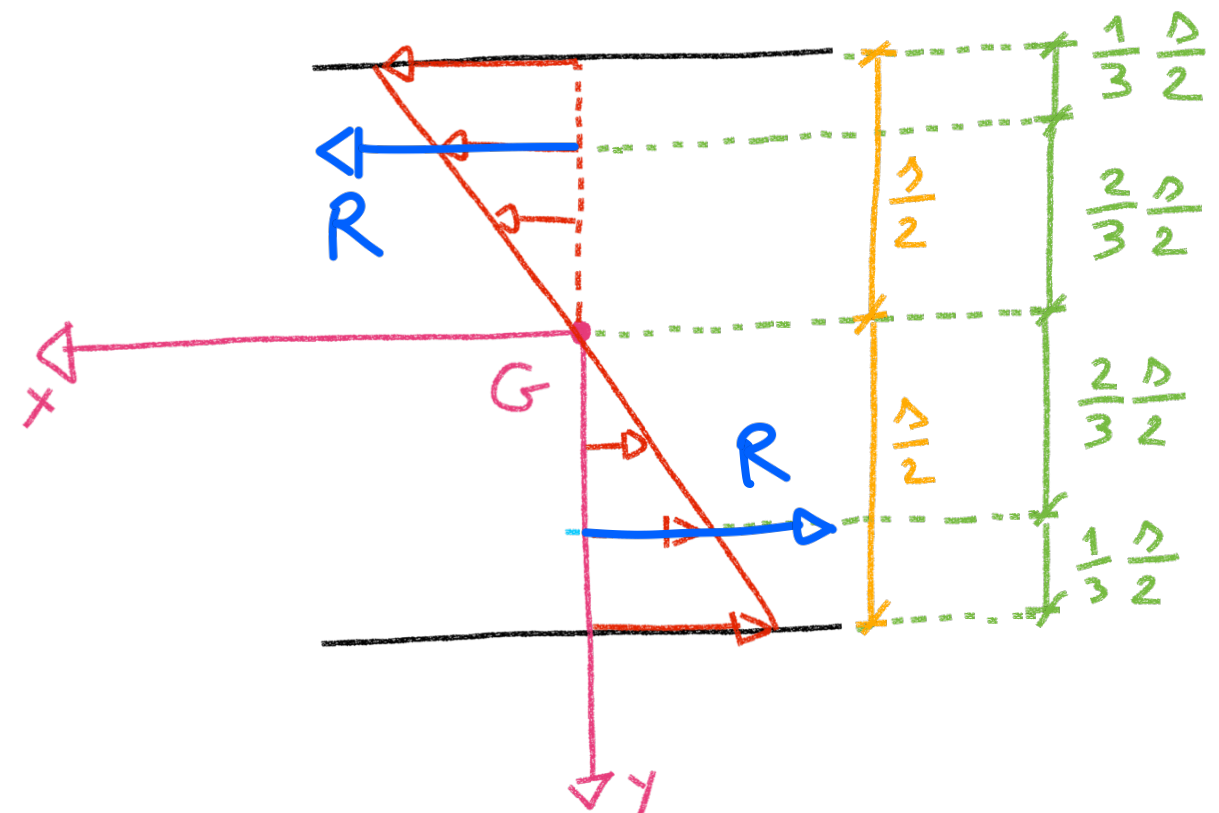
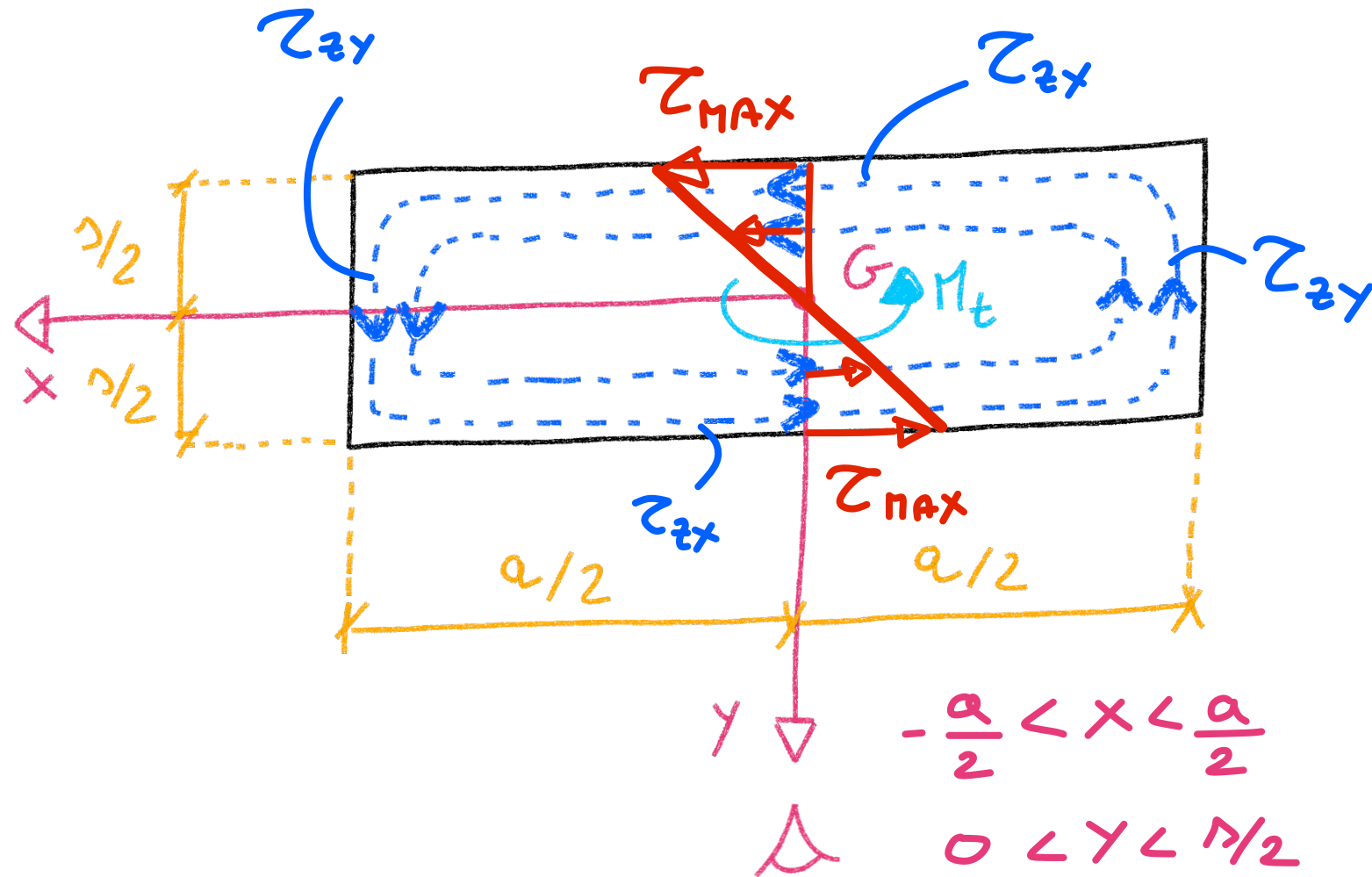
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}



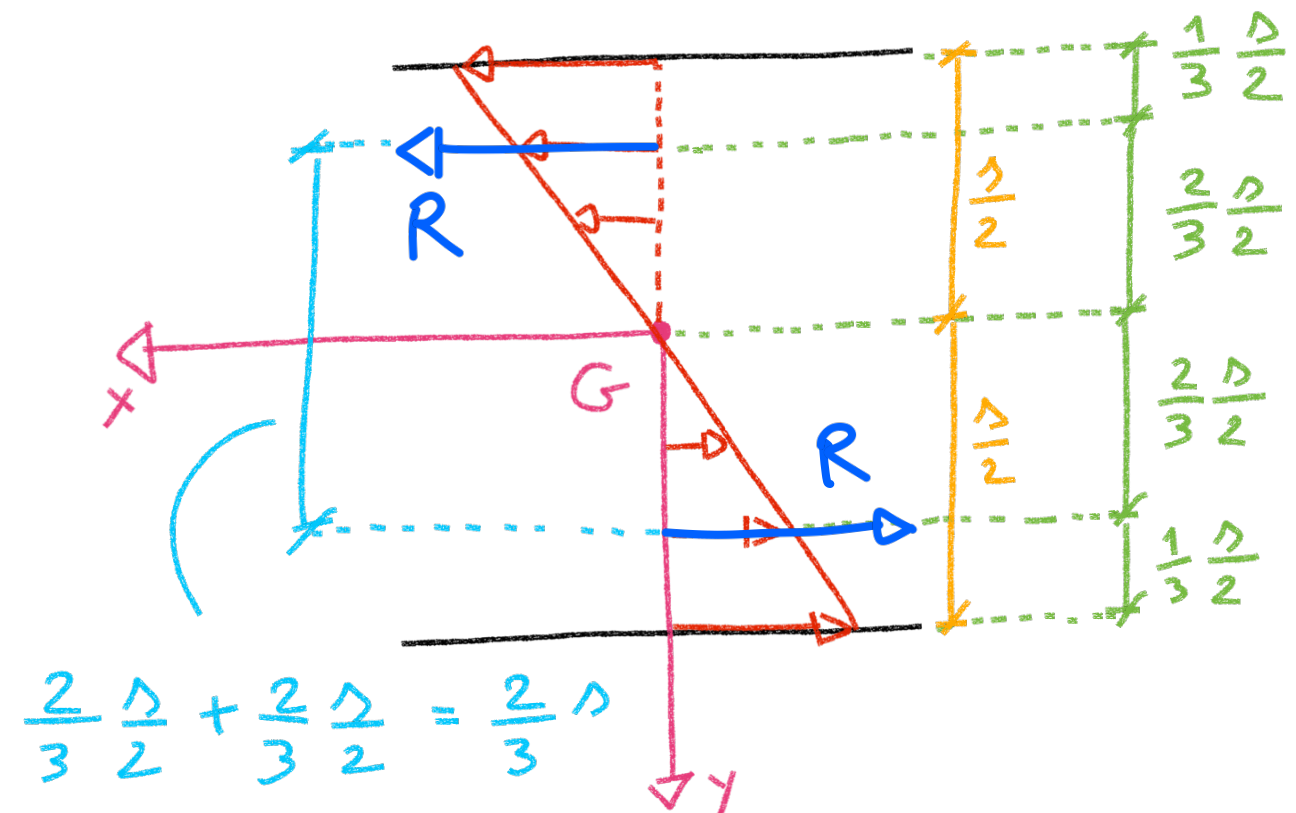
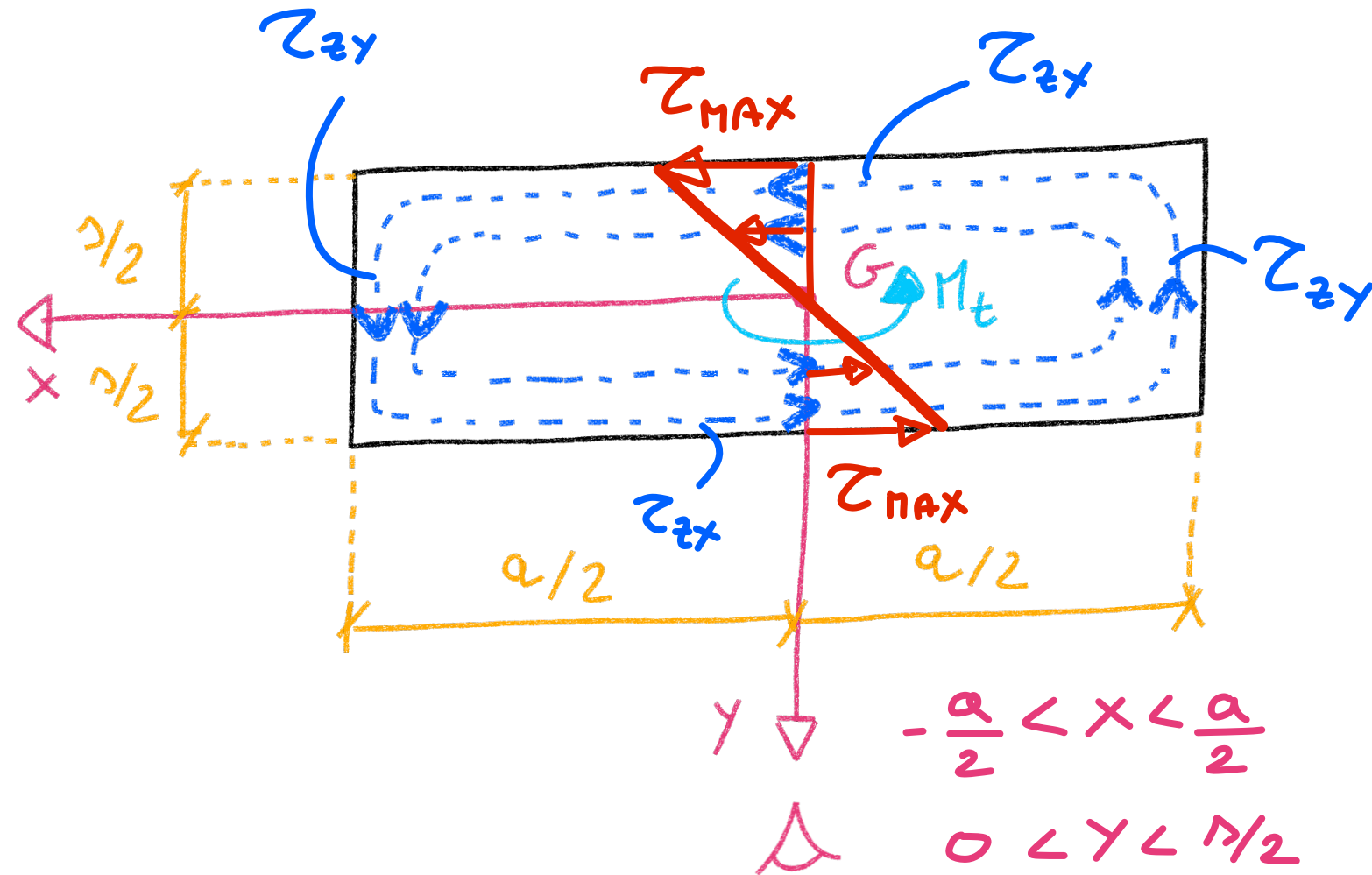
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC. INDOTTO DA Z_{zx}



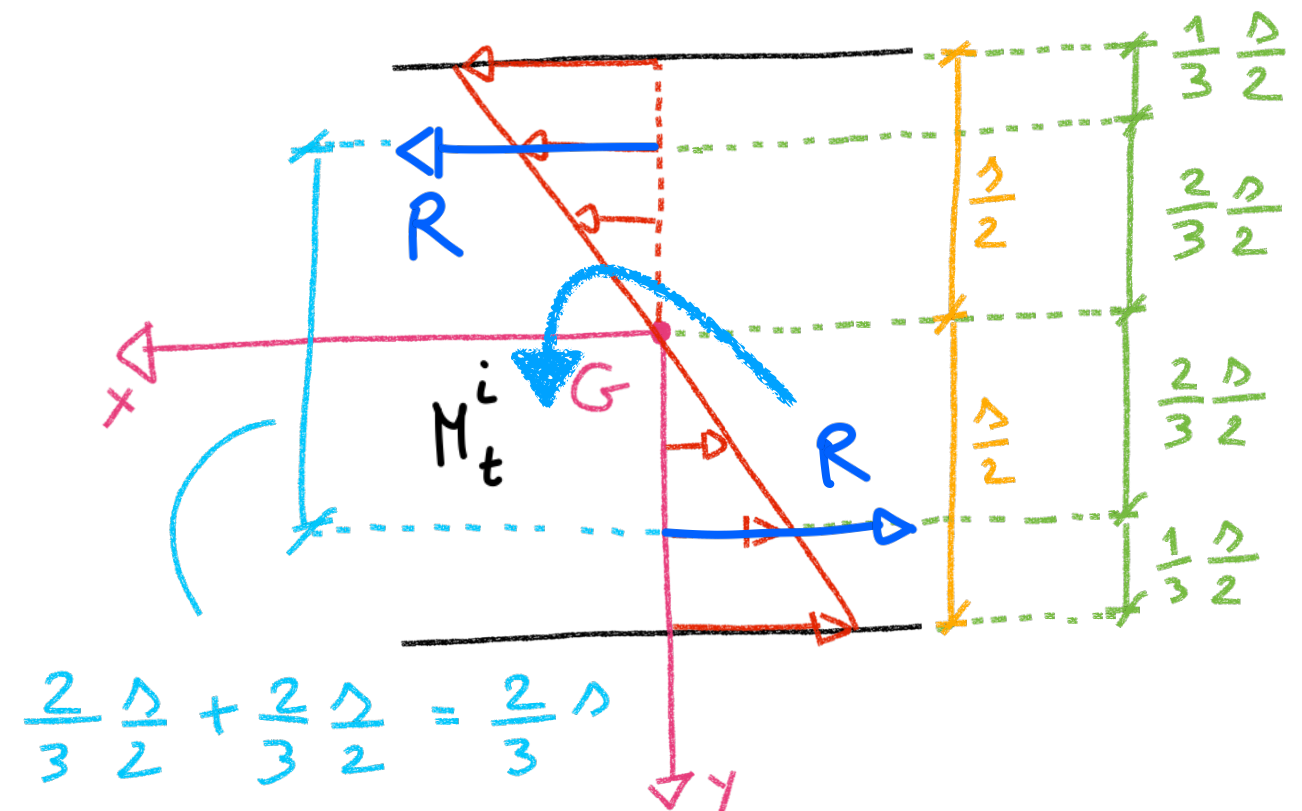
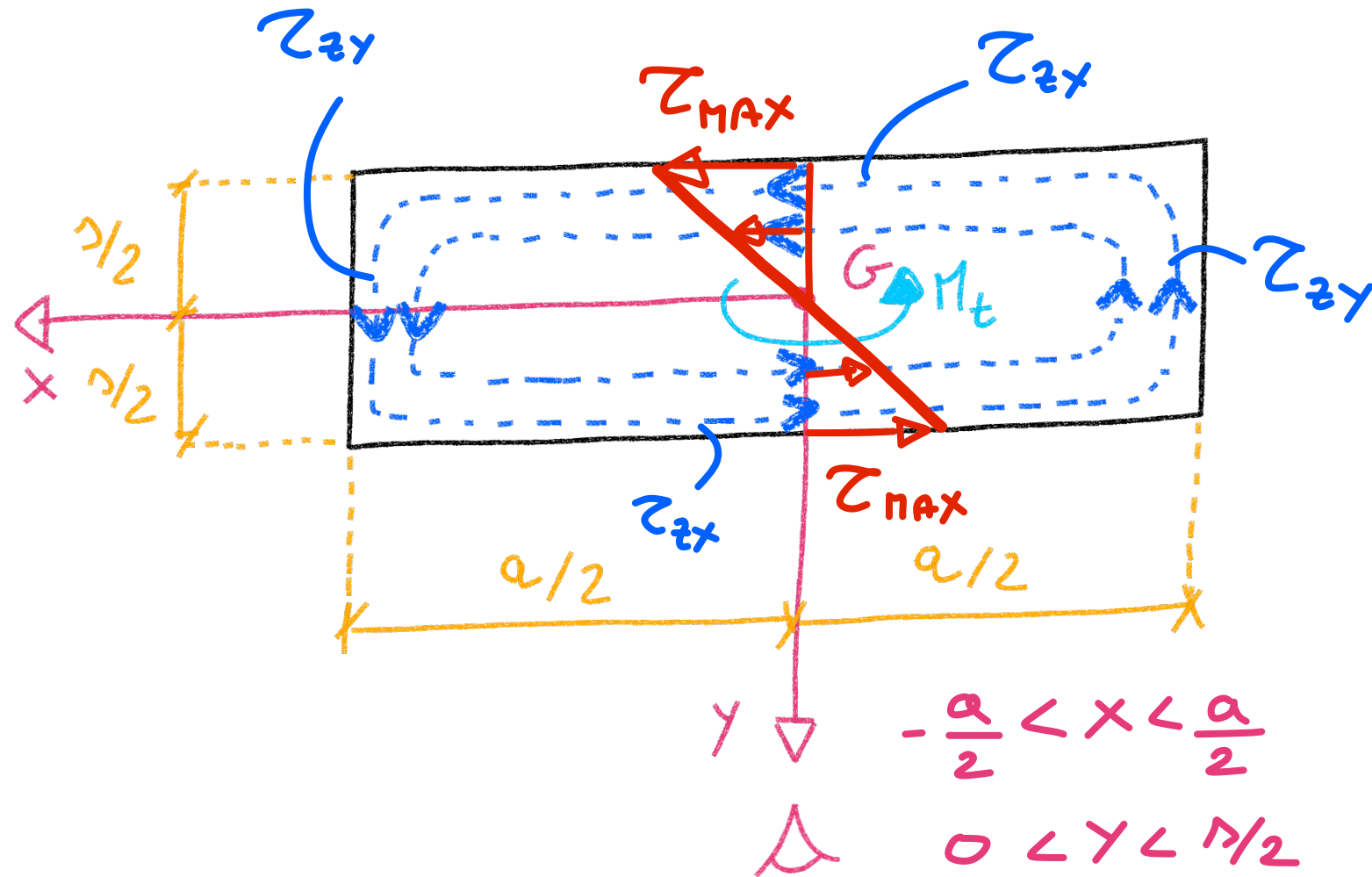
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}



SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC. INDOTTO DA Z_{zx}

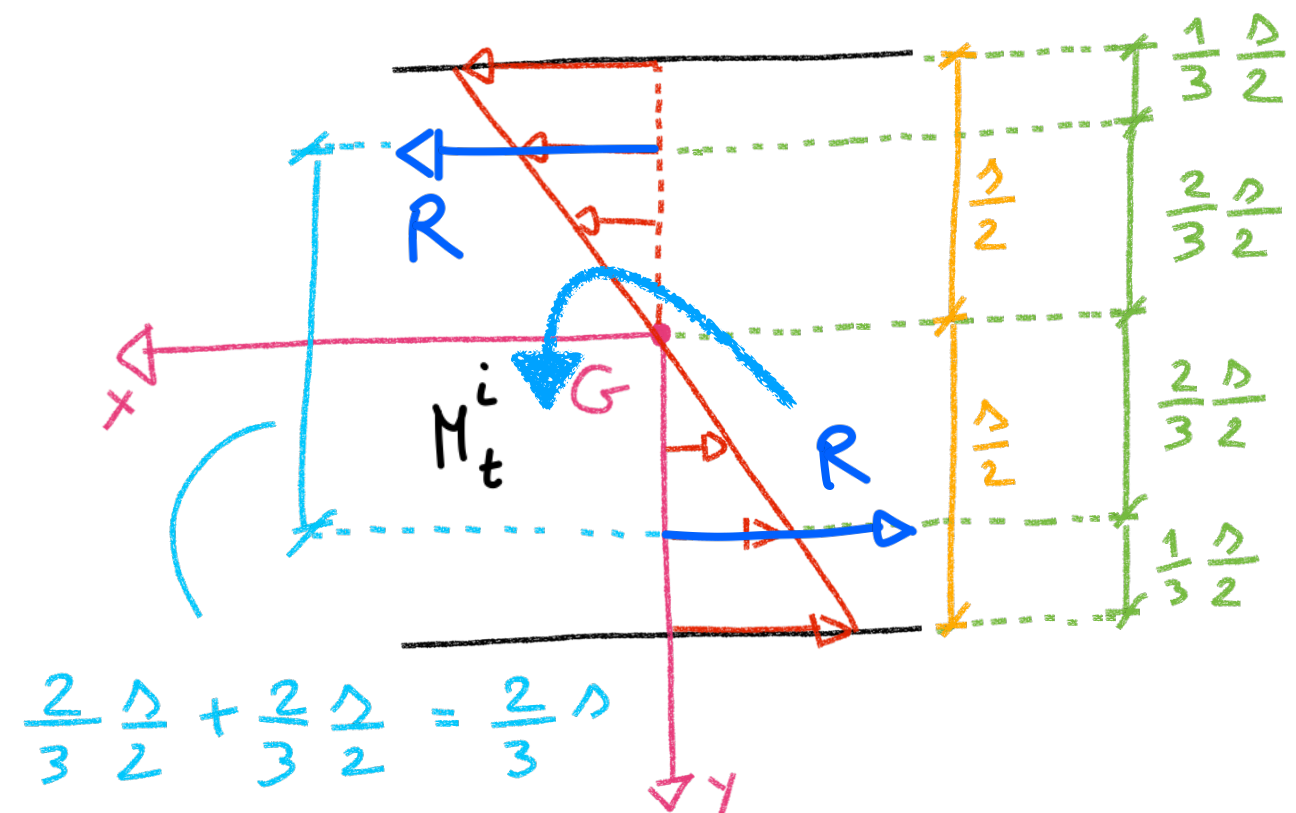
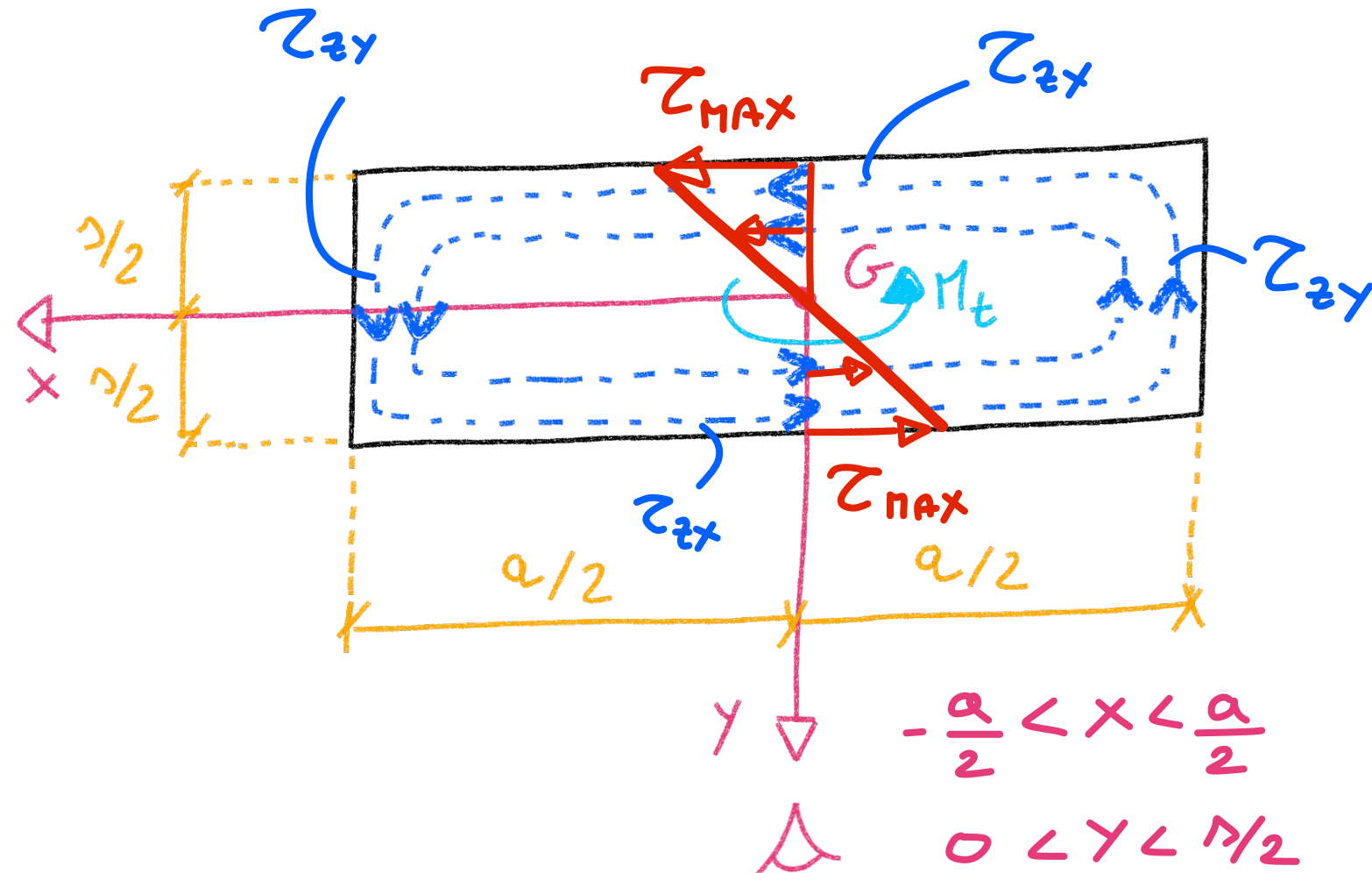


SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} b$$



$$\frac{2}{3} \frac{b}{2} + \frac{2}{3} \frac{b}{2} = \frac{2}{3} b$$

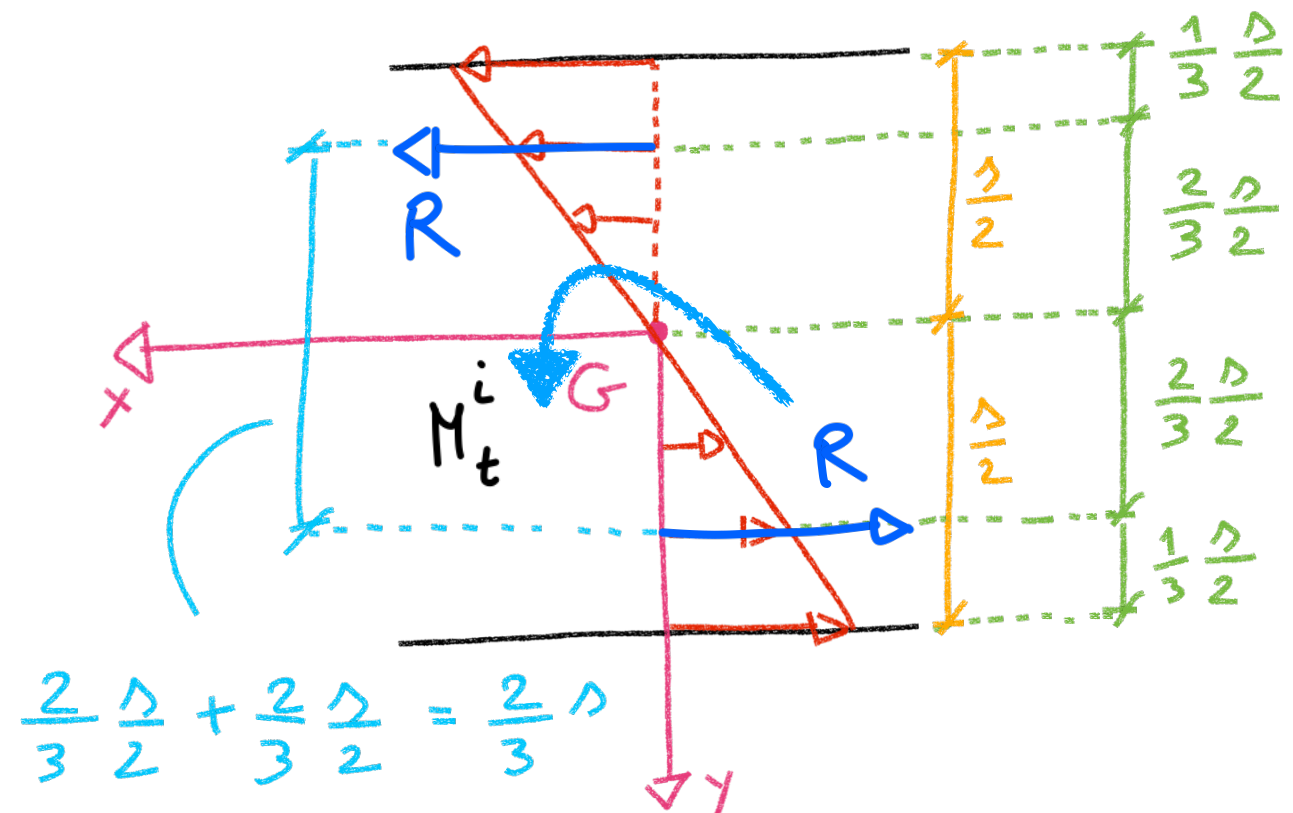
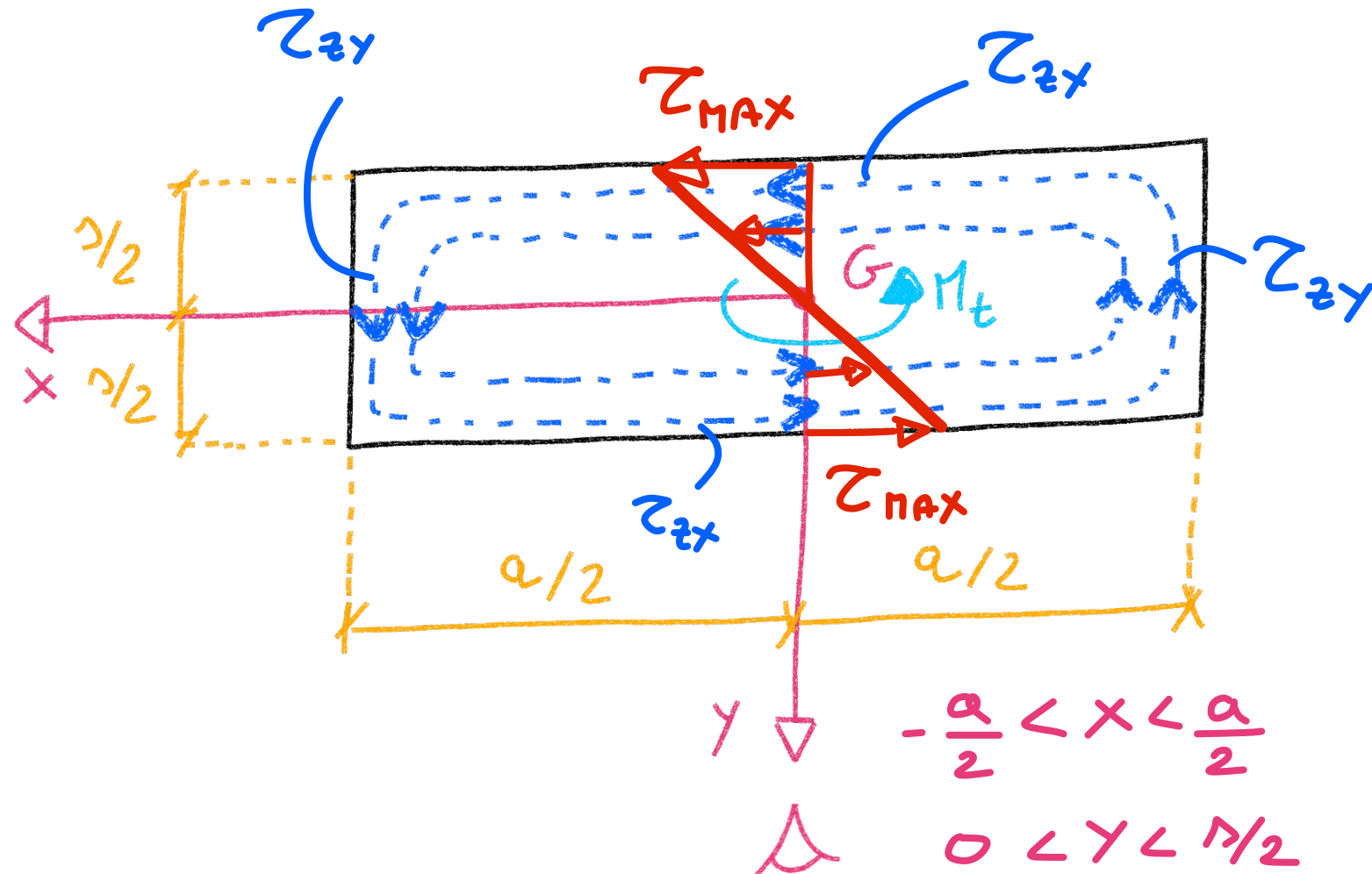
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} \rightarrow$$

$$= \frac{2}{3} \rightarrow \left| \int_{-a/2}^{a/2} \int_0^{b/2} Z_{zx} dx dy \right|$$



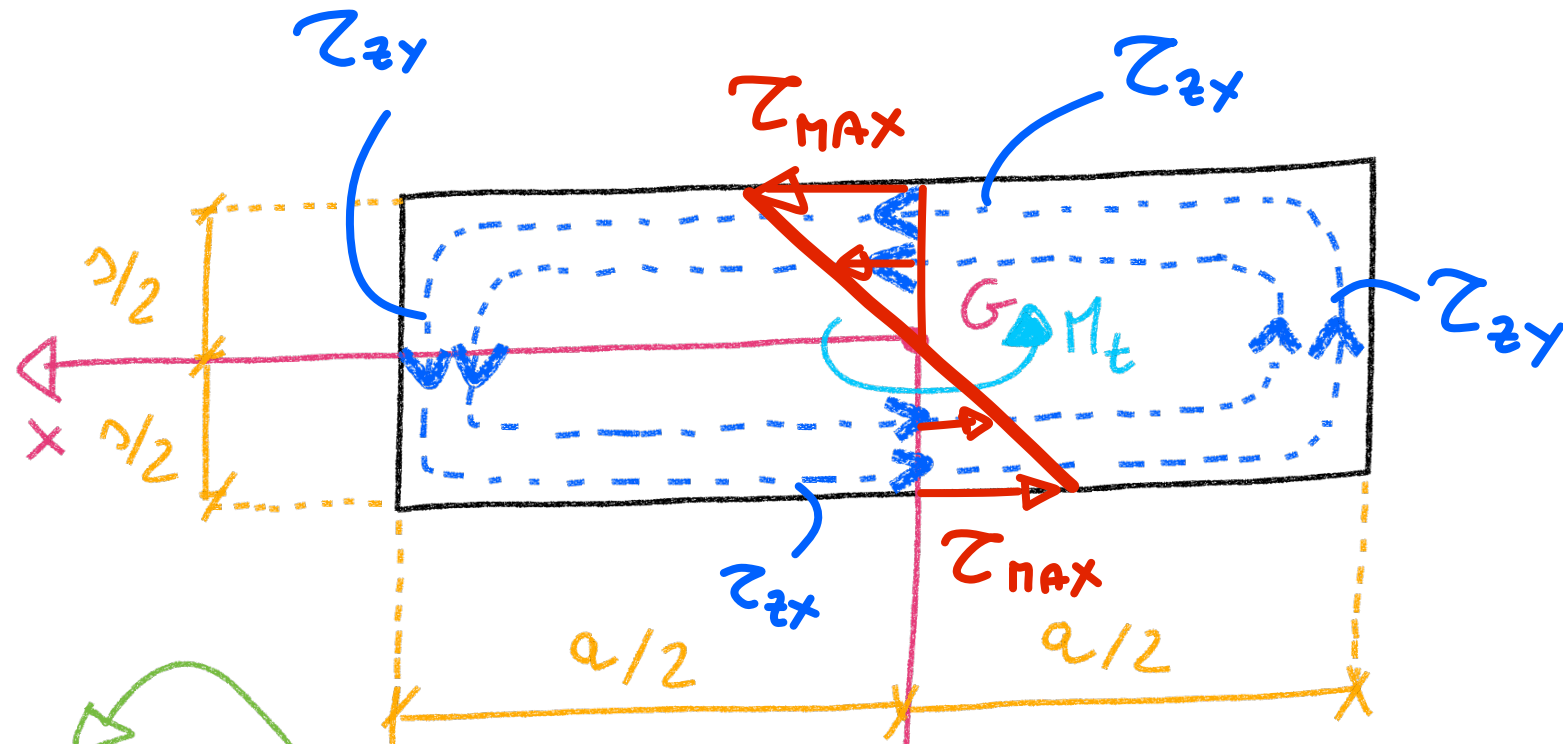
SEZIONE RETTANGOLARE SOTTILE

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INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} \rightarrow$$

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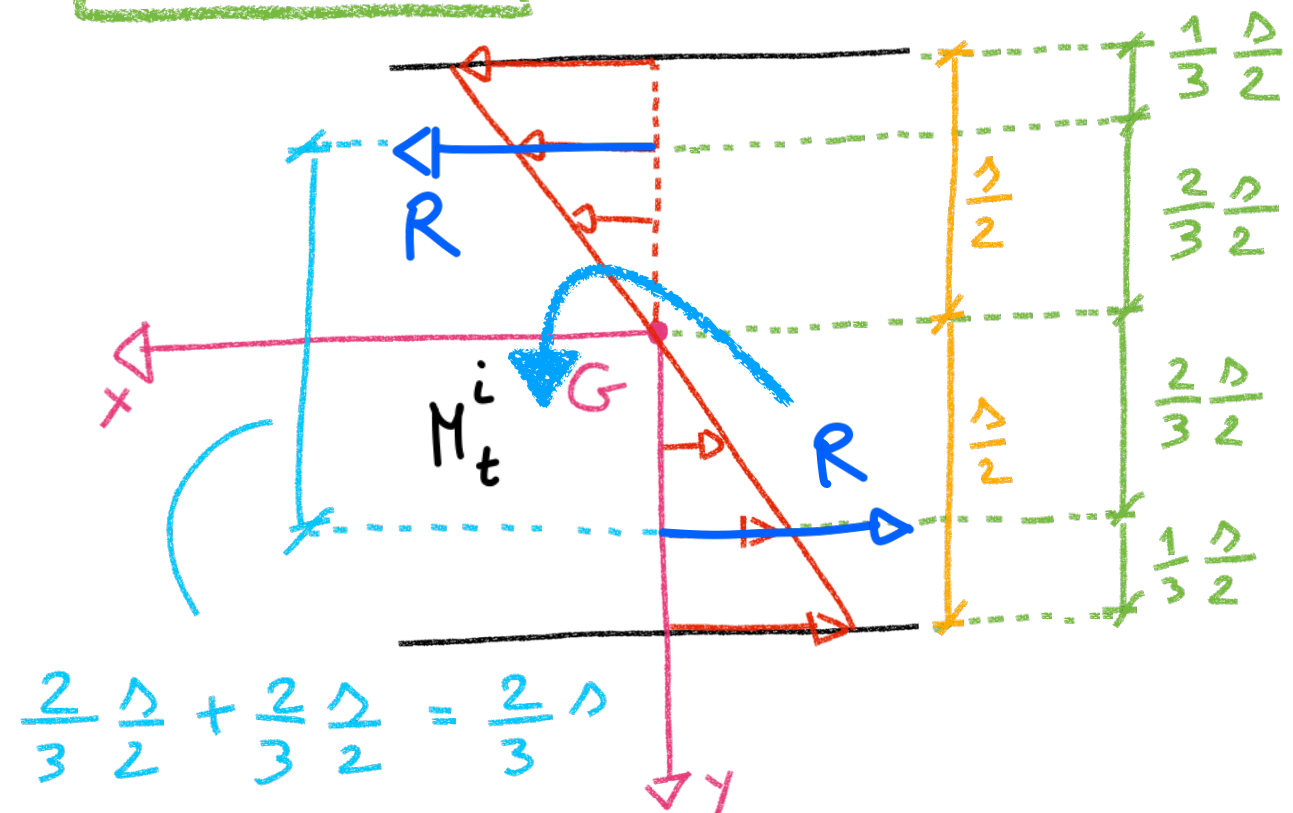


$$Z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a b^3$$

$$-\frac{a}{2} < x < \frac{a}{2}$$

$$0 < y < \frac{b}{2}$$



SEZIONE RETTANGOLARE SOTTILE

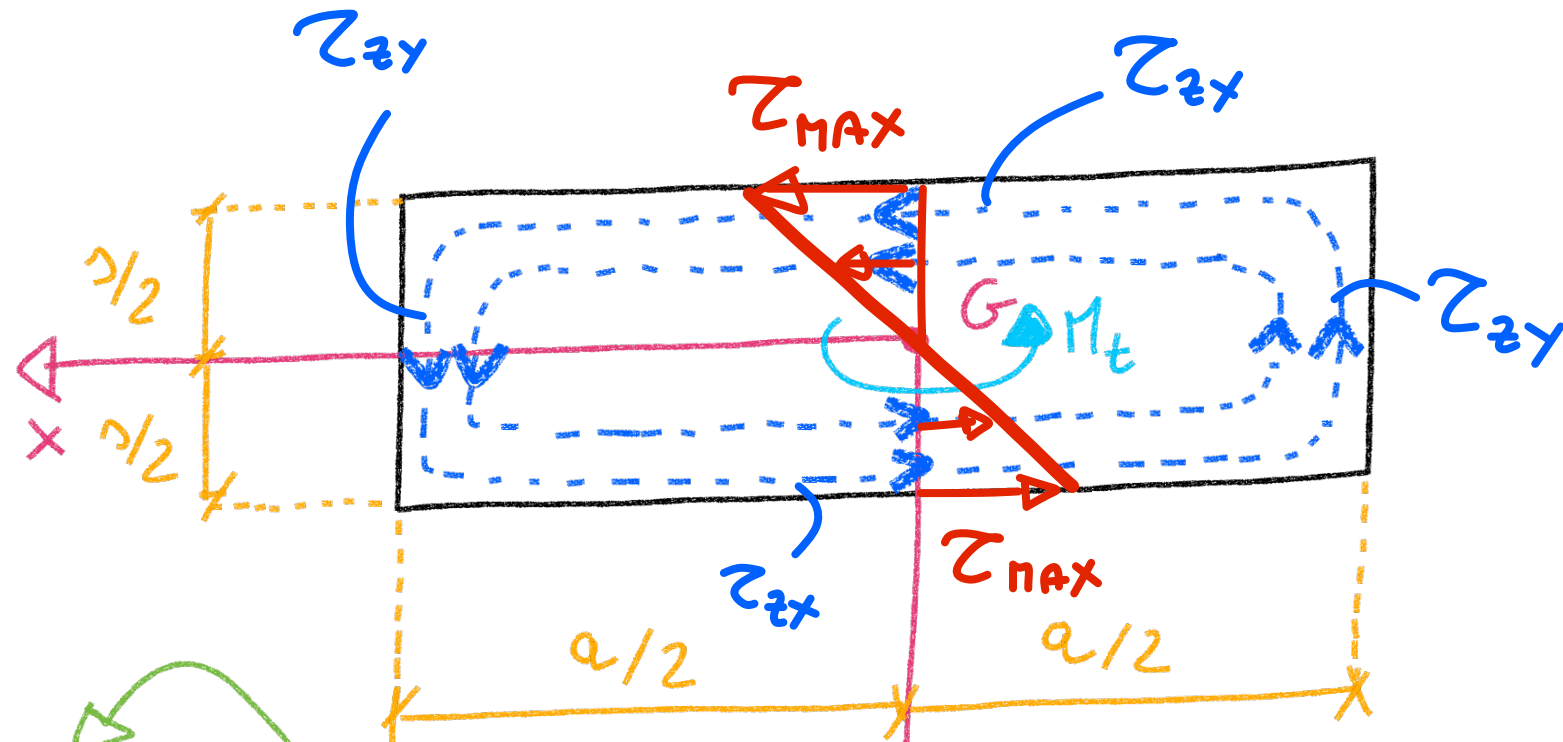
CALCOLO MOMENTO TORC.

INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} \rightarrow$$

$$= \frac{2}{3} \rightarrow \left| \int_{-a/2}^{a/2} \int_0^{b/2} Z_{zx} dx dy \right|$$

$$= \frac{2}{3} \rightarrow \left| \int_{-a/2}^{a/2} \int_0^{b/2} -2 \frac{M_t}{I_t} y dx dy \right|$$

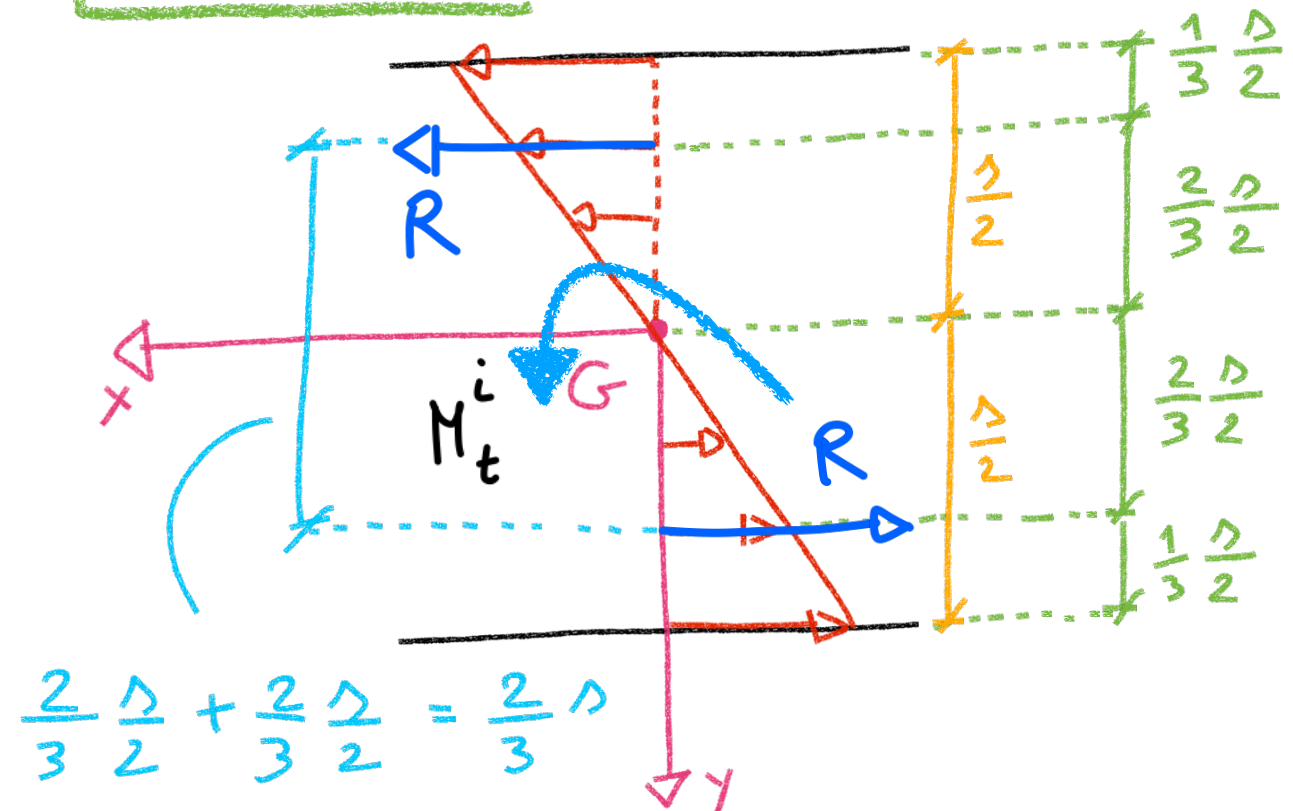


$$Z_{zx} = -2 \frac{M_t}{I_t} y$$

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$$-\frac{a}{2} < x < \frac{a}{2}$$

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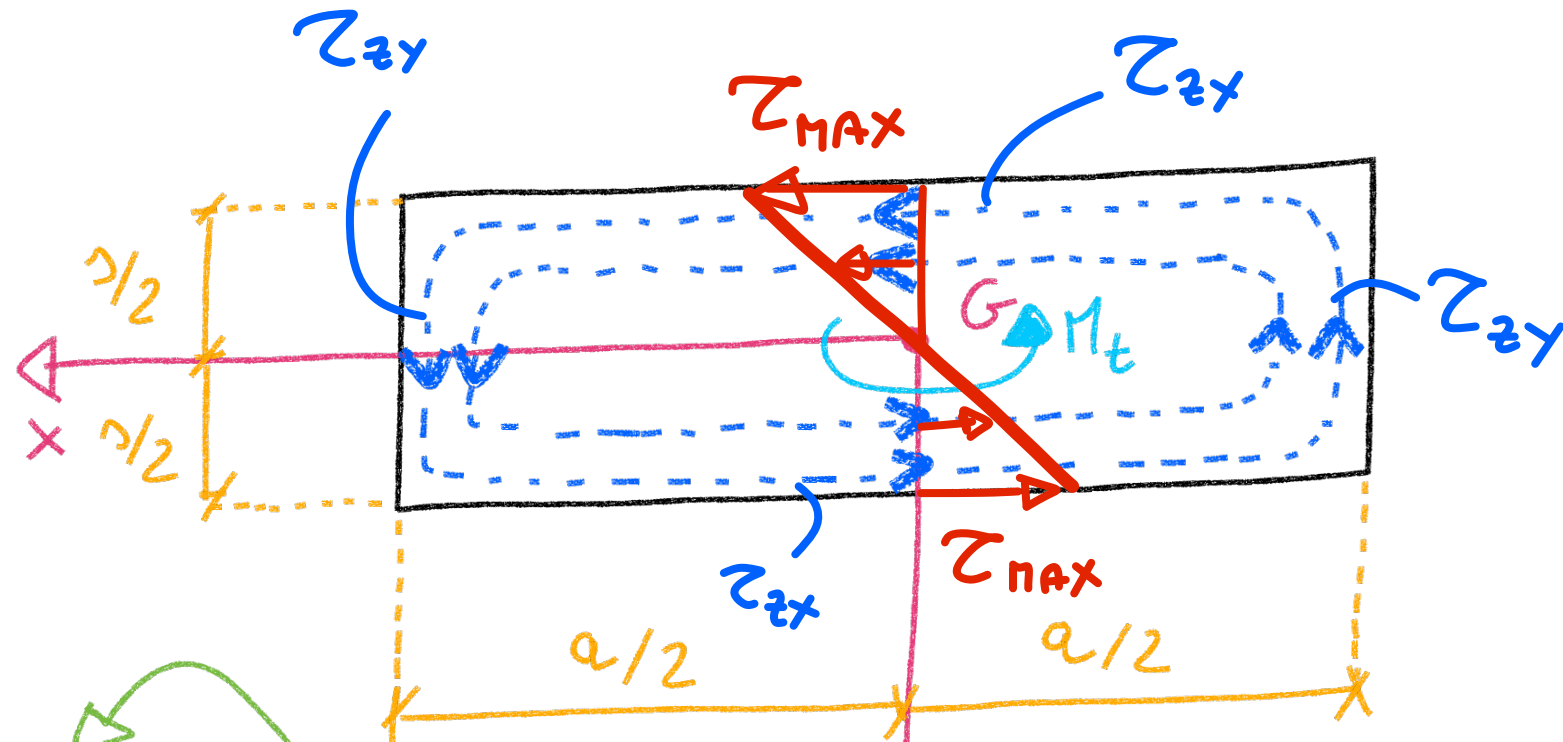


SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

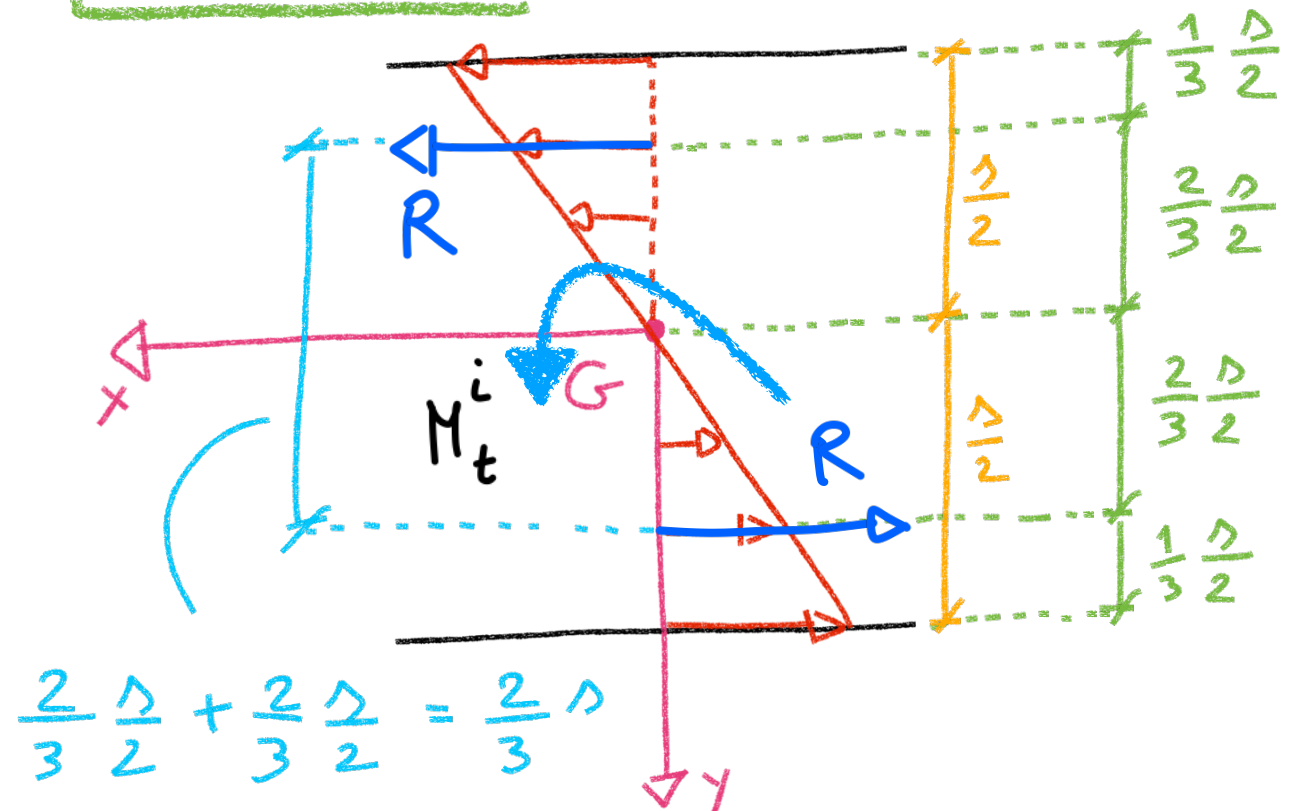
INDOTTO DA Z_{zx}

$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} \Delta \\
 &= \frac{2}{3} \Delta \left| \int_{-a/2}^{a/2} \int_0^{b/2} Z_{zx} dx dy \right| \\
 &= \frac{2}{3} \Delta \left| \int_{-a/2}^{a/2} \int_0^{b/2} -2 \frac{M_t}{I_t} y dx dy \right| \\
 &= \frac{2}{3} \Delta \left| \int_{-a/2}^{a/2} -2 \frac{M_t}{I_t} dx \int_0^{b/2} y dy \right|
 \end{aligned}$$



$$\begin{aligned}
 Z_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a b^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{b}{2}
 \end{aligned}$$

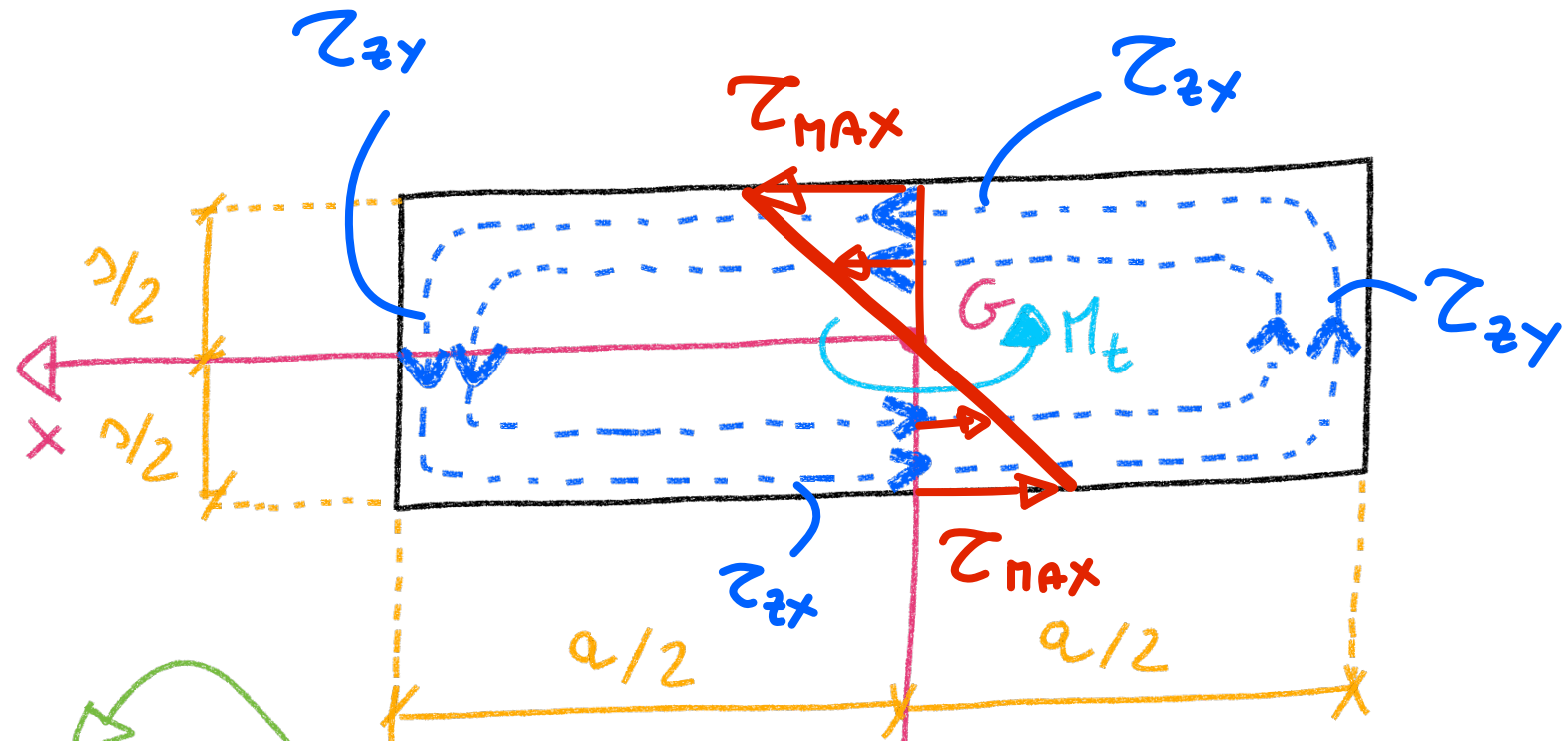


SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

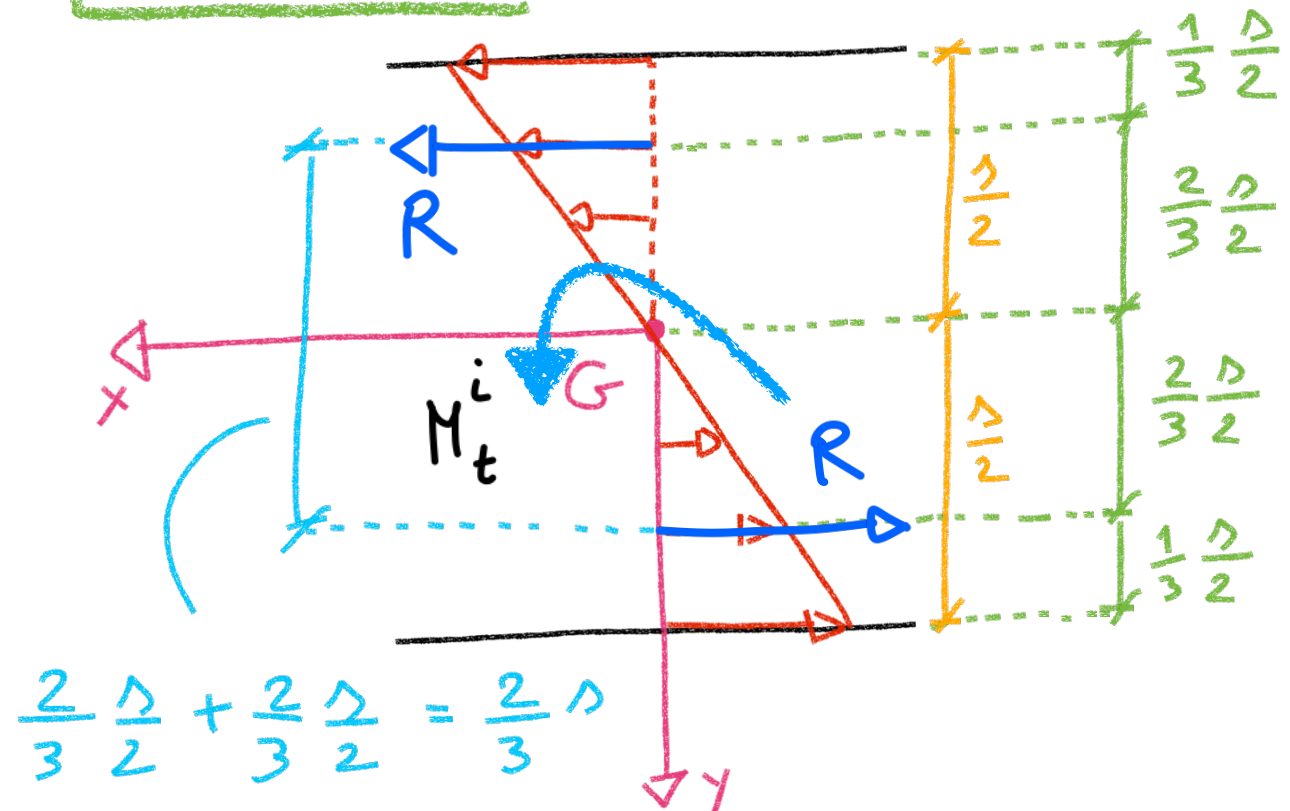
INDOTTO DA Z_{zx}

$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} \rightarrow \\
 &= \frac{2}{3} \rightarrow \left| \int_{-a/2}^{a/2} \int_0^{b/2} Z_{zx} dx dy \right| \\
 &= \frac{2}{3} \rightarrow \left| \int_{-a/2}^{a/2} \int_0^{b/2} -2 \frac{M_t}{I_t} y dx dy \right| \\
 &= \frac{2}{3} \rightarrow \left| \int_{-a/2}^{a/2} -2 \frac{M_t}{I_t} dx \int_0^{b/2} y dy \right| \\
 &= \frac{2}{3} \rightarrow \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{b/2}
 \end{aligned}$$



$$\begin{aligned}
 Z_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a b^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{b}{2}
 \end{aligned}$$

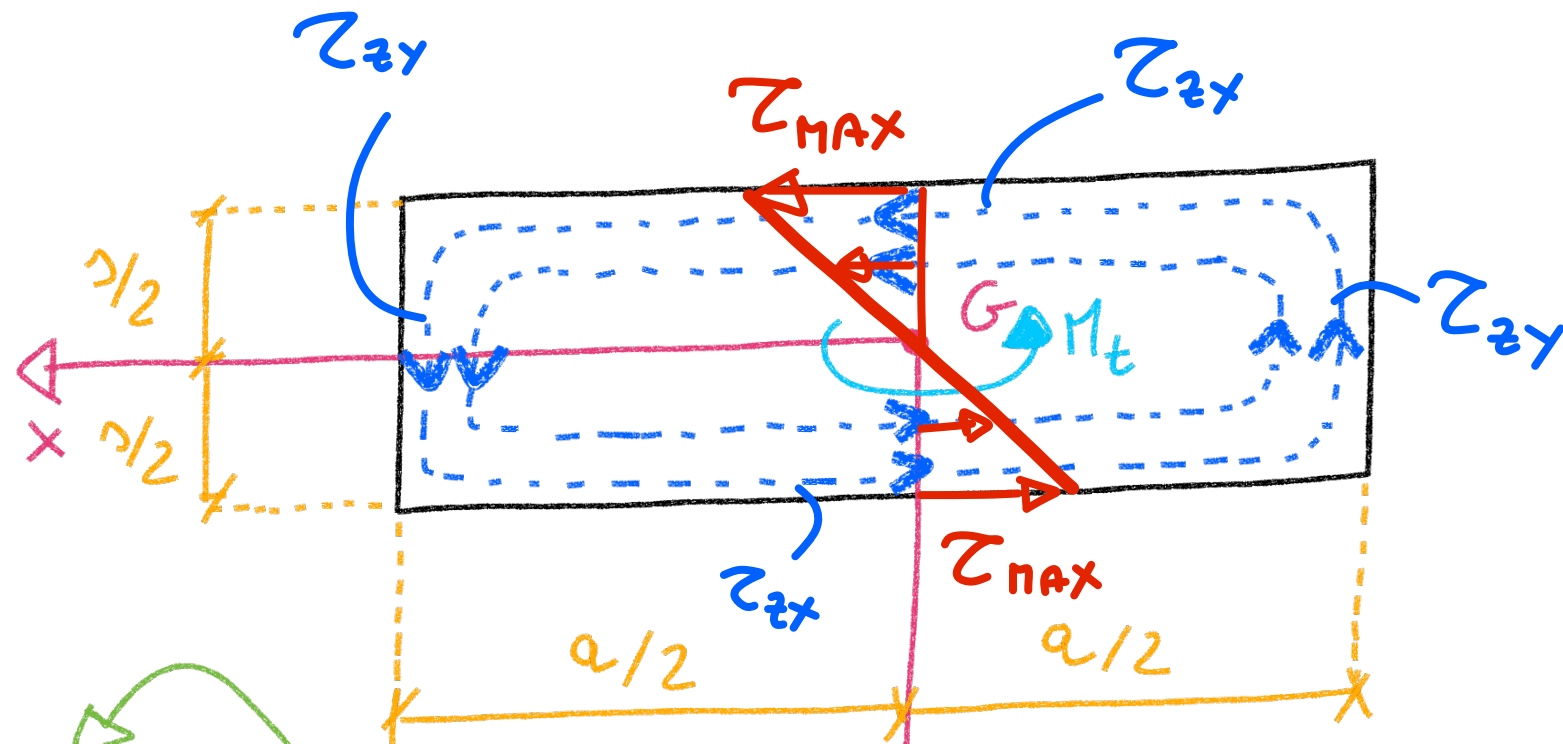


SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

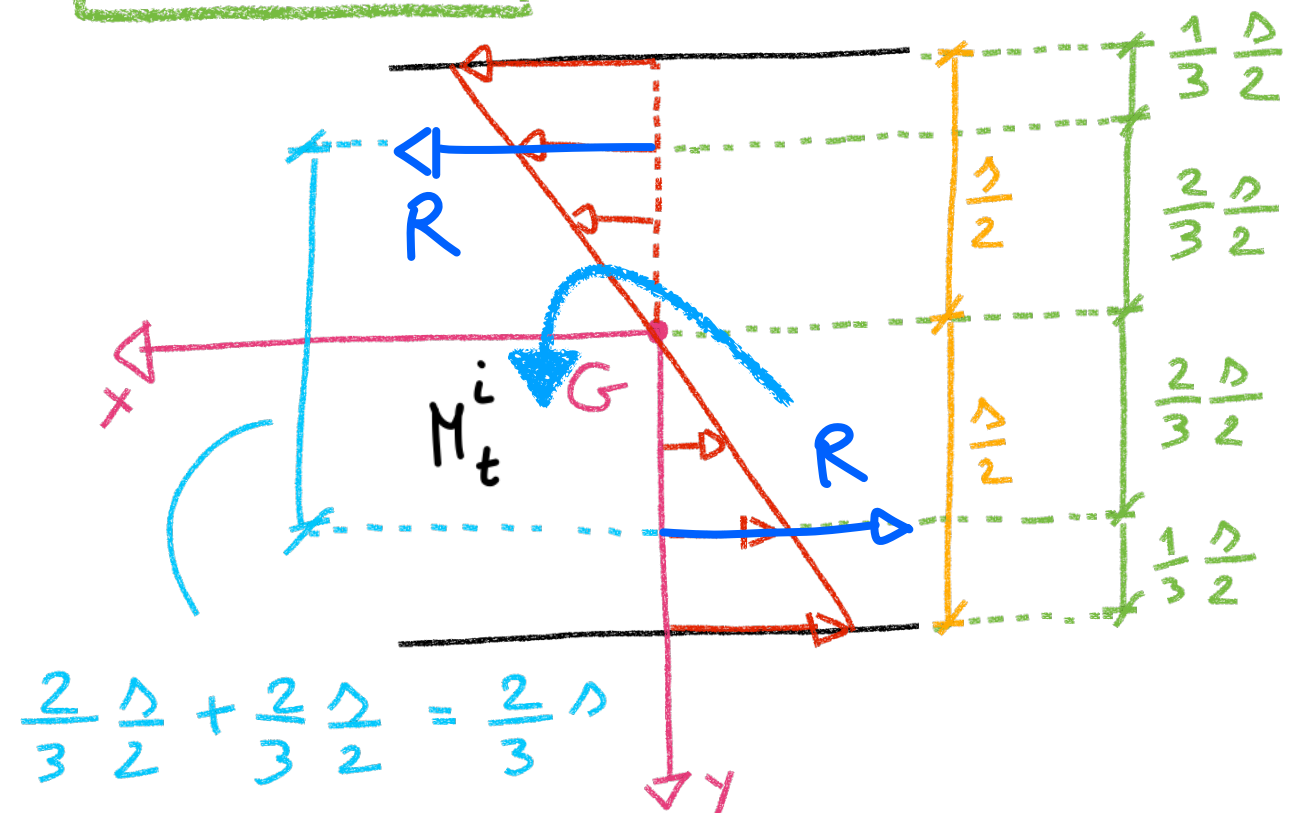
INDOTTO DA Z_{zx}

$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} \Delta \\
 &= \frac{2}{3} \Delta \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{\Delta/2} \\
 &= \frac{2}{3} \Delta \cdot 2 \frac{M_t}{\frac{1}{3} a \Delta^3} \cdot \frac{1}{2} \frac{\Delta^2}{4}
 \end{aligned}$$



$$\begin{aligned}
 \tau_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a \Delta^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{\Delta}{2}
 \end{aligned}$$



SEZIONE RETTANGOLARE SOTTILE

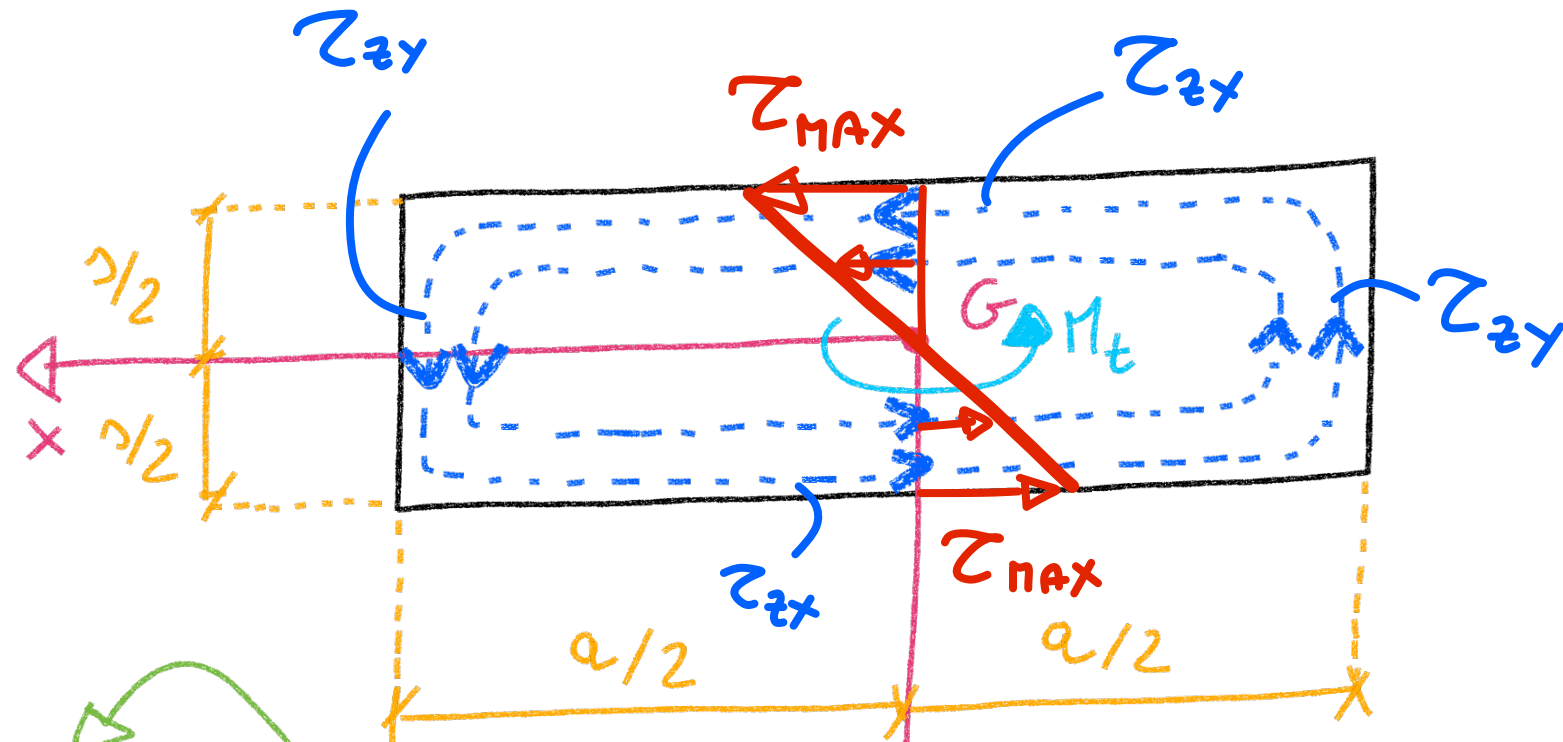
CALCOLO MOMENTO TORC.

INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} \Delta$$

$$= \frac{2}{3} \Delta \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{\Delta/2}$$

$$= \frac{2}{3} \Delta \cdot 2 \frac{M_t}{\frac{1}{3} a \Delta^3} \cdot \frac{1}{2} \frac{\Delta^2}{4}$$

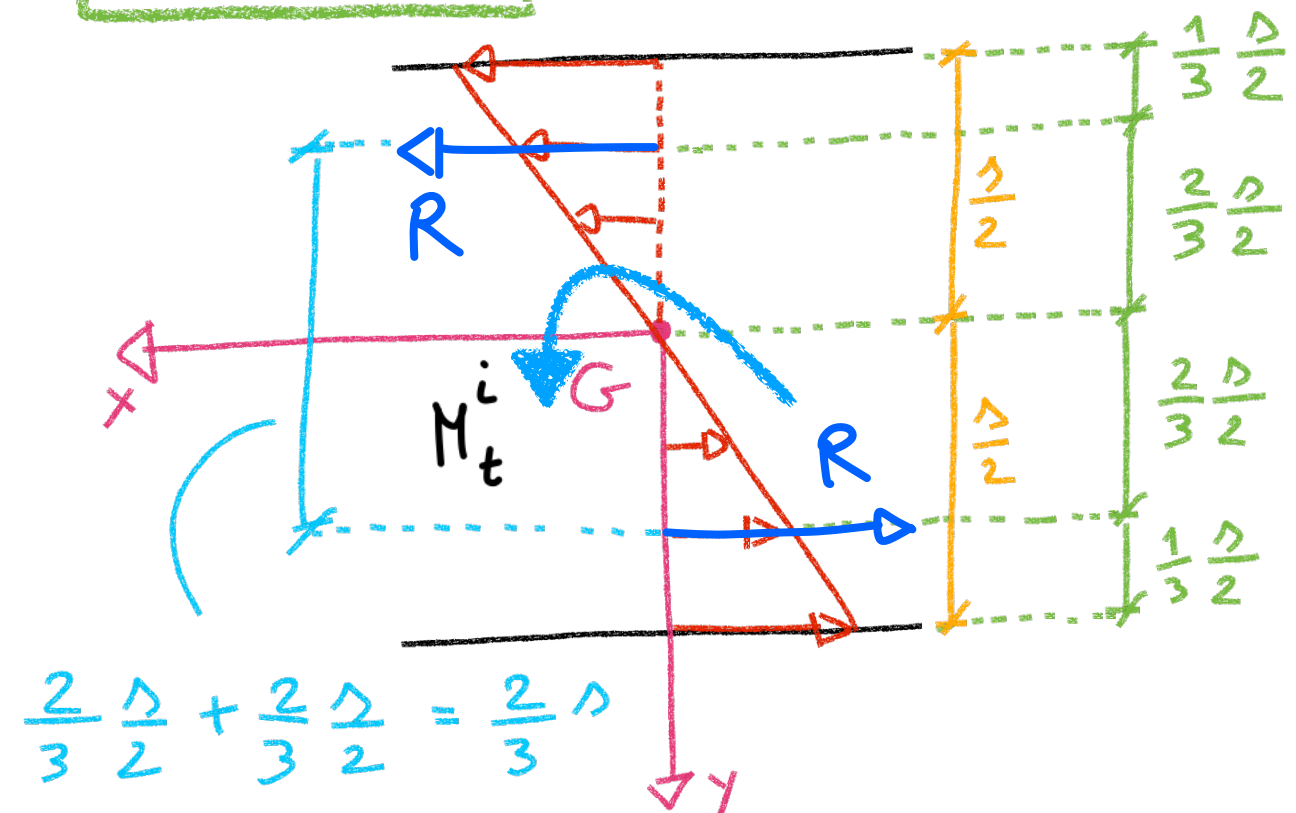


$$Z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a \Delta^3$$

$$-\frac{a}{2} < x < \frac{a}{2}$$

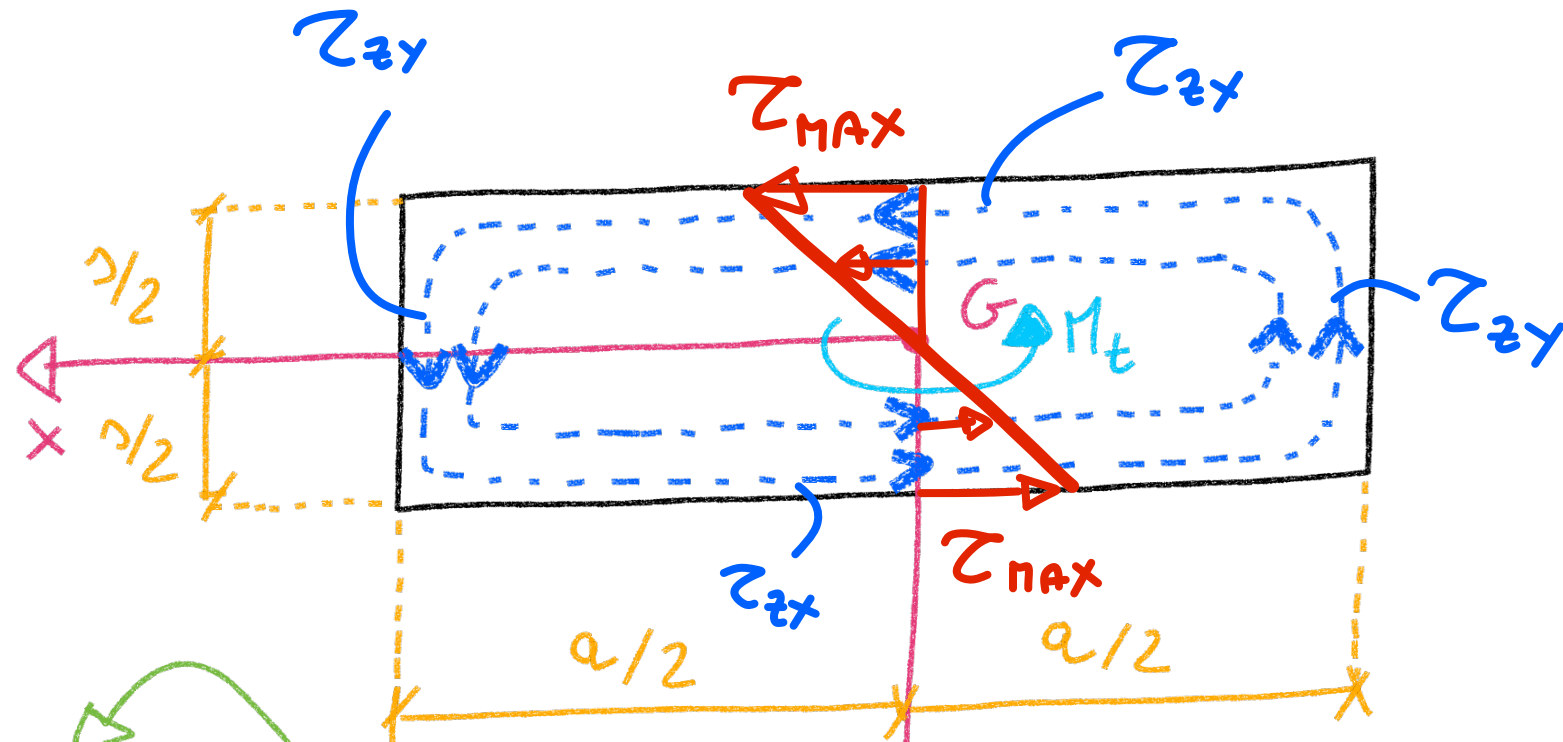
$$0 < y < \frac{\Delta}{2}$$



SEZIONE RETTANGOLARE SOTTILE

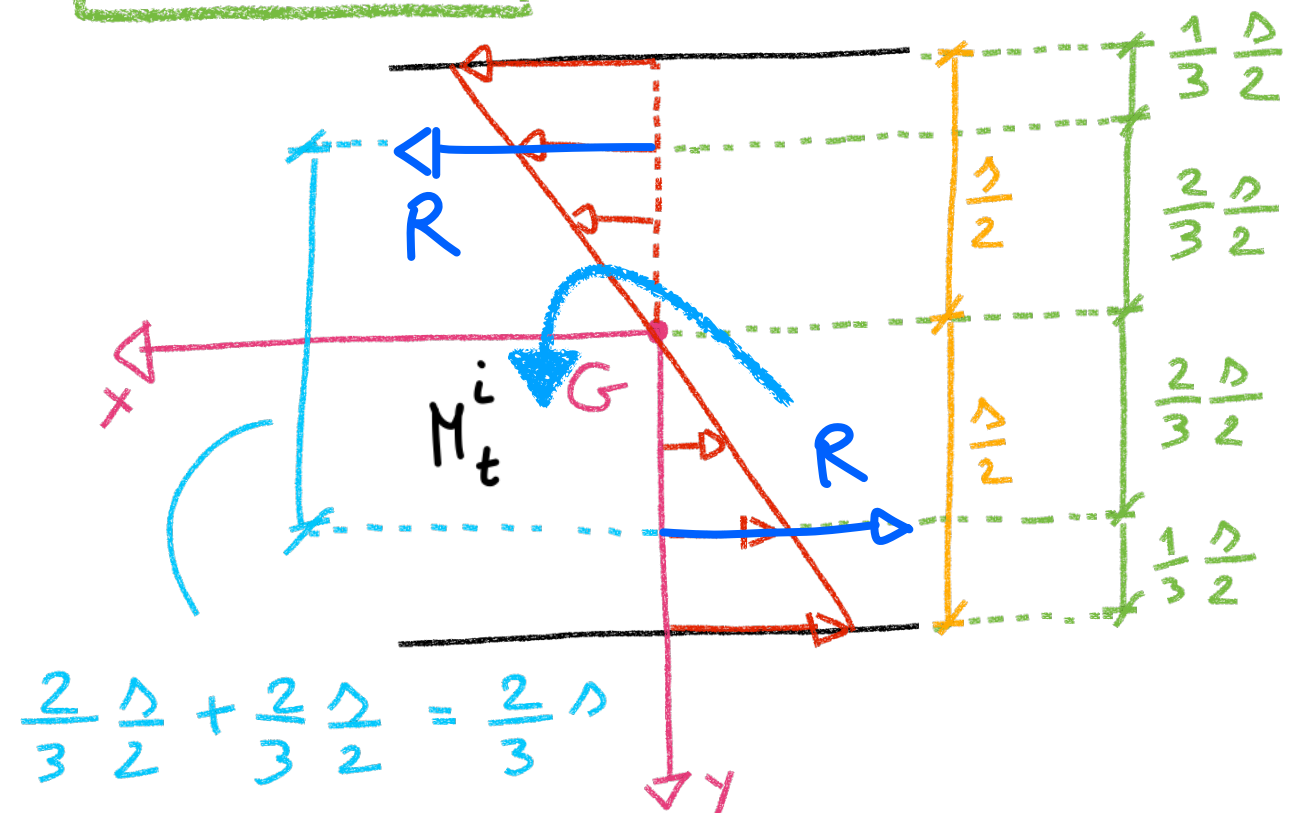
CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}

$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} a \\
 &= \frac{2}{3} a \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{a/2} \\
 &= \frac{2}{3} a \cdot 2 \frac{M_t}{\frac{1}{3} a^3} a \frac{1}{2} \frac{a^2}{4}
 \end{aligned}$$



$$\begin{aligned}
 Z_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{h}{2}
 \end{aligned}$$



$$\frac{2}{3} \frac{h}{2} + \frac{2}{3} \frac{h}{2} = \frac{2}{3} h$$

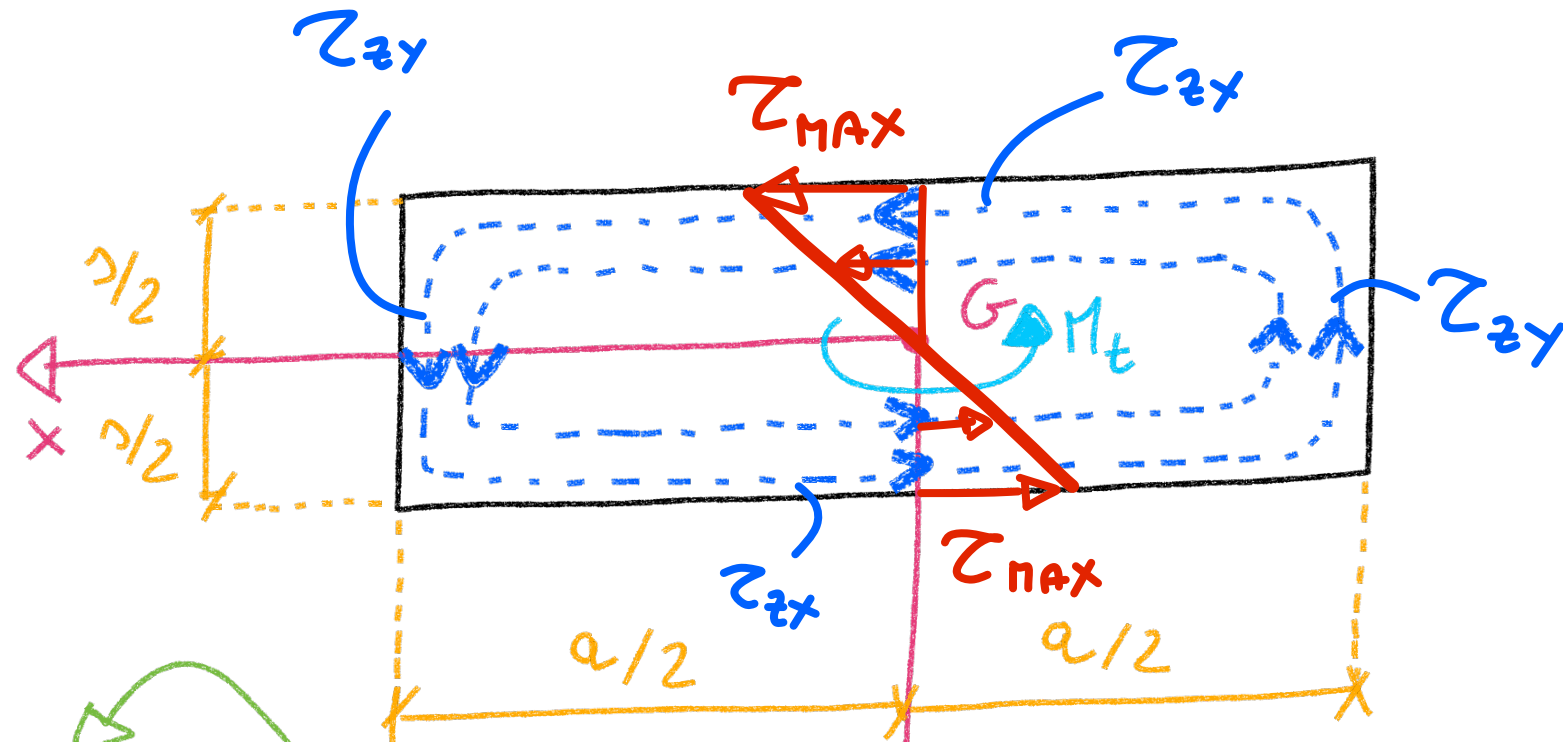
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.
INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} a$$

$$= \frac{2}{3} a \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{a/2}$$

$$= \frac{2}{3} a \cdot 2 \frac{M_t}{\frac{1}{3} a^3} a \frac{1}{2} \frac{a^2}{4}$$

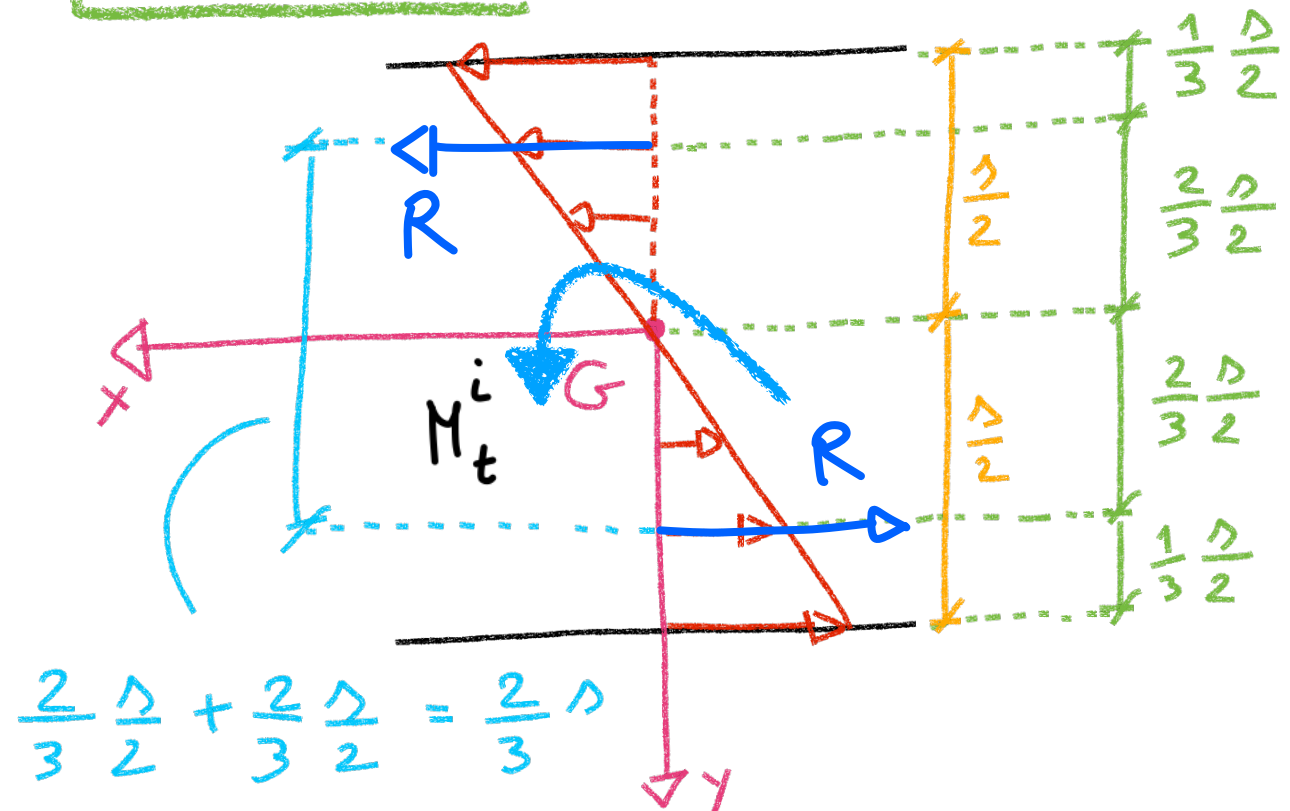


$$Z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a^3$$

$$-\frac{a}{2} < x < \frac{a}{2}$$

$$0 < y < \frac{h}{2}$$



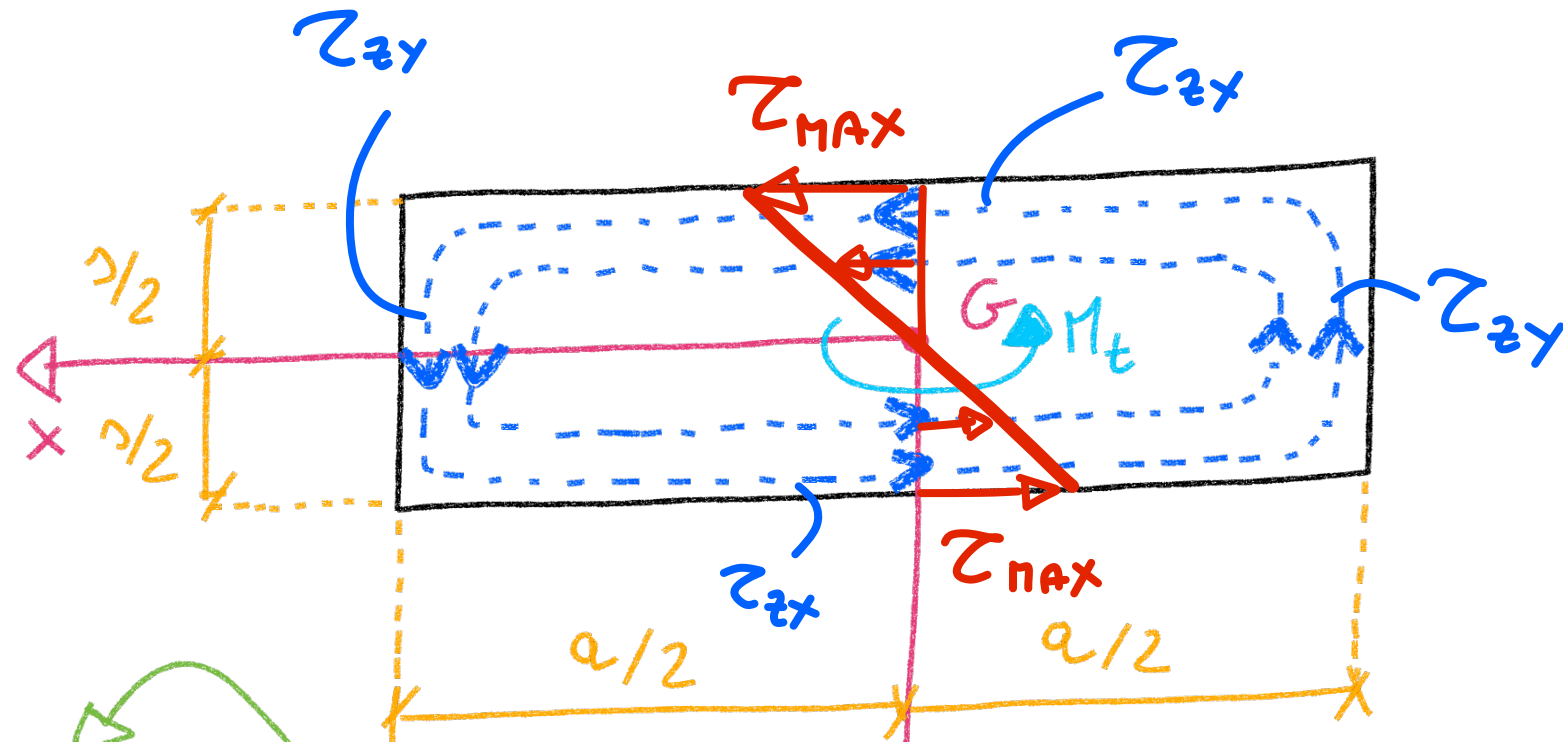
SEZIONE RETTANGOLARE SOTTILE

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INDOTTO DA Z_{zx}

$$M_t^i = |R| \frac{2}{3} a$$

$$= \frac{2}{3} a \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{a/2}$$

$$= \frac{2}{3} a \cdot 2 \frac{M_t}{\frac{1}{3} a^3} a \frac{1}{2} \frac{a^2}{4}$$

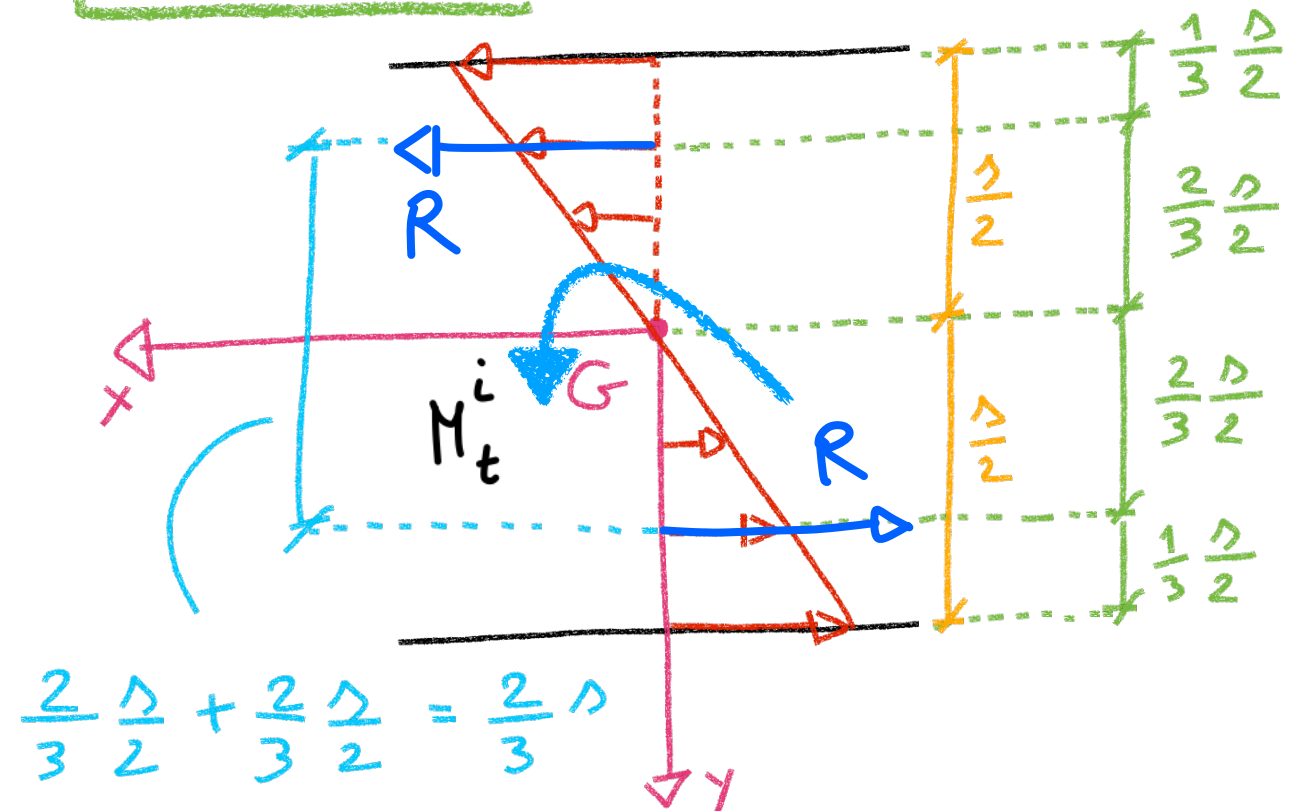


$$Z_{zx} = -2 \frac{M_t}{I_t} y$$

$$I_t = \frac{1}{3} a^3$$

$$-\frac{a}{2} < x < \frac{a}{2}$$

$$0 < y < \frac{h}{2}$$

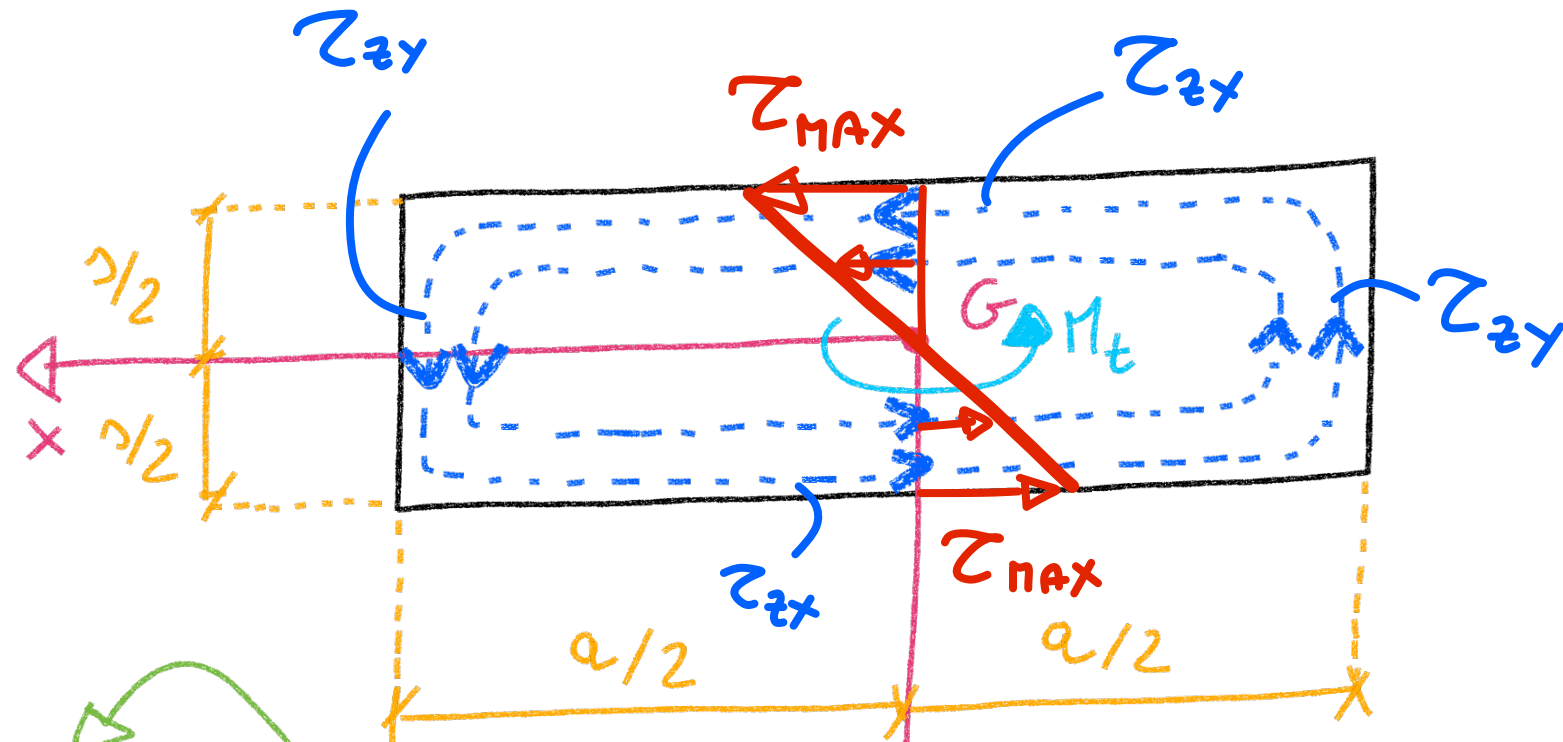


SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

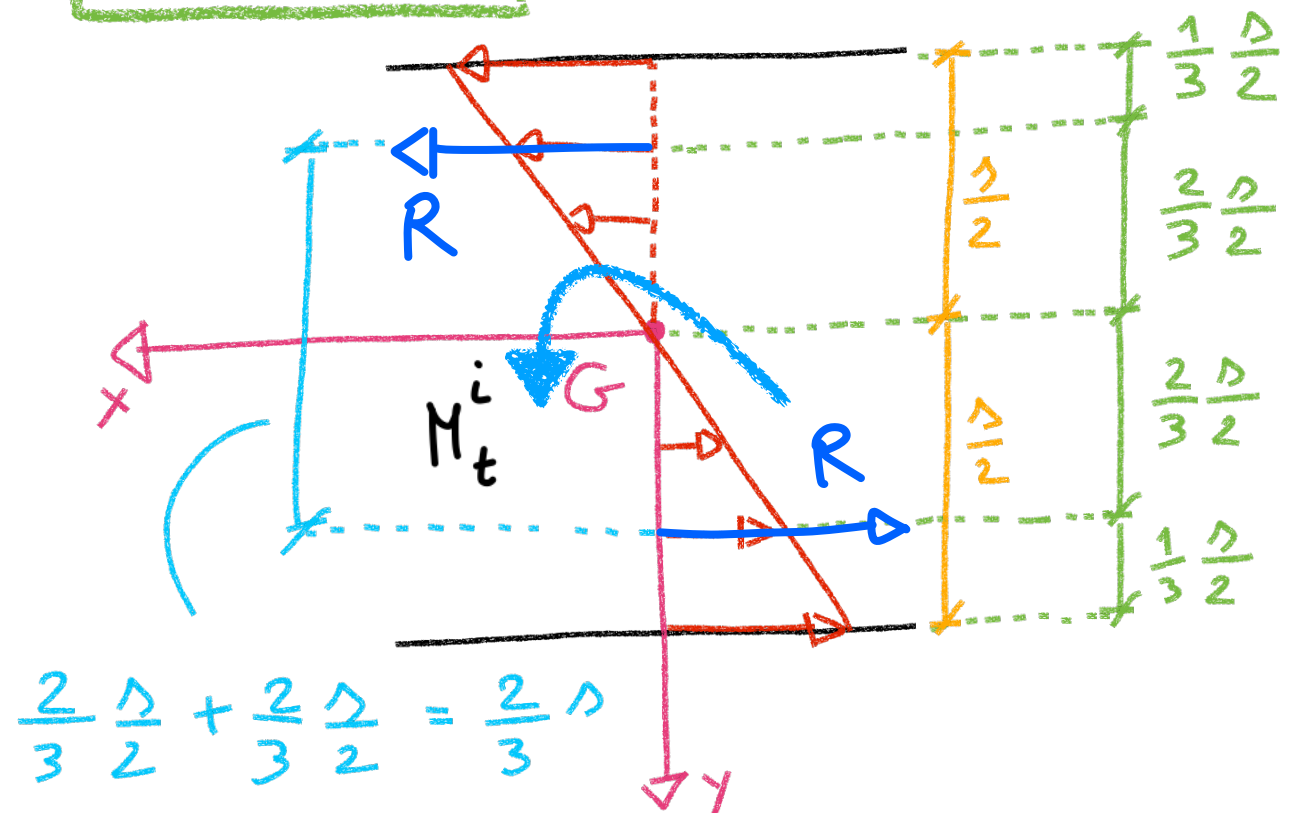
INDOTTO DA Z_{zx}

$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} a \\
 &= \frac{2}{3} a \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{a/2} \\
 &= \frac{2}{3} a \cdot 2 \frac{M_t}{\frac{1}{3} a^3} a \frac{1}{2} \frac{a^2}{4} \\
 &= \frac{M_t}{2}
 \end{aligned}$$



$$\begin{aligned}
 \tau_{zx} &= -2 \frac{M_t}{I_t} y \\
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 \end{aligned}$$

$$\begin{aligned}
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 0 < y < \frac{h}{2}
 \end{aligned}$$



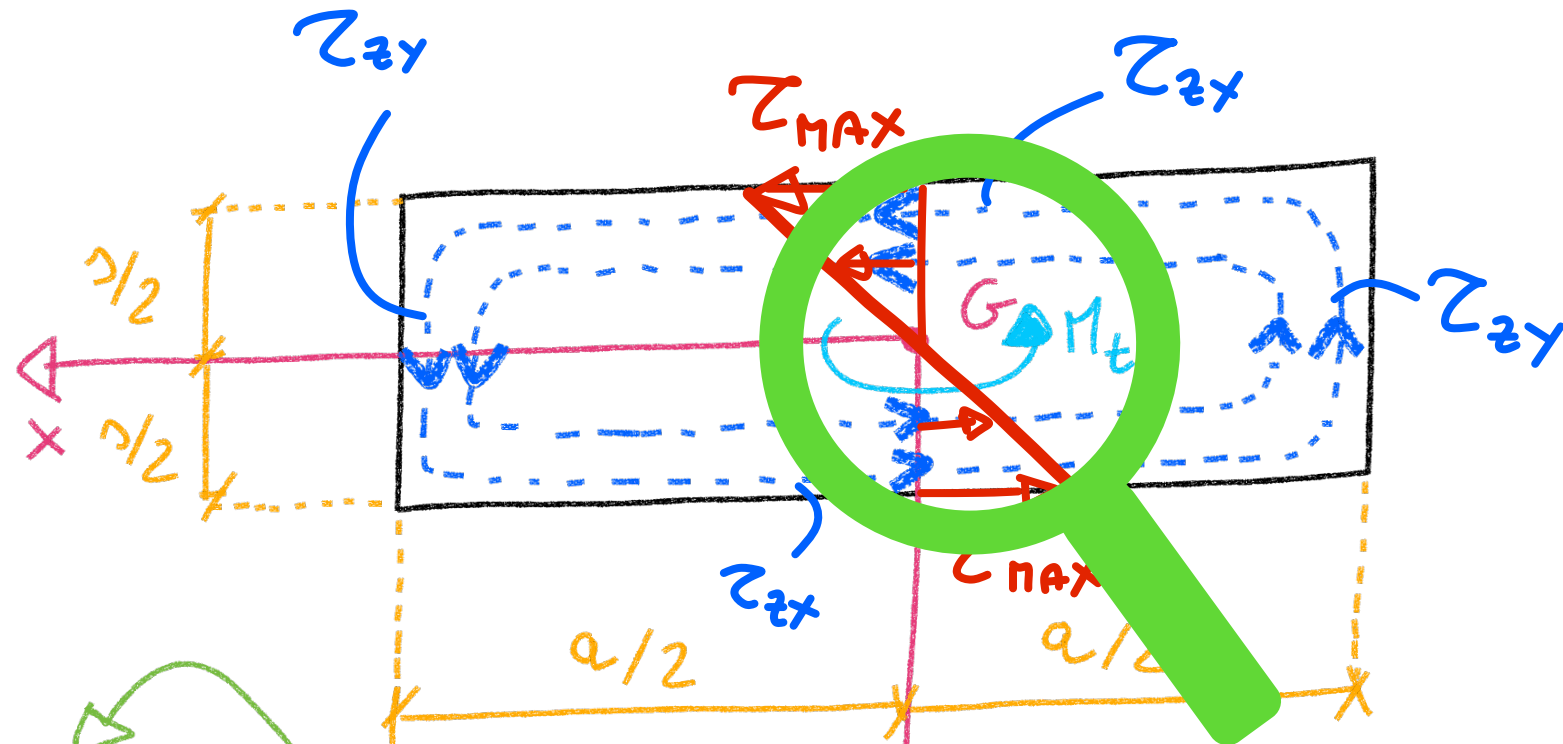
SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

INDOTTO DA Z_{zx}

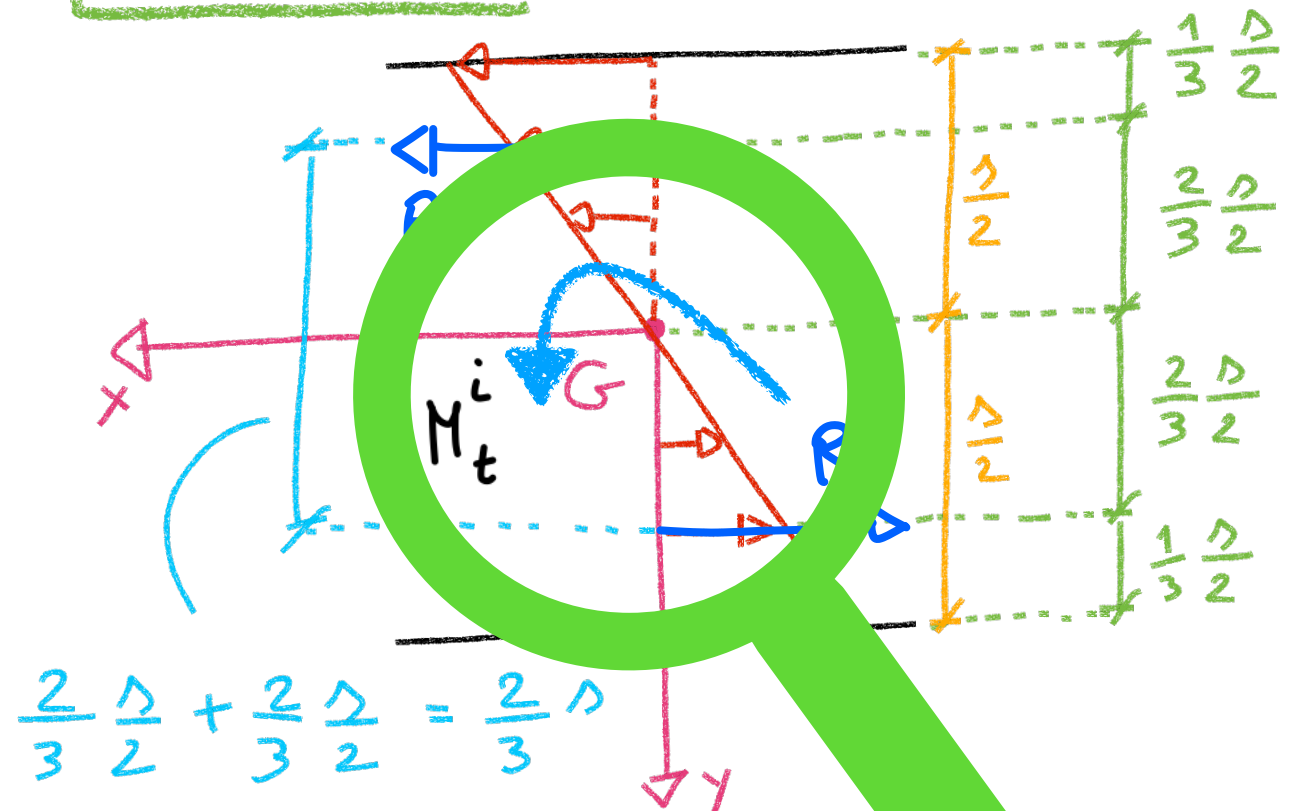
$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} \Delta \\
 &= \frac{2}{3} \Delta \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{\Delta/2} \\
 &= \frac{2}{3} \Delta \cdot 2 \frac{M_t}{\frac{1}{3} a \Delta^3} a \frac{1}{2} \frac{\Delta^2}{4} \\
 &= \frac{M_t}{2}
 \end{aligned}$$

DOVE È FINITA LA
RESTANTE METÀ ($M_t/2$) ???



$$\begin{aligned}
 Z_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a \Delta^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{\Delta}{2}
 \end{aligned}$$



SEZIONE RETTANGOLARE SOTTILE

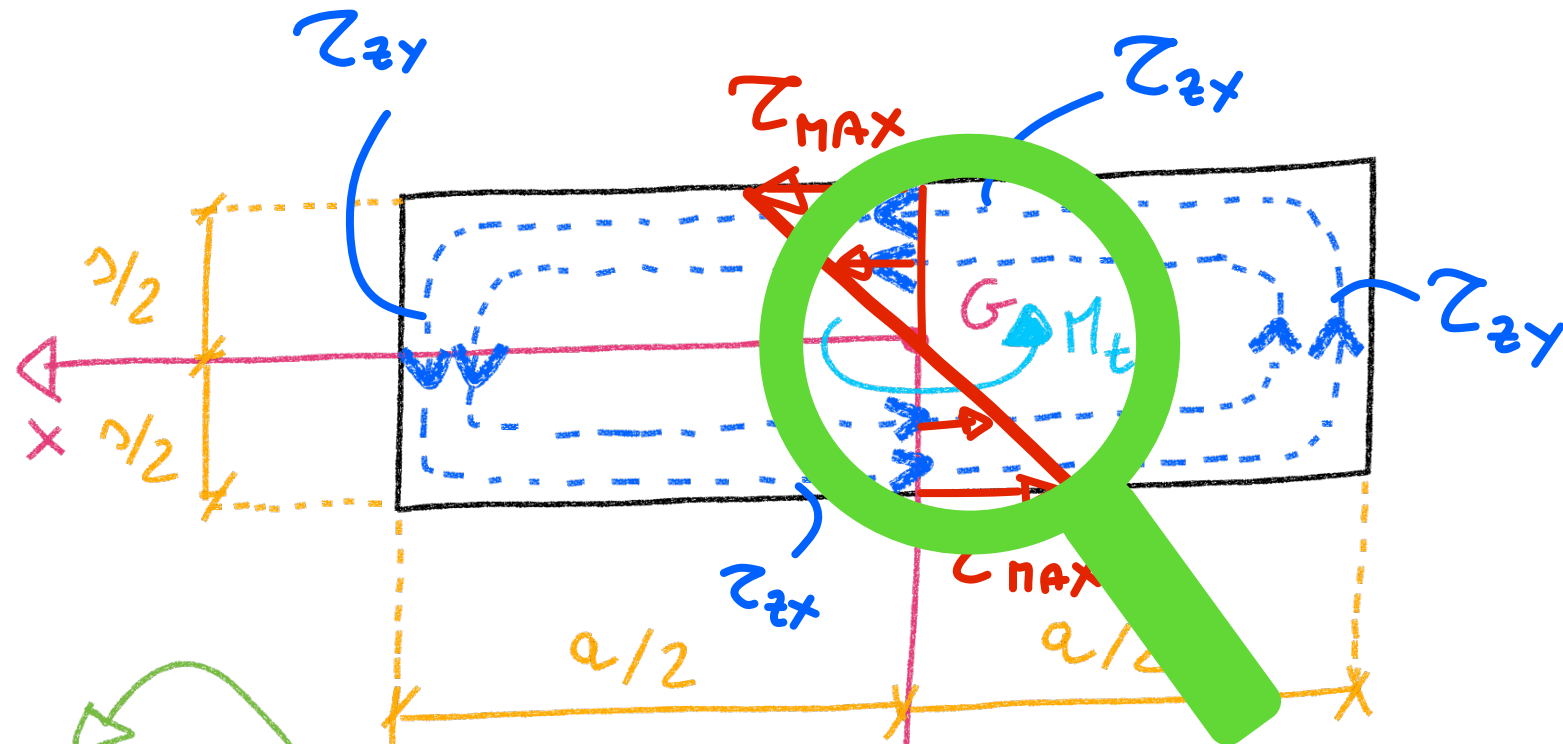
CALCOLO MOMENTO TORC.

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 &= \frac{2}{3} a \cdot 2 \frac{M_t}{\frac{1}{3} a^3} a \frac{1}{2} \frac{a^2}{4} \\
 &= \frac{M_t}{2}
 \end{aligned}$$

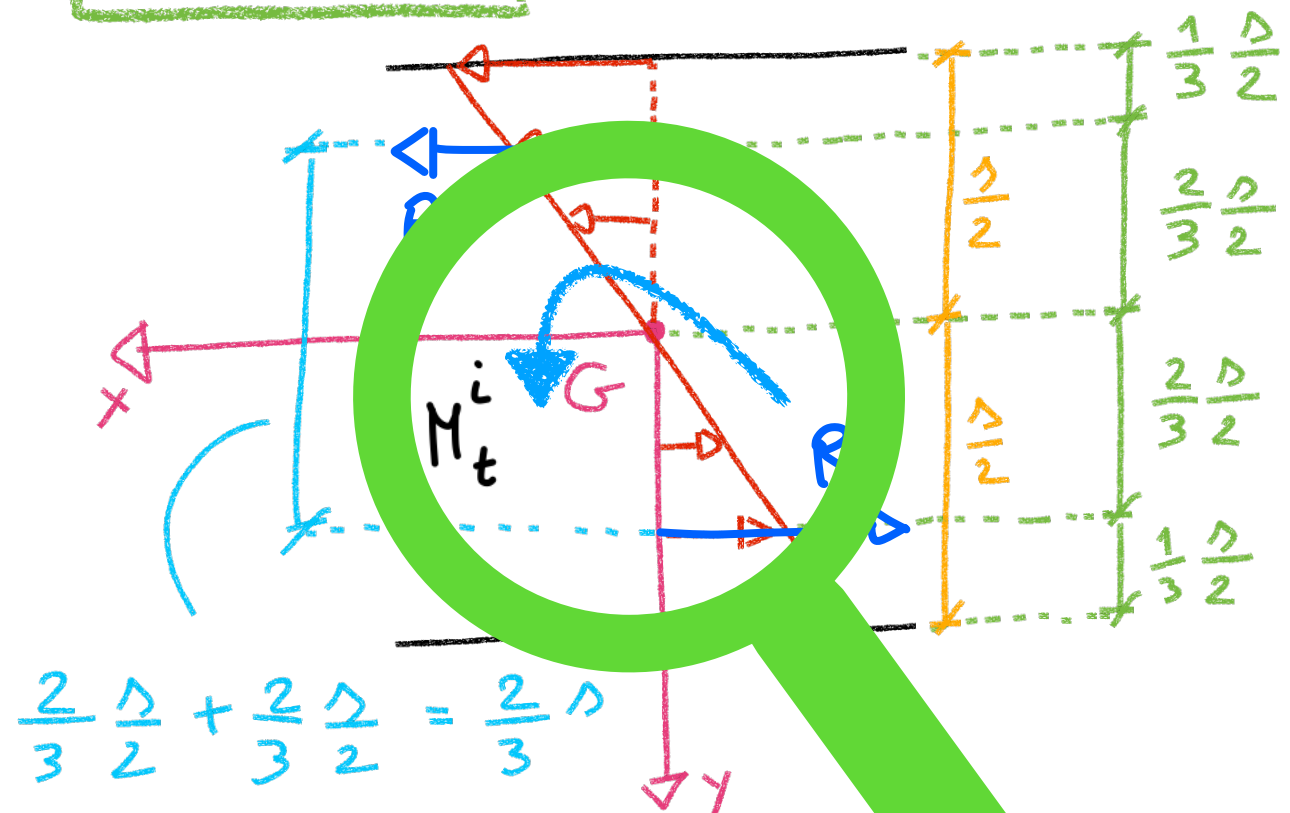
DOVE È FINITA LA
RESTANTE METÀ ($M_t/2$) ???

↳ Z_{zy} SUI LATI CORTI



$$\begin{aligned}
 Z_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{b}{2}
 \end{aligned}$$



$$\frac{2}{3} \frac{a}{2} + \frac{2}{3} \frac{a}{2} = \frac{2}{3} a$$

SEZIONE RETTANGOLARE SOTTILE

CALCOLO MOMENTO TORC.

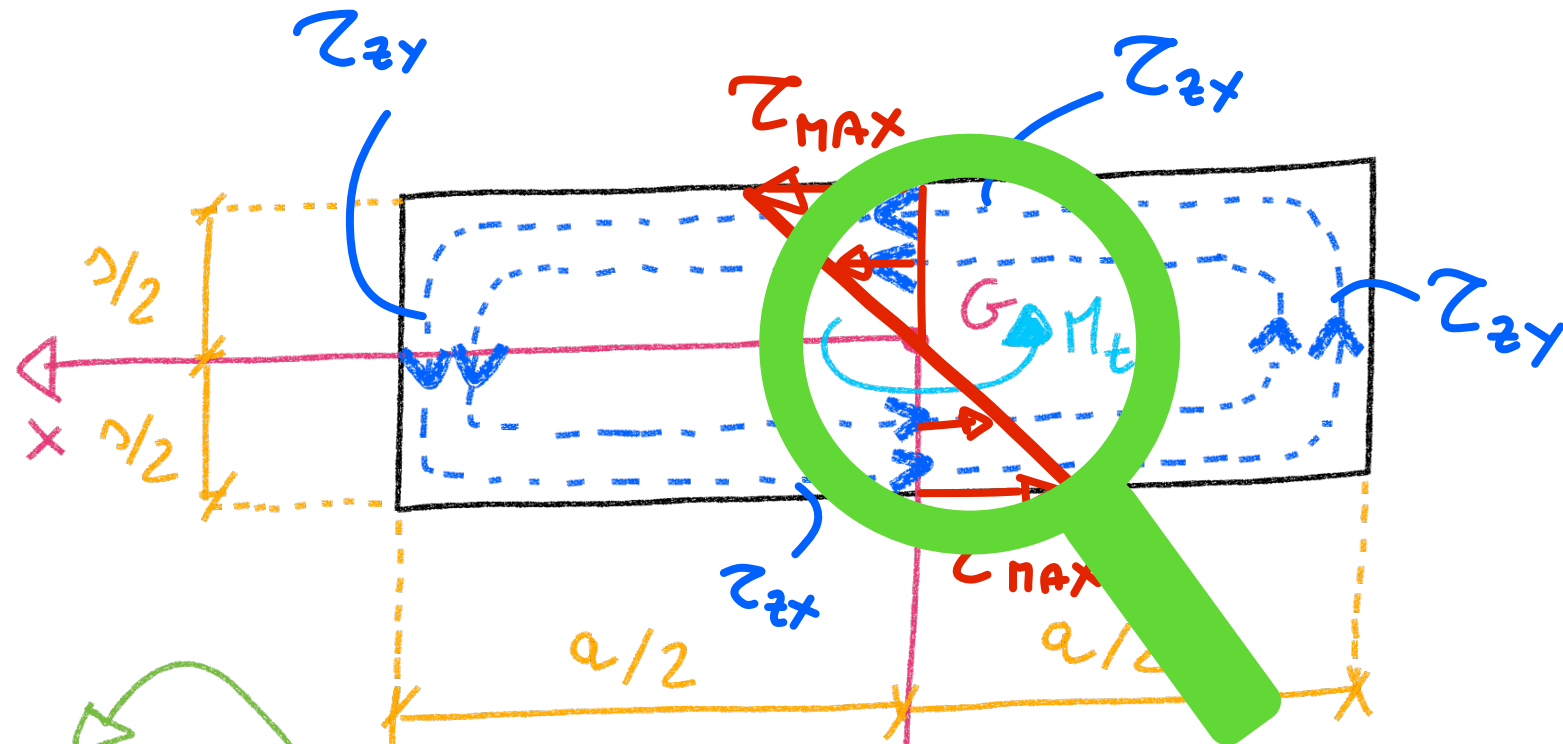
INDOTTO DA Z_{zx}

$$\begin{aligned}
 M_t^i &= |R| \frac{2}{3} \Delta \\
 &= \frac{2}{3} \Delta \left| -2 \frac{M_t}{I_t} a \frac{y^2}{2} \right|_0^{\Delta/2} \\
 &= \frac{2}{3} \Delta \cdot 2 \frac{M_t}{\frac{1}{3} a \Delta^3} a \frac{1}{2} \frac{\Delta^2}{4} \\
 &= \frac{M_t}{2}
 \end{aligned}$$

DOVE È FINITA LA
RESTANTE METÀ ($M_t/2$) ???

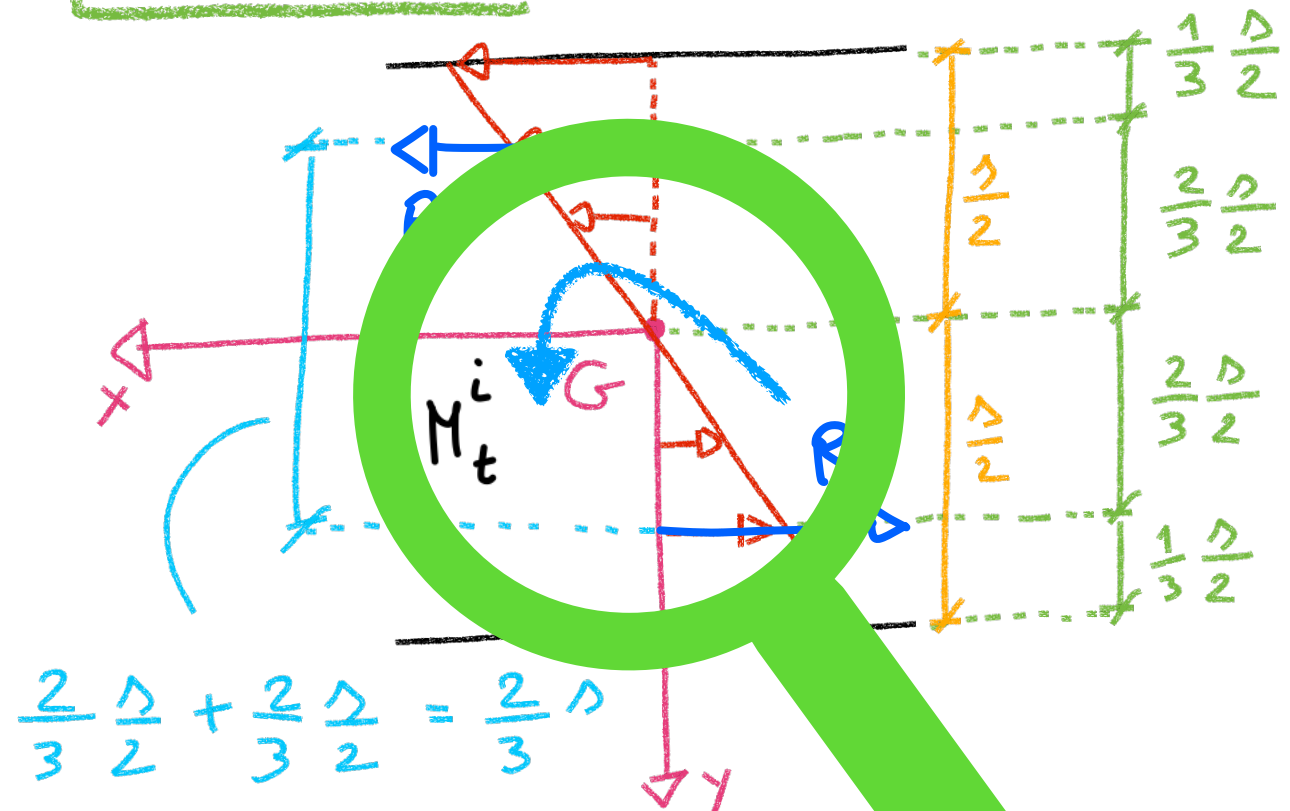
↳ Z_{zy} SUI LATI CORTI

↳ CONTRIBUTO NON TRASCURAB.
PERCHÉ $a/2 \gg \Delta$



$$\begin{aligned}
 Z_{zx} &= -2 \frac{M_t}{I_t} y \\
 I_t &= \frac{1}{3} a \Delta^3
 \end{aligned}$$

$$\begin{aligned}
 -\frac{a}{2} < x < \frac{a}{2} \\
 0 < y < \frac{\Delta}{2}
 \end{aligned}$$



$$\frac{2}{3} \frac{\Delta}{2} + \frac{2}{3} \frac{\Delta}{2} = \frac{2}{3} \Delta$$

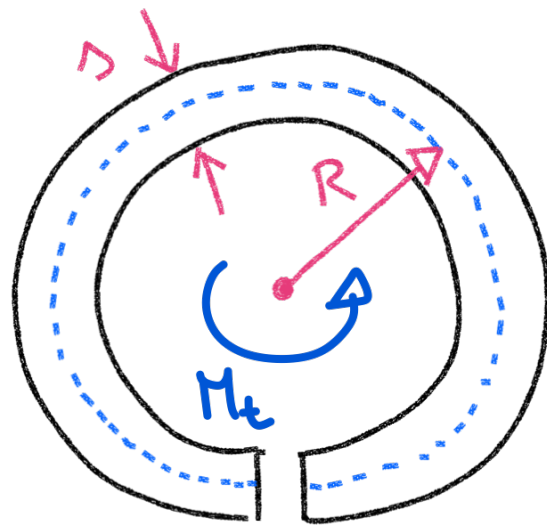
SEZIONI RICONDUCEBILI A RETT. SOTT.

SEZIONI RICONDUCIBILI A RETT. SOTT.

(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER
SEZIONE RETTANGOLARE SOTTILE (APERTE)

SEZIONI RICONDUCIBILI A RETT. SOTT.

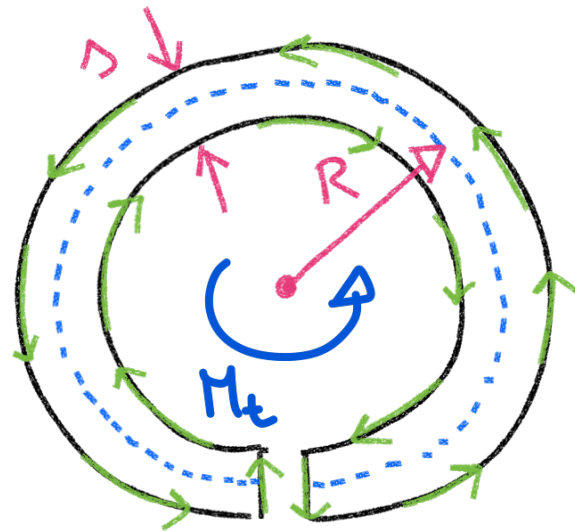
(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER SEZIONE RETTANGOLARE SOTTILE (APERTE)



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SEZIONI RICONDUCIBILI A RETT. SOTT.

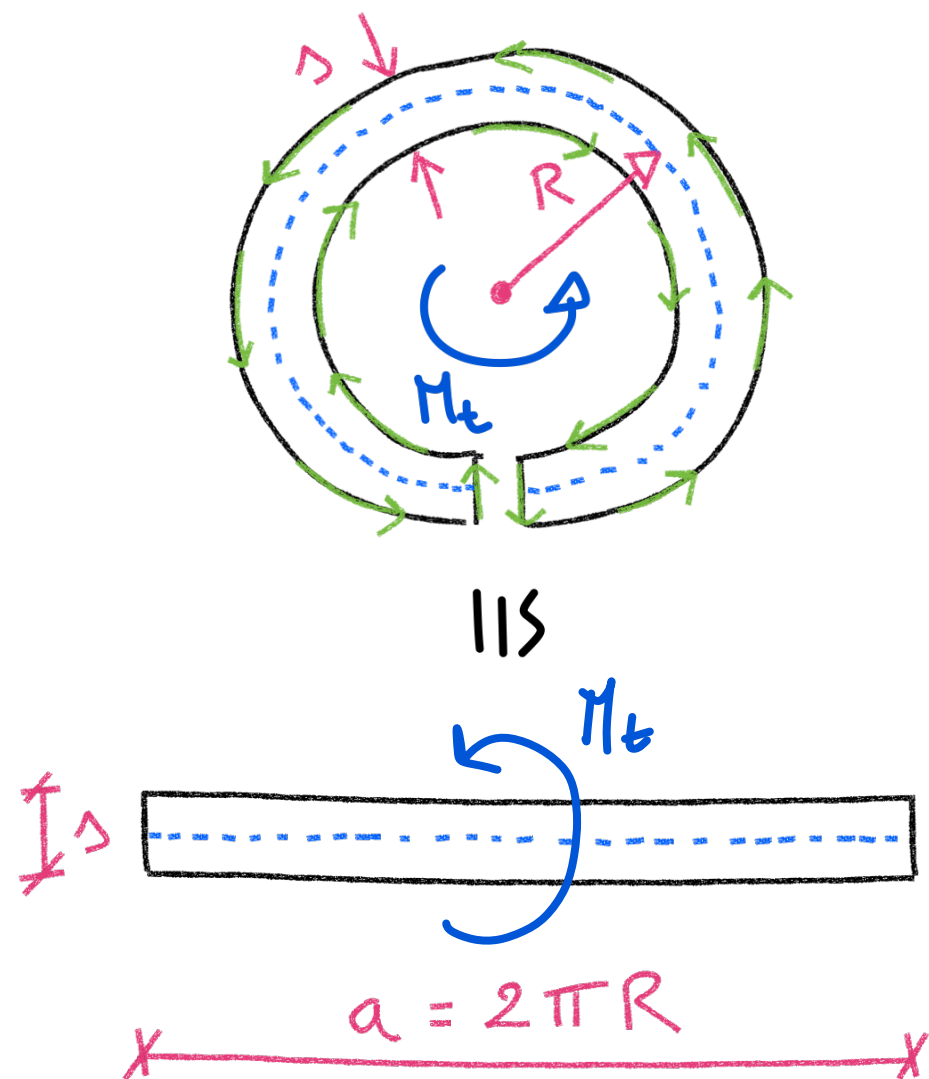
(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER SEZIONE RETTANGOLARE SOTTILE (APERTE)



115

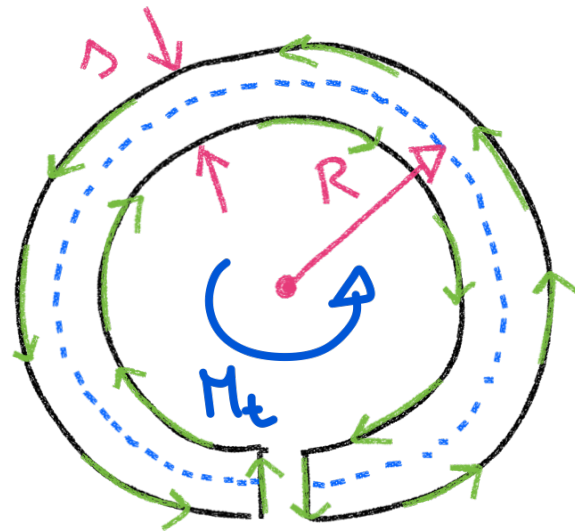
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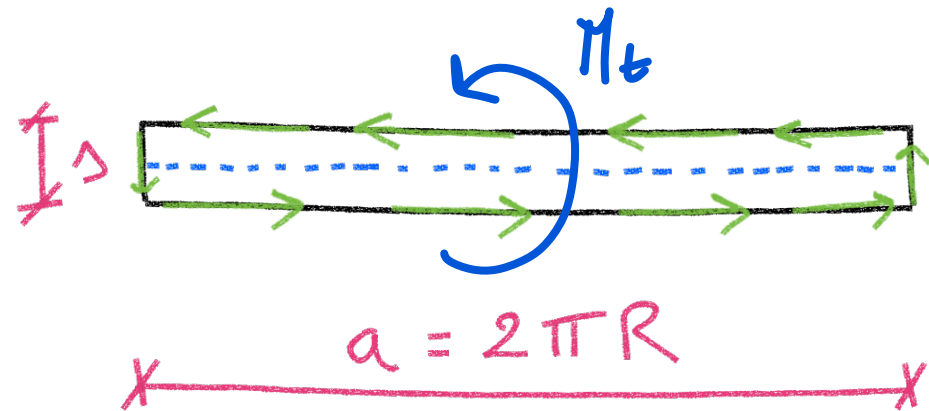


SEZIONI RICONDUCIBILI A RETT. SOTT.

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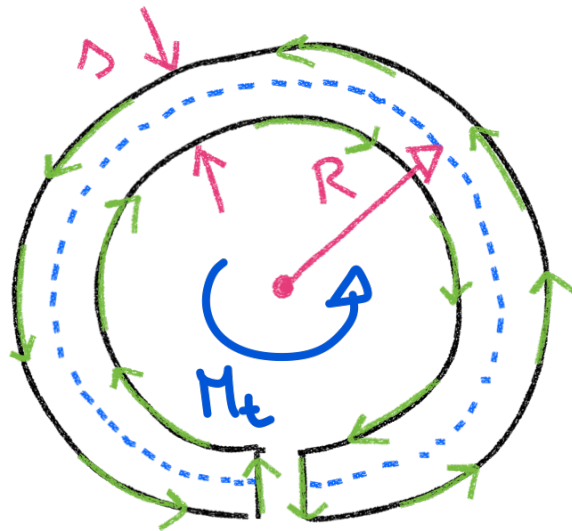


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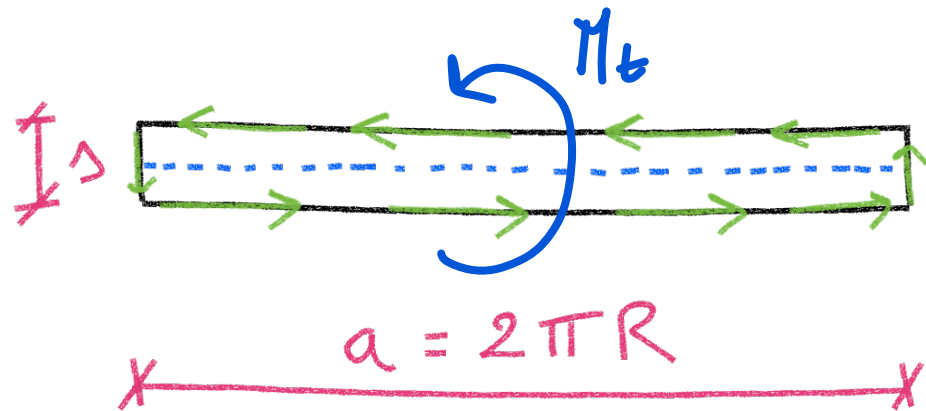


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(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER SEZIONE RETTANGOLARE SOTTILE (APERTE)



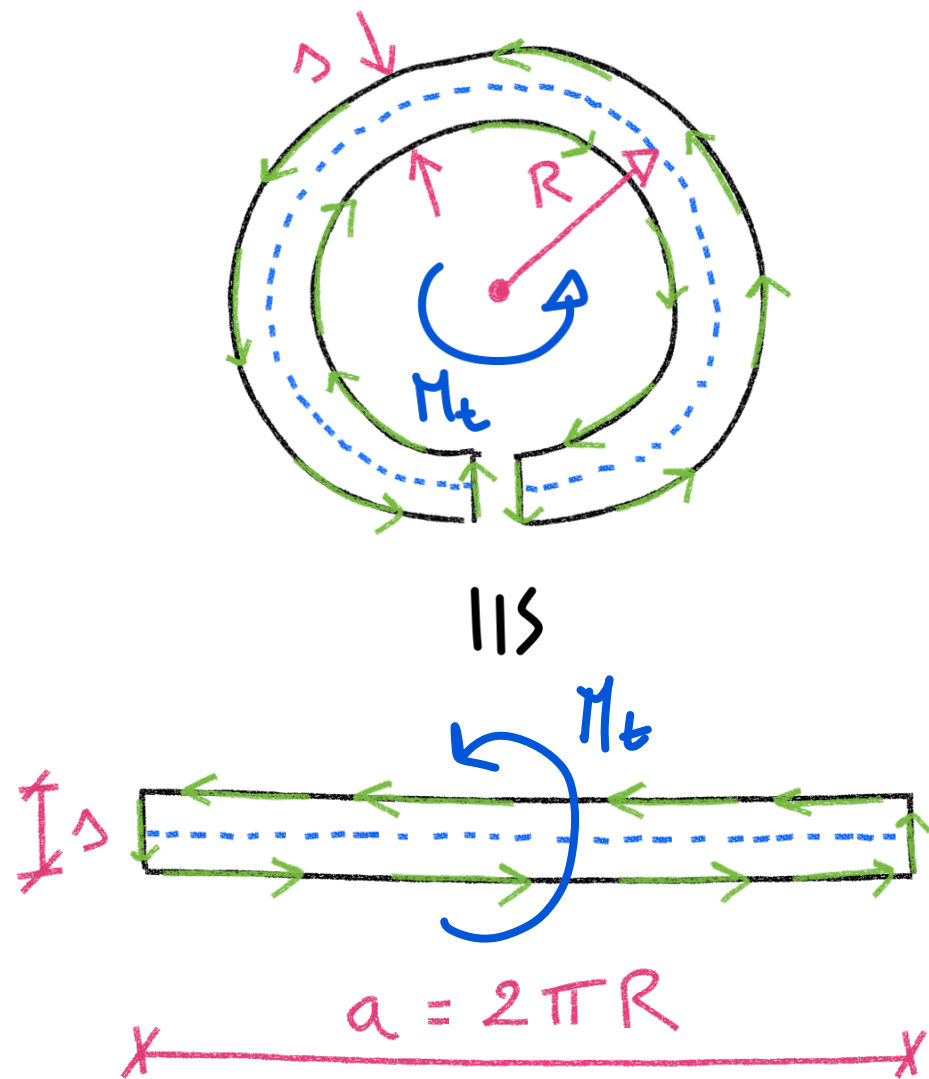
|||



$$I_t = \frac{1}{3} a \delta^3 = \frac{1}{3} (2\pi R) \delta^3$$

SEZIONI RICONDUCIBILI A RETT. SOTT.

(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER SEZIONE RETTANGOLARE SOTTILE (APERTE)

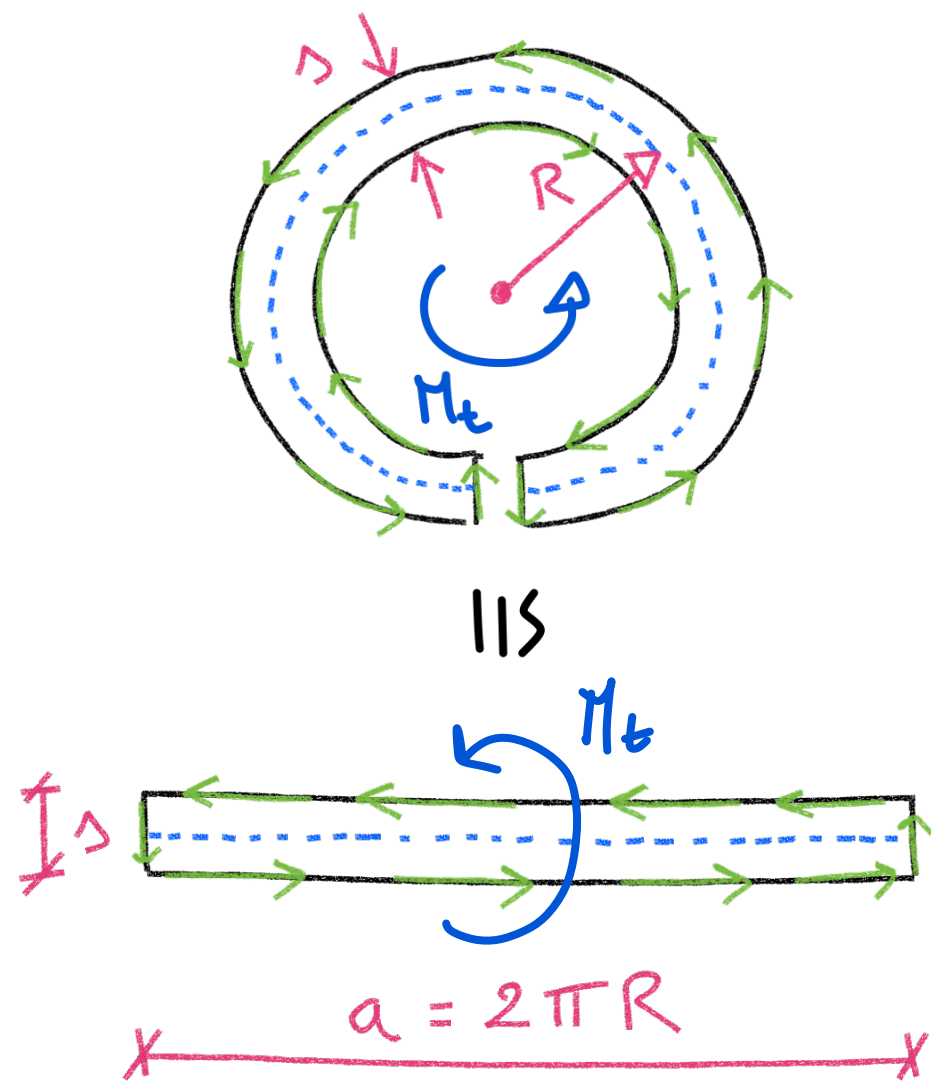


$$I_t = \frac{1}{3} a \Delta^3 = \frac{1}{3} (2\pi R) \Delta^3$$

$$Z_{MAX} = \frac{M_t}{I_t} \Delta = \frac{3}{2} \frac{M_t}{\pi R \Delta^2}$$

SEZIONI RICONDUCIBILI A RETT. SOTT.

(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER SEZIONE RETTANGOLARE SOTTILE (APERTE)

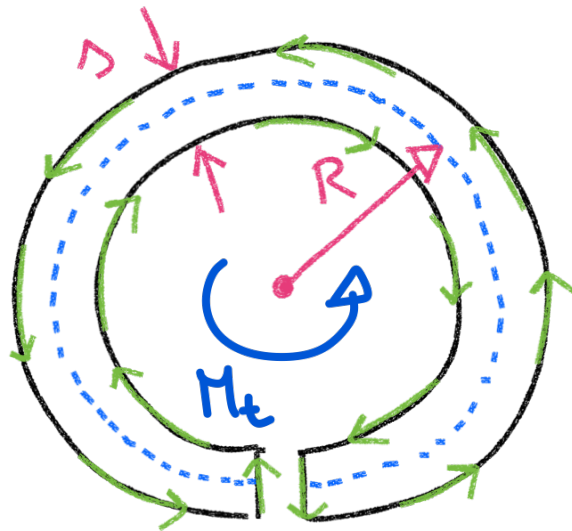


$$I_t = \frac{1}{3} a \delta^3 = \frac{1}{3} (2\pi R) \delta^3$$

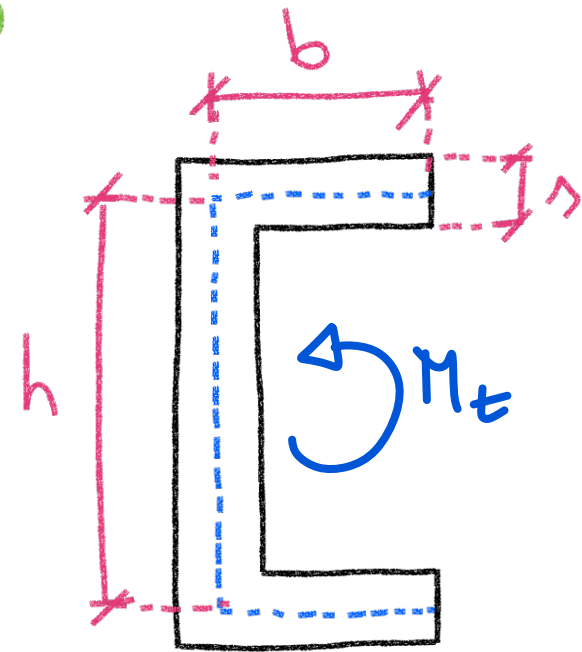
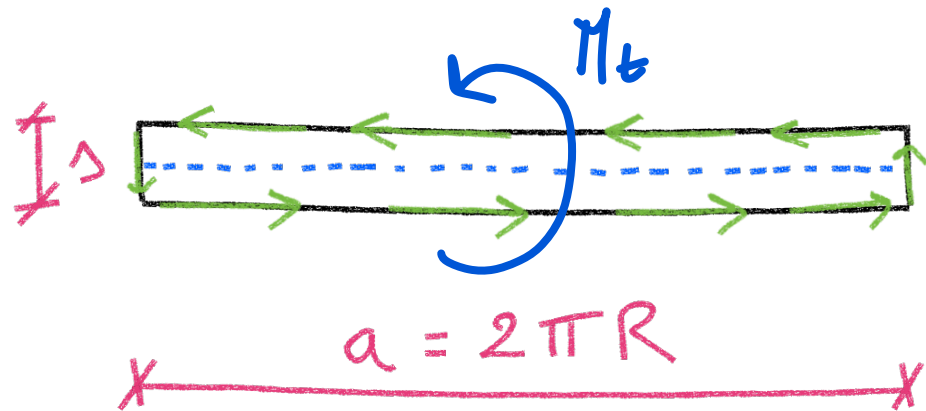
$$Z_{MAX} = \frac{M_t}{I_t} \delta = \frac{3}{2} \frac{M_t}{\pi R \delta^2} \dots \Theta = ..$$

SEZIONI RICONDUCIBILI A RETT. SOTT.

(*) APPROX. SEZIONI GENERICHE SOTTILI A RISULTATI OTTENUTI PER SEZIONE RETTANGOLARE SOTTILE (APERTE)



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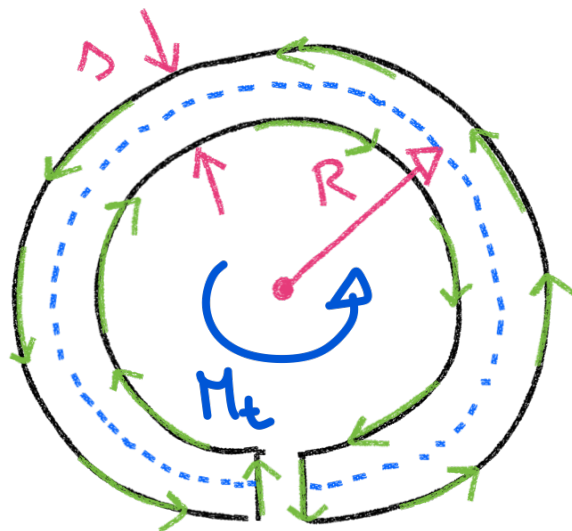


$$I_t = \frac{1}{3} a \Delta^3 = \frac{1}{3} (2\pi R) \Delta^3$$

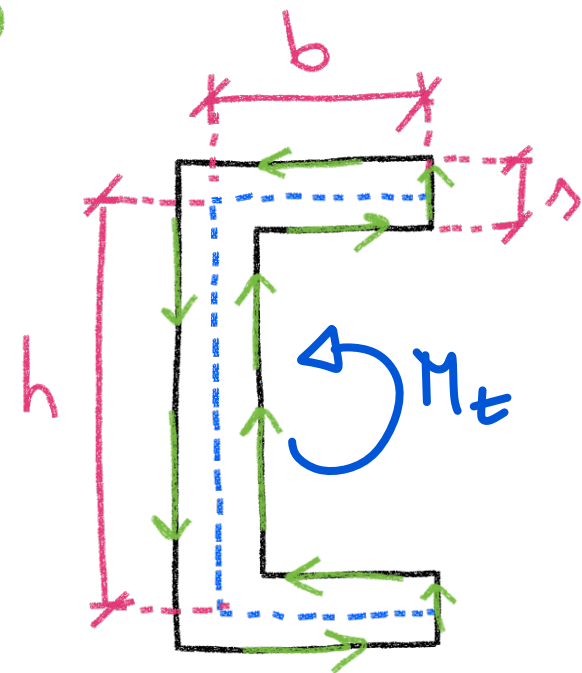
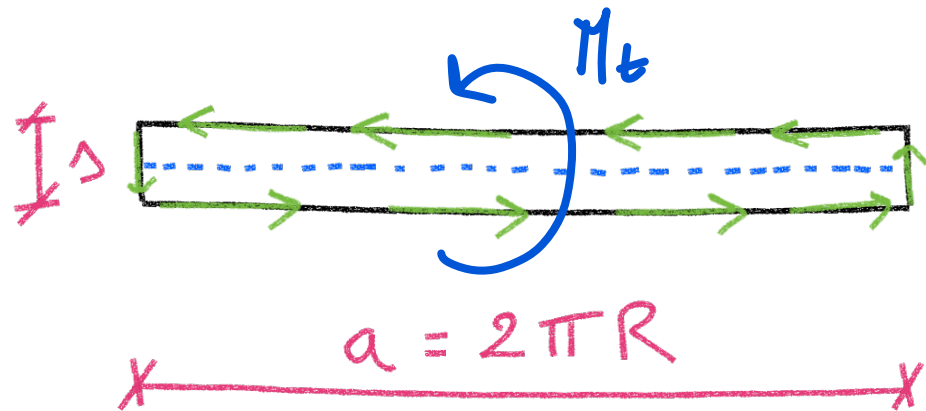
$$Z_{MAX} = \frac{M_t}{I_t} \Delta = \frac{3}{2} \frac{M_t}{\pi R \Delta^2} \dots \Theta = ..$$

SEZIONI RICONDUCIBILI A RETT. SOTT.

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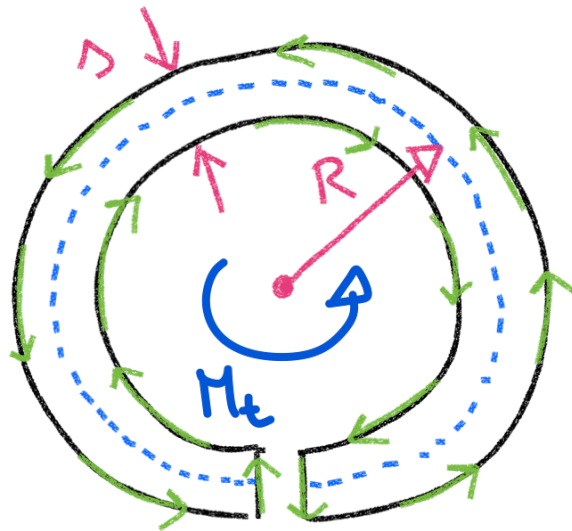


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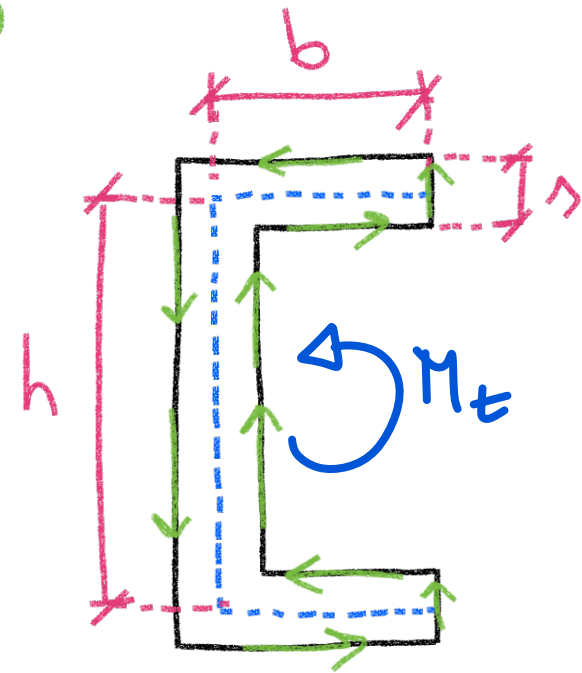
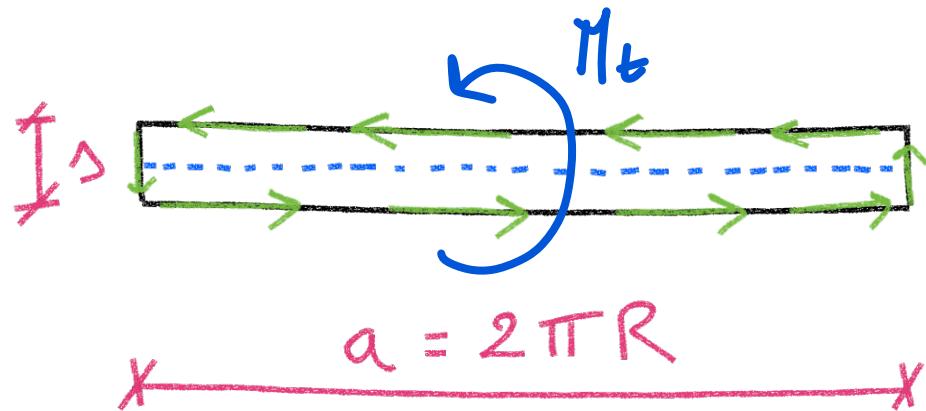
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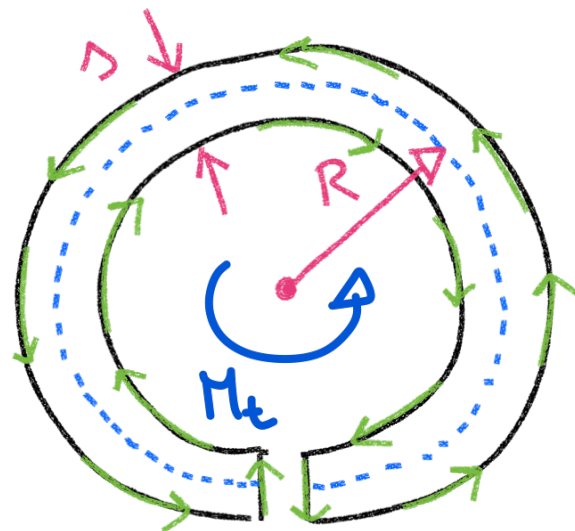
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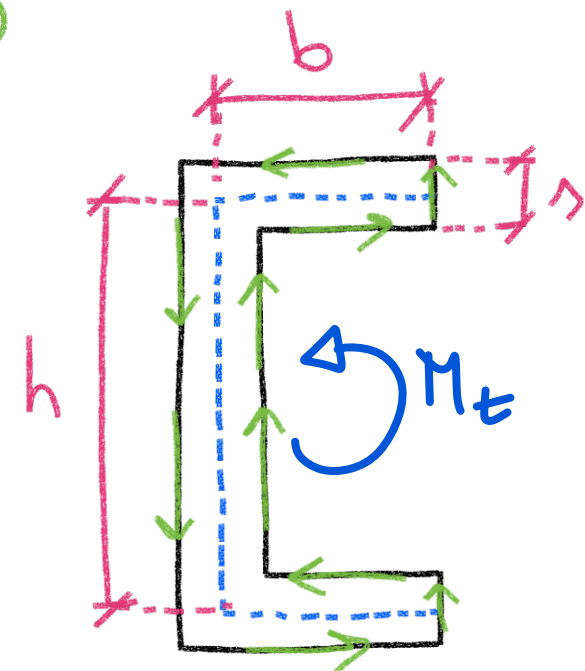
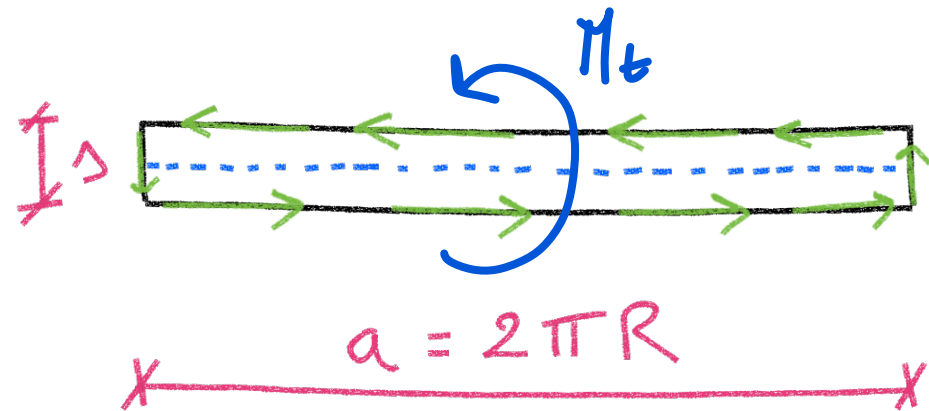
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SEZIONI RICONDUCIBILI A RETT. SOTT.

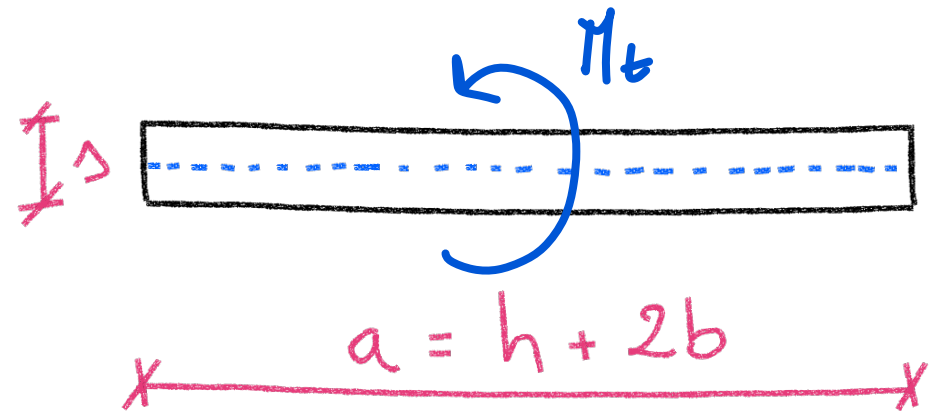
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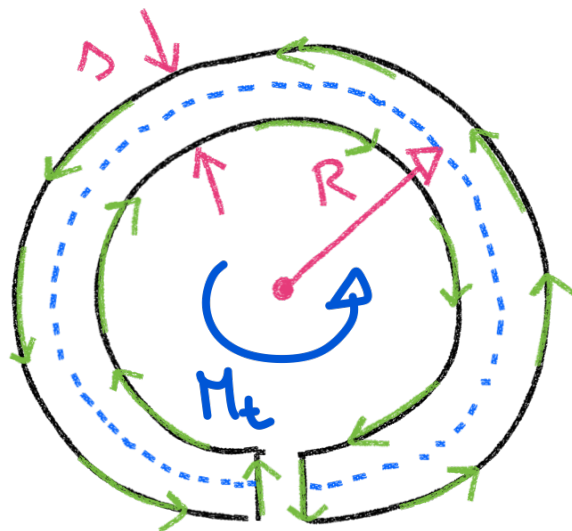


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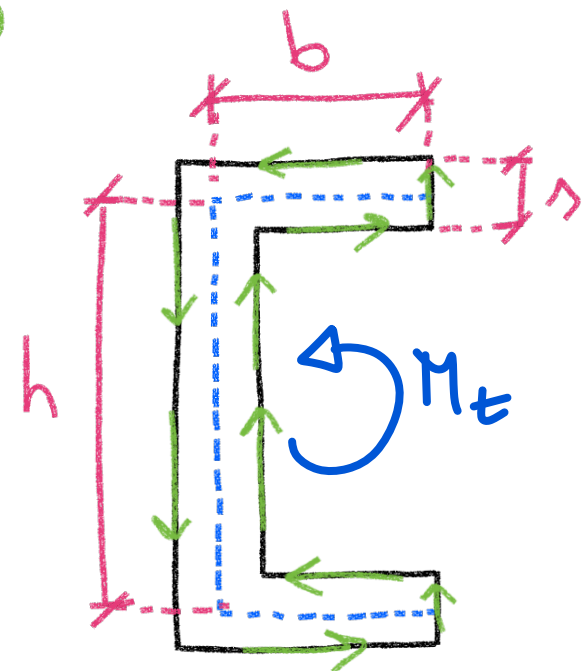
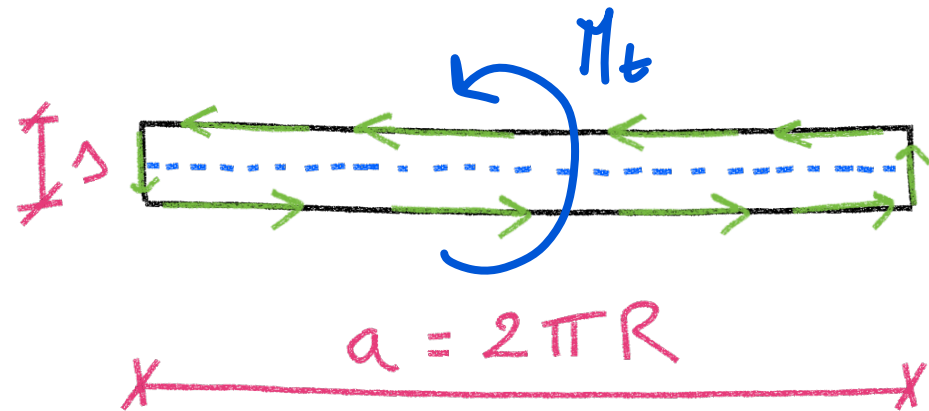
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SEZIONI RICONDUCIBILI A RETT. SOTT.

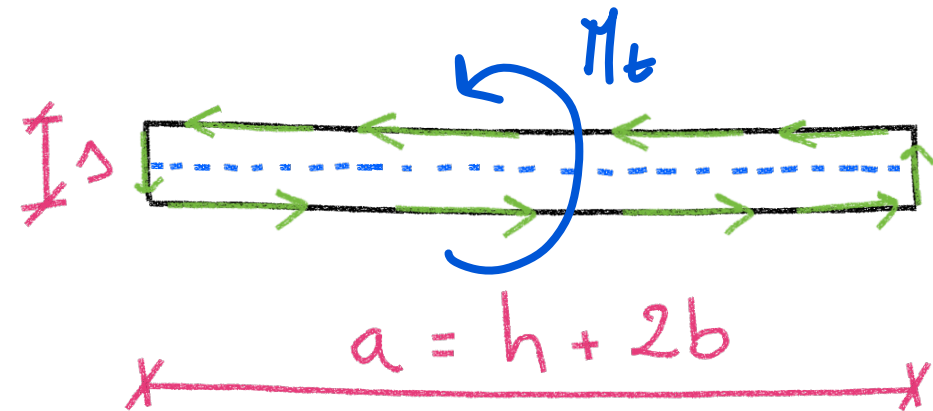
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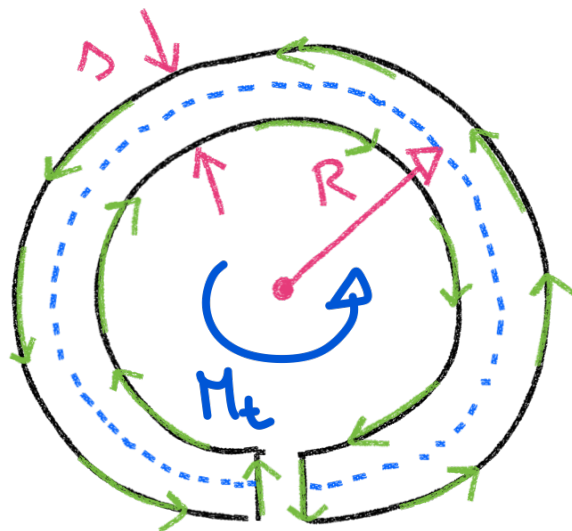


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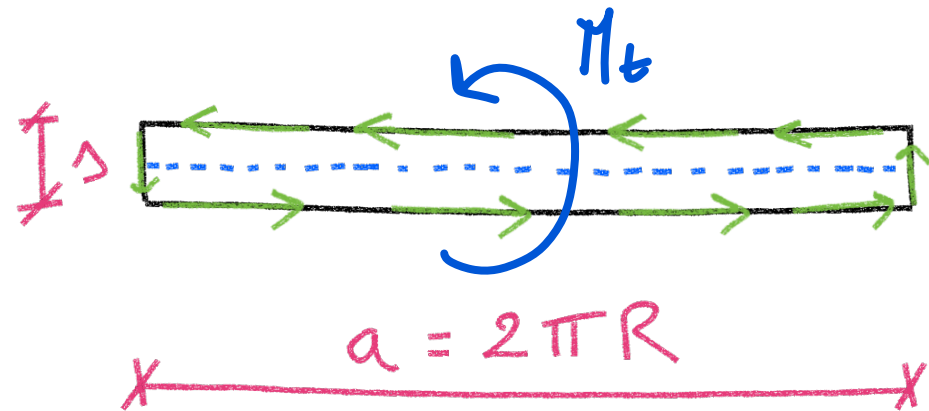
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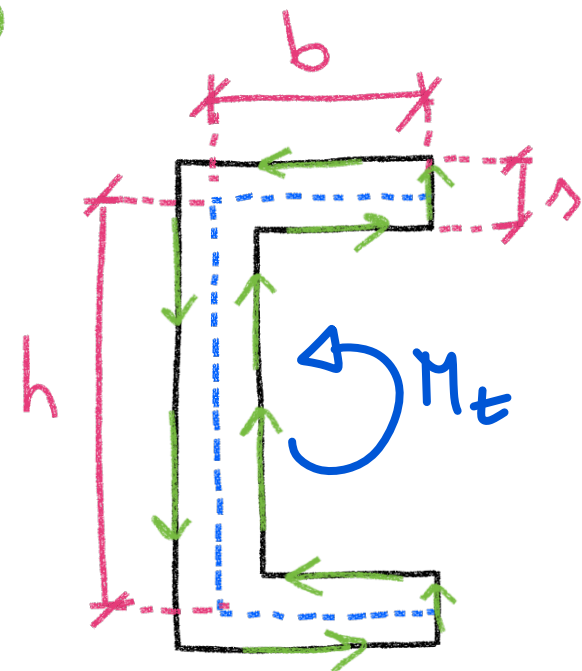


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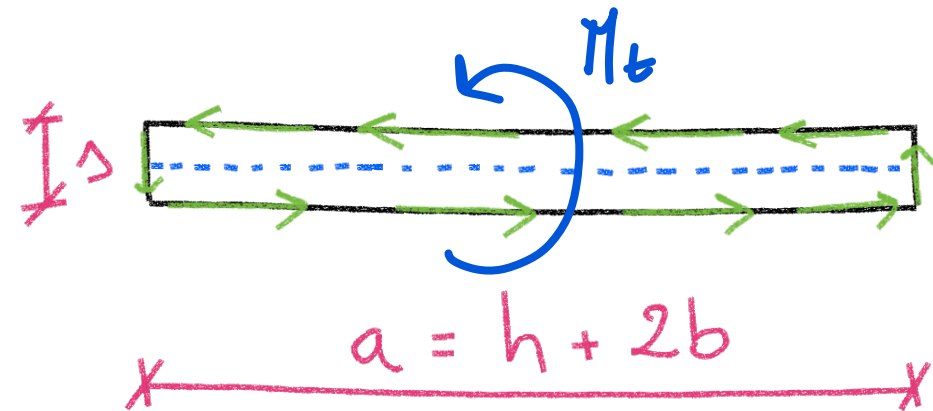


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$$I_t = \frac{1}{3} a \Delta^3 = \frac{1}{3} (h + 2b) \Delta^3$$

$$Z_{MAX} = \frac{M_t}{I_t} \Delta = \frac{3 M_t}{(h + 2b) \Delta^2}$$

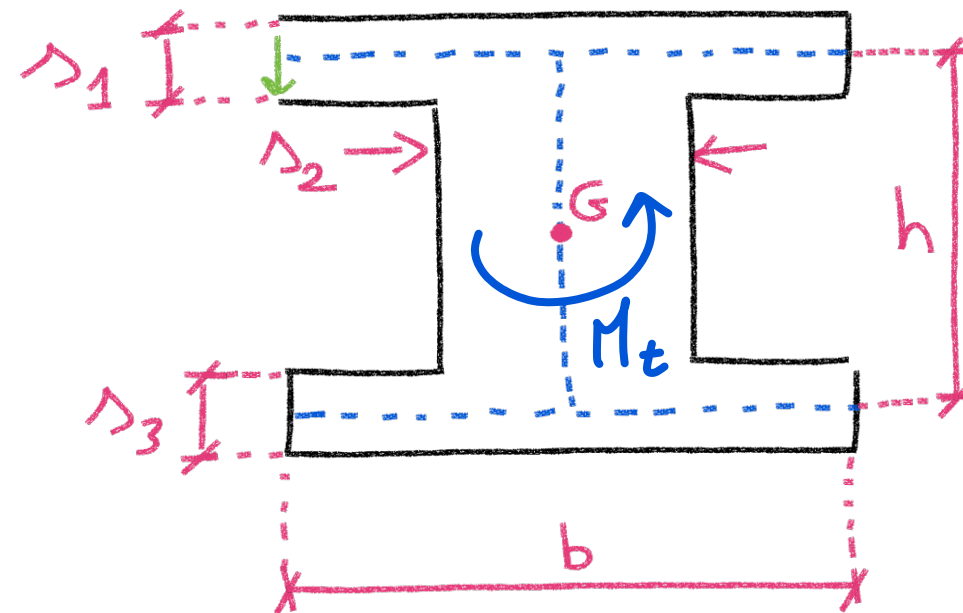
SEZIONI APERTE SOTTILI (Non Costante)

SEZIONI APERTE SOTTILI (Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

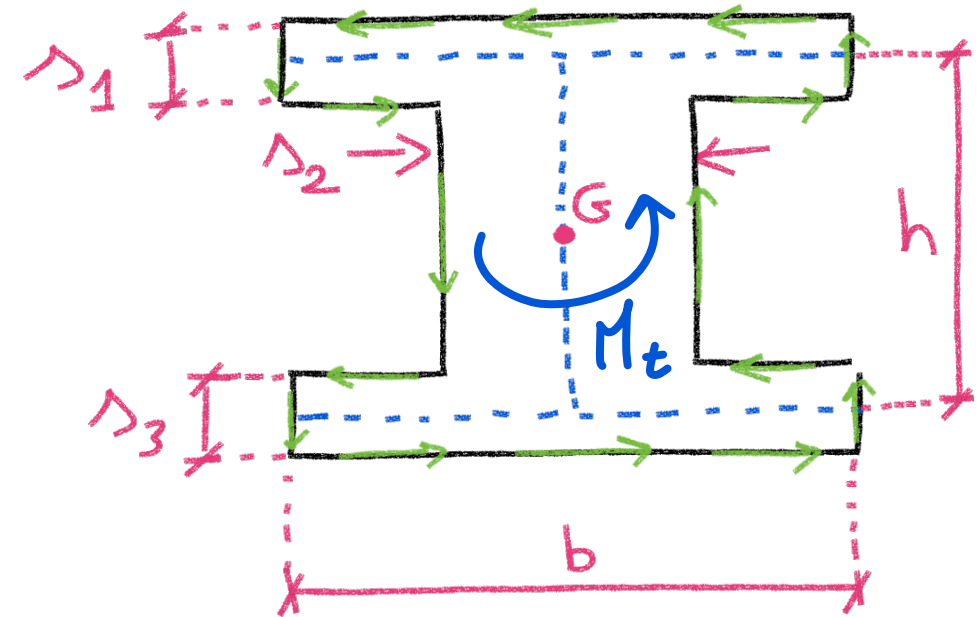
SEZIONI APERTE SOTTILI (γ Non Costante)

Ripartizione Momento Torcente



SEZIONI APERTE SOTTILI (γ Non Costante)

Ripartizione Momento Torcente



SEZIONI APERTE SOTTILI ($\rightarrow$ Non Costante)

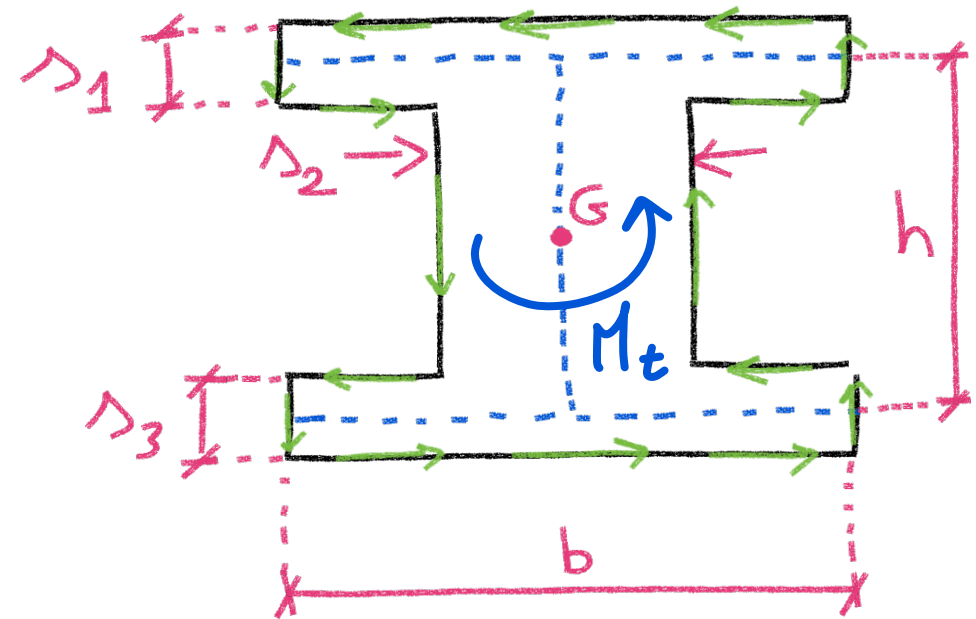
Ripartizione Momento Torcente

$$I_t = \sum_m I_{ti}$$

$\Uparrow$

1) m RETTANGOLI SOTTILI

$$\Delta_i, a_i \text{ con } i = 1 \dots m \Rightarrow I_{ti} = \frac{1}{3} a_i \Delta_i^3$$



SEZIONI APERTE SOTTILI ($\rightarrow$ Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

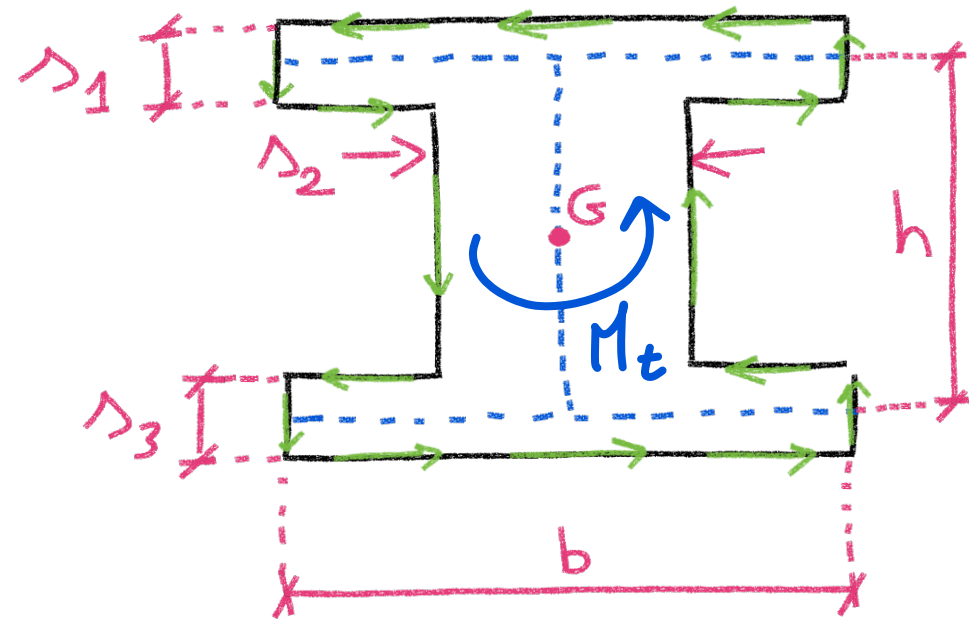
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2) CONGRUENZA : $\Theta_i = \Theta$
(stessa rotazione)



SEZIONI APERTE SOTTILI ($\rightarrow$ Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

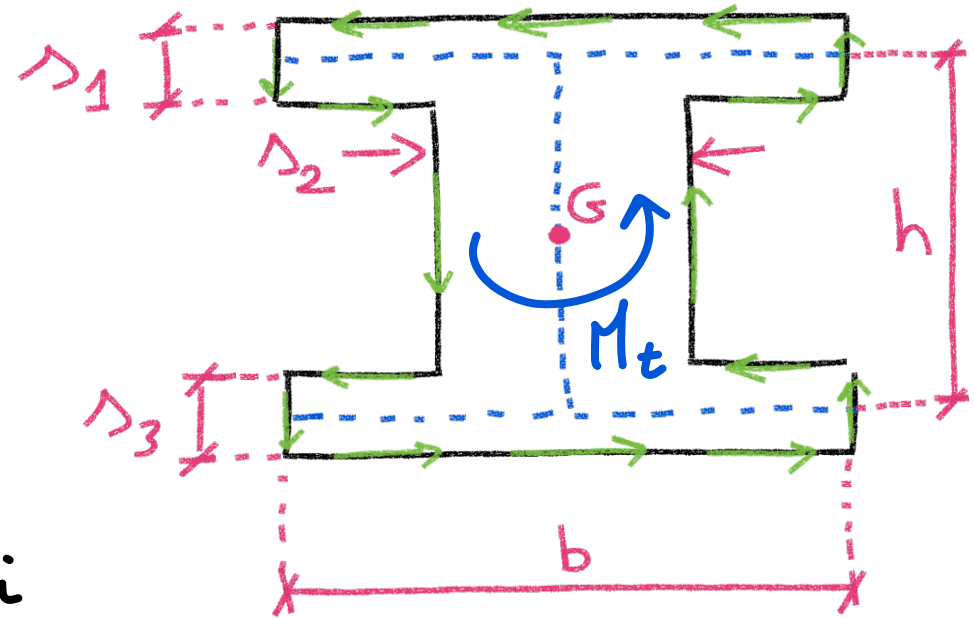
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SEZIONI APERTE SOTTILI ($\rightarrow$ Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

$$I_t = \sum_m I_{ti}$$

$\Uparrow$

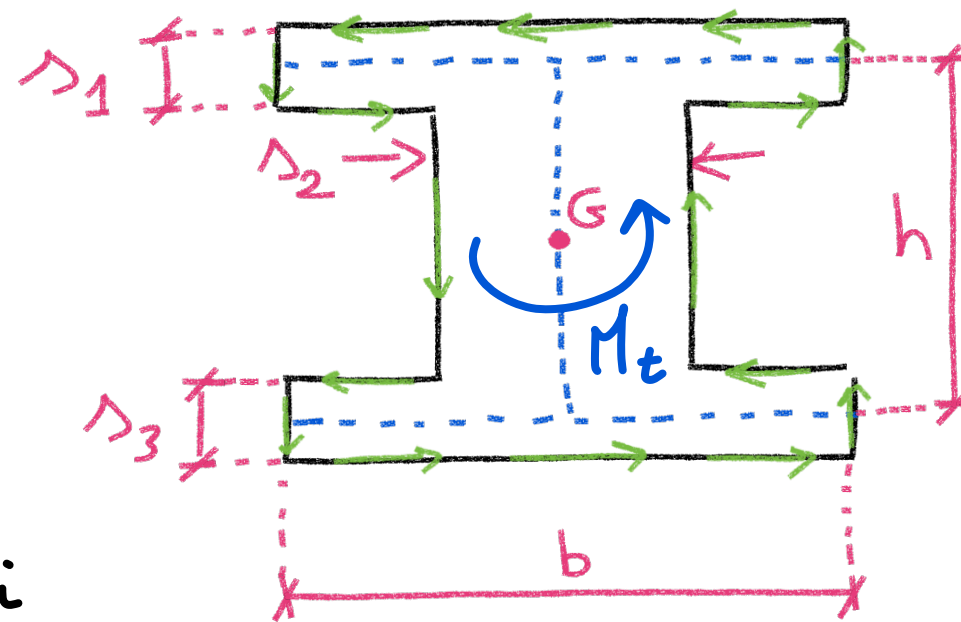
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4) LEGAME COSTITUTIVO (ELIO) : $\Theta_i = \frac{M_{ti}}{G I_{ti}}$



SEZIONI APERTE SOTTILI ($\rightarrow$ Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

$$I_t = \sum_m I_{ti}$$

$\Uparrow$

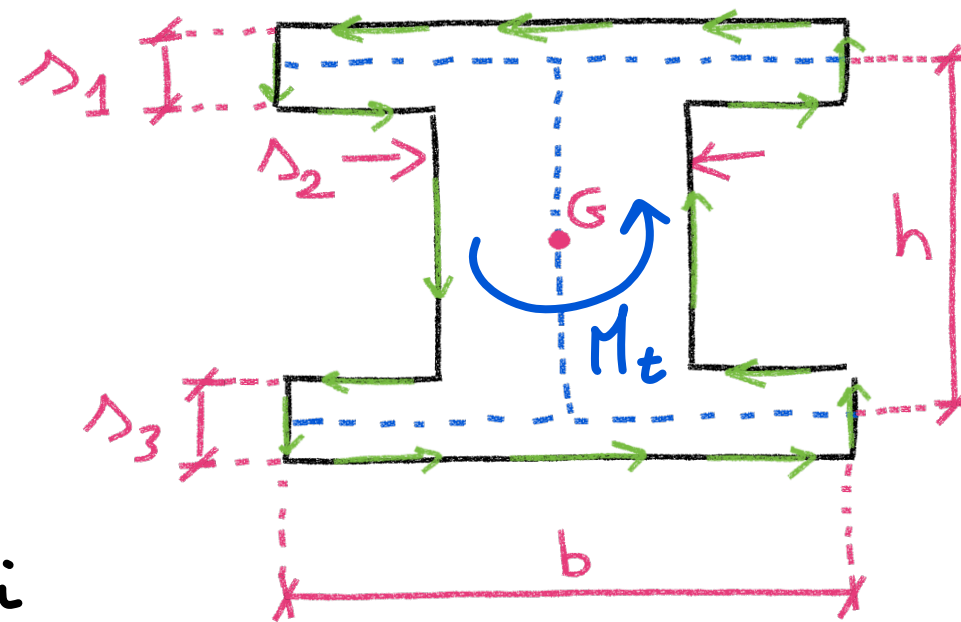
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SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

$$I_t = \sum_m I_{ti}$$

$\Uparrow$

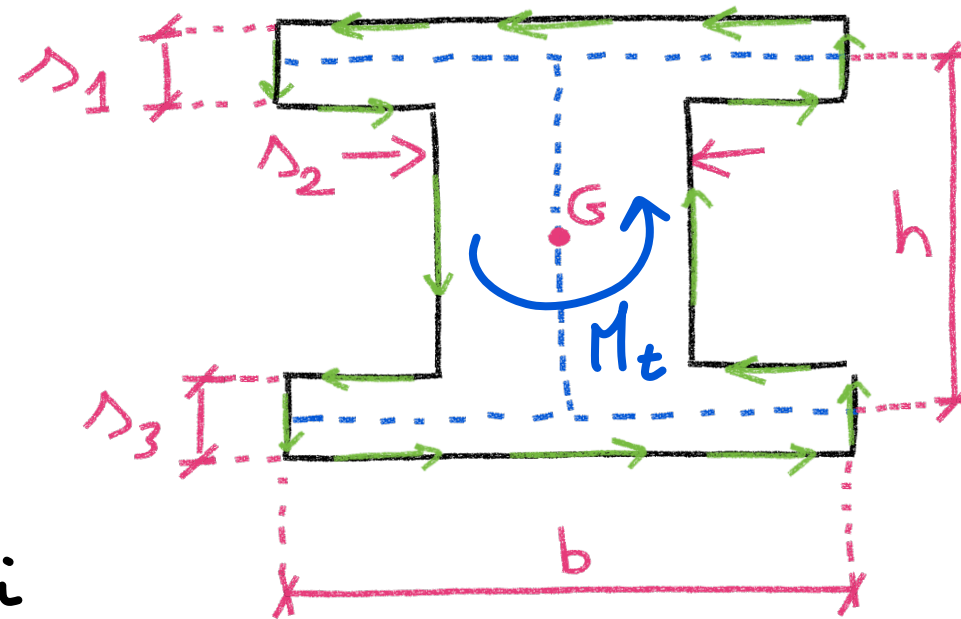
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$\hookrightarrow M_t = \sum_m M_{ti}$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

$$I_t = \sum_m I_{ti}$$

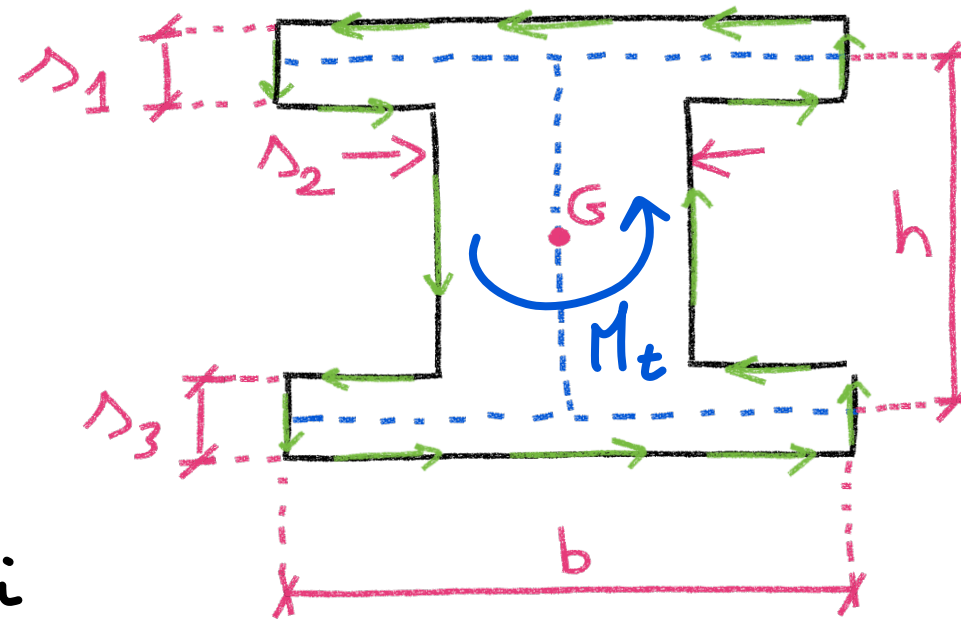
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$$M_t = \sum_m M_{ti} = \sum_m G \Theta_i I_{ti}$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

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$\Uparrow$

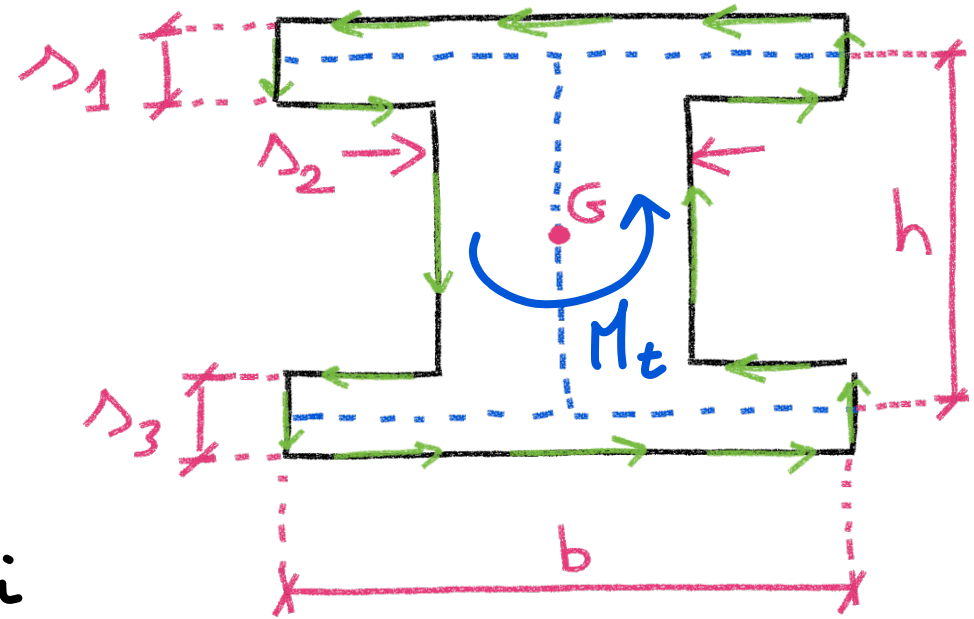
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$$\rightarrow M_t \stackrel{(3)}{=} \sum_m M_{ti} \stackrel{(4)}{=} \sum_m G \Theta_i I_{ti} \stackrel{(2)}{=} \sum_m G \Theta I_{ti}$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

$$I_t = \sum_m I_{ti}$$

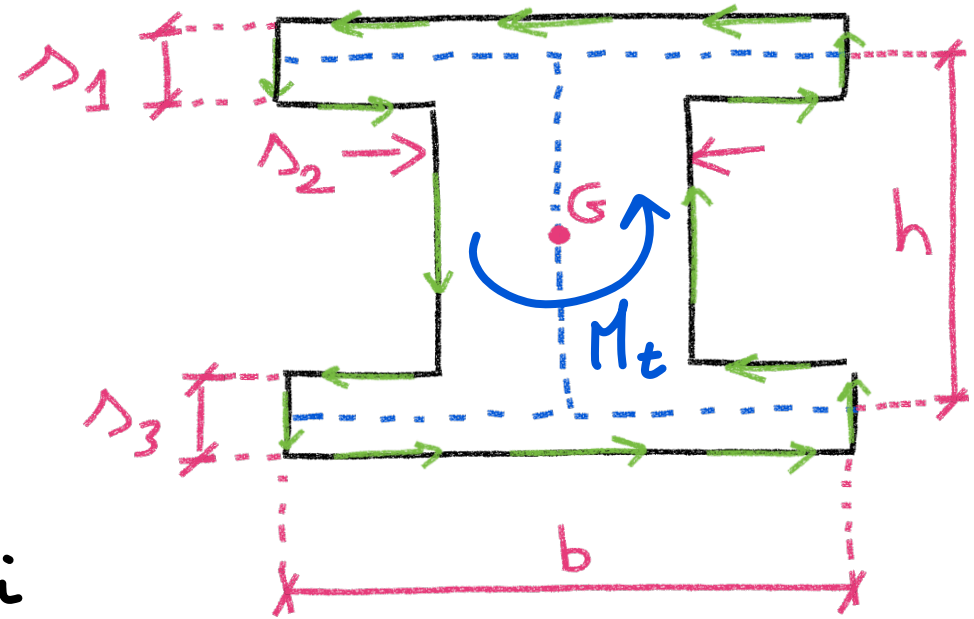
① m RETTANGOLI SOTTILI

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(stessa rotazione)

③ EQUIVALENZA STATICA : $M_t = \sum_m M_{ti}$

④ LEGAME COSTITUTIVO (ELIO) : $\Theta_i = \frac{M_{ti}}{G I_{ti}} \rightarrow M_{ti} = G \Theta_i I_{ti}$



$$\rightarrow M_t \stackrel{\textcircled{3}}{=} \sum_m M_{ti} \stackrel{\textcircled{4}}{=} \sum_m G \Theta_i I_{ti} \stackrel{\textcircled{2}}{=} \sum_m G \Theta I_{ti} \stackrel{\textcircled{1}}{=} G \Theta I_t$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

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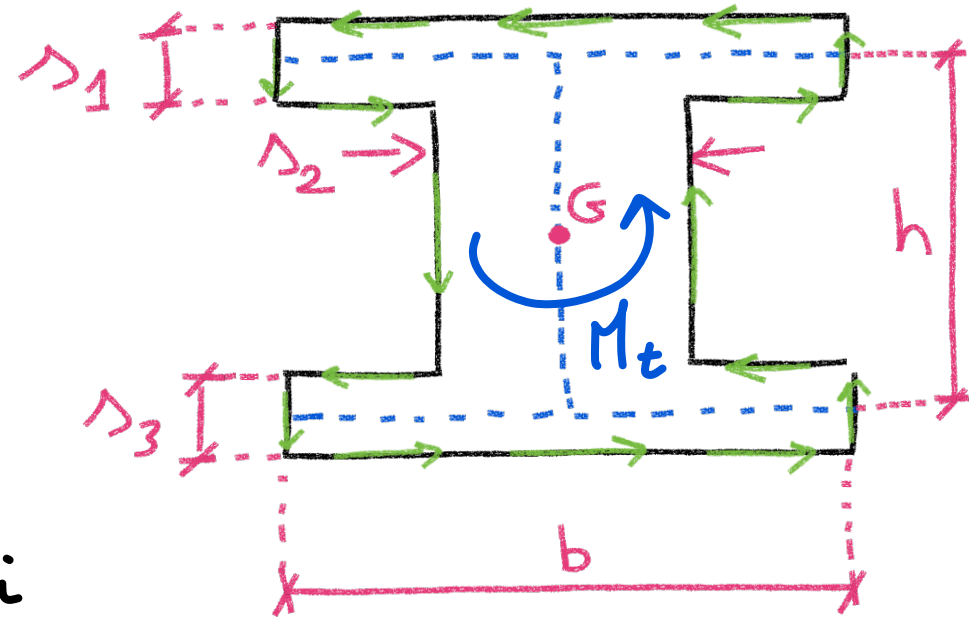
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$$M_t \stackrel{③}{=} \sum_m M_{ti} \stackrel{④}{=} \sum_m G \Theta_i I_{ti} \stackrel{②}{=} \sum_m G \Theta I_{ti} \stackrel{①}{=} G \Theta I_t$$

$$\stackrel{②}{\rightarrow} \Theta_i = \frac{M_{ti}}{G I_{ti}}$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

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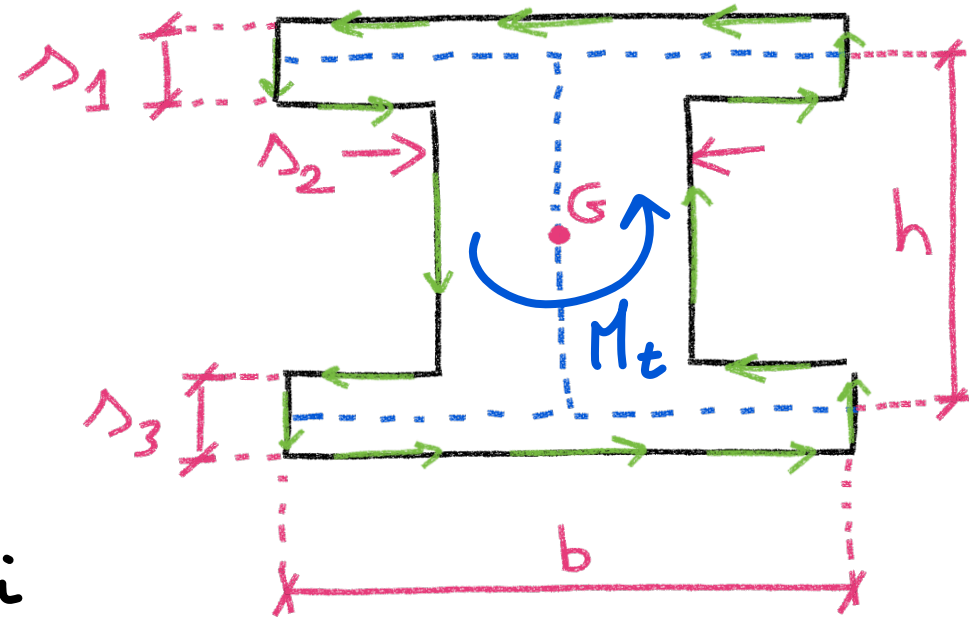
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$$M_t \stackrel{③}{=} \sum_m M_{ti} \stackrel{④}{=} \sum_m G \Theta_i I_{ti} \stackrel{②}{=} \sum_m G \Theta I_{ti} \stackrel{①}{=} G \Theta I_t$$

$$\stackrel{②}{\Rightarrow} \Theta_i = \frac{M_{ti}}{G I_{ti}} = \Theta = \frac{M_t}{G I_t}$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

$$I_t = \sum_m I_{ti}$$

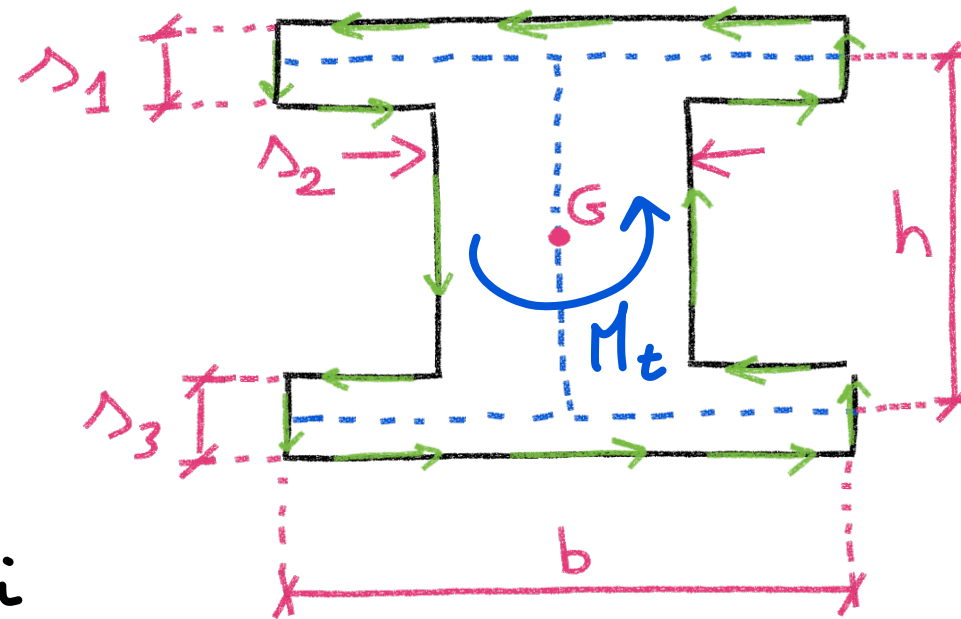
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2) CONGRUENZA : $\Theta_i = \Theta$
(stessa rotazione)

3) EQUIVALENZA STATICA : $M_t = \sum_m M_{ti}$

4) LEGAME COSTITUTIVO (ELIO) : $\Theta_i = \frac{M_{ti}}{G I_{ti}} \rightarrow M_{ti} = G \Theta_i I_{ti}$



$$M_t \stackrel{(3)}{=} \sum_m M_{ti} \stackrel{(4)}{=} \sum_m G \Theta_i I_{ti} \stackrel{(2)}{=} \sum_m G \Theta I_{ti} \stackrel{(1)}{=} G \Theta I_t$$

$$\stackrel{(2)}{\Theta_i} = \frac{M_{ti}}{G I_{ti}} = \Theta = \frac{M_t}{G I_t} \Rightarrow \frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t} \Rightarrow M_{ti} = \frac{I_{ti}}{I_t} M_t$$

SEZIONI APERTE SOTTILI (Non Costante)

RIPARTIZIONE MOMENTO TORCENTE

$$I_t = \sum_m I_{ti}$$

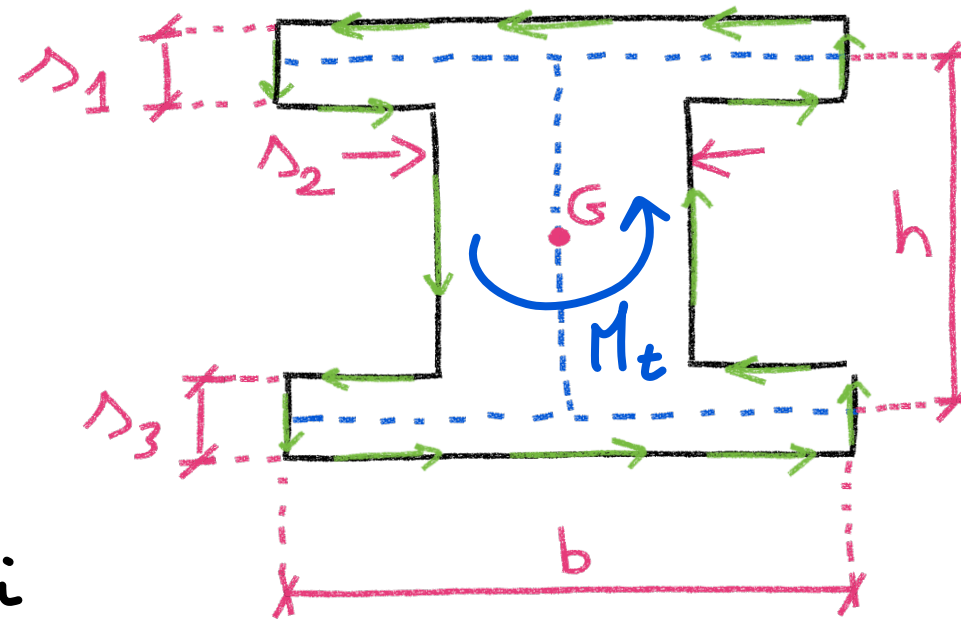
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(stessa rotazione)

3) EQUIVALENZA STATICA : $M_t = \sum_m M_{ti}$

4) LEGAME COSTITUTIVO (ELIO) : $\Theta_i = \frac{M_{ti}}{G I_{ti}} \rightarrow M_{ti} = G \Theta_i I_{ti}$



$$M_t = \sum_m M_{ti} = \sum_m G \Theta_i I_{ti} = \sum_m G \Theta I_{ti} = G \Theta I_t$$

$$\Theta_i = \frac{M_{ti}}{G I_{ti}} = \Theta = \frac{M_t}{G I_t} \Rightarrow \frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t} \Rightarrow M_{ti} = \frac{I_{ti}}{I_t} M_t$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

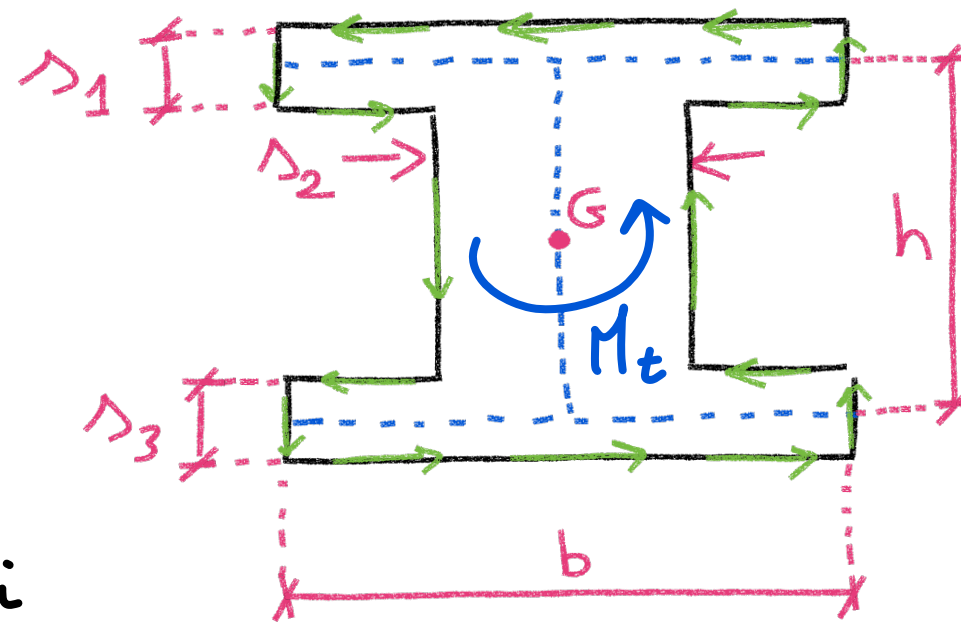
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$$\Delta_i, a_i \text{ con } i = 1 \dots m \Rightarrow I_{ti} = \frac{1}{3} a_i \Delta_i^3$$

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$$\frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t}$$



$$z_{max,i} = \frac{M_{ti}}{I_{ti}} \Delta_i = \frac{M_t}{I_t} \Delta_i$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

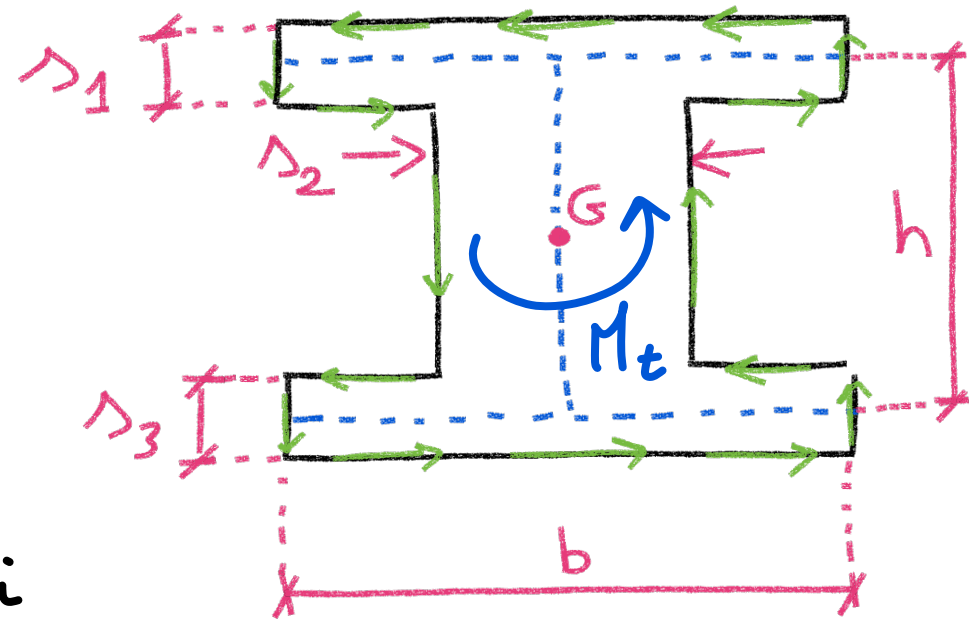
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$$\frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t}$$



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SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

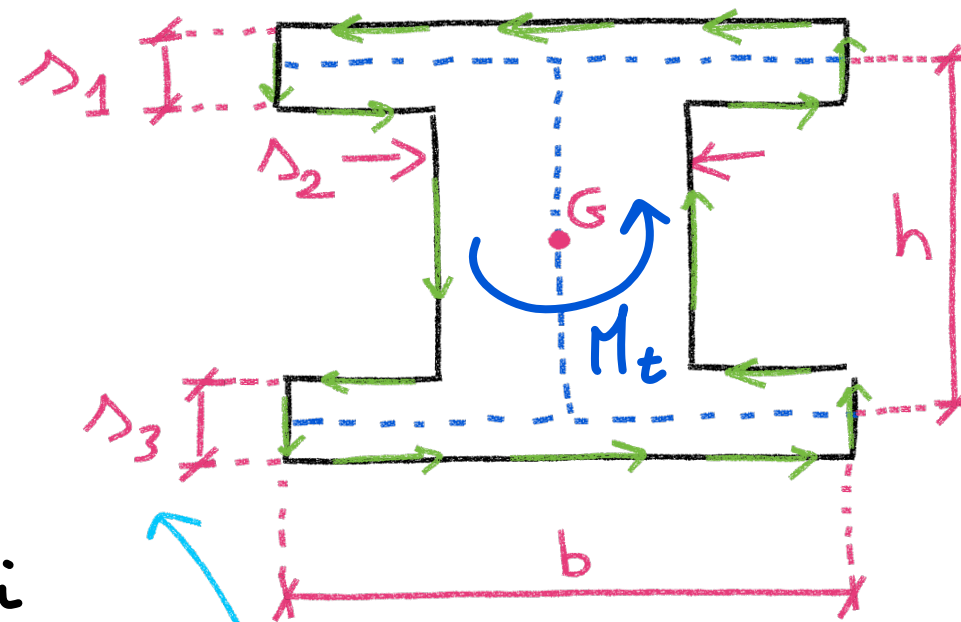
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3) EQUIVALENZA STATICA : $M_t = \sum_m M_{ti}$

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$$\frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t}$$



$$\tau_{\max,i} = \frac{M_{ti}}{I_{ti}} \Delta_i = \frac{M_t}{I_t} \Delta_i$$

SEZIONI APERTE SOTTILI ($\hookrightarrow$ Non Costante)

Ripartizione Momento Torcente

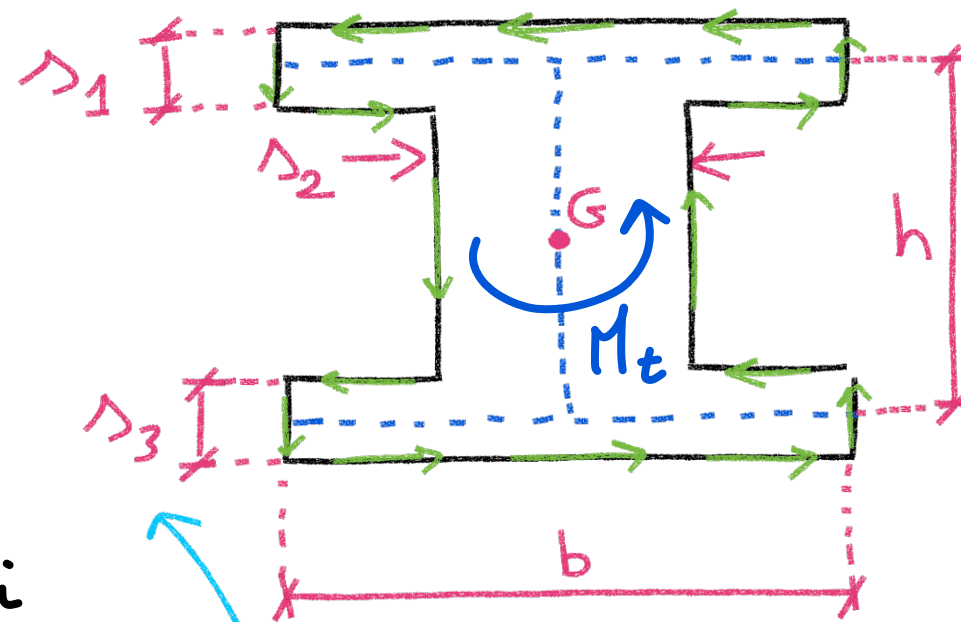
1) m RETTANGOLI SOTTILI

$$\Delta_i, a_i \text{ con } i = 1 \dots m \Rightarrow I_{ti} = \frac{1}{3} a_i \Delta_i^3$$

2) CONGRUENZA : $\Theta_i = \Theta$
(stessa rotazione)

3) EQUIVALENZA STATICA : $M_t = \sum_m M_{ti}$

4) LEGAME COSTITUTIVO (ELIO) : $\Theta_i = \frac{M_{ti}}{G I_{ti}}$



$$\frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t}$$



$$\tau_{\max, i} = \frac{M_{ti}}{I_{ti}} \Delta_i = \frac{M_t}{I_t} \Delta_i$$

$$\Delta \gg \Rightarrow \tau_{\max} \gg$$



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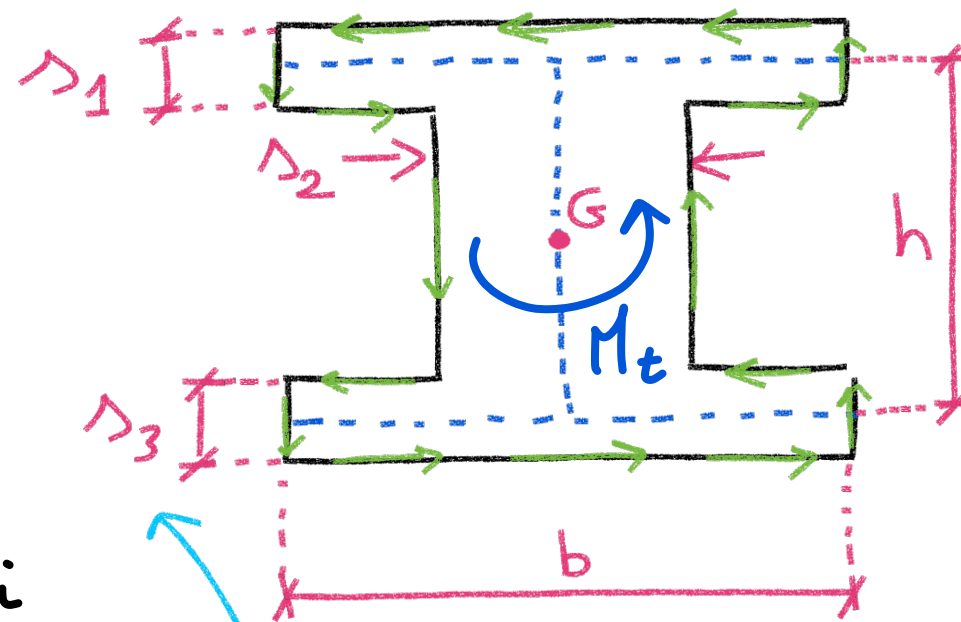
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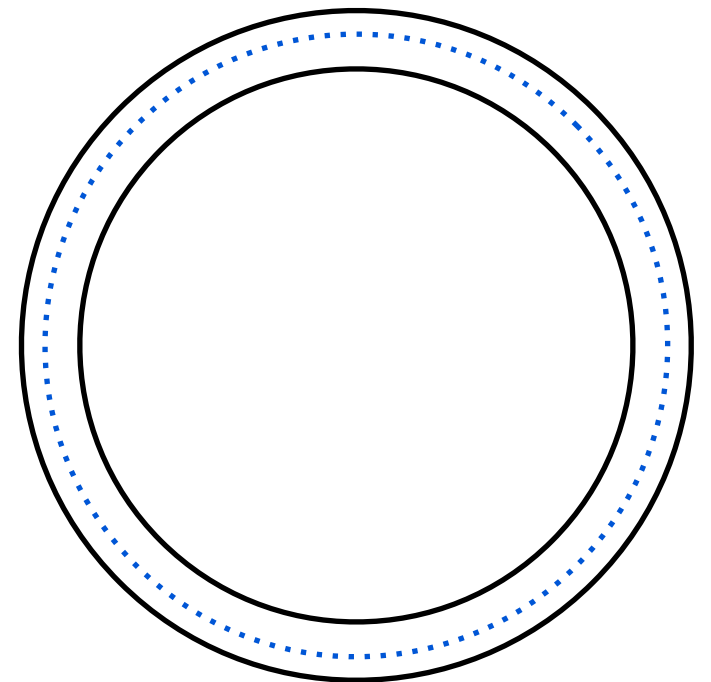
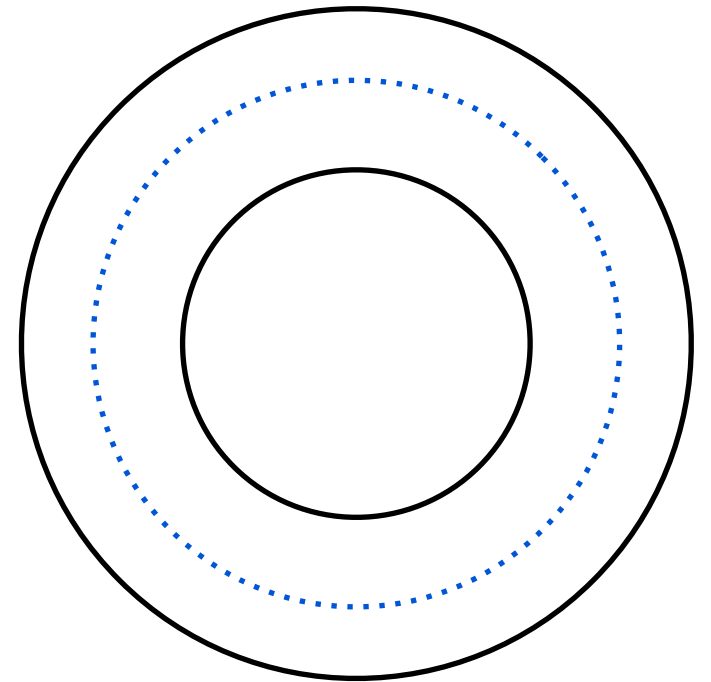


$$\tau_{\max, i} = \frac{M_{ti}}{I_{ti}} \Delta_i = \frac{M_t}{I_t} \Delta_i$$

$$\Delta \gg \Rightarrow \tau_{\max} \gg$$

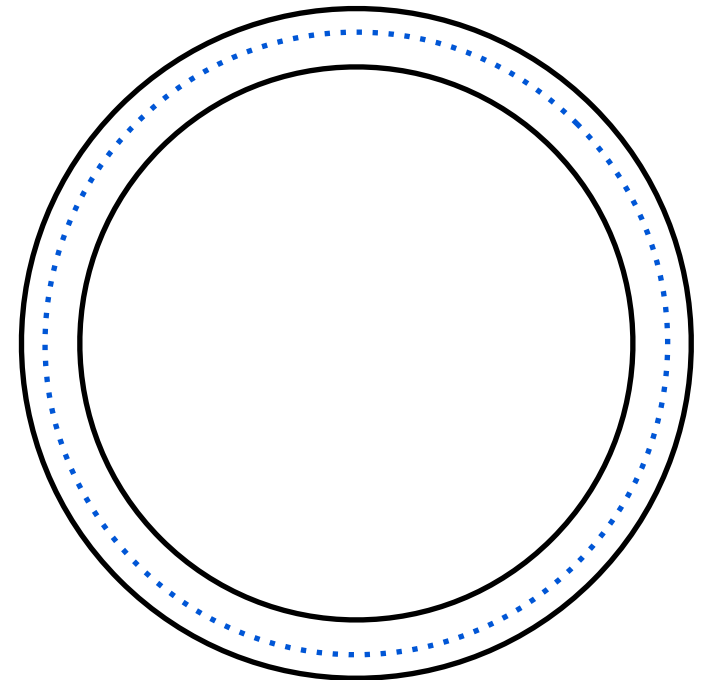
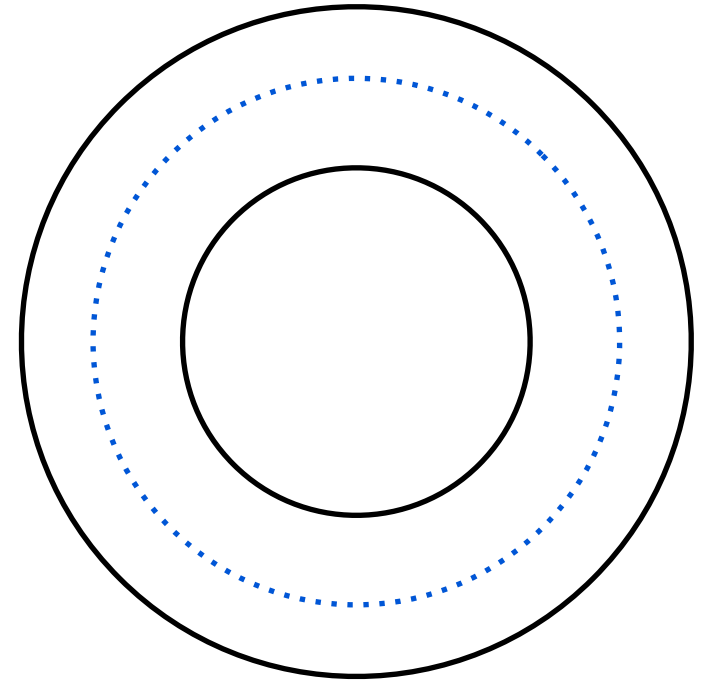
Dipende dallo
Spessore

SEZIONI CAVE IN PARETE SOTTILE



SEZIONI CAVE IN PARETE SOTTILE

SEZIONE CIRCOLARE CAVA: $\delta = R_e - R_i$

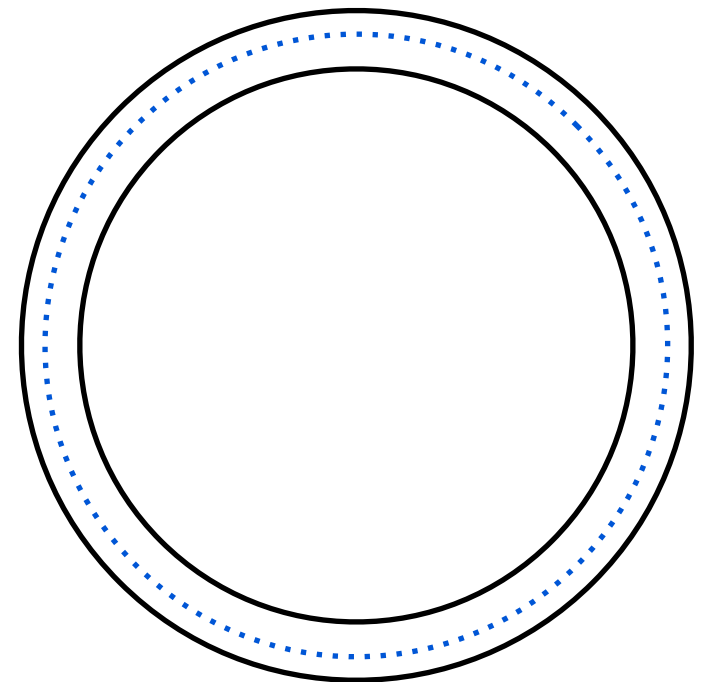
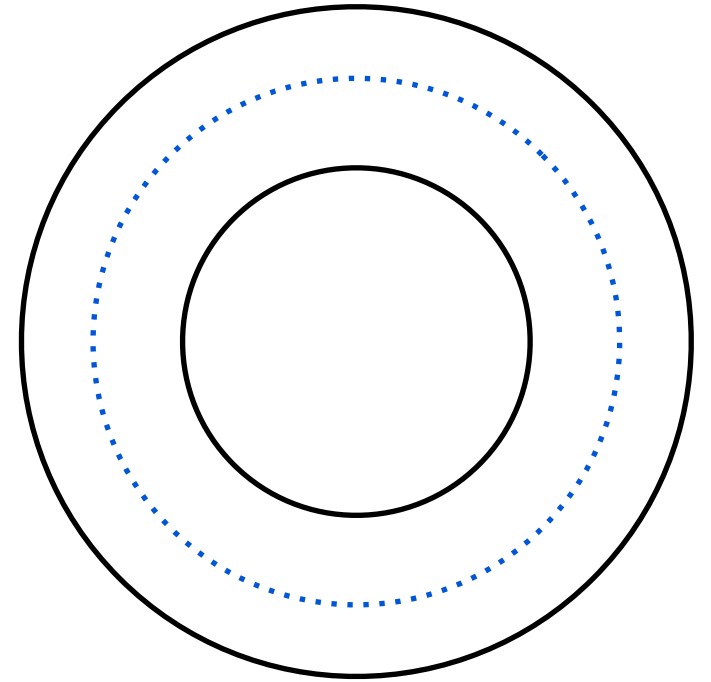


SEZIONI CAVE IN PARETE SOTTILE

SEZIONE CIRCOLARE CAVA: $\Delta = R_e - R_i$

DA
SEZIONE
CIRCOL.
COMPATTA

$$\begin{cases} \tau_{zx} = -\frac{M_t}{I_p} y \\ \tau_{zy} = \frac{M_t}{I_p} x \end{cases}$$

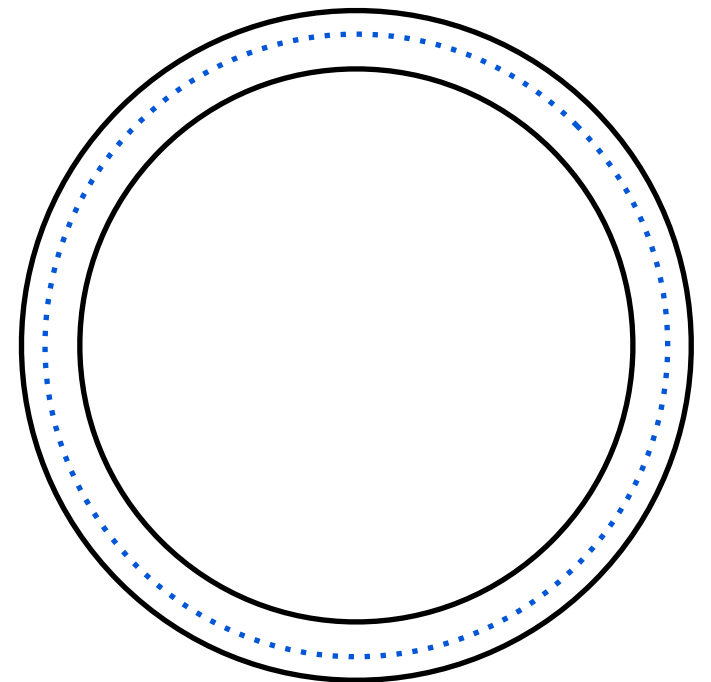
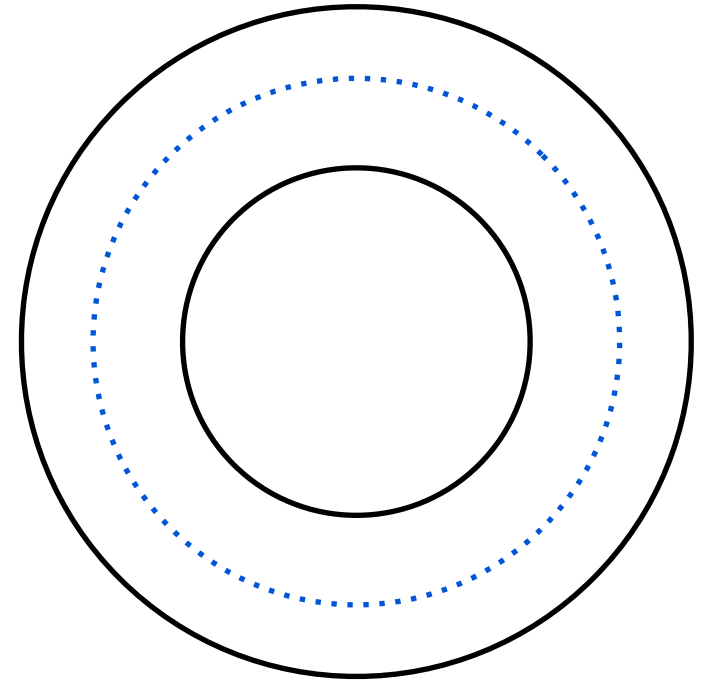


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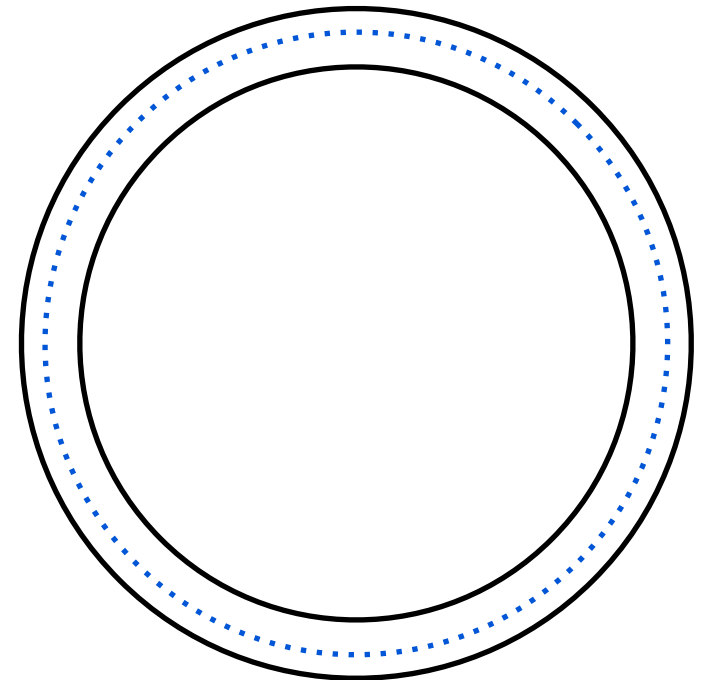
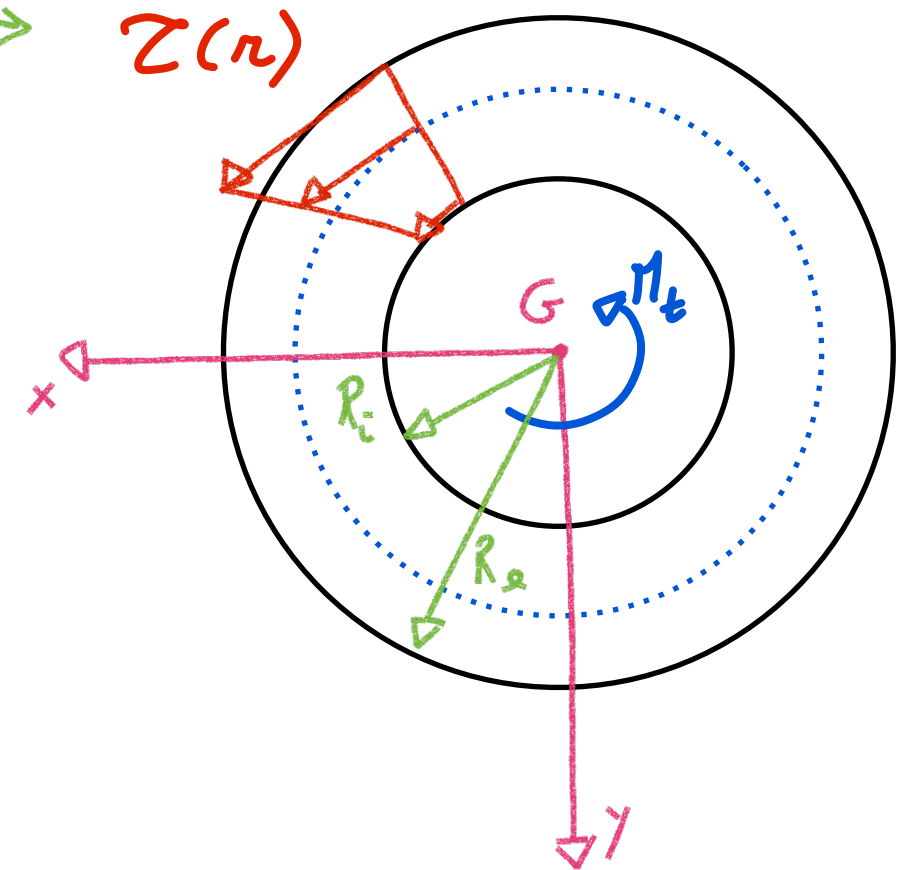
$$\begin{cases} \tau_{zx} = -\frac{M_t}{I_p} y \\ \tau_{zy} = \frac{M_t}{I_p} x \end{cases} \Rightarrow \tau_z(r) = \frac{M_t}{I_p} r$$



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SEZIONI CAVE IN PARETE SOTTILE

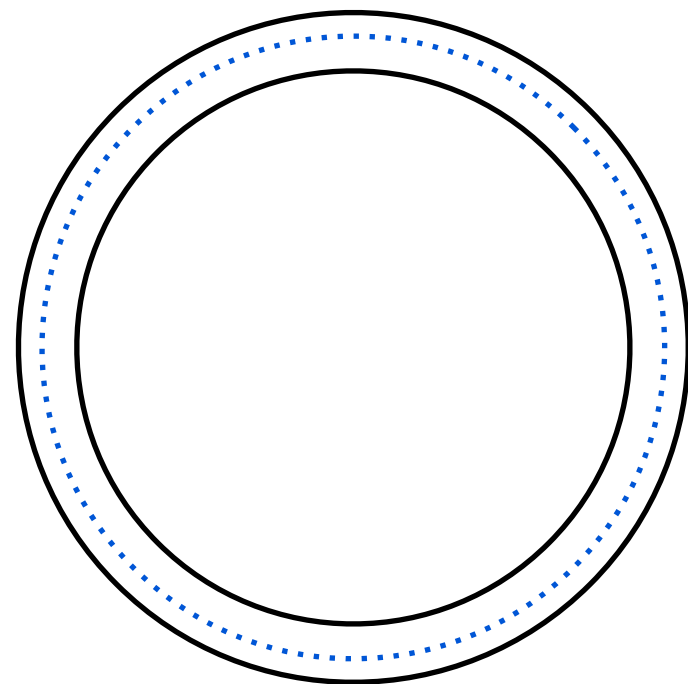
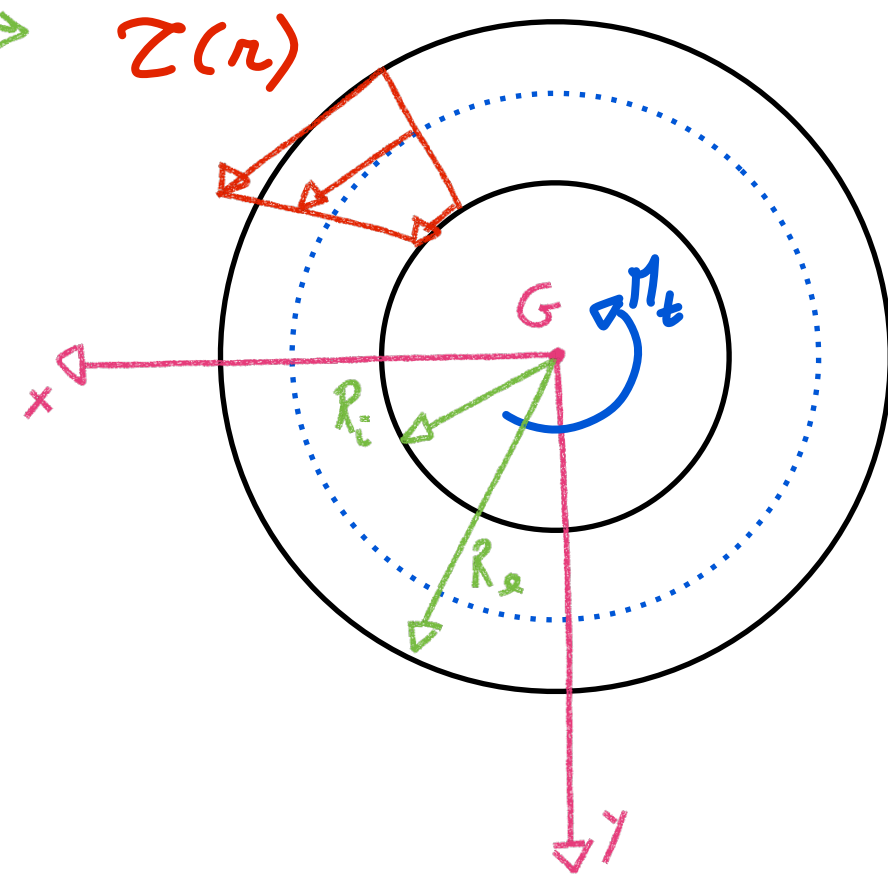
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con $I_p \equiv I_t = I_{pe} - I_{pi} = \frac{\pi}{2} (R_e^4 - R_i^4)$

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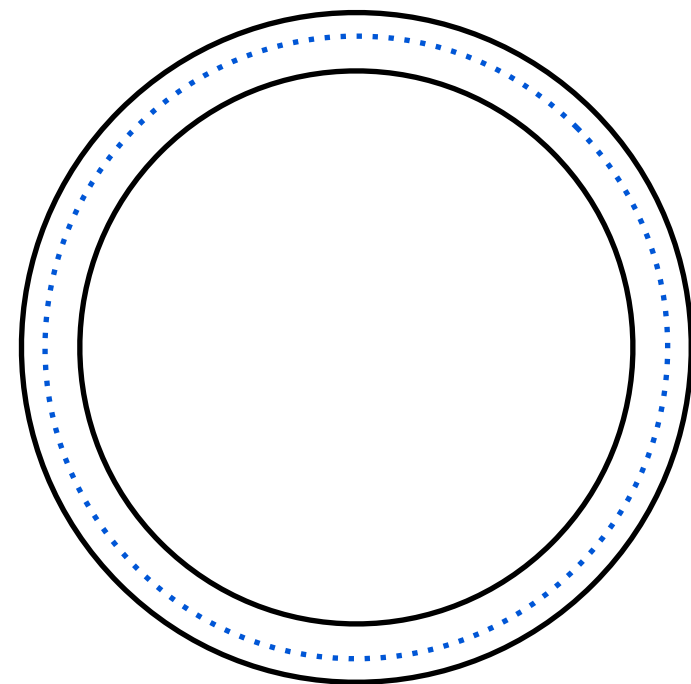
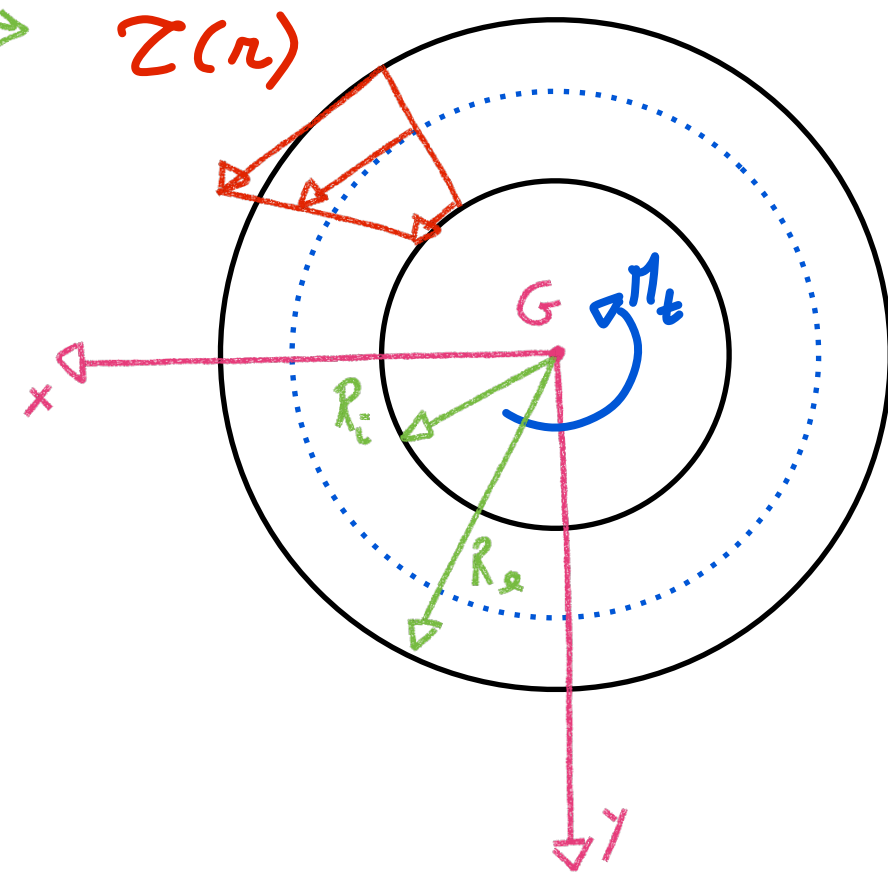
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(*) SE SOTTILE: $R = \frac{R_i + R_e}{2} \Rightarrow I_p \approx 2\pi R^3 \Delta$



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DA SEZIONE CIRCOL. COMPATTA

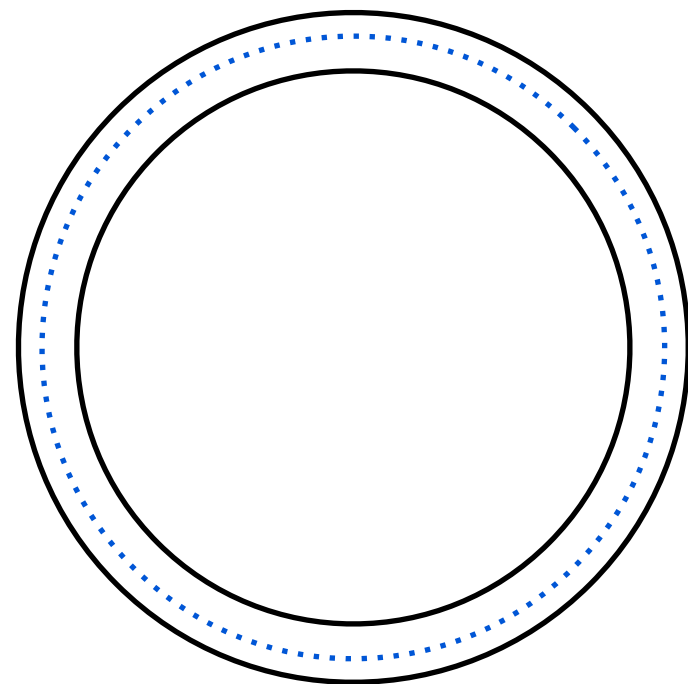
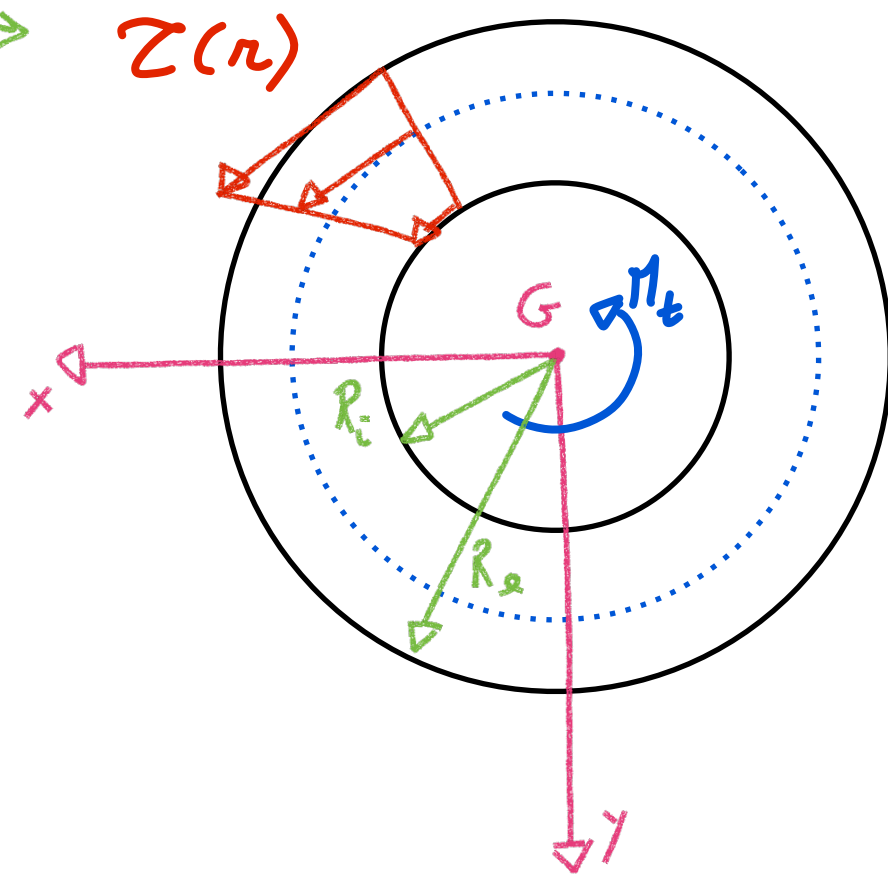
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$\text{L} \rightarrow \text{APPROX } \underline{\tau(r) = \tau_{\text{MEDIA}} = \text{cost}}$



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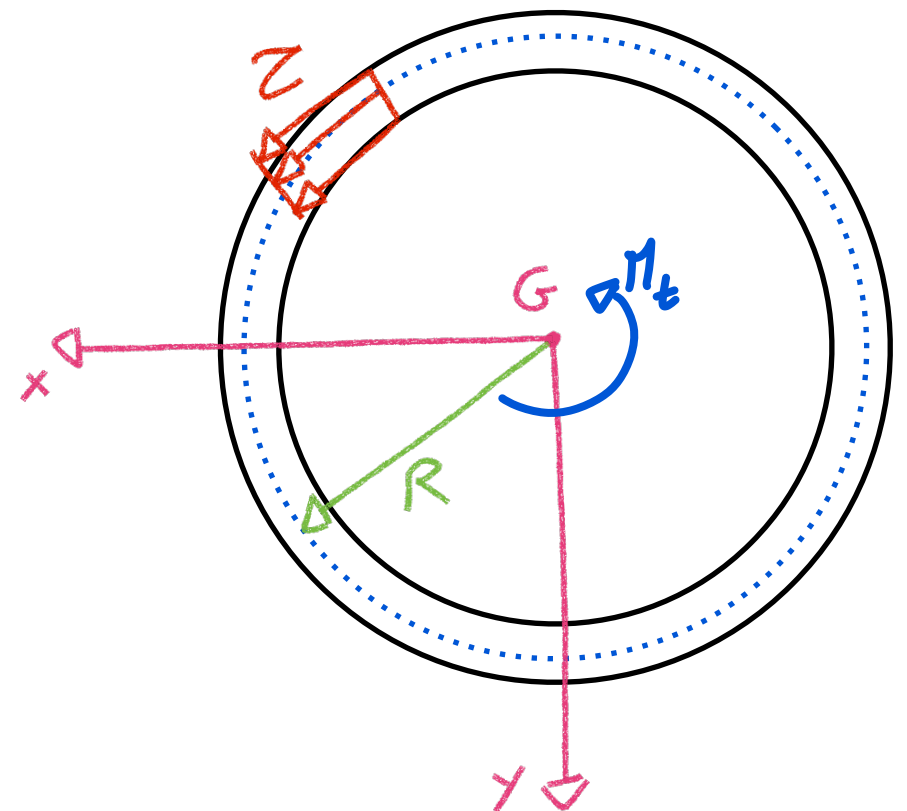
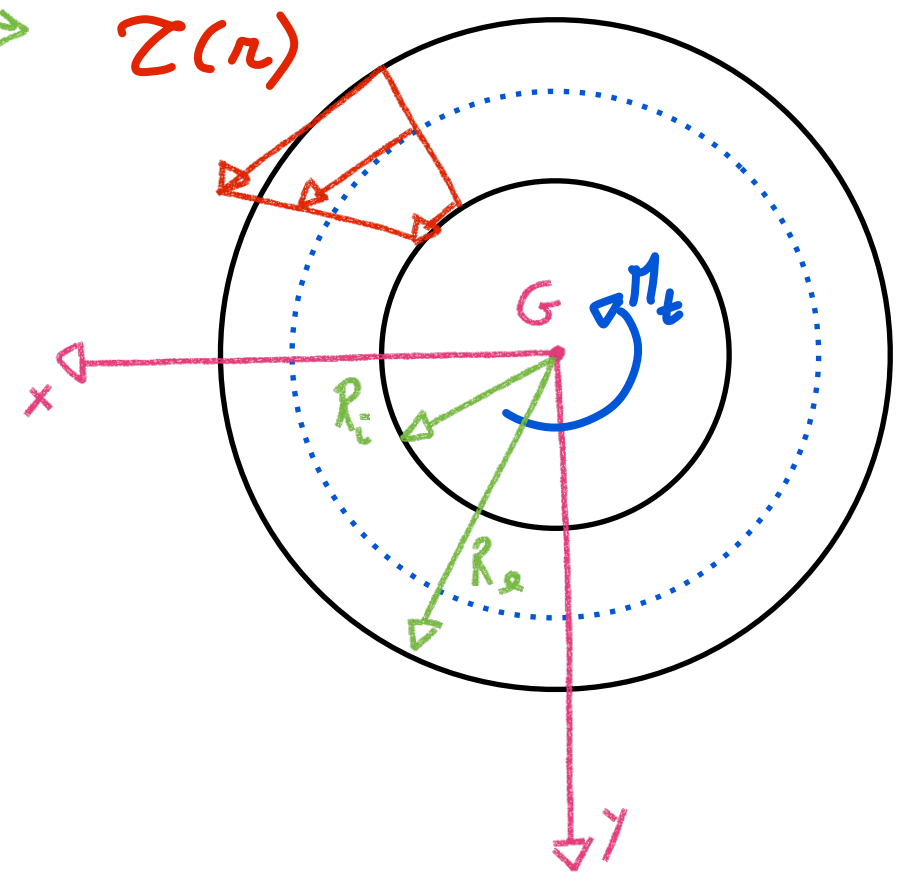
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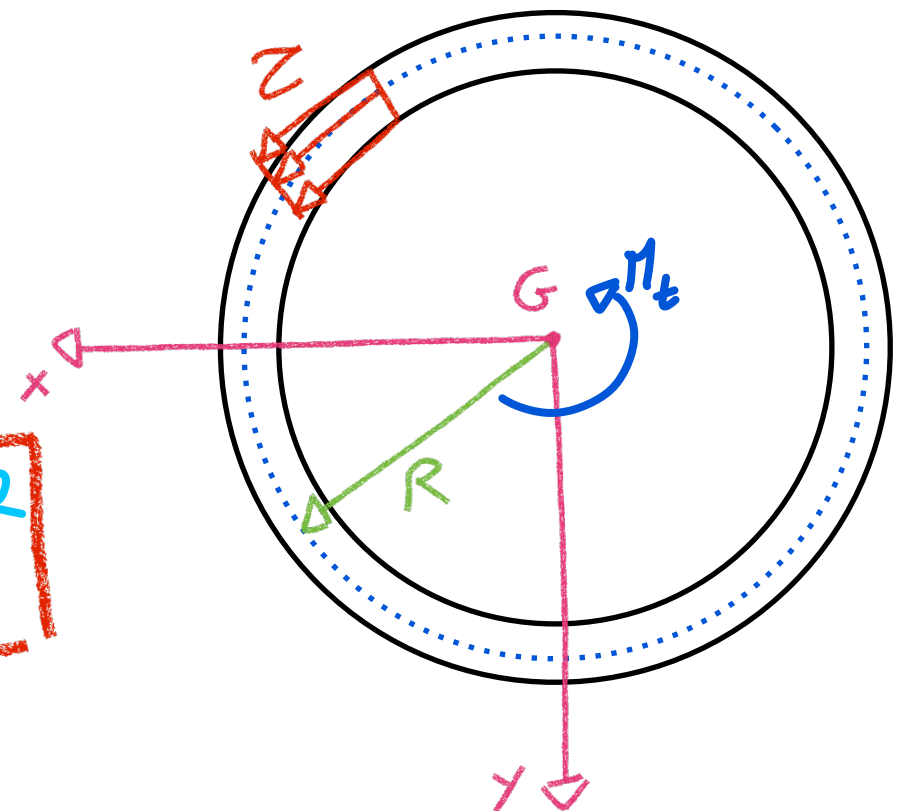
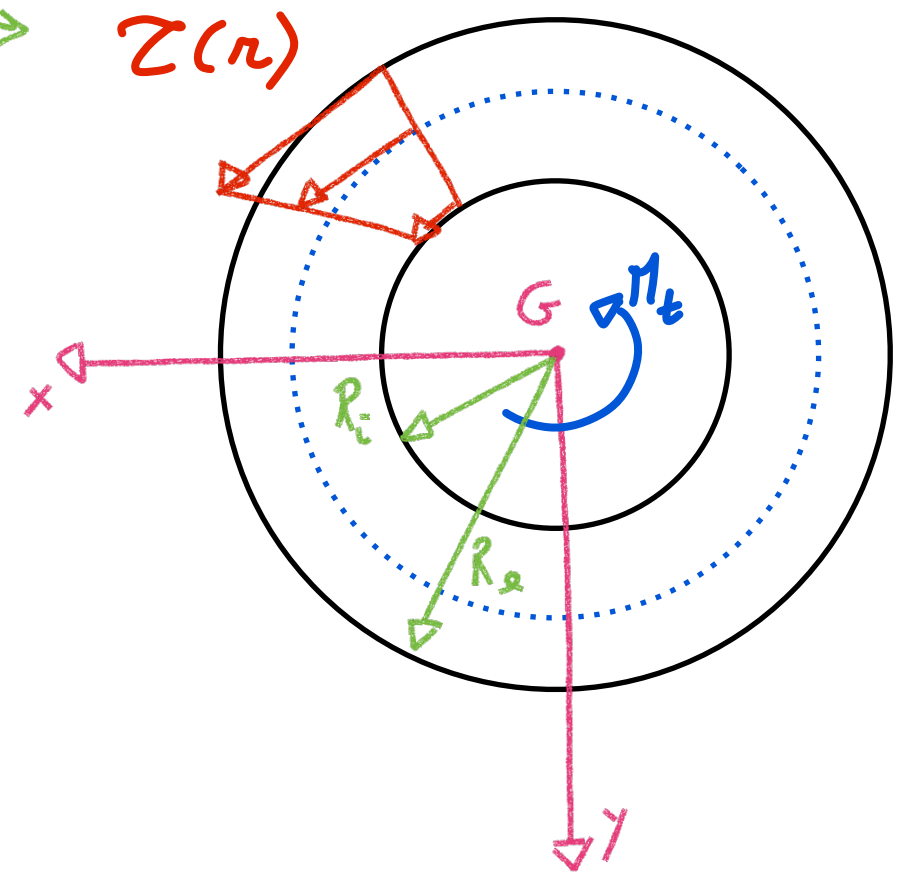
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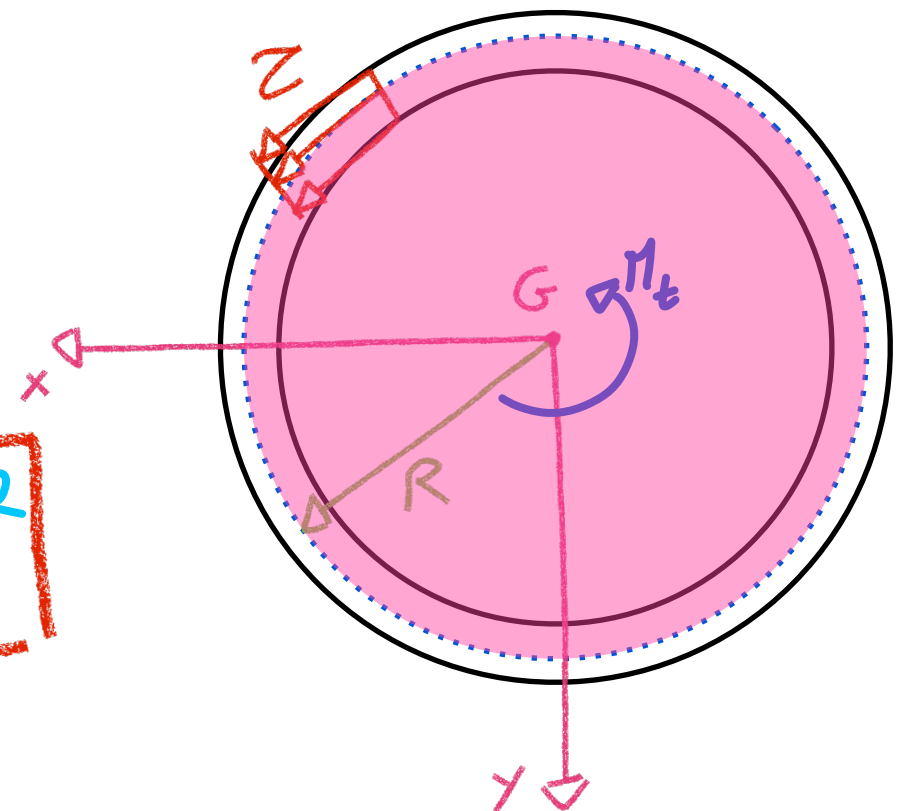
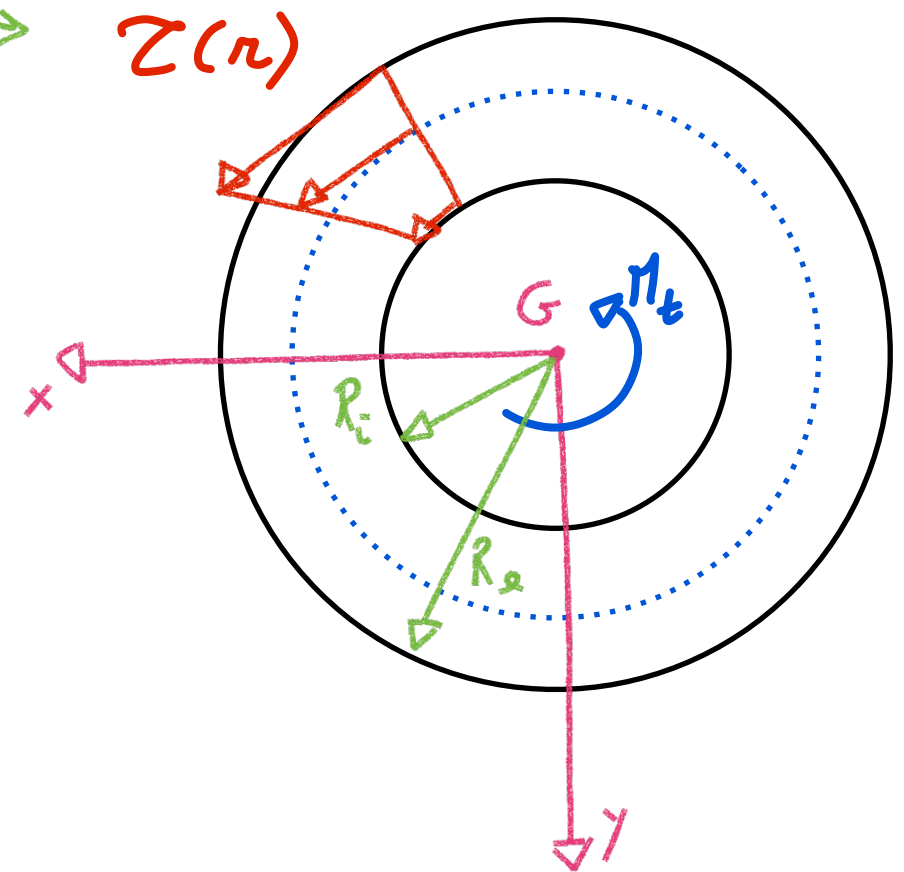
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TEORIA APPROSSIMATA DI BREDT

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- FORMA CAVA GENERICA
- SPESSORE VARIABILE $s(t)$
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PICCOLO
SPESSORE

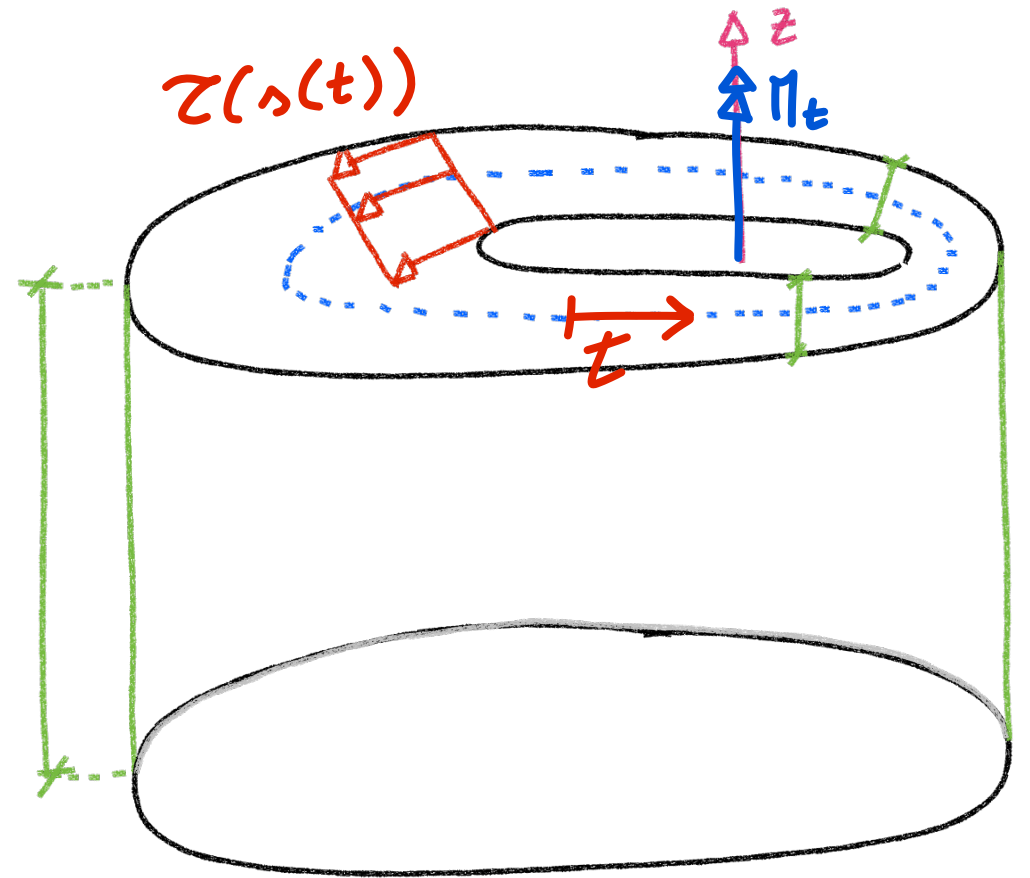
$$Z(t) = \text{cost} \quad m \quad s(t)$$

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PICCOLO
SPESSORE

$$Z(t) = \text{cost} \quad \text{in} \quad s(t)$$

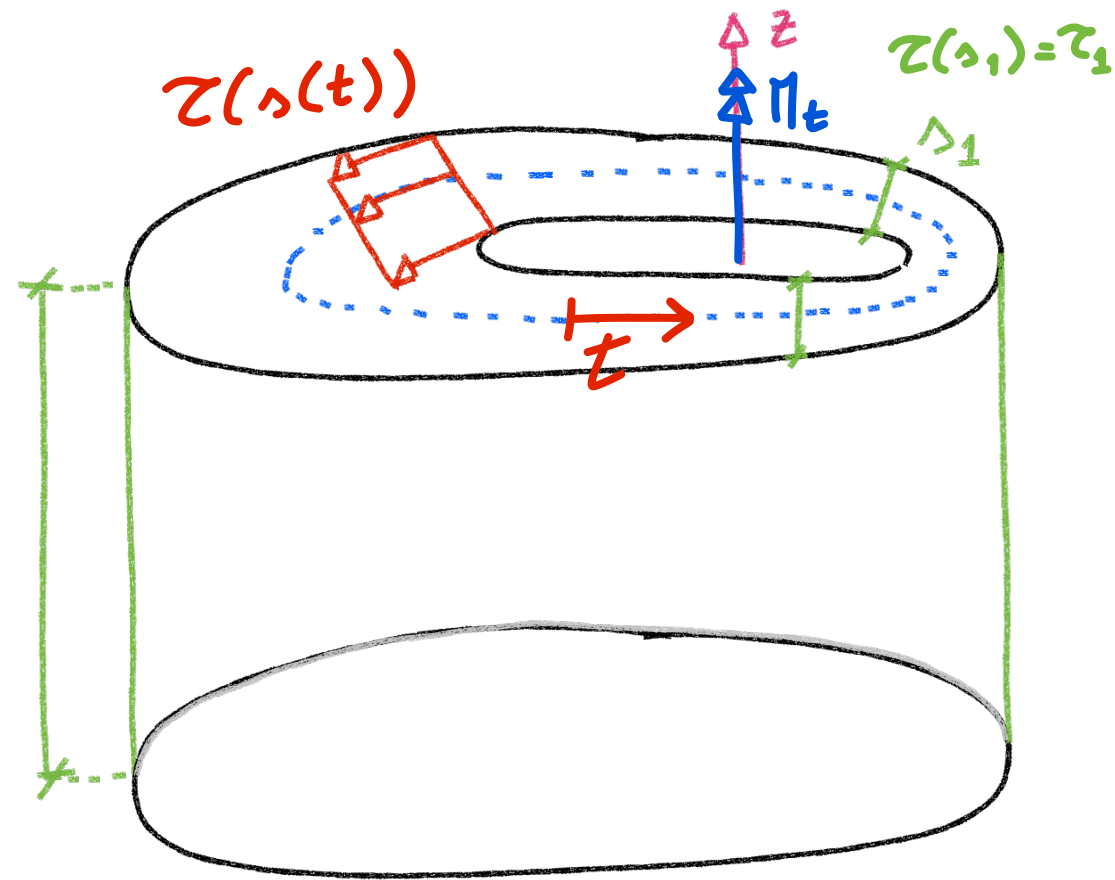


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PICCOLO
SPESSORE

$$Z(t) = \cos t \quad m \quad s(t)$$

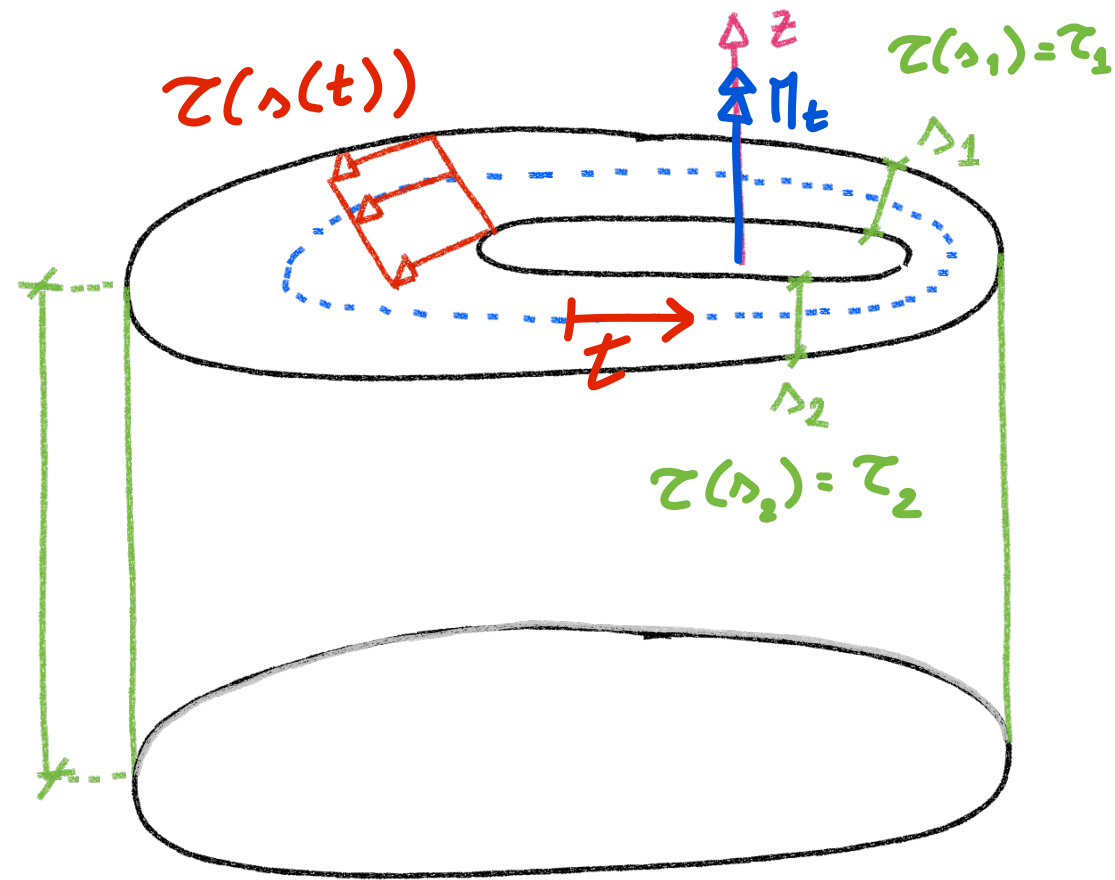


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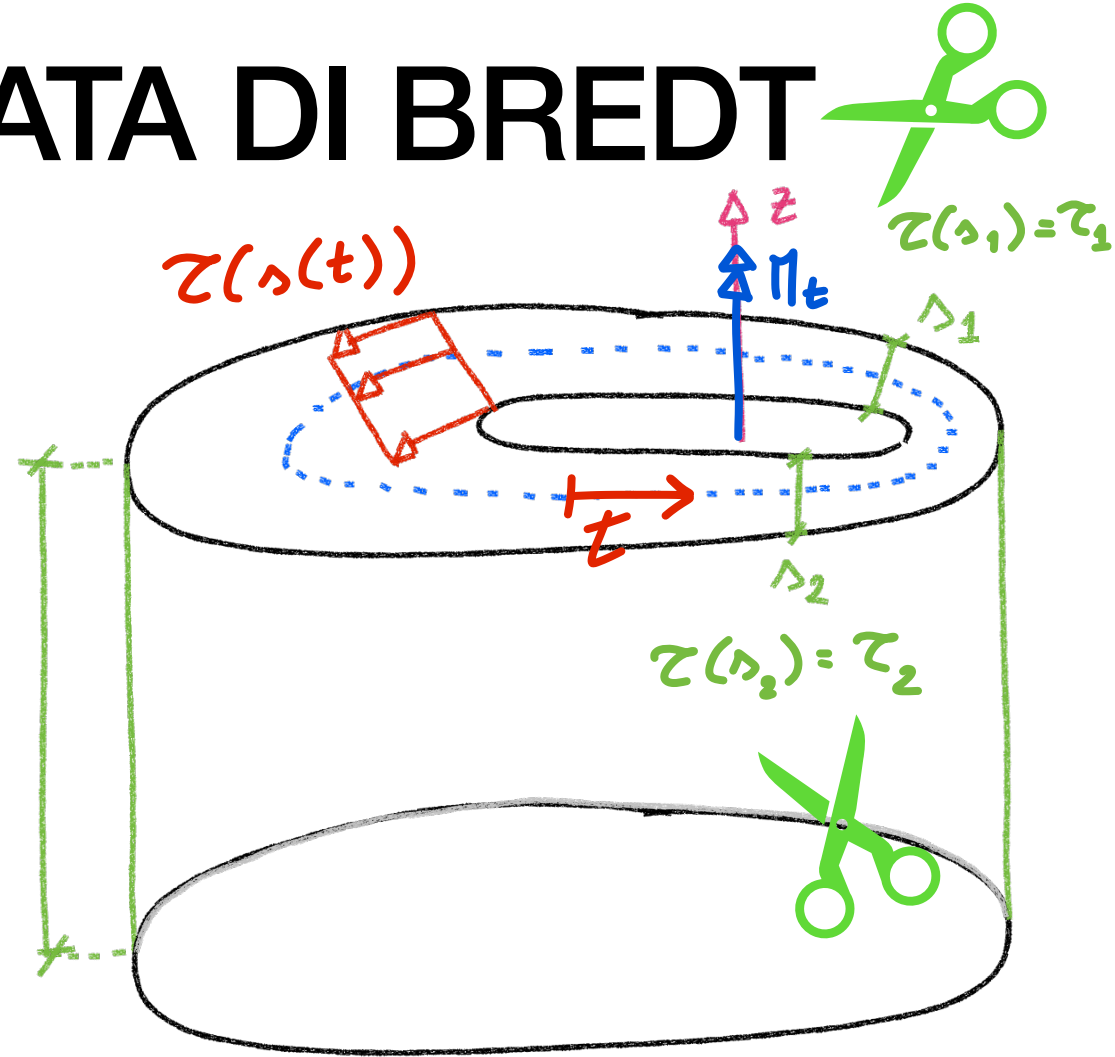


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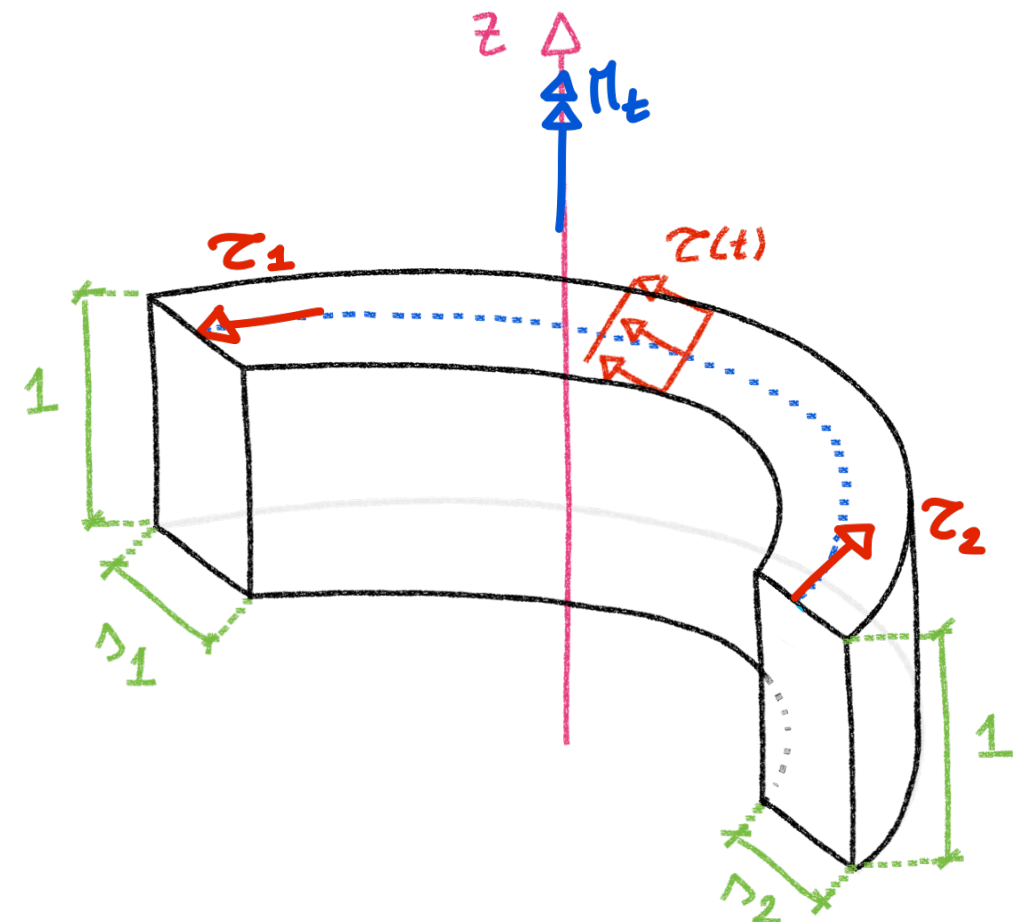
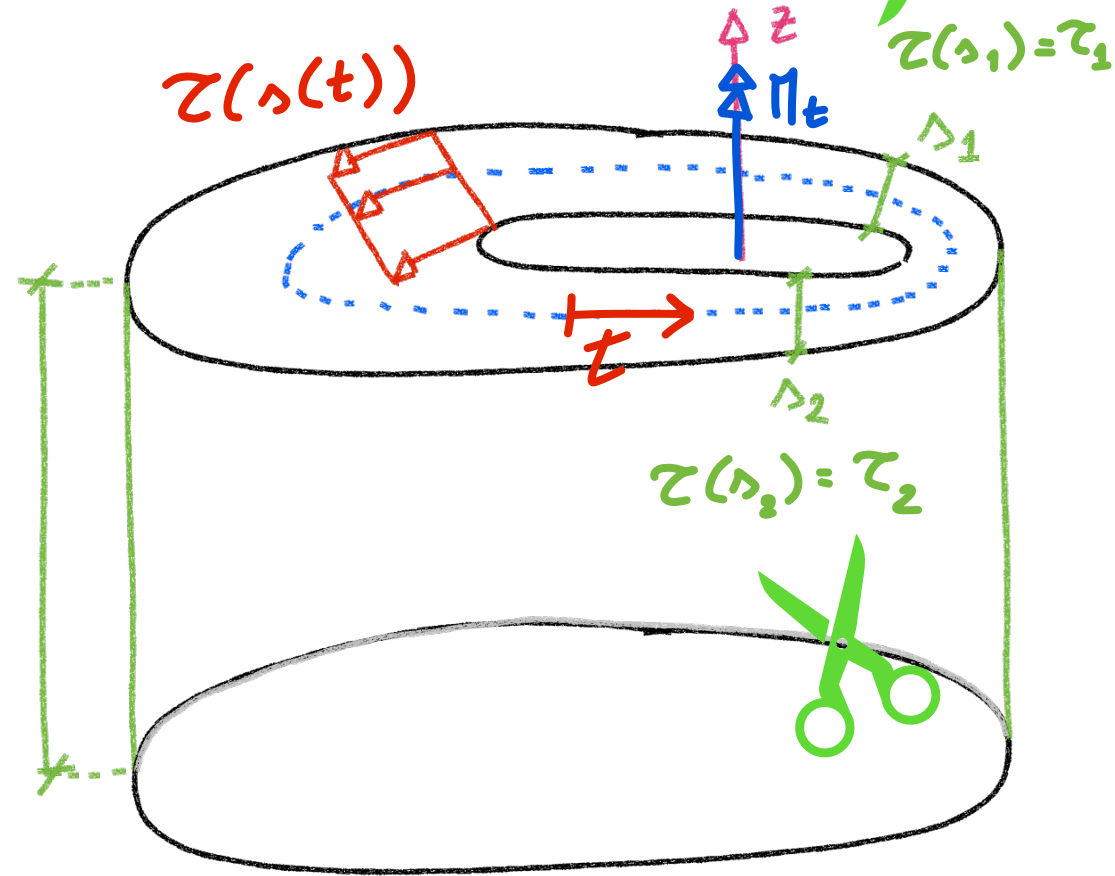
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PICCOLO
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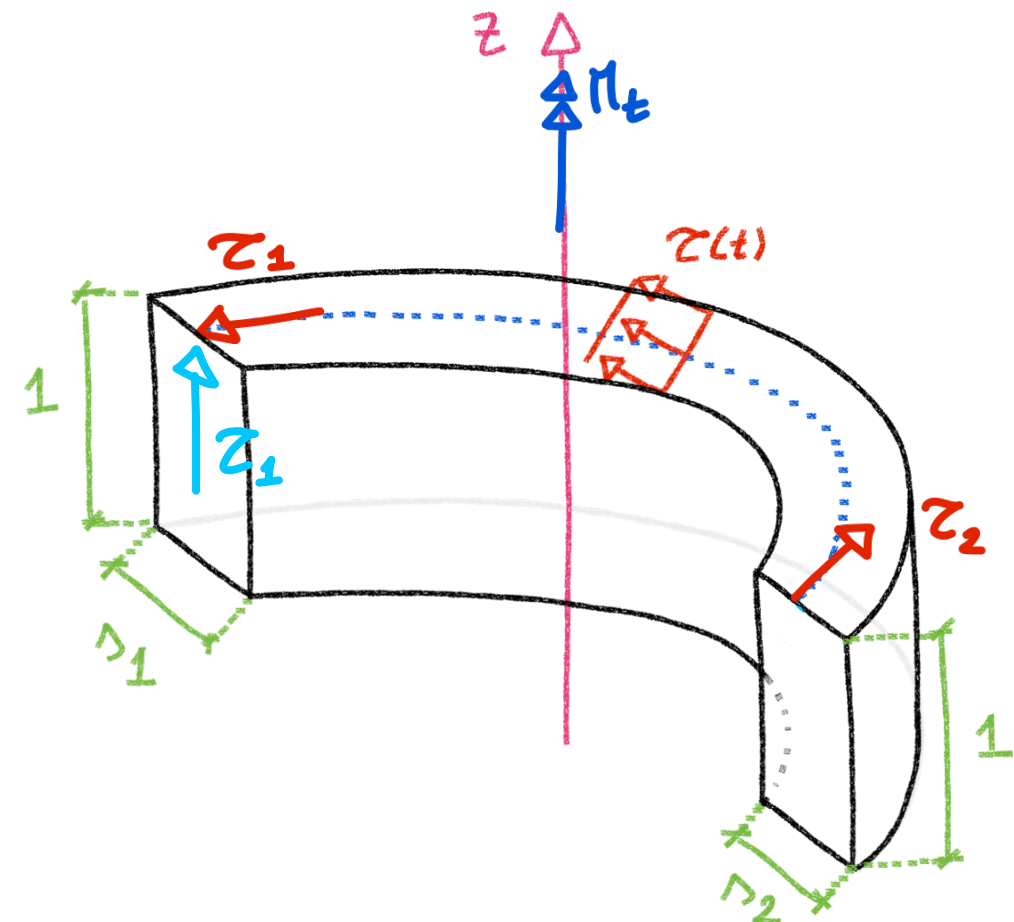
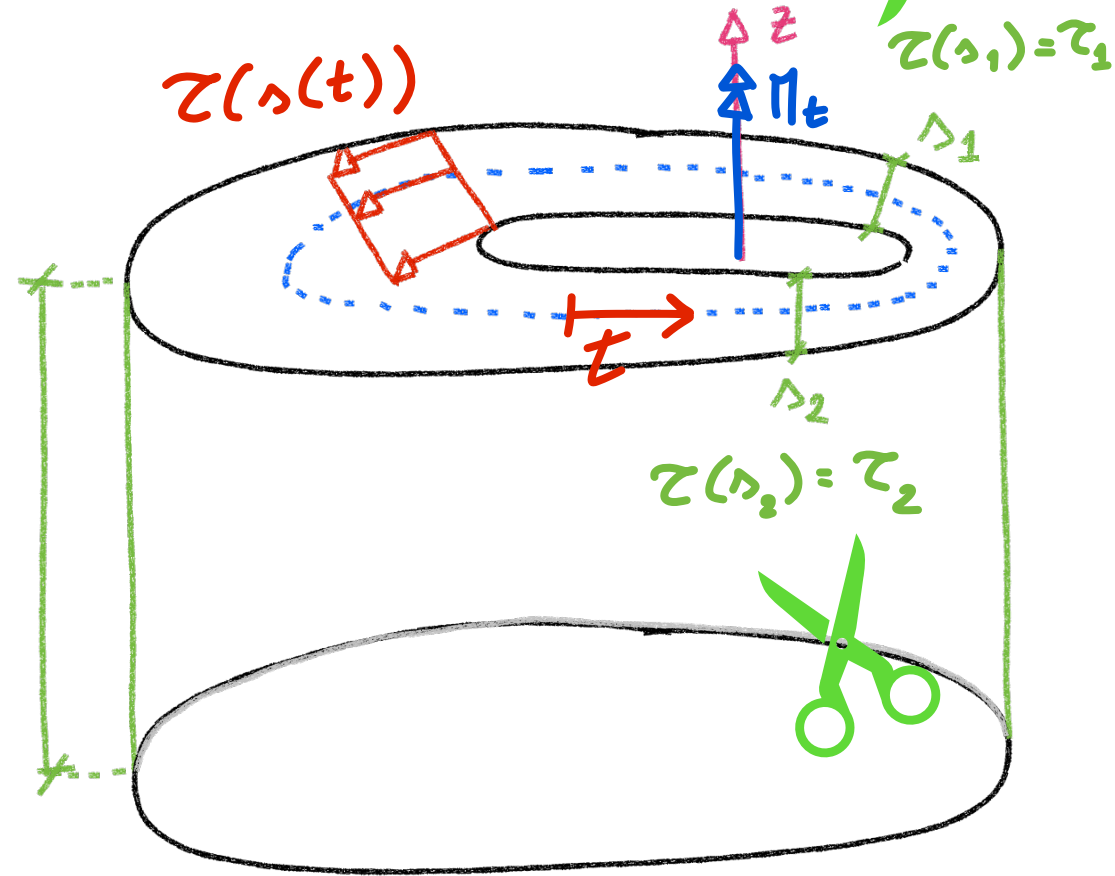
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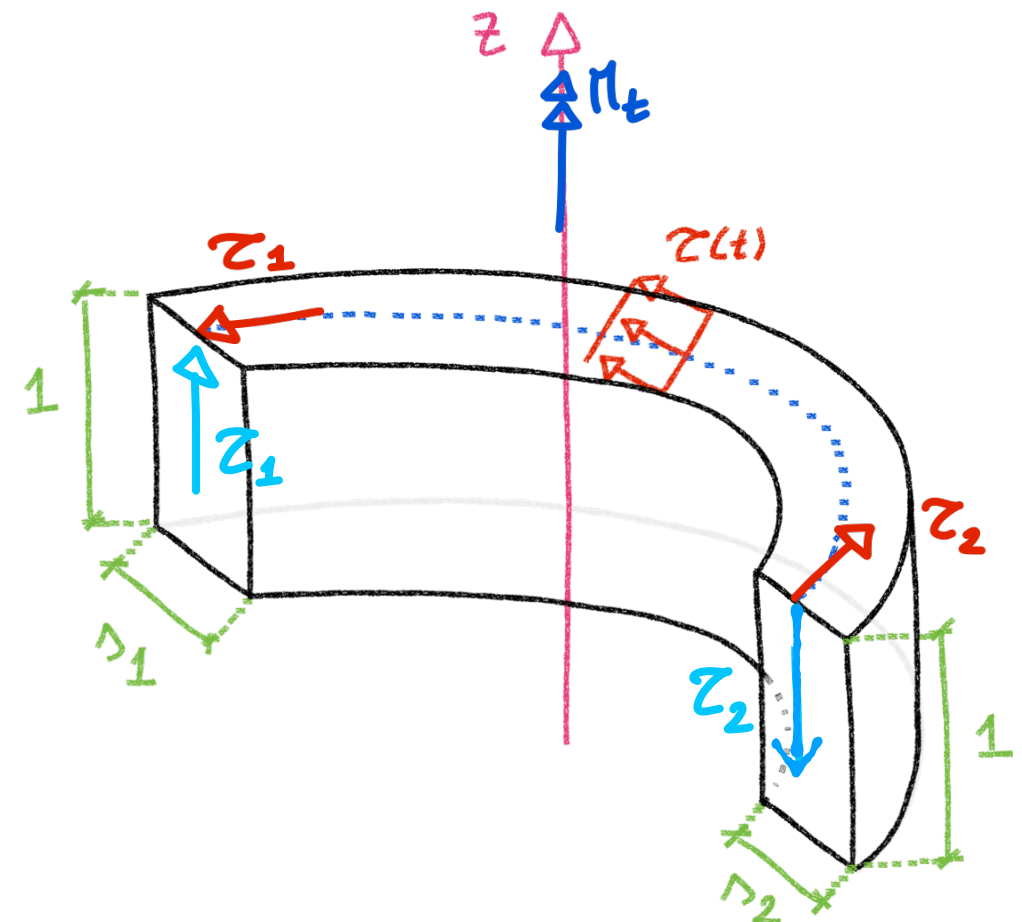
PICCOLO
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- PICCOLO
SPESSORE

$$Z(t) = \cos t \quad \text{in} \quad \mathcal{S}(t)$$



TEORIA APPROSSIMATA DI BREDT



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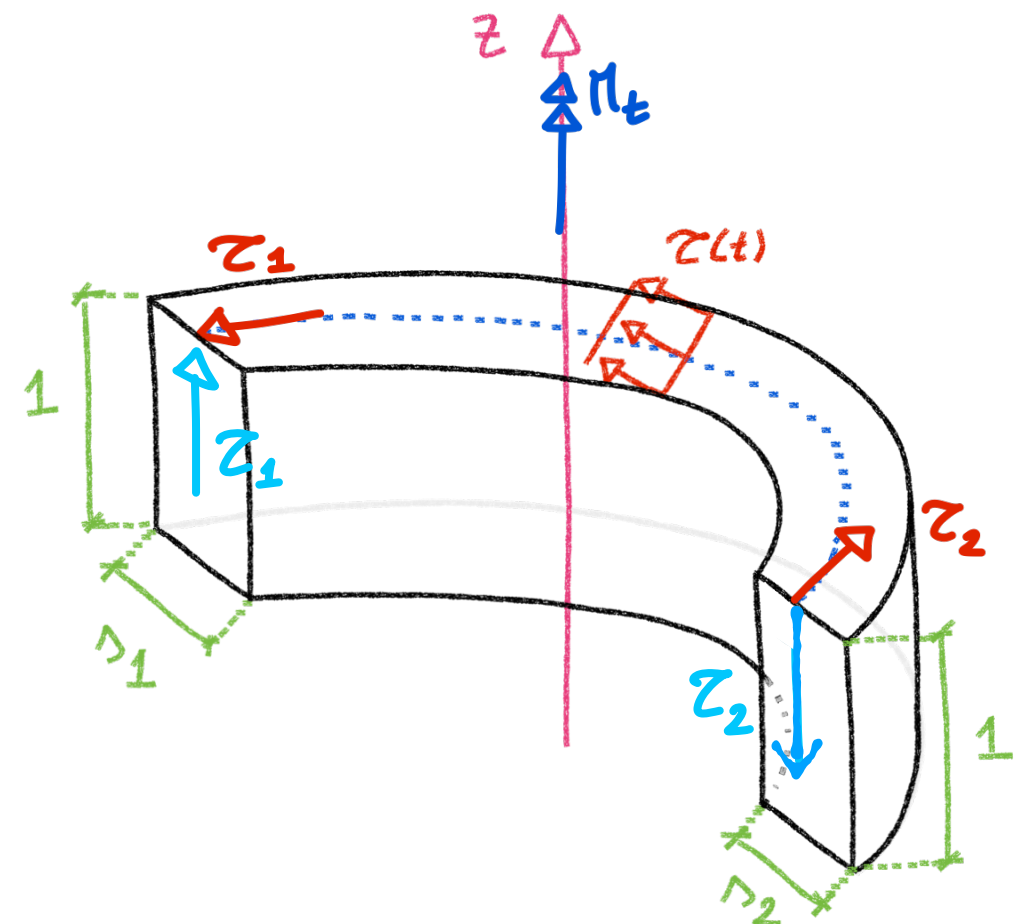
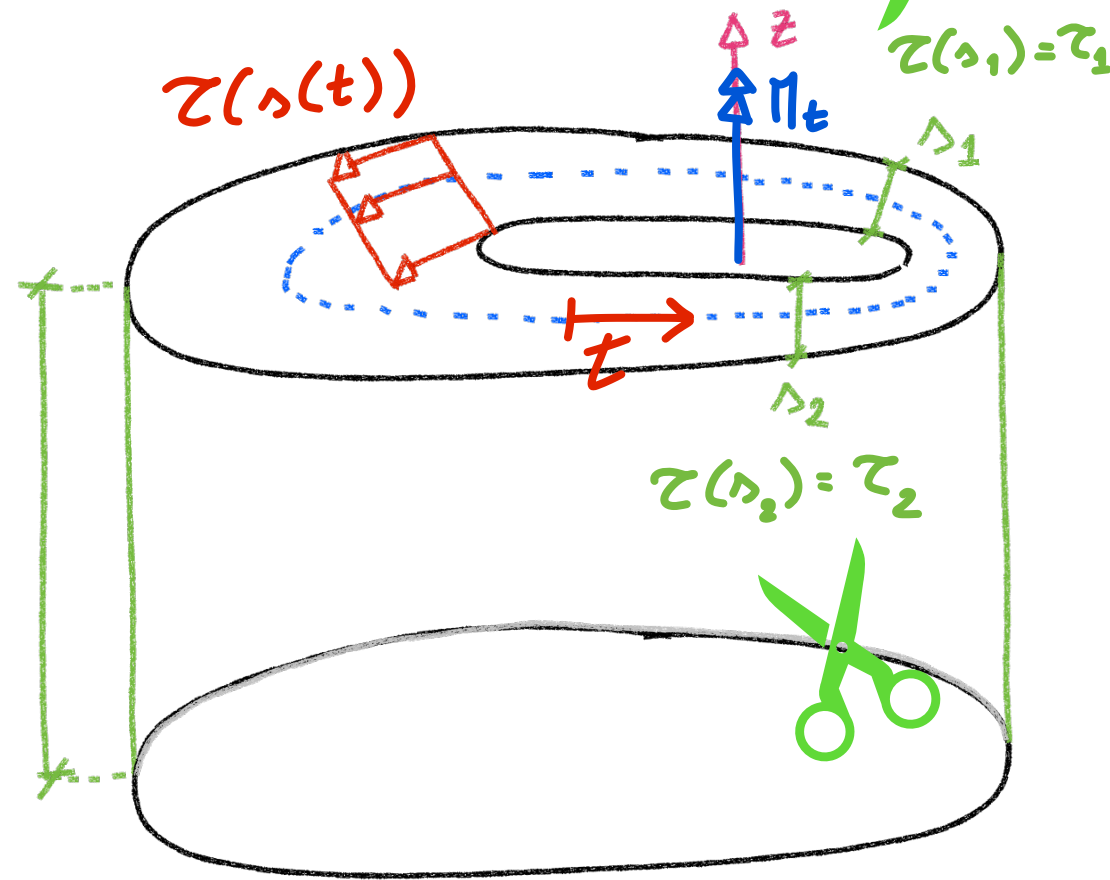
PICCOLO
SPESSORE

$$\tau(t) = \text{cost} \quad \text{in} \quad \lambda(t)$$

* PRINC. RECIPROCA' TENS. TANG.

$$\begin{cases} \tau_1 \equiv \tau_1 \\ \tau_2 \equiv \tau_2 \end{cases}$$

- FORZA SU λ_1 : $\tau_1 \lambda_1 \cdot 1$
- FORZA SU λ_2 : $\tau_2 \lambda_2 \cdot 1$



TEORIA APPROSSIMATA DI BREDT



- FORMA CAVA GENERICA
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PICCOLO
SPESSORE

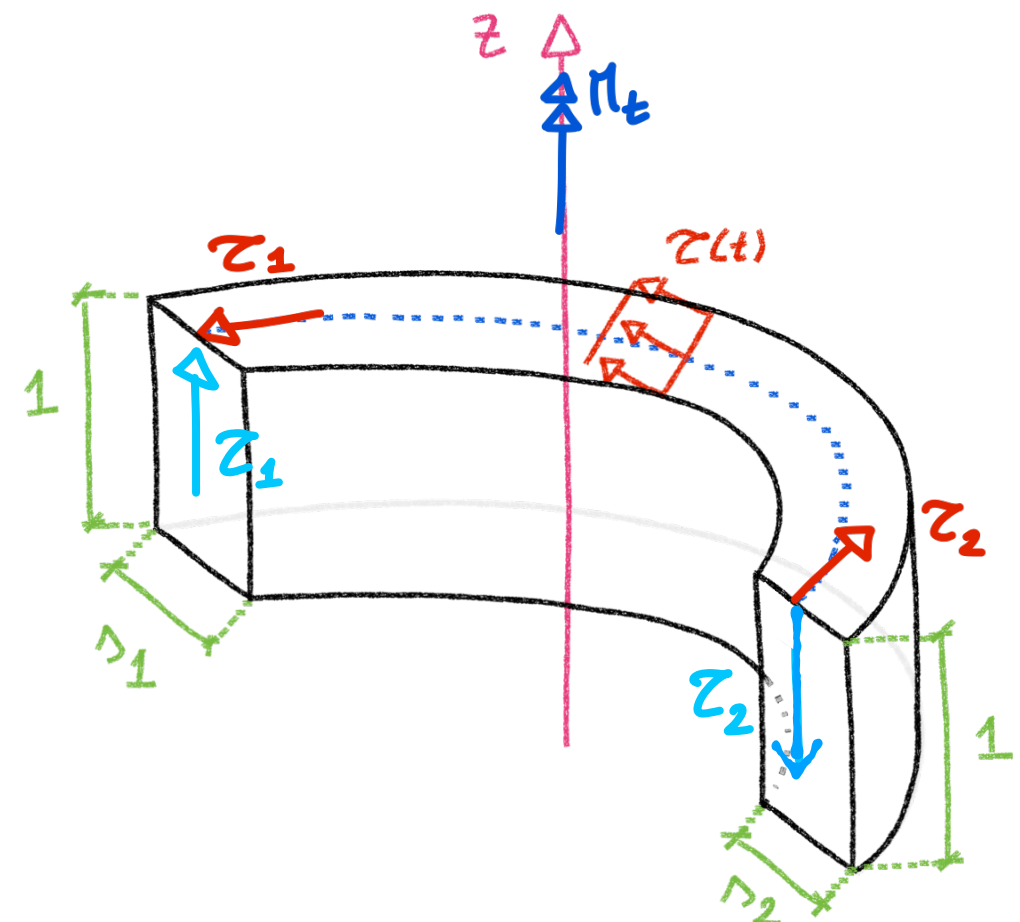
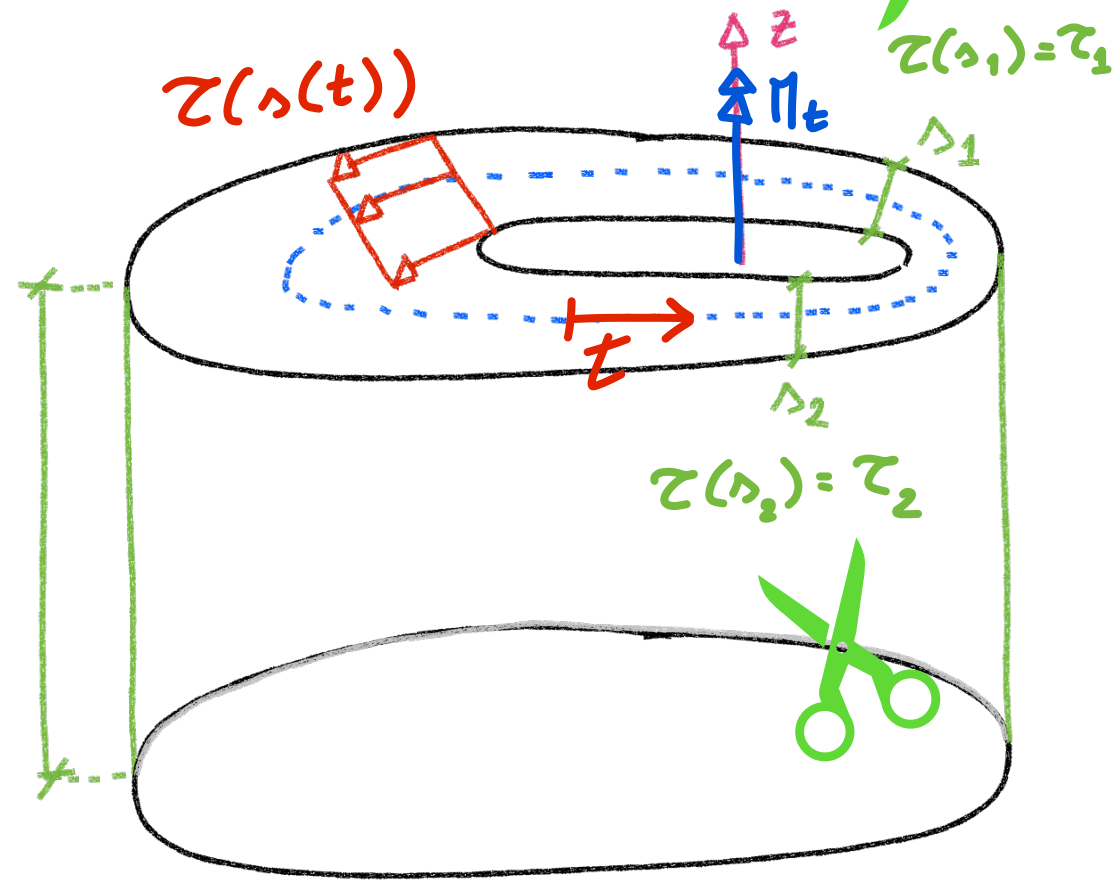
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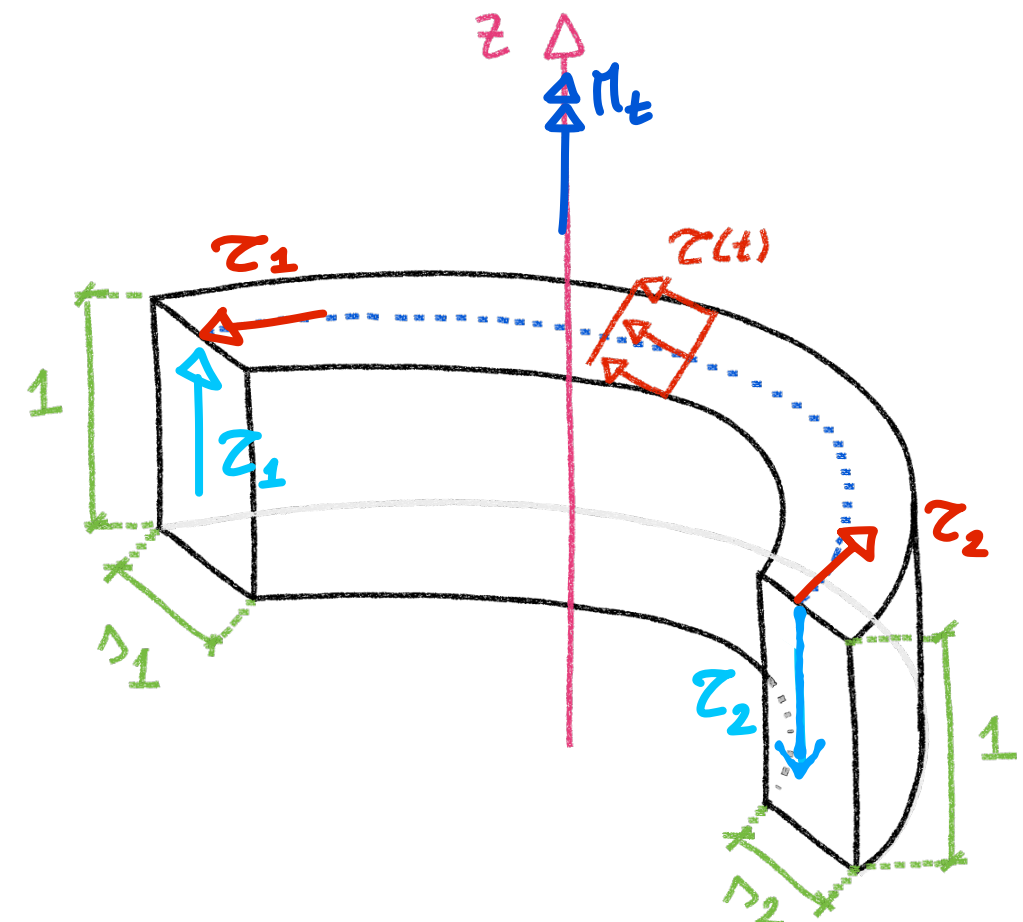
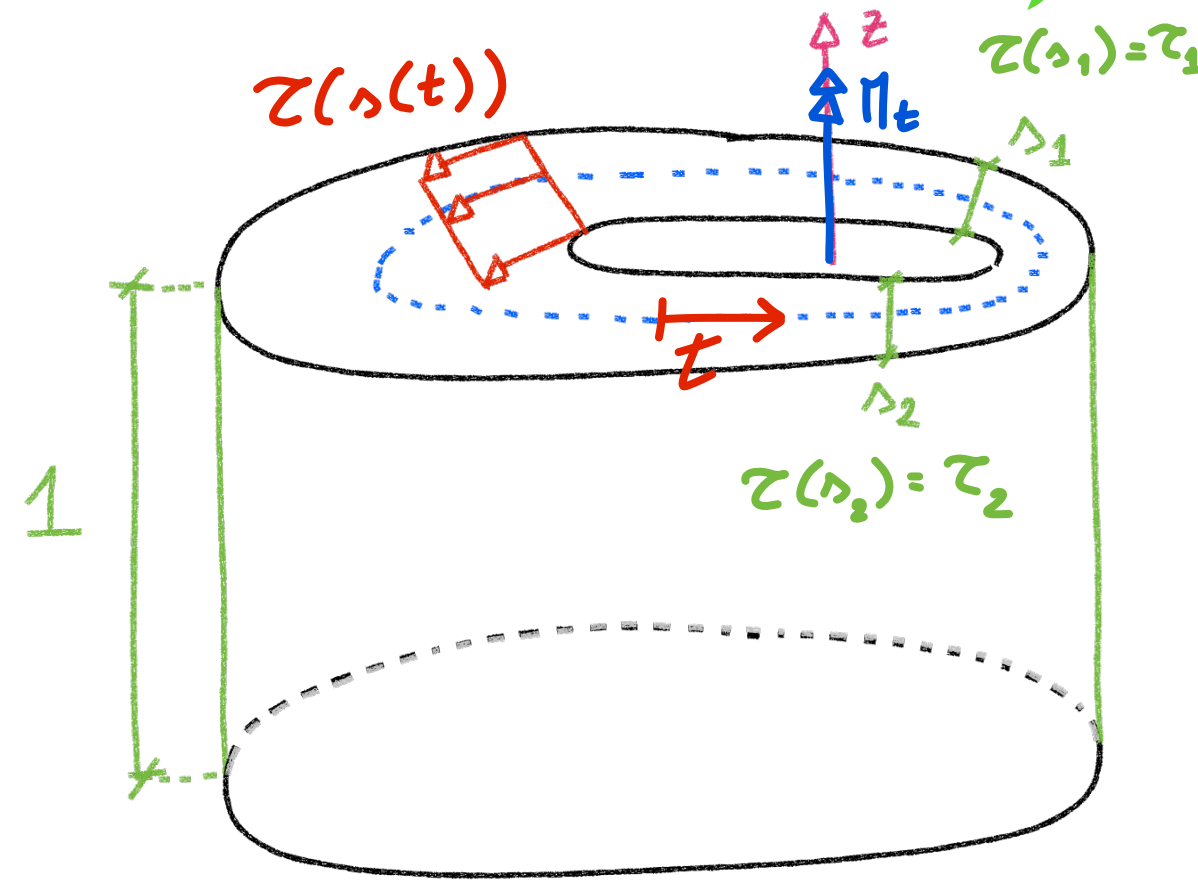
$$\begin{cases} \tau_1 \equiv \tau_1 \\ \tau_2 \equiv \tau_2 \end{cases}$$

- FORZA SU λ_1 : $\tau_1 \lambda_1 \cdot 1$
- FORZA SU λ_2 : $\tau_2 \lambda_2 \cdot 1$

→ "FLUSSO DELLE TENSIONI TANGENZ." "



TEORIA APPROSSIMATA DI BREDT



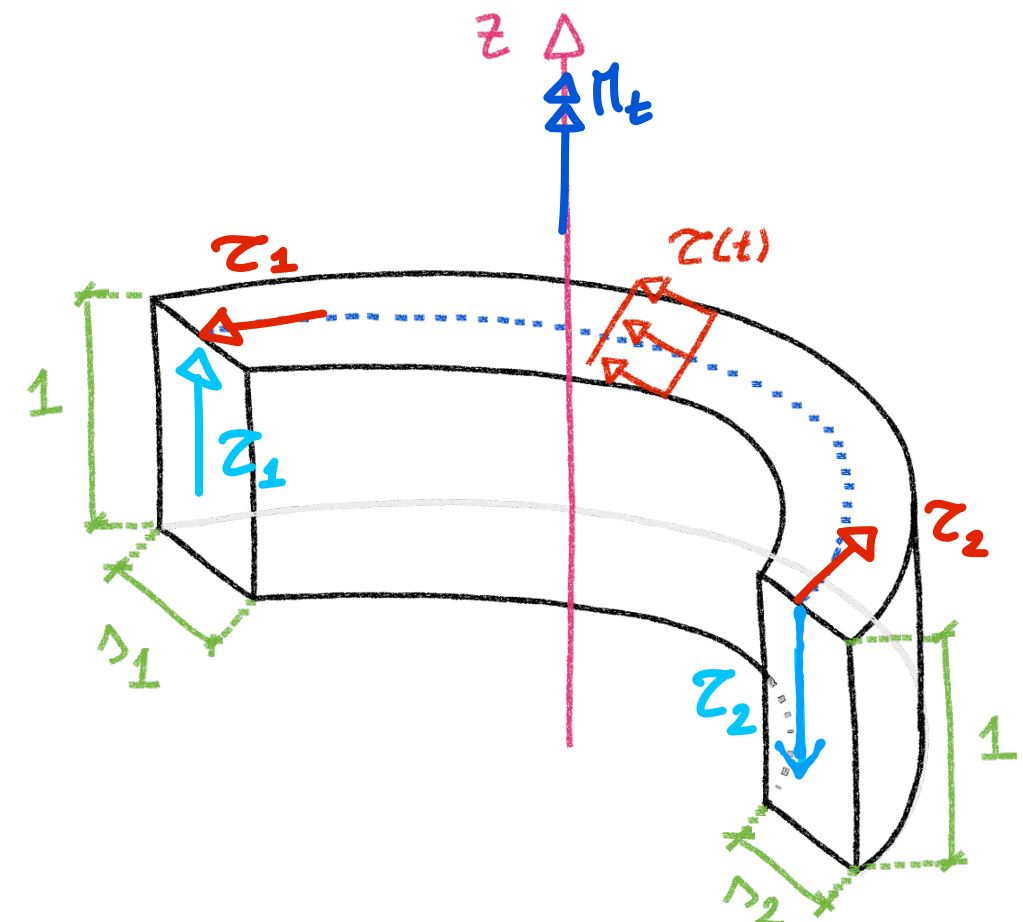
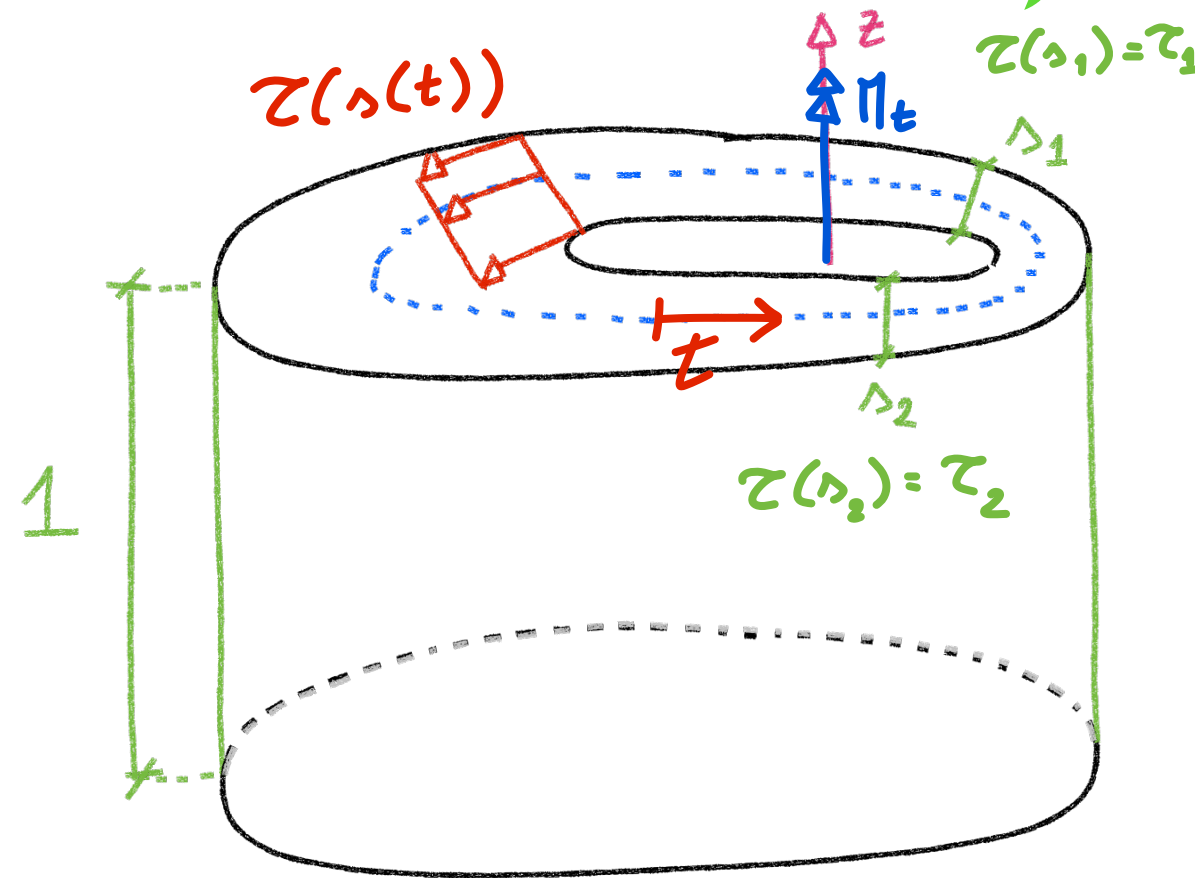
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TEORIA APPROSSIMATA DI BREDT



"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot s(t)$$



- FORZA SU s_1 : $z_1 s_1 \cdot 1$
- FORZA SU s_2 : $z_2 s_2 \cdot 1$

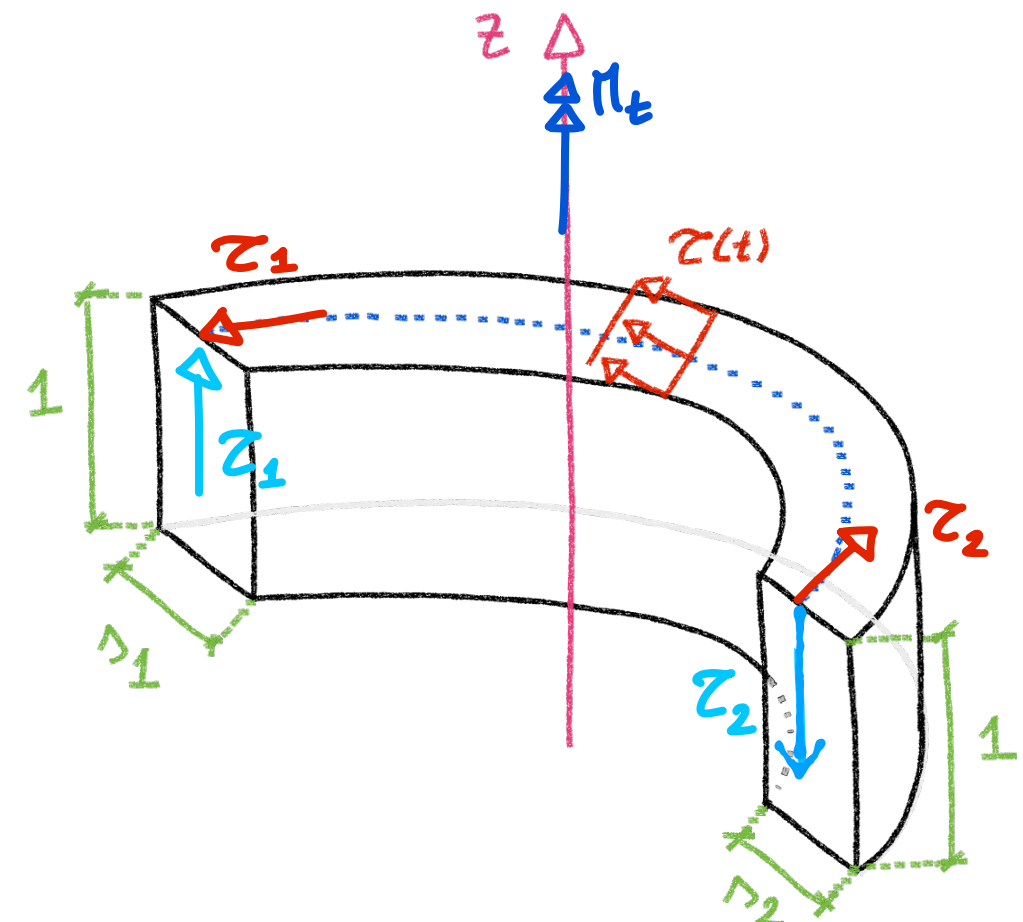
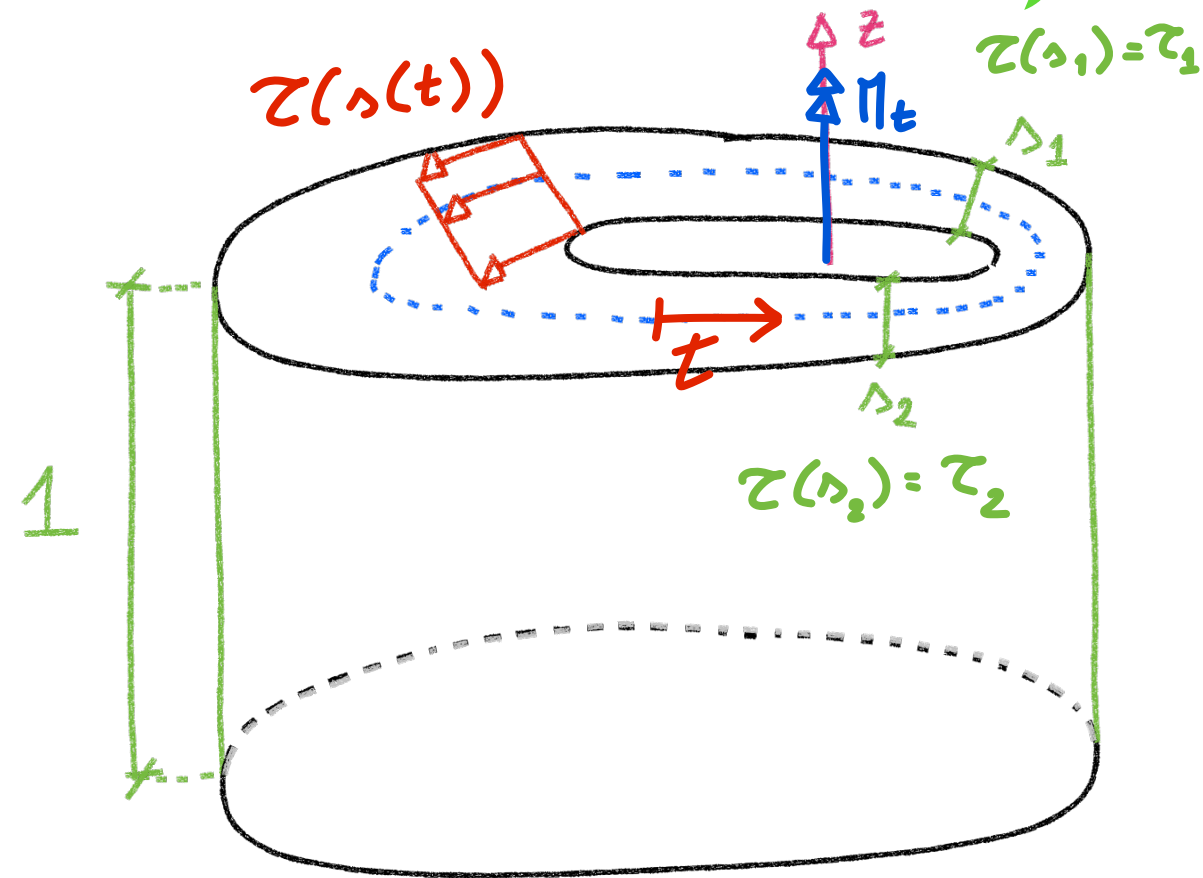
TEORIA APPROSSIMATA DI BREDT



"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot s(t)$$

• EQUILIBRIO ALLA TRASL. SU z:



• FORZA SU s1 : $z_1 s_1 \cdot 1$

• FORZA SU s2 : $z_2 s_2 \cdot 1$

TEORIA APPROSSIMATA DI BREDT

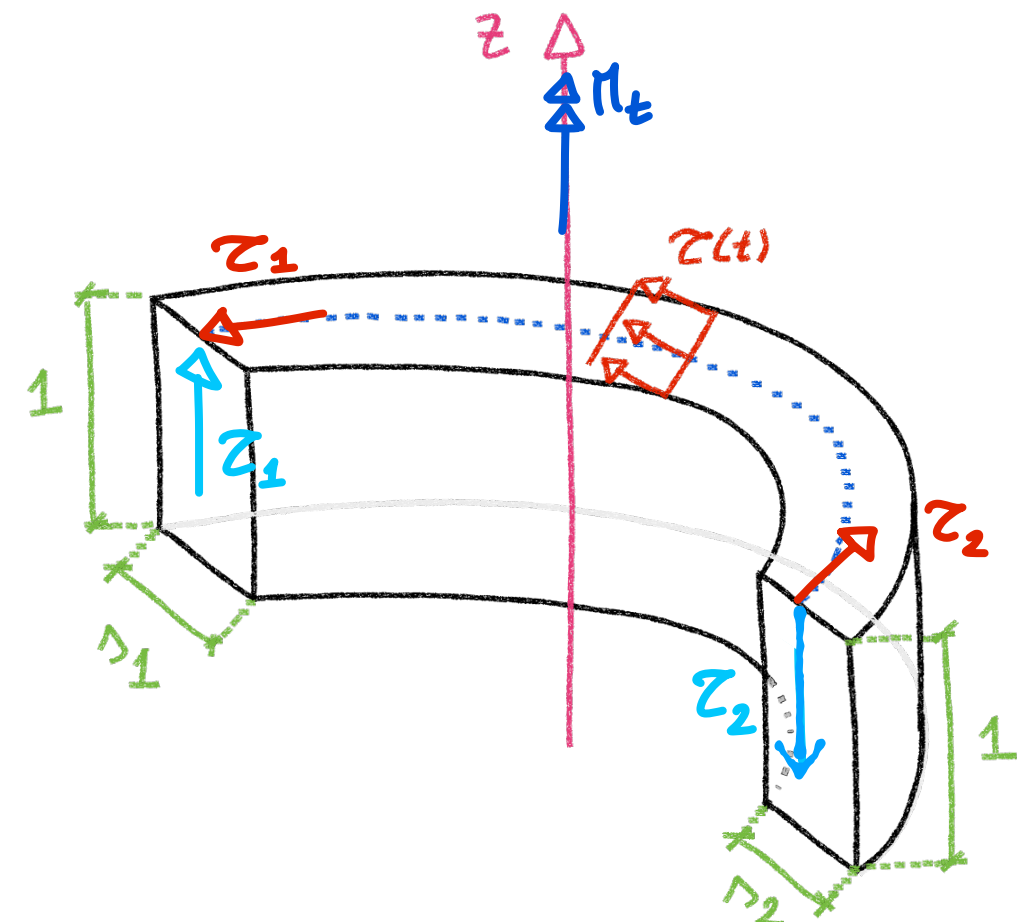
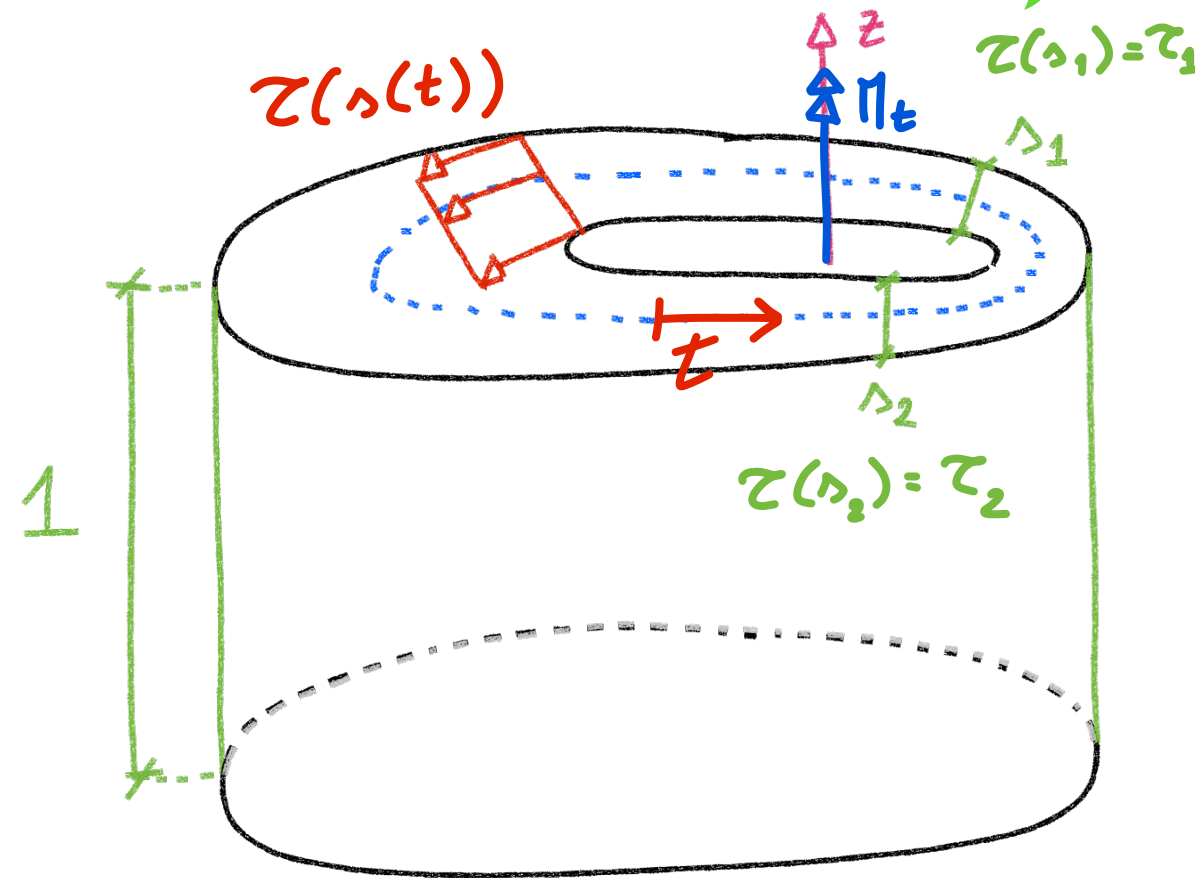


"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot s(t)$$

• EQUILIBRIO ALLA TRASL. SU z:

$$z_1 s_1 \cdot 1 - z_2 s_2 \cdot 1 = 0$$



• FORZA SU s1 : $z_1 s_1 \cdot 1$

• FORZA SU s2 : $z_2 s_2 \cdot 1$

TEORIA APPROSSIMATA DI BREDT



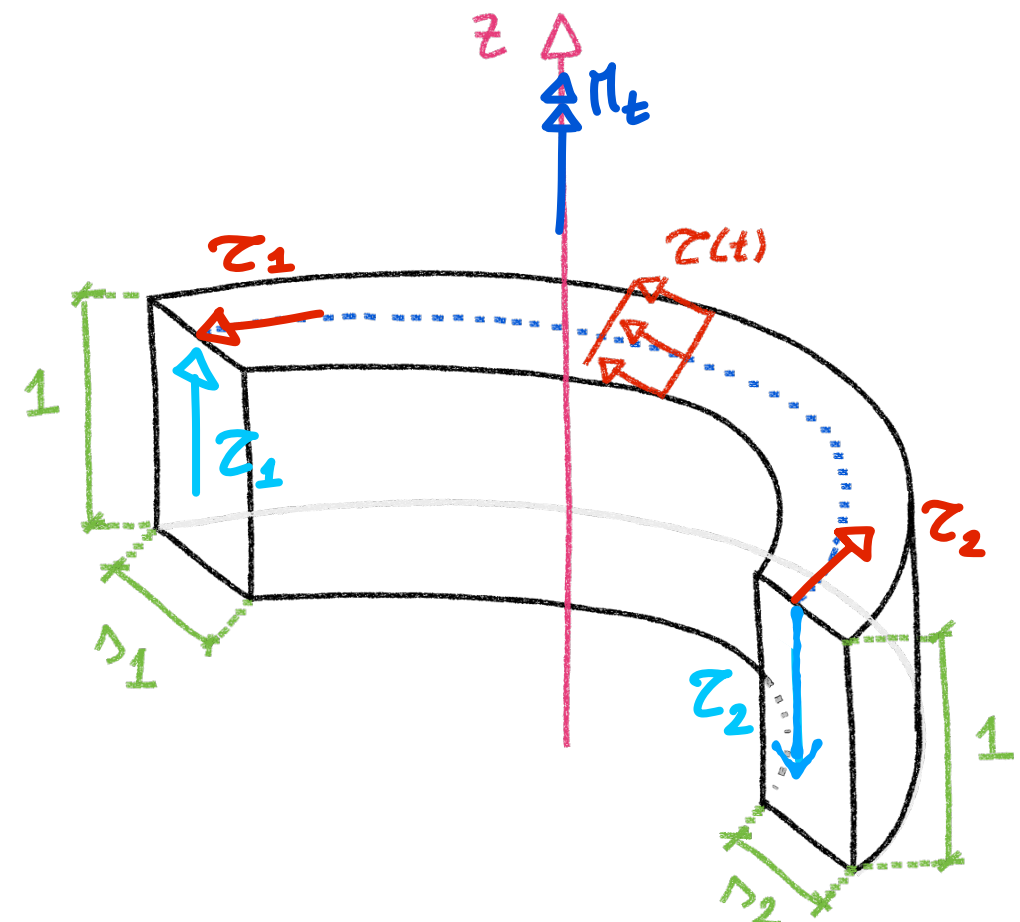
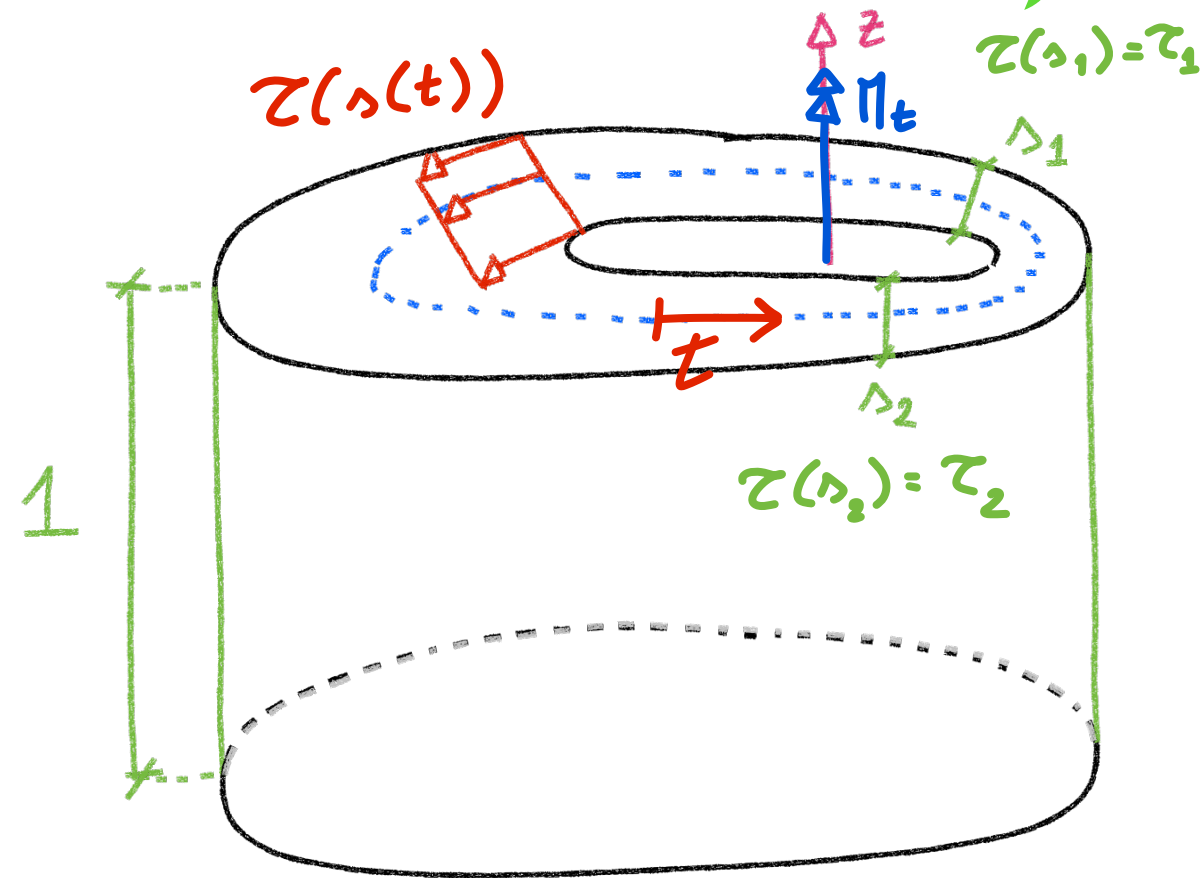
"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot s(t)$$

• EQUILIBRIO ALLA TRASL. SU z:

$$\tau_1 s_1 \cdot 1 - \tau_2 s_2 \cdot 1 = 0$$

$$\tau_1 s_1 = \tau_2 s_2$$



• FORZA SU s_1 : $\tau_1 s_1 \cdot 1$

• FORZA SU s_2 : $\tau_2 s_2 \cdot 1$

TEORIA APPROSSIMATA DI BREDT



"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot s(t)$$

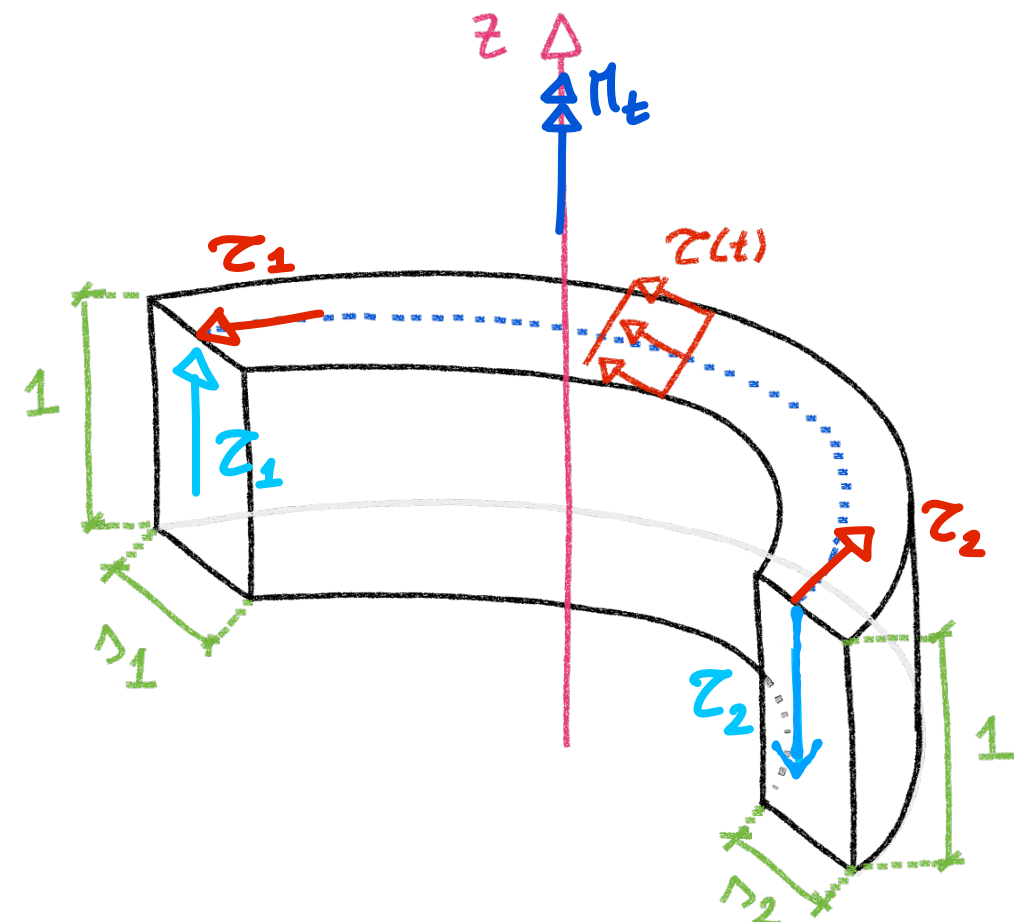
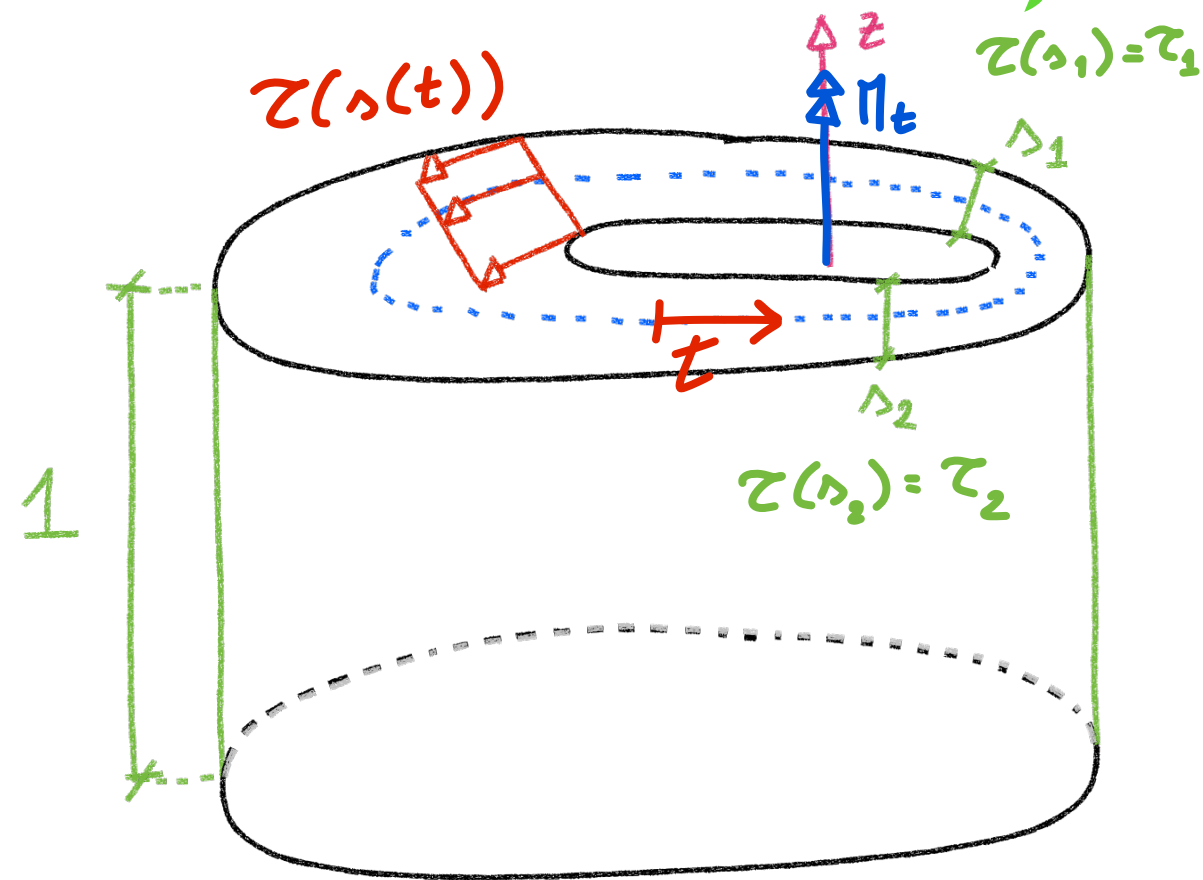
• EQUILIBRIO ALLA TRASL. SU z:

$$\tau_1 s_1 \cdot 1 - \tau_2 s_2 \cdot 1 = 0$$

$$\tau_1 s_1 = \tau_2 s_2$$

VALIDO $\forall$ CORDA $\Rightarrow$

$$q = \omega r t$$



• FORZA SU s1 : $\tau_1 s_1 \cdot 1$

• FORZA SU s2 : $\tau_2 s_2 \cdot 1$

TEORIA APPROSSIMATA DI BREDT



"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot s(t)$$

EQUILIBRIO ALLA TRASL. SU z:

$$z_1 s_1 \cdot 1 - z_2 s_2 \cdot 1 = 0$$

$$z_1 s_1 = z_2 s_2$$

VALIDO $\forall$ CORDA $\Rightarrow$

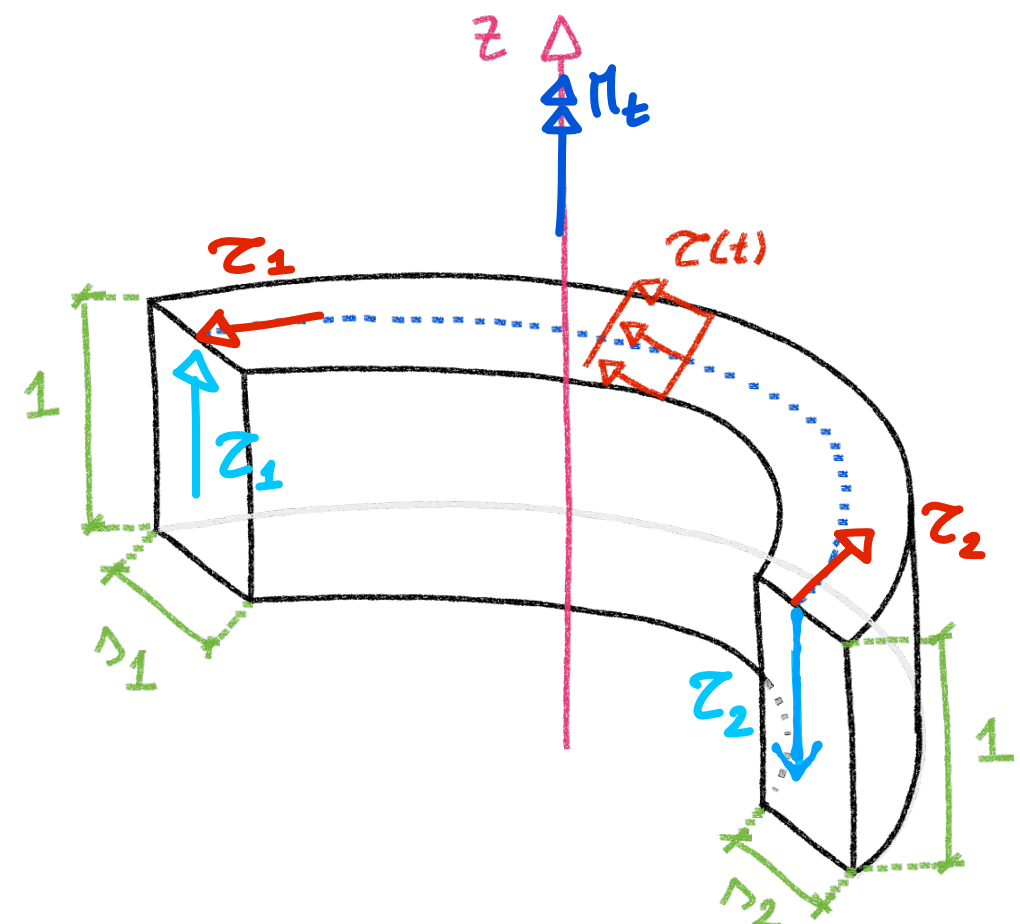
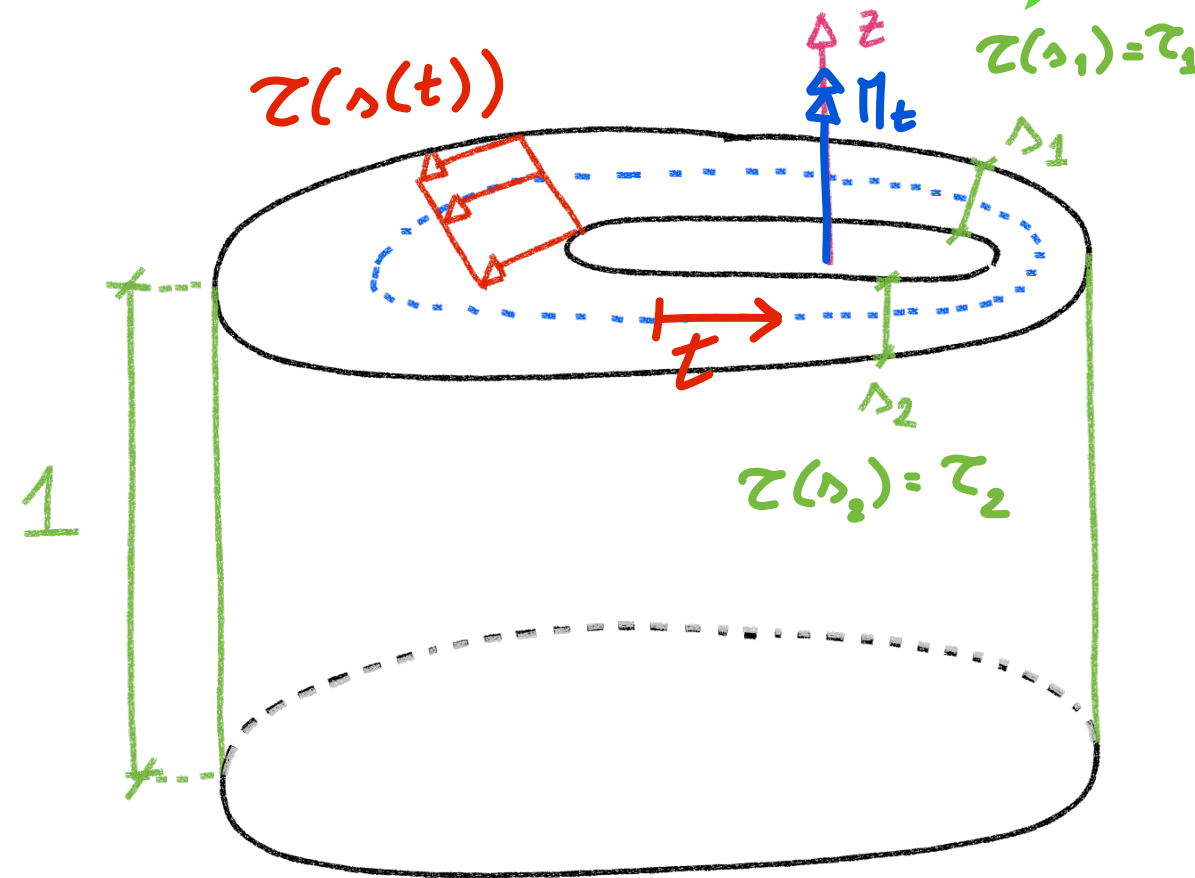
$$q = \omega r t$$

$$z(t) = \frac{q}{s(t)}$$

$z \gg$ per
 $s \ll$

• FORZA SU s_1 : $z_1 s_1 \cdot 1$

• FORZA SU s_2 : $z_2 s_2 \cdot 1$



TEORIA APPROSSIMATA DI BREDT



"FLUSSO DELLE TENSIONI TANGENZ."

$$q = z(t) \cdot \Delta(t)$$

EQUILIBRIO ALLA TRASL. SU z:

$$\tau_1 \Delta_1 \cdot 1 - \tau_2 \Delta_2 \cdot 1 = 0$$

$$\tau_1 \Delta_1 = \tau_2 \Delta_2$$

VALIDO $\forall$ CORDA $\Rightarrow$

$$q = \omega r t$$

$$\tau(t) = \frac{q}{\Delta(t)}$$

$\tau \gg$ per

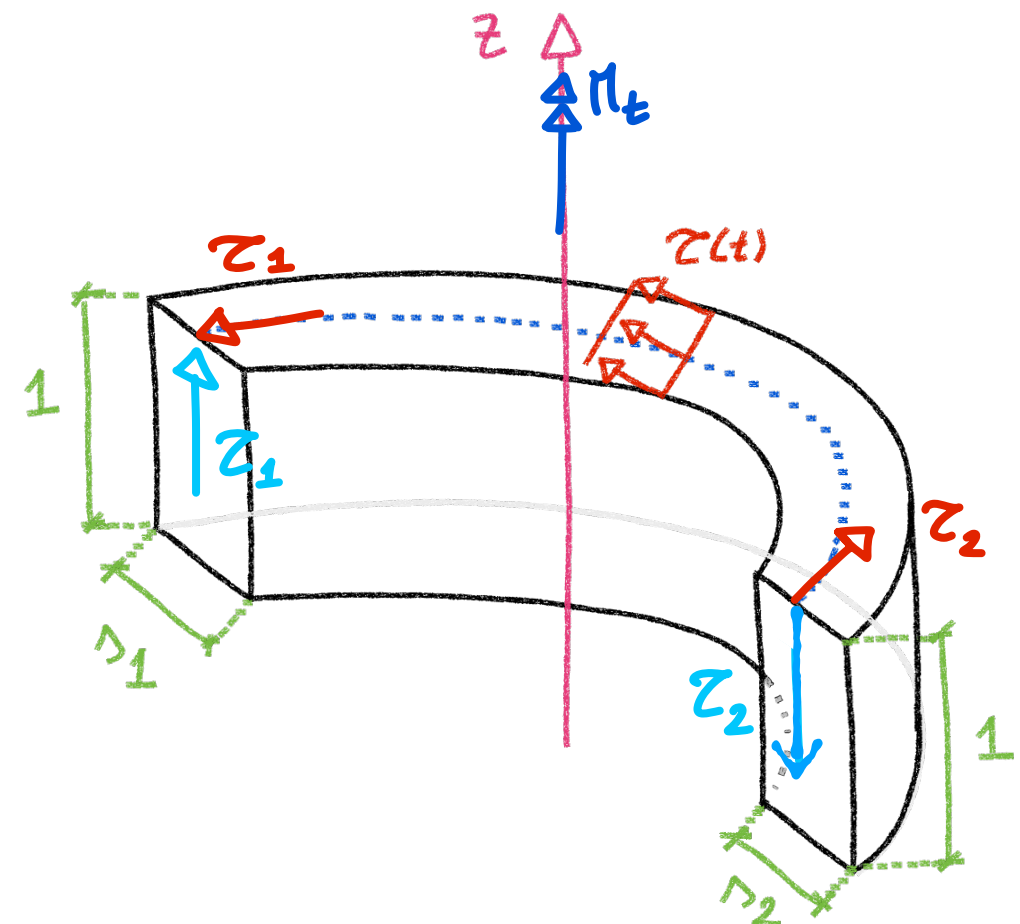
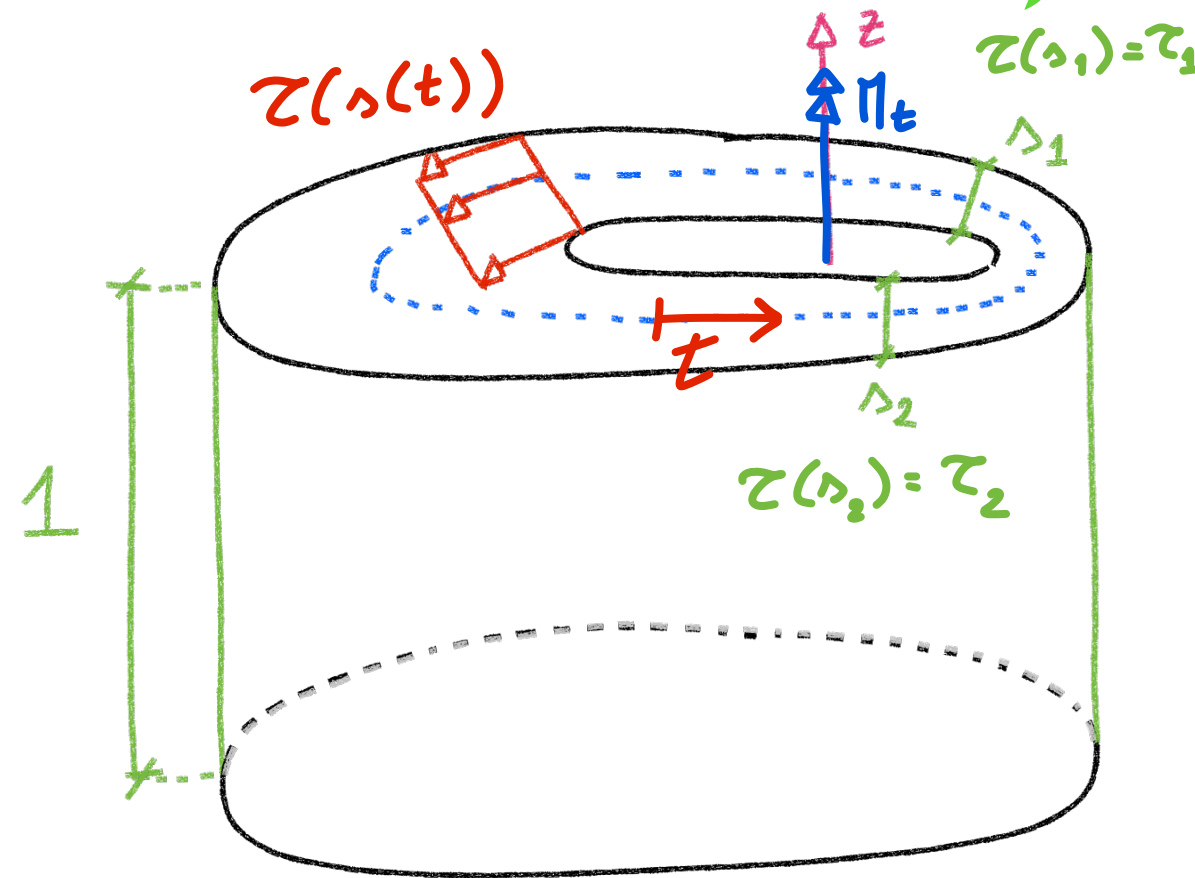
$\Delta \ll$

? CHI CI FORNISCE

q

• FORZA SU Δ_1 : $\tau_1 \Delta_1 \cdot 1$

• FORZA SU Δ_2 : $\tau_2 \Delta_2 \cdot 1$



TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $z(t)$?

"FLUSSO DELLE TENSIONI TANGENZ."
 $q = z(t) \cdot s(t)$

* EQUIVALENZA STATICA DI M_t RISPETTO

• INCREMENTO dt

↳ FORZA ELEMENTARE:

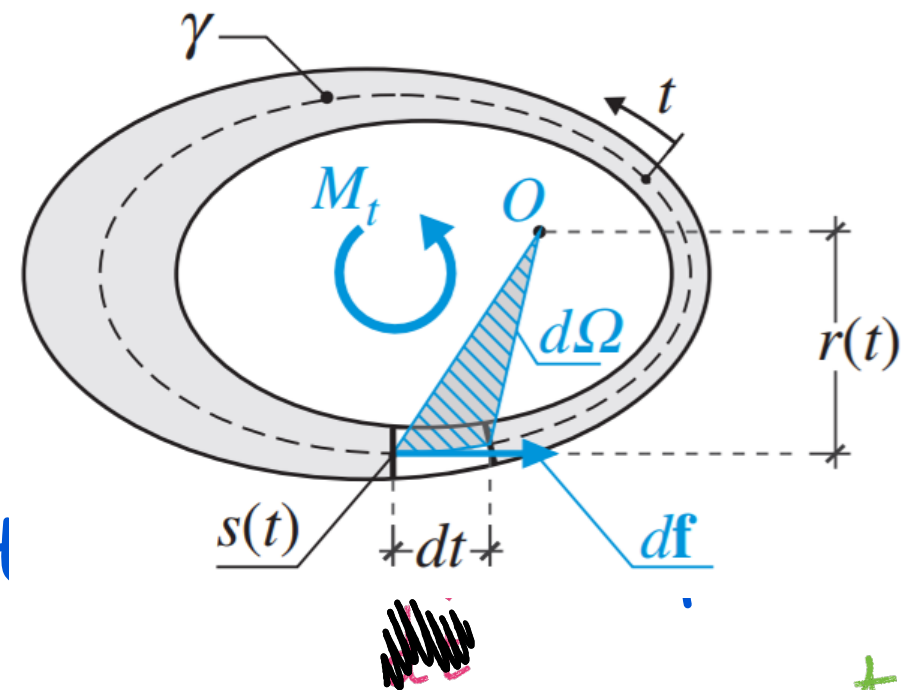
$$df(t) = q(t) dt = z(t) s(t) dt$$

↳ MOMENTO DI $df(t)$:

$$dm(t) = r(t) df(t) = \int r(t) z(t) s(t) dt$$

• SOMMATORIA SU TUTTA LA LINEA MEDIA (a) $\Rightarrow$ INTEGRALE

$$M_t = \int_0^a dm(t) = \int_0^a r(t) z(t) s(t) dt = \int_0^a r(t) q(t) dt = q \int_0^a r(t) dt$$



$q = \text{cost}$

chi e' -

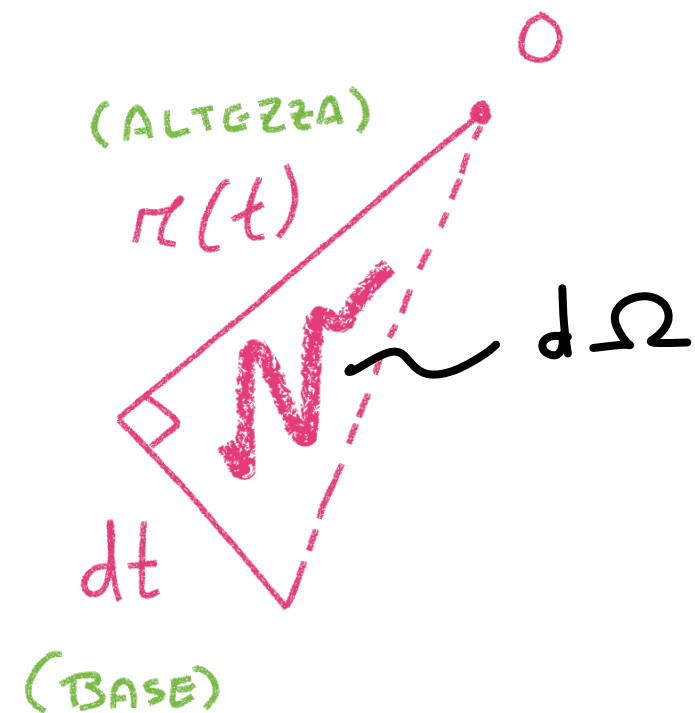
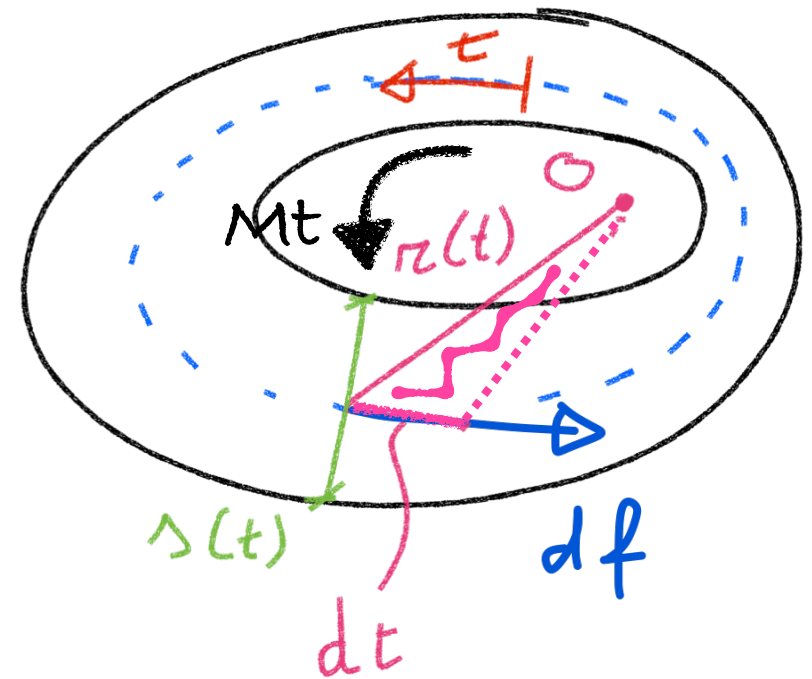
TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $z(t)$?

⊛ EQUIVALENZA STATICA DI M_t RISPETTO ○

• INCREMENTO dt

"FLUSSO DELLE TENSIONI TANGENZ."
|| $q = z(t) \cdot s(t)$ ||



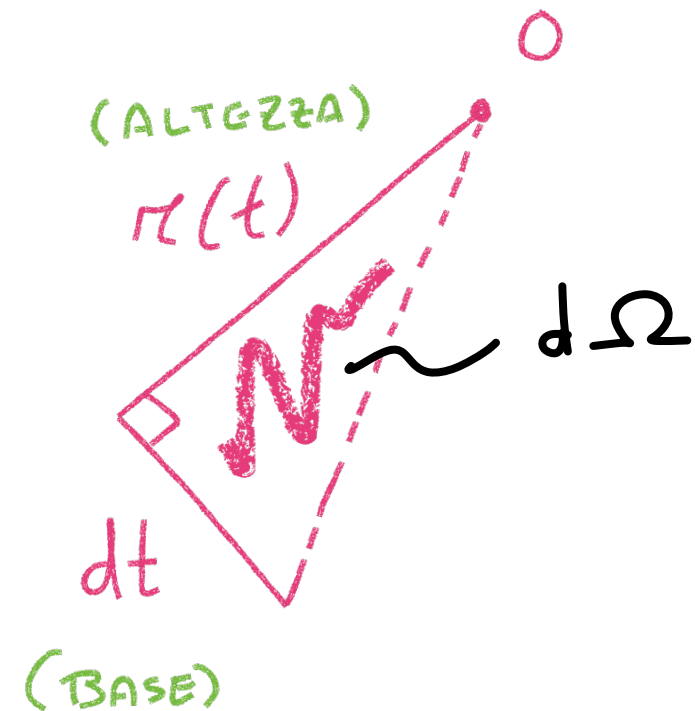
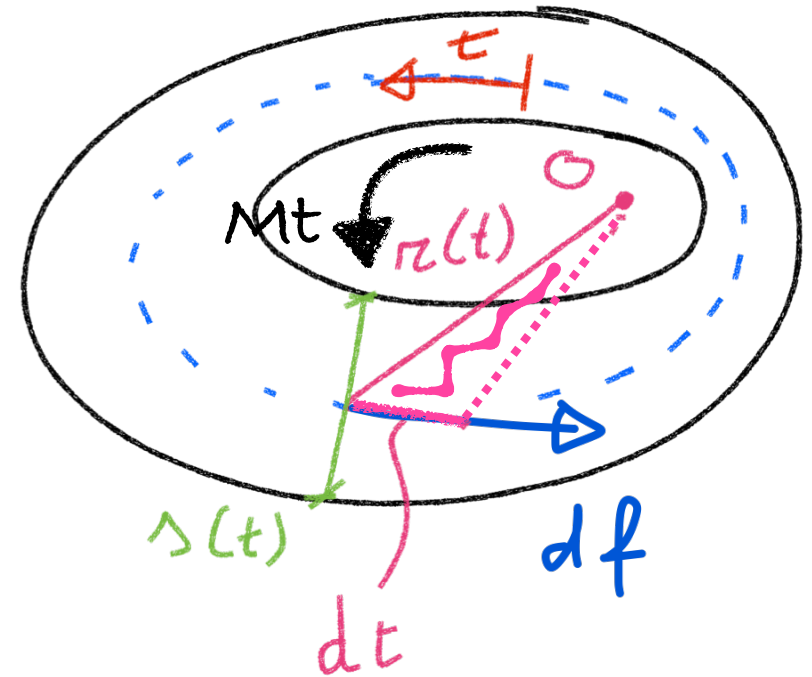
TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $z(t)$?

"FLUSSO DELLE TENSIONI TANGENZ."
|| $q = z(t) \cdot s(t)$ ||

⊛ EQUIVALENZA STATICA DI M_t RISPETTO O

- INCREMENTO dt
- INTEGRALE $M_t = q \int_0^a r(t) dt$



TEORIA APPROSSIMATA DI BREDT

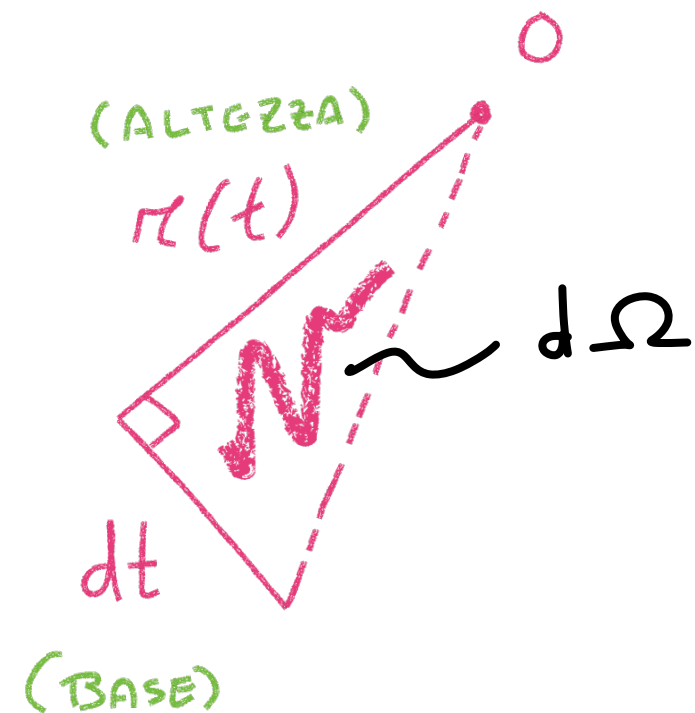
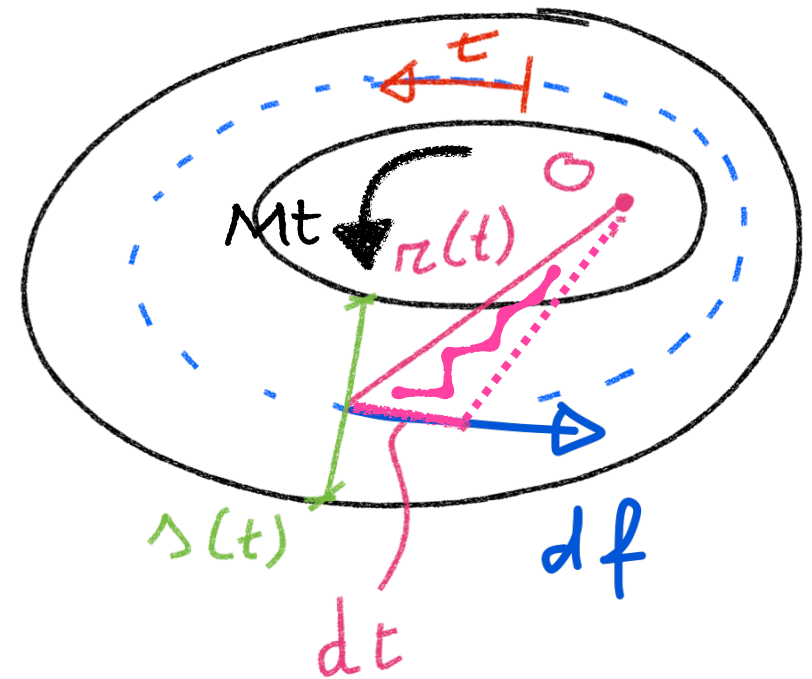
FORMULA DI BREDT: come variano $z(t)$?

"FLUSSO DELLE TENSIONI TANGENZ."
|| $q = z(t) \cdot s(t)$ ||

⊛ EQUIVALENZA STATICA DI M_t RISPETTO ○

- INCREMENTO dt
- INTEGRALE $M_t = q \int_0^a r(t) dt$

$$d\Omega = \frac{1}{2} r(t) dt$$



TEORIA APPROSSIMATA DI BREDT

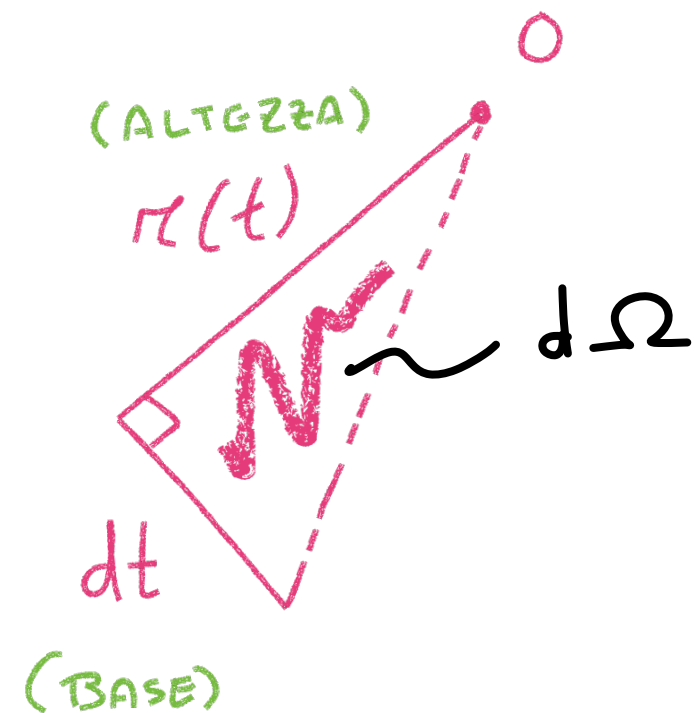
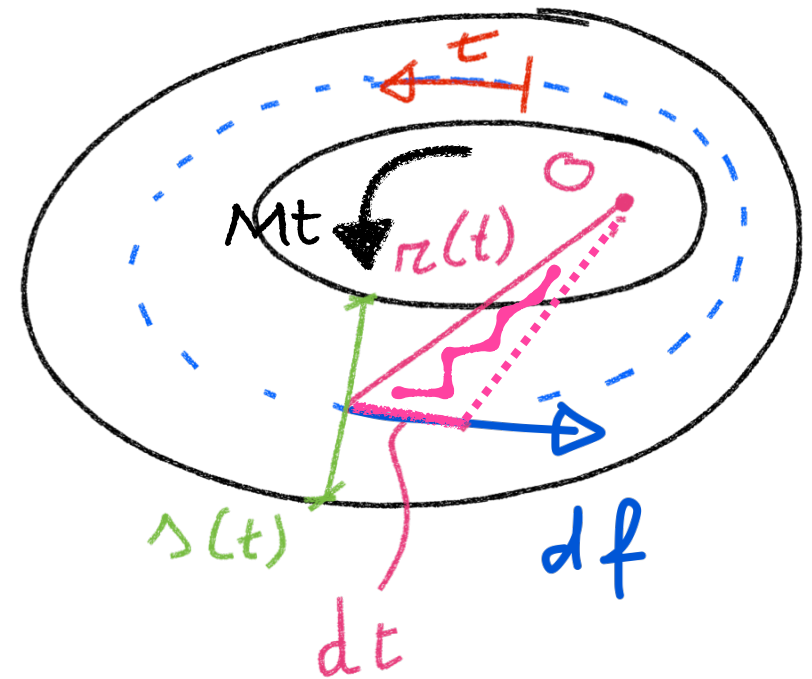
FORMULA DI BREDT: come variano $z(t)$?

"FLUSSO DELLE TENSIONI TANGENZ."
 $q = z(t) \cdot s(t)$

* EQUIVALENZA STATICA DI M_t RISPETTO O

- INCREMENTO dt
- INTEGRALE $M_t = q \int_0^a r(t) dt$

$$d\Omega = \frac{1}{2} r(t) dt \rightarrow 2d\Omega = r(t) dt$$



TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $z(t)$?

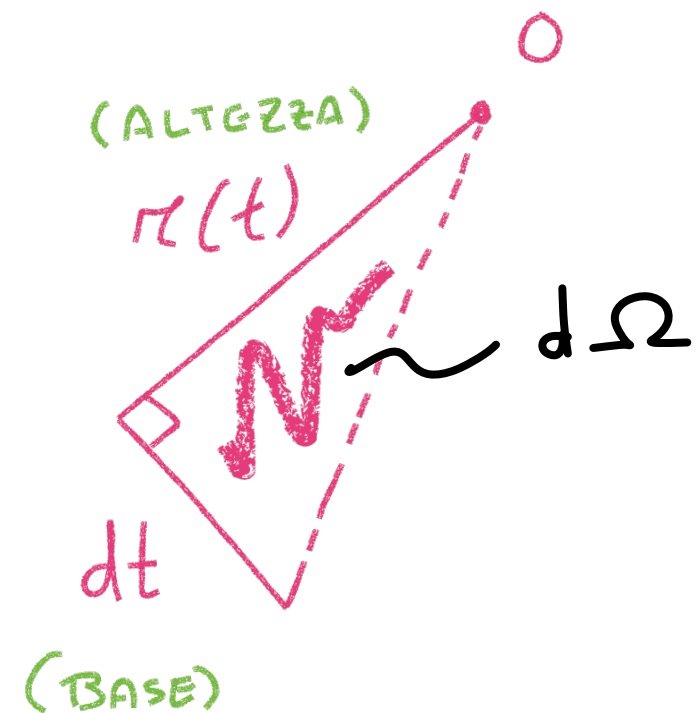
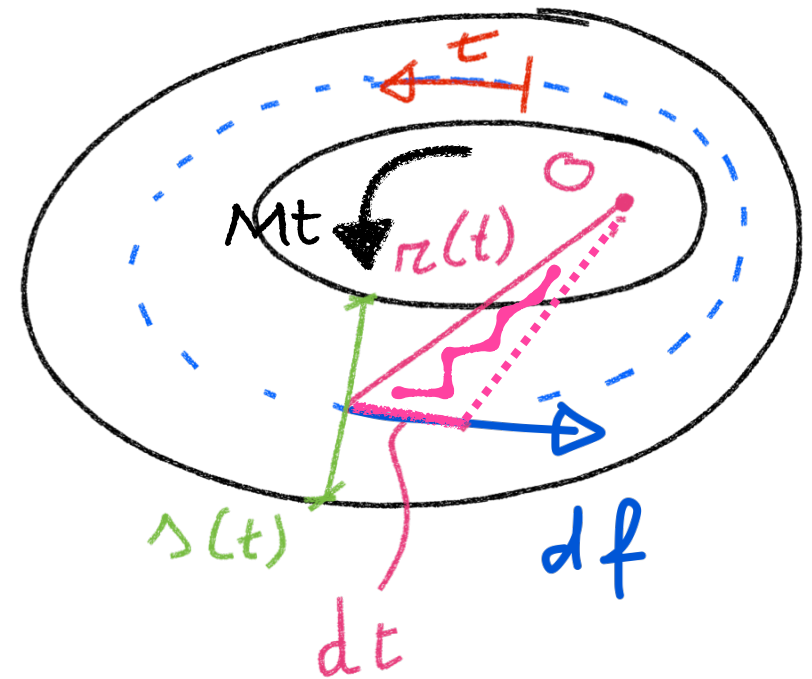
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* EQUIVALENZA STATICA DI M_t RISPETTO

- INCREMENTO dt
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$$d\Omega = \frac{1}{2} r(t) dt \rightarrow 2d\Omega = r(t) dt$$

$$\rightarrow \int_0^a r(t) dt = \int_0^a 2d\Omega = 2\Omega$$



TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $z(t)$?

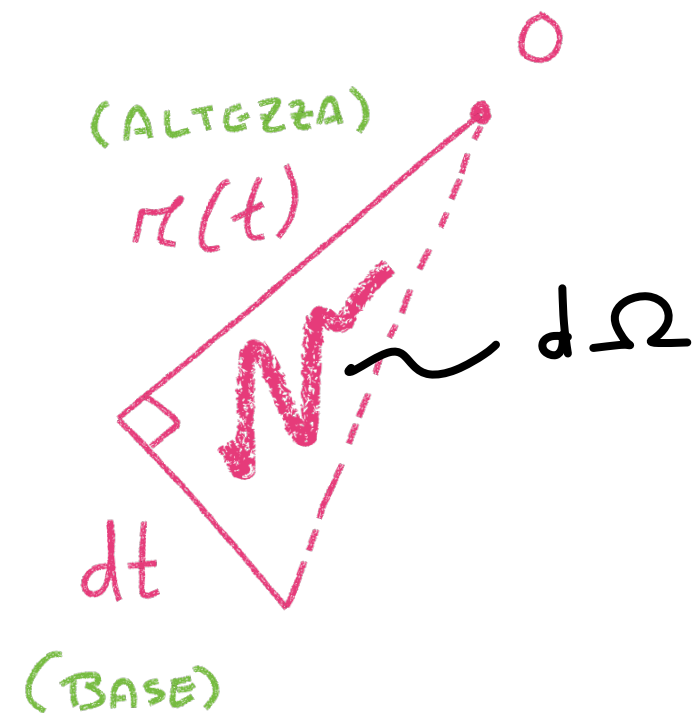
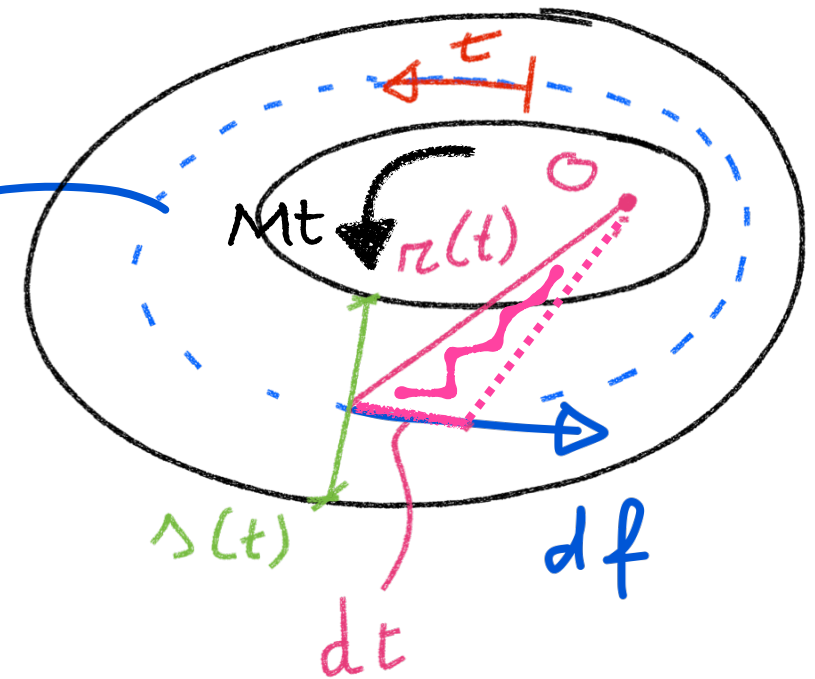
"FLUSSO DELLE TENSIONI TANGENZ."
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* EQUIVALENZA STATICA DI M_t RISPETTO

- INCREMENTO dt
- INTEGRALE $M_t = q \int_0^a r(t) dt$
- AREA RACCHIUSA DA LINEA MEDIA: Ω

$$d\Omega = \frac{1}{2} r(t) dt \rightarrow 2d\Omega = r(t) dt$$

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TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $z(t)$?

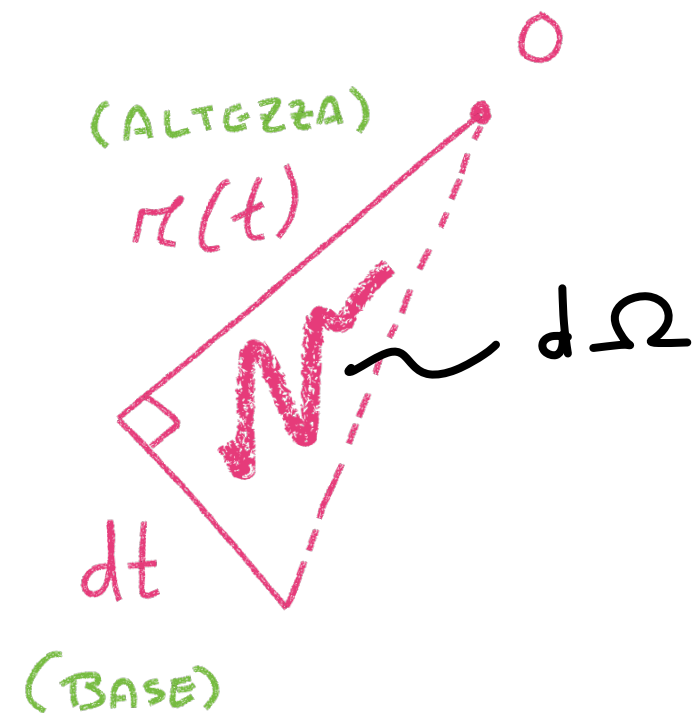
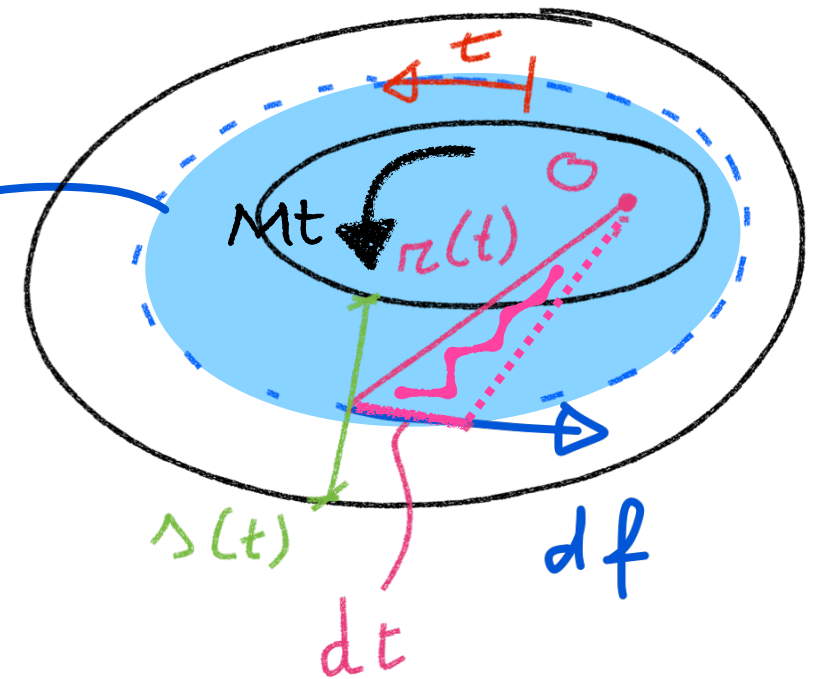
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TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $\tau(t)$?

"FLUSSO DELLE TENSIONI TANGENZ."
 $q = \tau(t) \cdot s(t)$

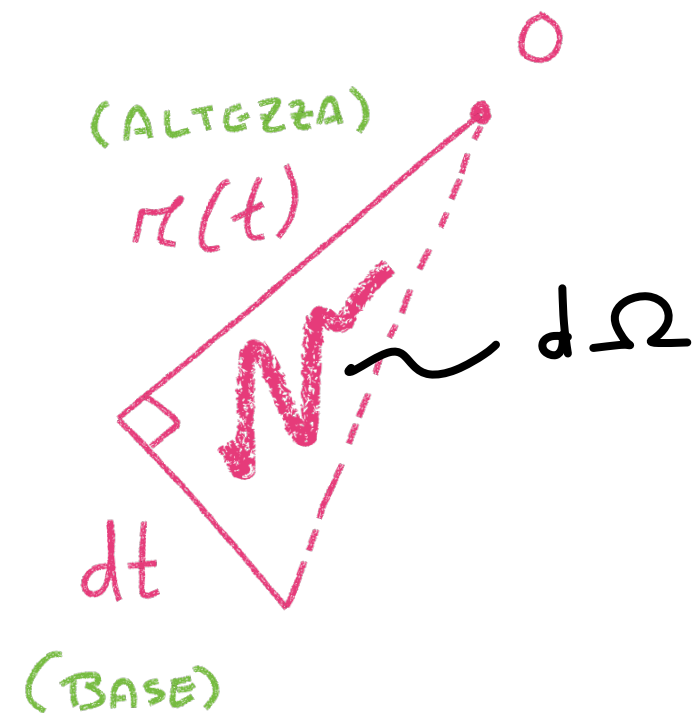
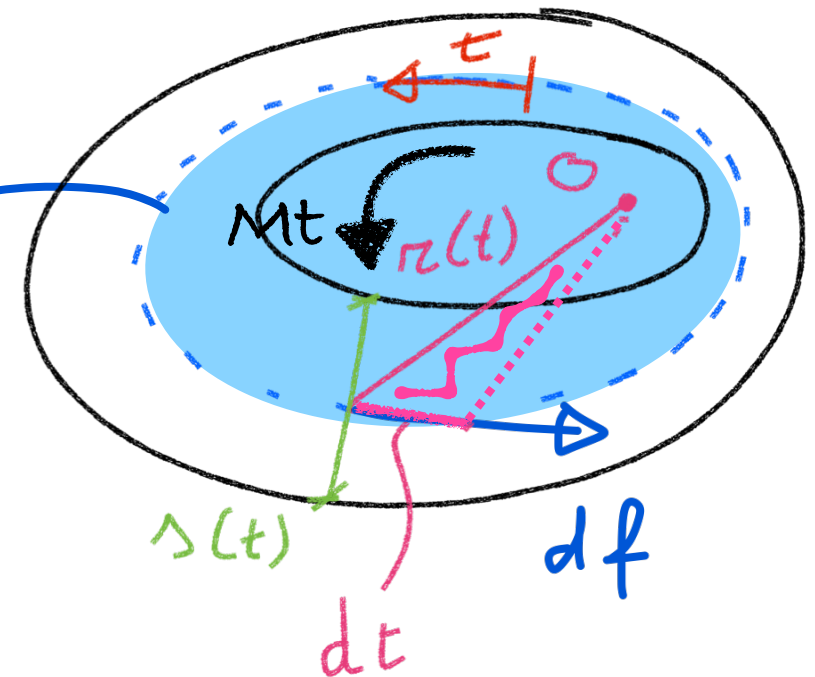
* EQUIVALENZA STATICA DI M_t RISPETTO

- INCREMENTO dt
- INTEGRALE $M_t = q \int_0^a \tau(t) dt$
- AREA RACCHIUSA DA LINEA MEDIA: Ω

$$d\Omega = \frac{1}{2} \tau(t) dt \rightarrow 2d\Omega = \tau(t) dt$$

$$\rightarrow \int_0^a \tau(t) dt = \int_0^a 2d\Omega = 2\Omega$$

$$\rightarrow M_t = q \cdot 2\Omega = \tau(t) s(t) \cdot 2\Omega$$



TEORIA APPROSSIMATA DI BREDT

FORMULA DI BREDT: come variano $\tau(t)$?

"FLUSSO DELLE TENSIONI TANGENZ."
 $q = \tau(t) \cdot s(t)$

* EQUIVALENZA STATICA DI M_t RISPETTO

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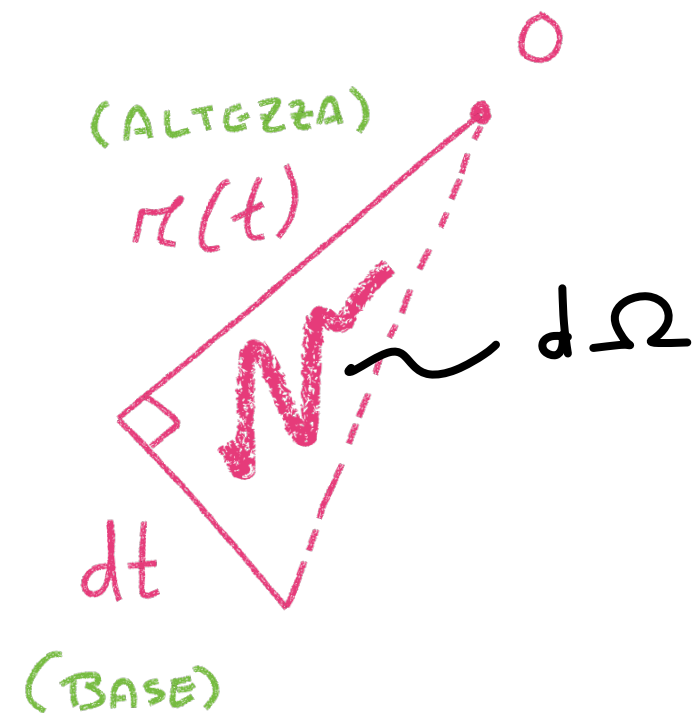
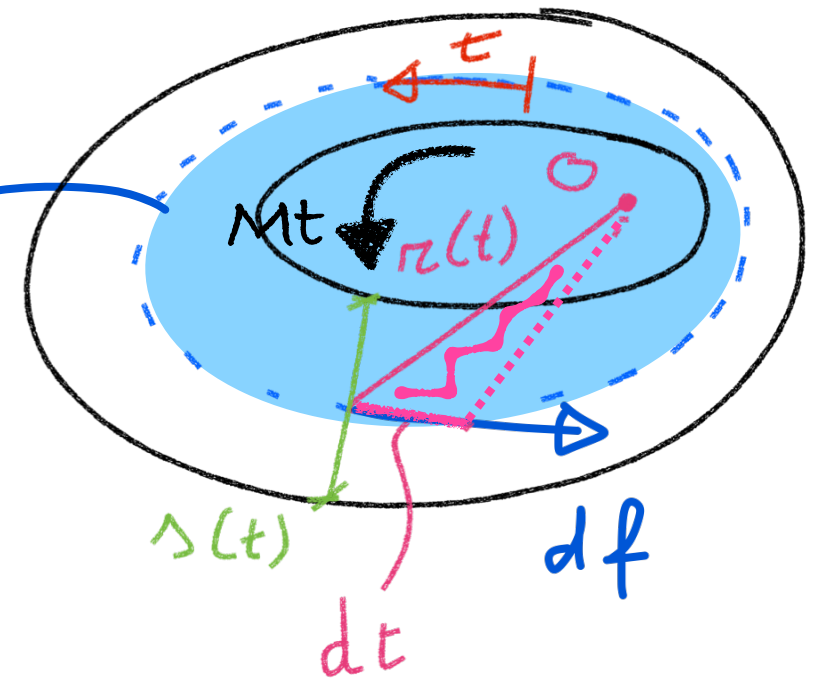
$$d\Omega = \frac{1}{2} \tau(t) dt \rightarrow 2d\Omega = \tau(t) dt$$

$$\rightarrow \int_0^a \tau(t) dt = \int_0^a 2d\Omega = 2\Omega$$

$$\rightarrow M_t = q \cdot 2\Omega = \tau(t) s(t) \cdot 2\Omega$$

$$\rightarrow \tau(t) = \frac{M_t}{2\Omega s(t)}$$

PRIMA
FORMULA
DI
BREDT



TEORIA APPROSSIMATA DI BREDT

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases}$$

TEORIA APPROSSIMATA DI BREDT

Calcolo Angolo Unitario di Torsione

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

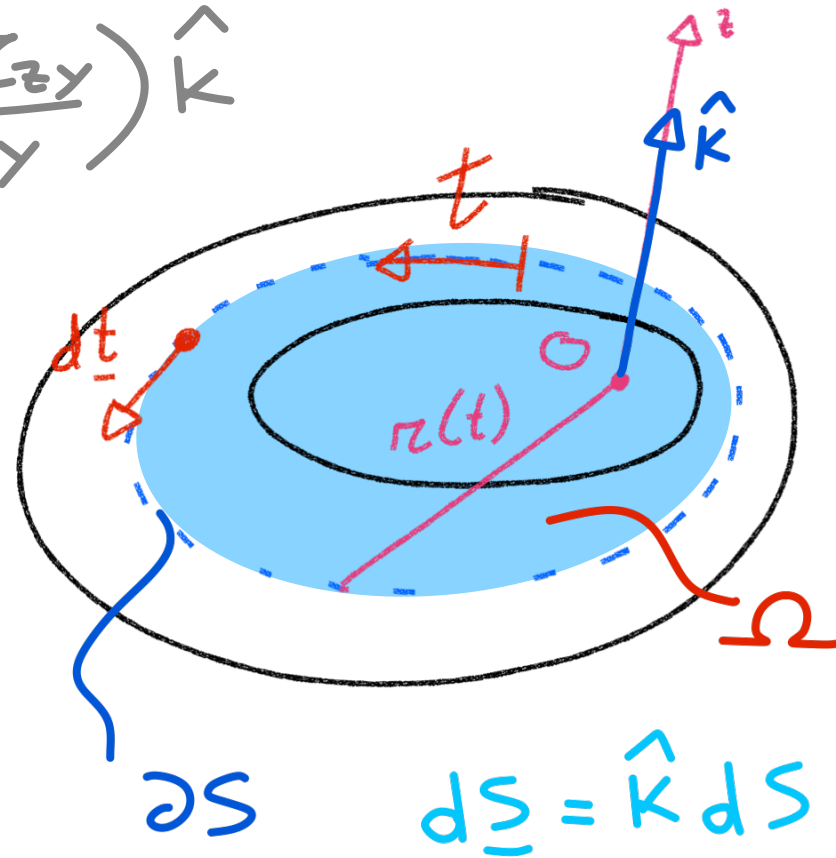
$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}(\underline{\tau})} = 2G\Theta \hat{k}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}}(\underline{\tau}) = 2G\Theta \hat{k}$$



$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$dt = \hat{n} dt$$

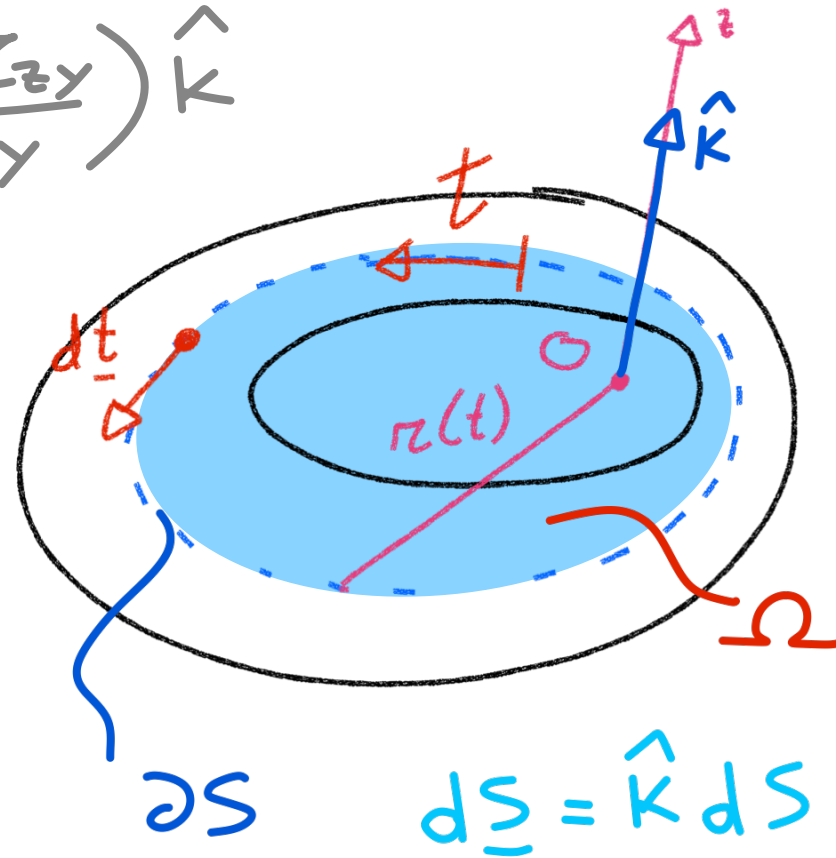
TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}}(\underline{\tau}) = \underline{2G\Theta \hat{k}}$$

$$\int_S \text{rot}(\underline{\tau}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$



$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$dt = \hat{n} dt$$

TEORIA APPROSSIMATA DI BREDT

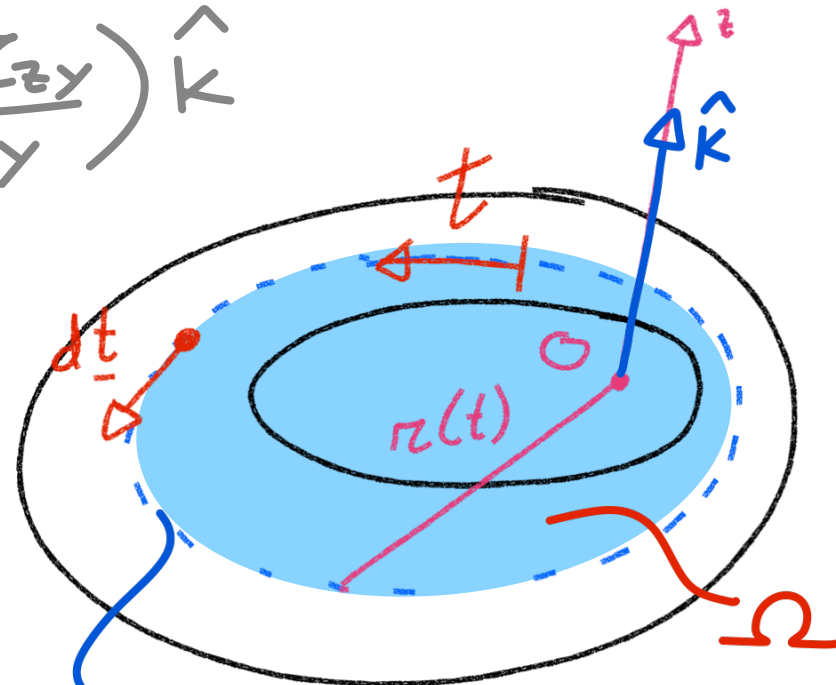
CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

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$$\int_S \text{rot}(\underline{\tau}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{\tau}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$



$$\partial S \rightarrow d\underline{\Sigma} = \hat{k} dS$$

$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$d\underline{t} = \hat{m} dt$$

TEORIA APPROSSIMATA DI BREDT

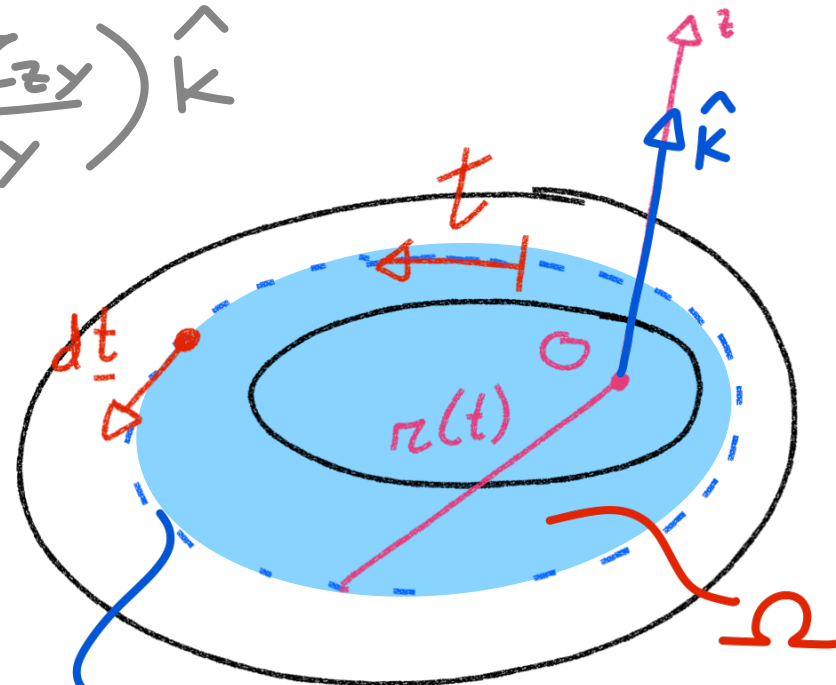
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$$\int_S \text{rot}(\underline{\tau}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{\tau}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$



$$\partial S \rightarrow d\underline{\Sigma} = \hat{k} dS$$

$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

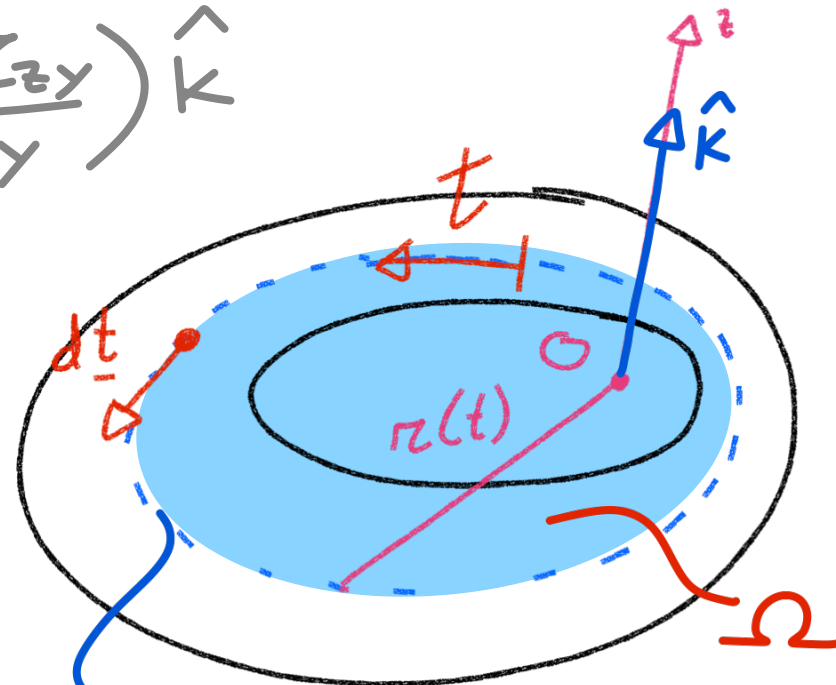
$$d\underline{t} = \hat{m} dt$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{z} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{z}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}}(\underline{z}) = 2G\Theta \hat{k}$$



$$\int_S \text{rot}(\underline{z}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$

$$\partial S \rightarrow d\underline{\Sigma} = \hat{k} dS$$

$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$d\underline{t} = \hat{m} dt$$

*** TEOREMA DI STOKES (CIRCVITAZIONE)**

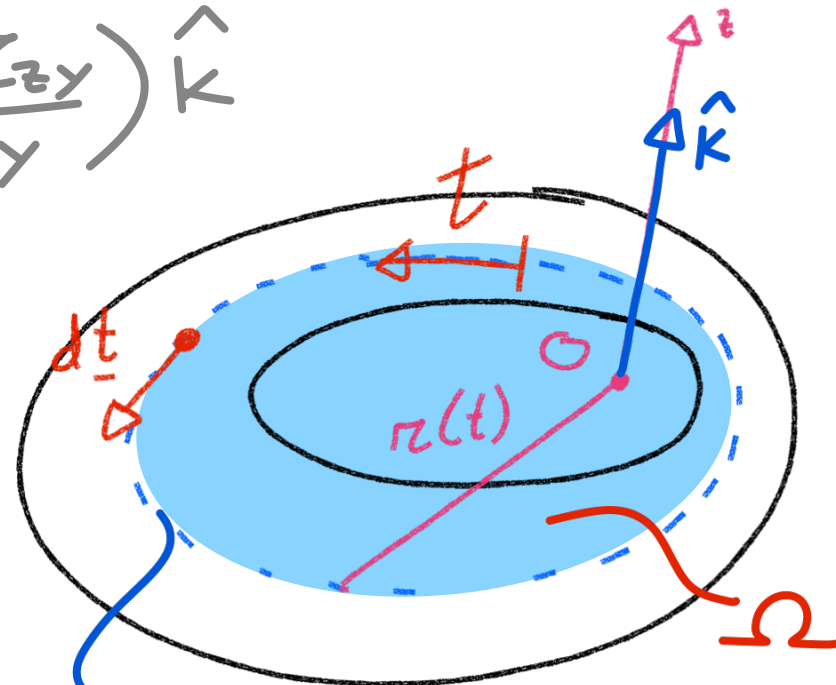
$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \oint_{\partial S} \underline{z} \cdot d\underline{t}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{\tau} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{\tau}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}}(\underline{\tau}) = 2G\Theta \hat{k}$$



$$\int_S \text{rot}(\underline{\tau}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{\tau}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$

I

$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$d\underline{t} = \hat{n} dt$$

* TEOREMA DI STOKES (CIRCVITAZIONE)

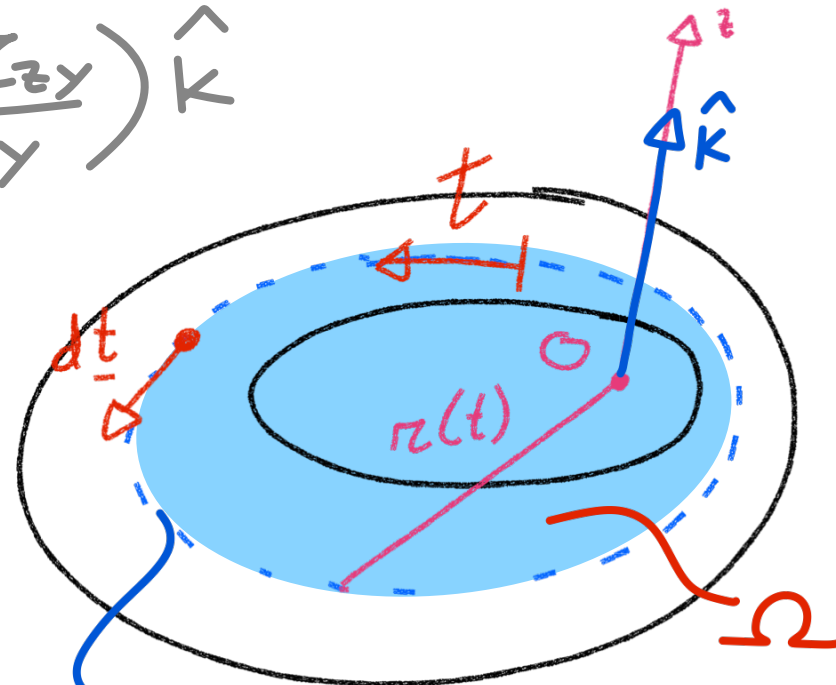
$$\int_S \text{rot}(\underline{\tau}) d\underline{\Sigma} = \oint_{\partial S} \underline{\tau} \cdot d\underline{t}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{z} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{z}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}(\underline{z})} = 2G\Theta \hat{k}$$



$$\int_S \text{rot}(\underline{z}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$

I

$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$d\underline{t} = \hat{n} dt$$

* TEOREMA DI STOKES (CIRCVITAZIONE)

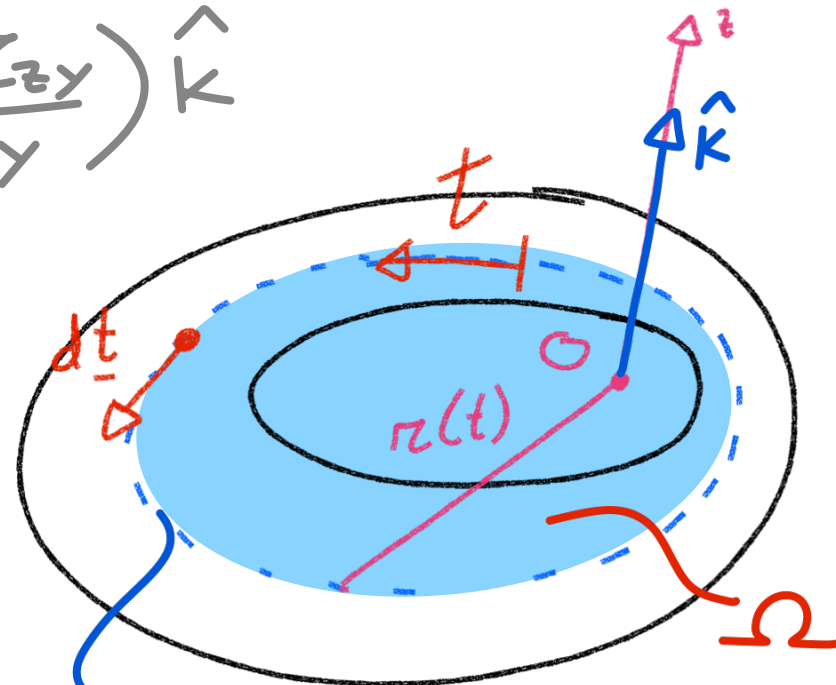
$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \oint_{\partial S} \underline{z} \cdot d\underline{t} = \oint_{\partial S} \underline{z} \cdot \hat{n} dt = \oint_{\partial S} z(t) dt$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{z} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{z}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}}(\underline{z}) = 2G\Theta \hat{k}$$



$$\int_S \text{rot}(\underline{z}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$

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$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

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* TEOREMA DI STOKES (CIRCVITAZIONE)

$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \oint_{\partial S} \underline{z} \cdot d\underline{t} = \oint_{\partial S} \underline{z} \cdot \hat{n} dt = \oint_{\partial S} z(t) dt = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

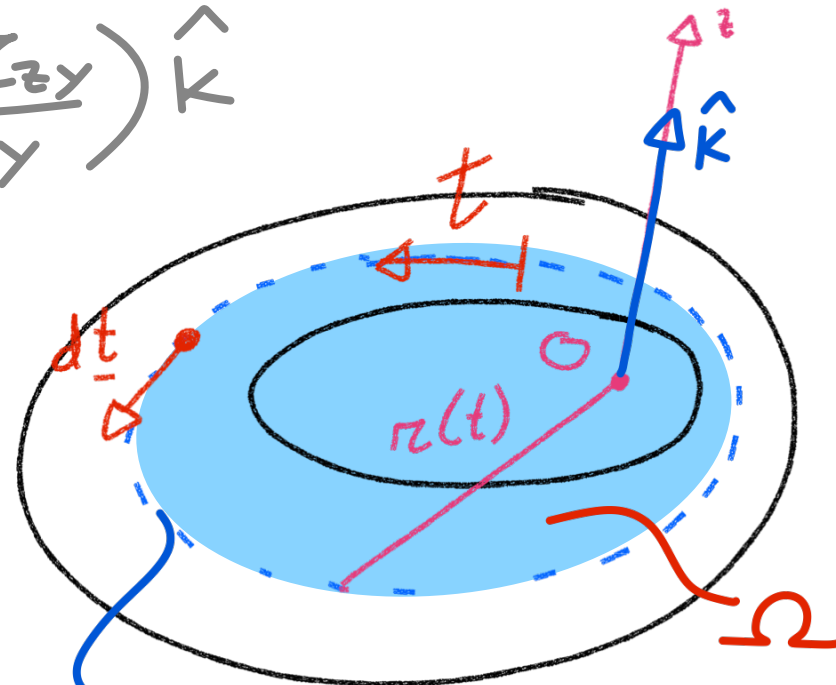
$$z(t) = M_t / (2\Omega s) \quad 1^a \text{ F.B.}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\underline{z} = [\tau_{zx} \ \tau_{zy} \ 0]^T \Rightarrow \text{rot}(\underline{z}) = \left(\frac{\partial \tau_{zx}}{\partial x} - \frac{\partial \tau_{zy}}{\partial y} \right) \hat{k}$$

$$\begin{cases} \tau_{zx} = G\Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G\Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases} \Rightarrow \underline{\text{rot}}(\underline{z}) = 2G\Theta \hat{k}$$



$$\int_S \text{rot}(\underline{z}) \cdot \hat{k} dS = \int_S 2G\Theta \hat{k} \cdot \hat{k} dS$$

$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \int_S 2G\Theta dS = 2G\Theta \Omega$$

$$\partial S \rightarrow d\underline{\Sigma} = \hat{k} dS$$

$$\Omega = \int_S dS; \quad a = \int_{\partial S} dt$$

$$d\underline{t} = \hat{n} dt$$

*** TEOREMA DI STOKES (CIRCVITAZIONE) II**

$$\int_S \text{rot}(\underline{z}) d\underline{\Sigma} = \oint_{\partial S} \underline{z} \cdot d\underline{t} = \oint_{\partial S} \underline{z} \cdot \hat{n} dt = \oint_{\partial S} z(t) dt = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$z(t) = M_t / (2\Omega s) \quad \text{1° F.B.}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\int_S \text{rot}(z) dS = \int_S 2G\Theta dS = 2G\Theta \Omega$$

Da Neumann

$$\int_S \text{rot}(z) dS = \frac{M_t}{2\Omega} \oint \frac{1}{\rho(t)} dt$$

Da 1a F. Bredt

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\int_S \text{rot}(z) dS = \int_S 2G\Theta dS = 2G\Theta \Omega$$

Da Neumann

$$\int_S \text{rot}(z) dS = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

Da 1a F. Bredt

$$\Downarrow$$
$$2G\Theta \Omega = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\int_S \text{rot}(z) dS = \int_S 2G\Theta dS = 2G\Theta \Omega$$

Da Neumann

$$\int_S \text{rot}(z) dS = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

Da 1a F. Bredt

$$2G\Theta \Omega = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$\Theta = \frac{M_t}{4G\Omega^2} \oint_{\partial S} \frac{1}{s(t)} dt$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\int_S \text{rot}(z) dS = \int_S 2G\Theta dS = 2G\Theta \Omega$$

Da Neumann

$$\int_S \text{rot}(z) dS = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

Da 1a F. Bredt

$$2G\Theta \Omega = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$\Theta = \frac{M_t}{4G\Omega^2} \oint_{\partial S} \frac{1}{s(t)} dt$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

INERZIA TORSIONALE
GENERALIZZATA

$$\int_S \text{rot}(\underline{z}) d\underline{S} = \int_S 2G\Theta dS = 2G\Theta \Omega$$

$$\int_S \text{rot}(\underline{z}) d\underline{S} = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$2G\Theta \Omega = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$\Theta = \frac{M_t}{4G\Omega^2} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{s(t)}}$$

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\int_S \text{rot}(z) dS = \int_S 2G\Theta dS = 2G\Theta \Omega$$

$$\int_S \text{rot}(z) dS = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$2G\Theta \Omega = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$\Theta = \frac{M_t}{4G\Omega^2} \oint_{\partial S} \frac{1}{s(t)} dt$$

INERZIA TORSIONALE
GENERALIZZATA

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{s(t)}}$$

$$\Theta = \frac{M_t}{GI_t}$$

SECONDA FORMULA
DI
BREDT

TEORIA APPROSSIMATA DI BREDT

CALCOLO ANGOLO UNITARIO DI TORSIONE

$$\int_S \text{rot}(z) dS = \int_S 2G\Theta dS = 2G\Theta \Omega$$

$$\int_S \text{rot}(z) dS = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$2G\Theta \Omega = \frac{M_t}{2\Omega} \oint_{\partial S} \frac{1}{s(t)} dt$$

$$\Theta = \frac{M_t}{4G\Omega^2} \oint_{\partial S} \frac{1}{s(t)} dt$$

INERZIA TORSIONALE
GENERALIZZATA

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{s(t)}}$$

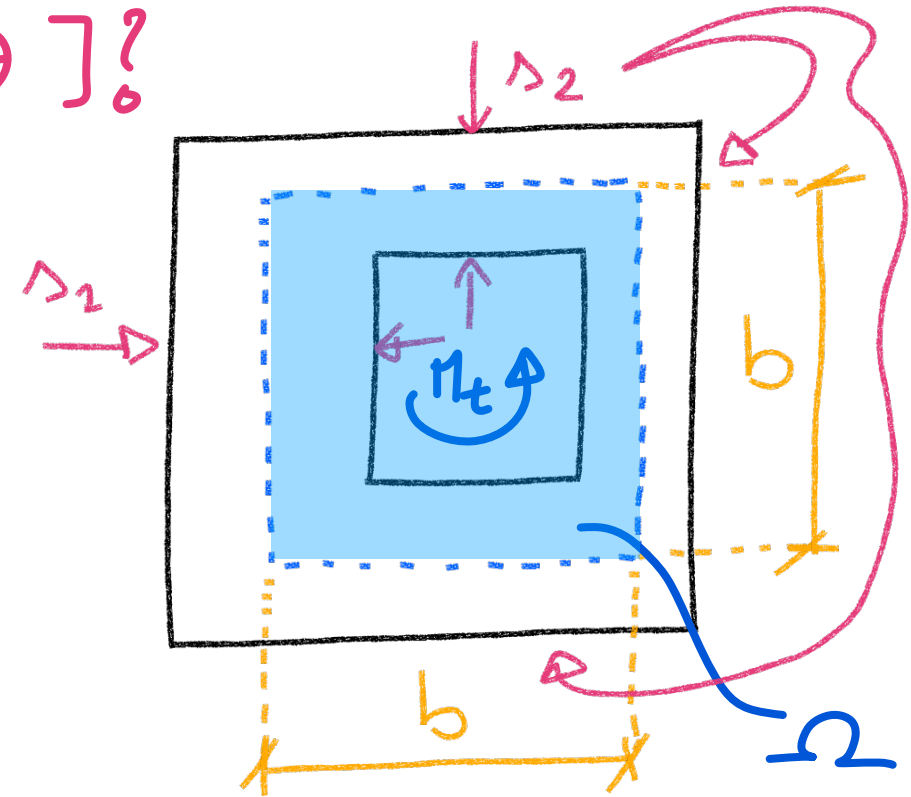
$$\Theta = \frac{M_t}{GI_t}$$

SECONDA FORMULA
DI
BREDT

(*) SE $s(t) = s = \text{cost} \Rightarrow I_t = 4\Omega^2 \frac{s}{a}$

ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \lambda_1 = 2\lambda_2 \equiv 2\lambda \ll b \quad [\gamma, \Theta]?$$

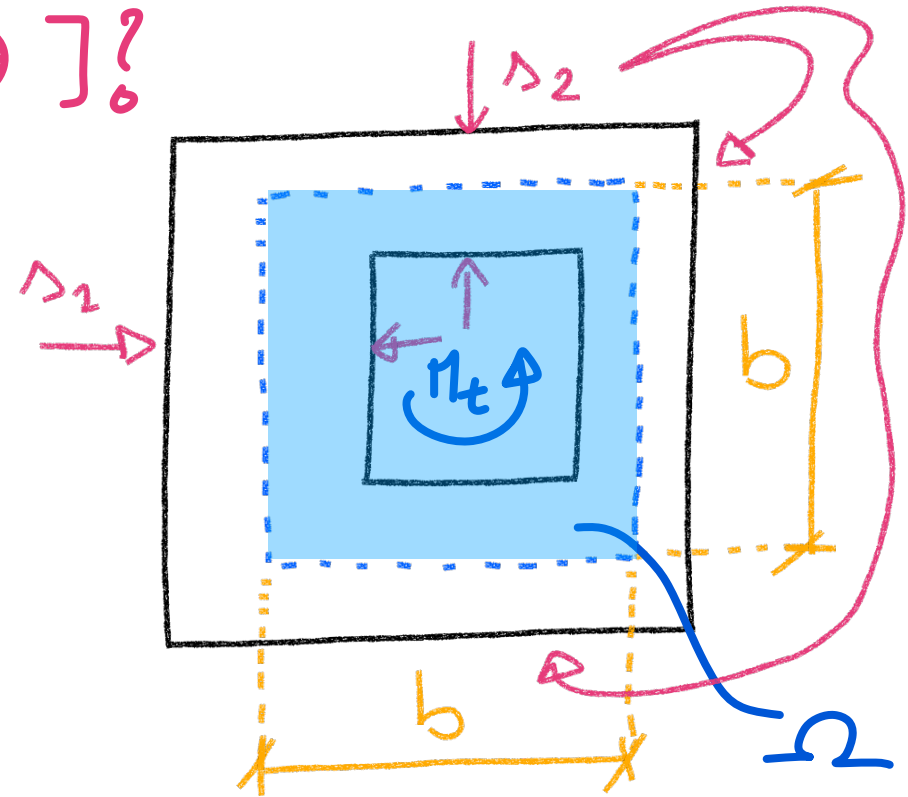


ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \Delta_1 = 2\Delta_2 \equiv 2\Delta \ll b \quad [\tau, \Theta]?$$

FLUSSO DI τ :

$$q = \frac{M_t}{2\Omega} = \frac{M_t}{2b^2} \quad ; \quad (\tau = q\Delta)$$

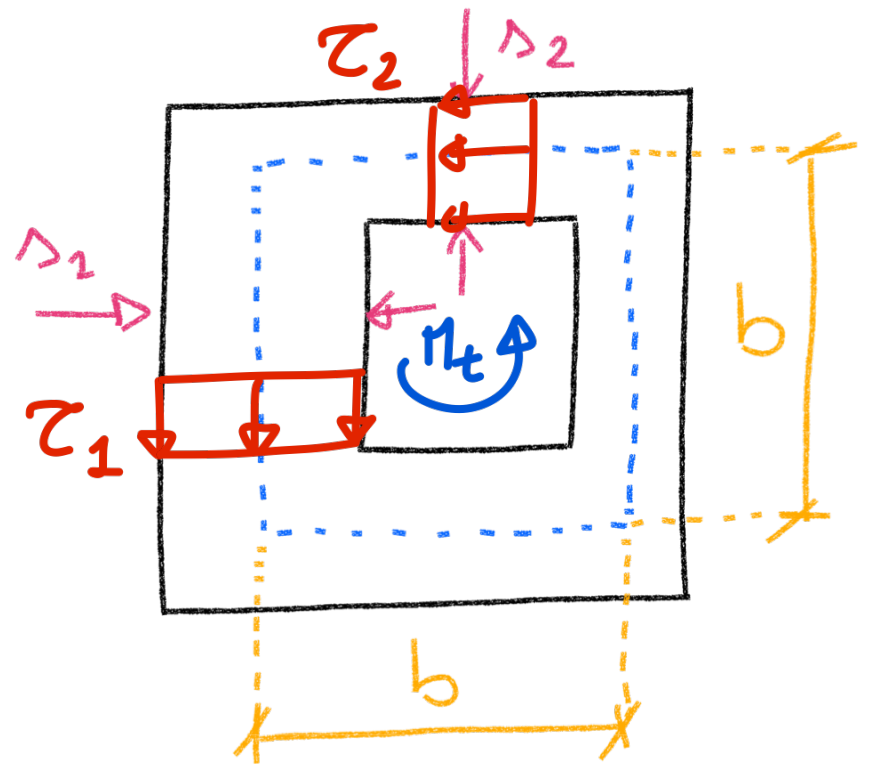
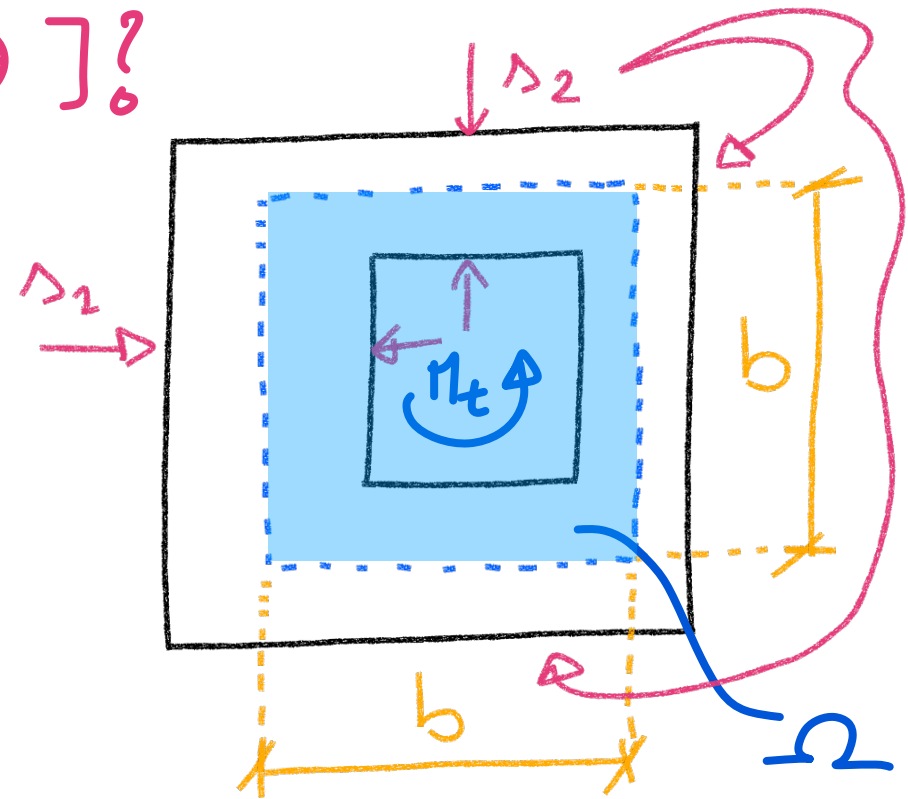


ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \Delta_1 = 2\Delta_2 \equiv 2\Delta < b \quad [\tau, \Theta]?$$

FLUSSO DI τ :

$$q = \frac{M_t}{2\Omega} = \frac{M_t}{2b^2} \quad ; \quad (\tau = q\Delta)$$



ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \Delta_1 = 2\Delta_2 = 2\Delta < b \quad [\tau, \Theta]?$$

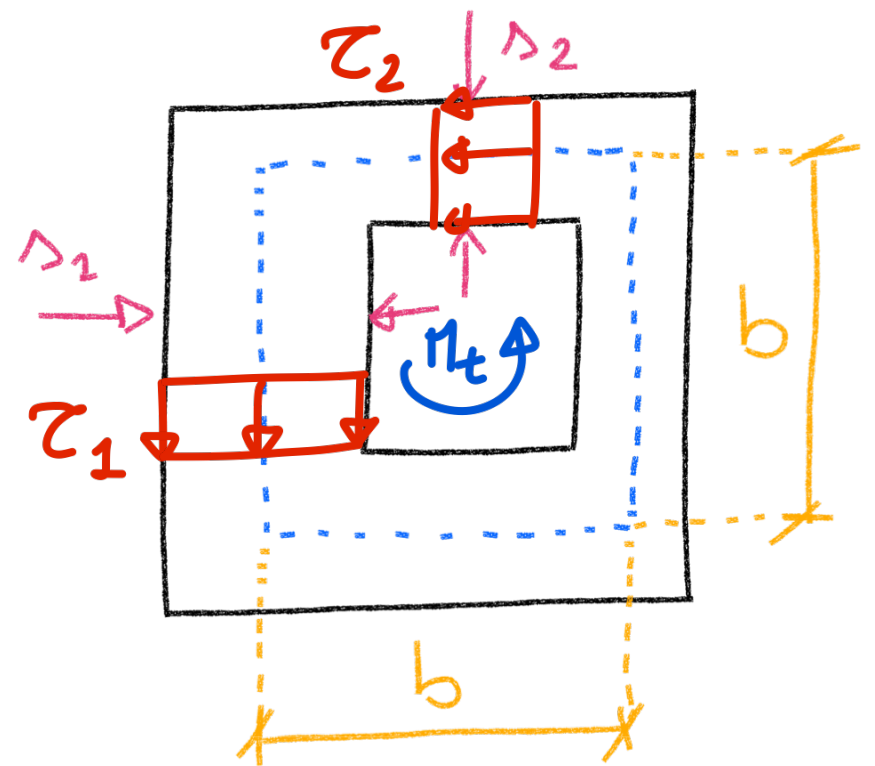
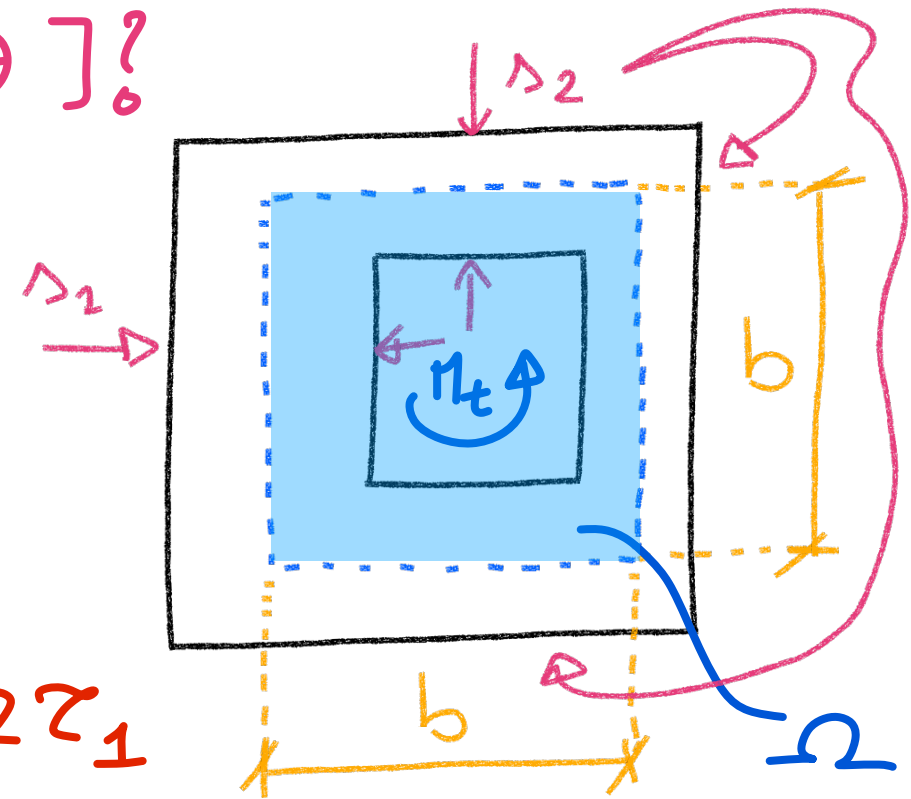
FLUSSO DI τ :

$$q = \frac{M_t}{2\Omega} = \frac{M_t}{2b^2} \quad ; \quad (\tau = q\Delta)$$

$$\hookrightarrow \tau_1 = \frac{q}{\Delta_1} = \frac{M_t}{2b^2 \cdot 2\Delta} = \frac{M_t}{4b^2\Delta}$$

$$\hookrightarrow \tau_2 = \frac{q}{\Delta_2} = \frac{M_t}{2b^2 \cdot \Delta} = \frac{M_t}{2b^2\Delta}$$

$$\tau_2 = 2\tau_1$$



ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \Delta_1 = 2\Delta_2 = 2\Delta < b \quad [\tau, \Theta]?$$

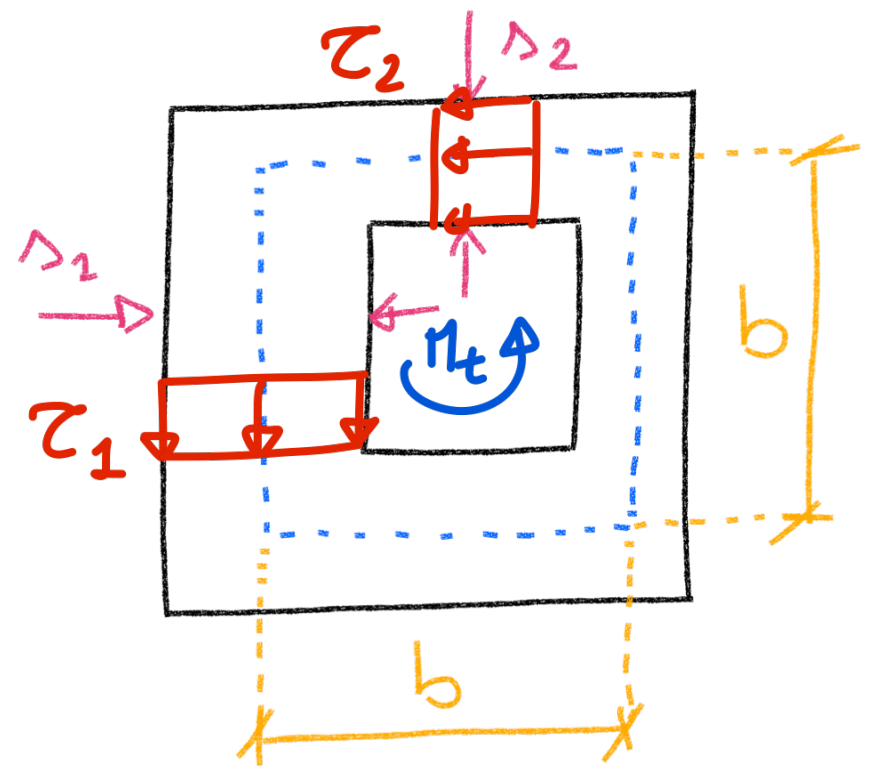
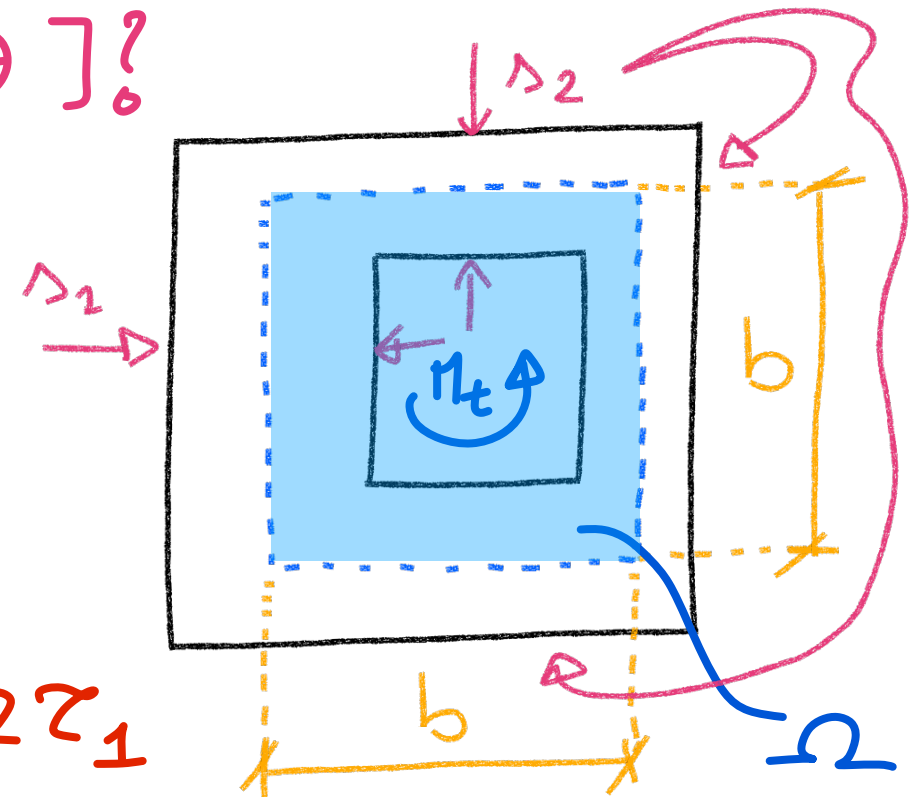
FLUSSO DI τ :

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$$\tau_2 = 2\tau_1$$



INERZIA TORSIONALE:

$$\int_0^a \frac{dt}{\Delta(t)} = \int_0^b \frac{dt}{\Delta_1} + \int_0^{3b} \frac{dt}{\Delta_2} = \frac{b}{2\Delta} + \frac{3b}{\Delta} = \frac{7}{2} \frac{b}{\Delta}$$

ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \Delta_1 = 2\Delta_2 = 2\Delta < b \quad [\tau, \Theta]?$$

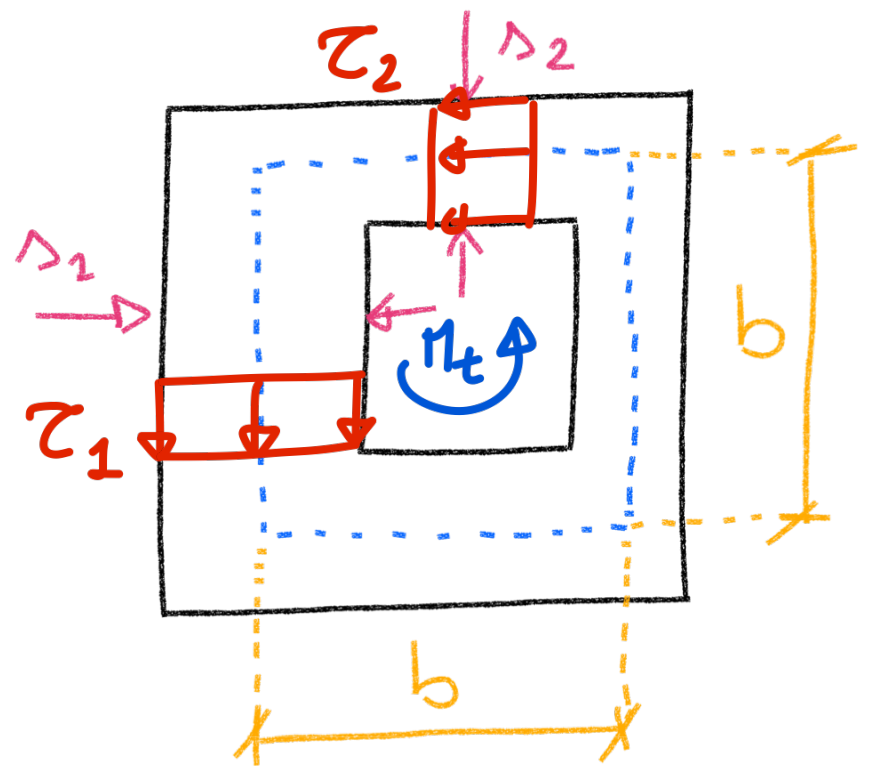
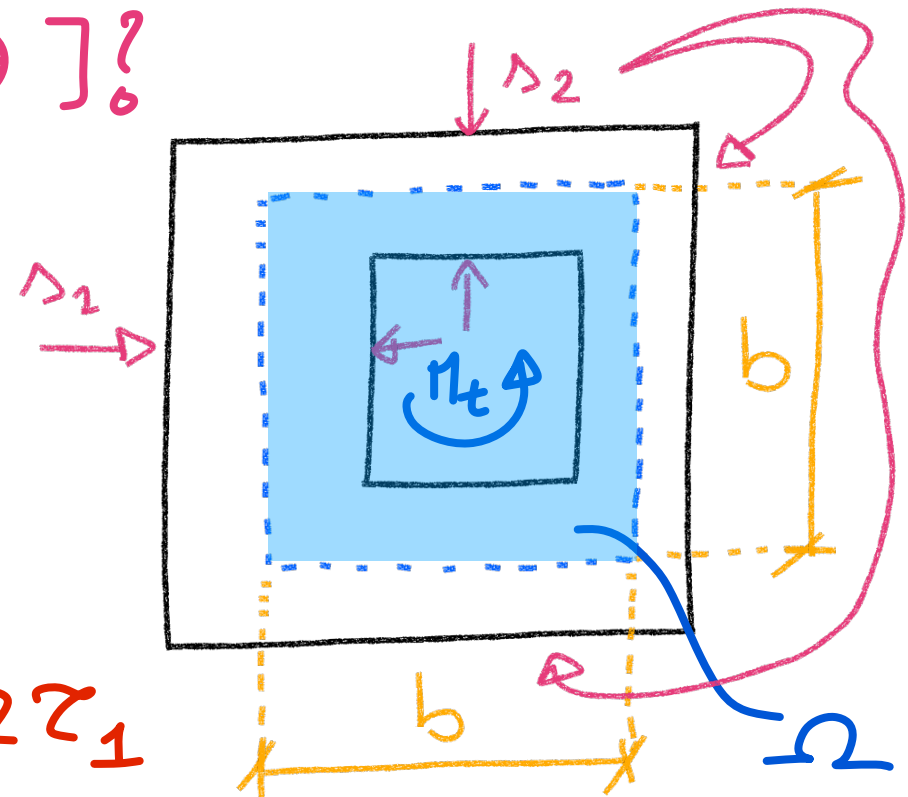
FLUSSO DI τ :

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$$\hookrightarrow \tau_2 = \frac{q}{\Delta_2} = \frac{M_t}{2b^2 \cdot \Delta} = \frac{M_t}{2b^2\Delta}$$

$$\tau_2 = 2\tau_1$$



INERZIA TORSIONALE:

$$\int_0^a \frac{dt}{\Delta(t)} = \int_0^b \frac{dt}{\Delta_1} + \int_0^{3b} \frac{dt}{\Delta_2} = \frac{b}{2\Delta} + \frac{3b}{\Delta} = \frac{7}{2} \frac{b}{\Delta}$$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\Delta(t)}} = \frac{4b^4}{\frac{7}{2} \frac{b}{\Delta}} = \frac{8}{7} b^3 \Delta$$

ESEMPIO FORMULA DI BREDT

$$b, \Omega = b^2, \quad \Delta_1 = 2\Delta_2 = 2\Delta < b \quad [\tau, \Theta]?$$

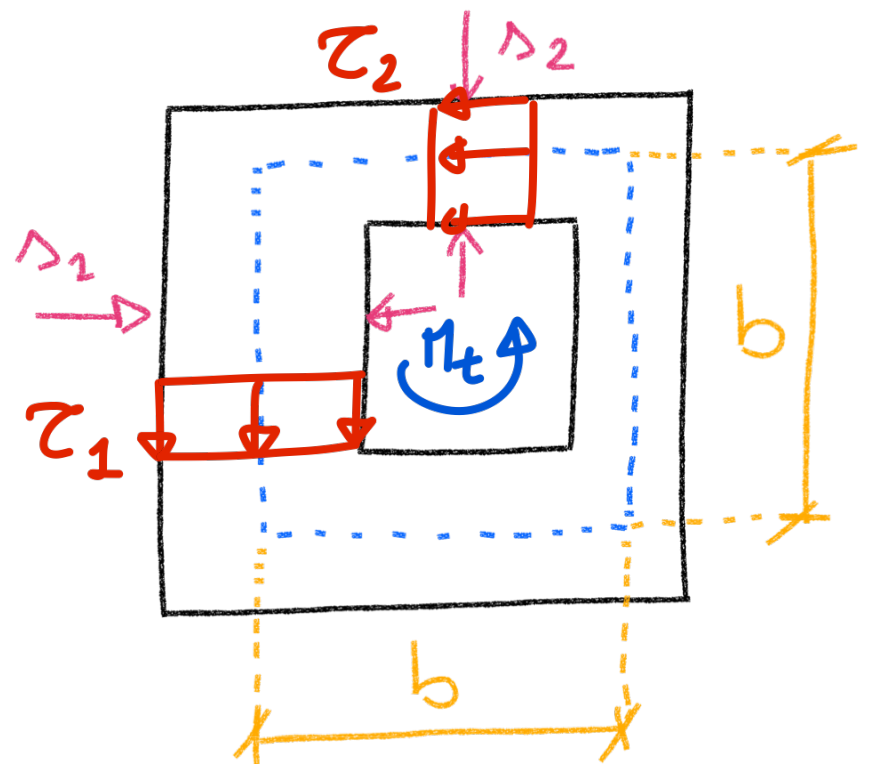
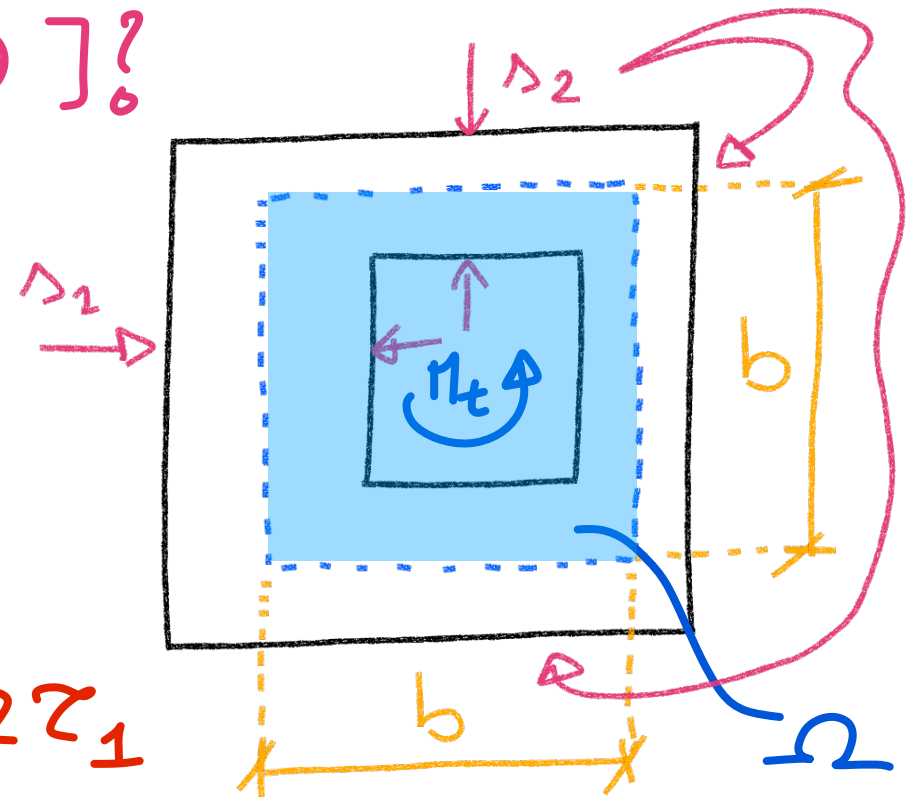
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$$\tau_2 = 2\tau_1$$



INERZIA TORSIONALE:

$$\int_0^a \frac{dt}{\Delta(t)} = \int_0^b \frac{dt}{\Delta_1} + \int_0^{3b} \frac{dt}{\Delta_2} = \frac{b}{2\Delta} + \frac{3b}{\Delta} = \frac{7}{2} \frac{b}{\Delta}$$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\Delta(t)}} = \frac{4b^4}{\frac{7}{2} \frac{b}{\Delta}} = \frac{8}{7} b^3 \Delta \quad \left. \vphantom{\frac{4b^4}{\frac{7}{2} \frac{b}{\Delta}}} \right\} \Theta = \frac{M_t}{GI_t} = \frac{M_t}{G} \frac{7}{8b^3\Delta}$$

SEZIONI SOTTILI APERTE VS. CHIUSE

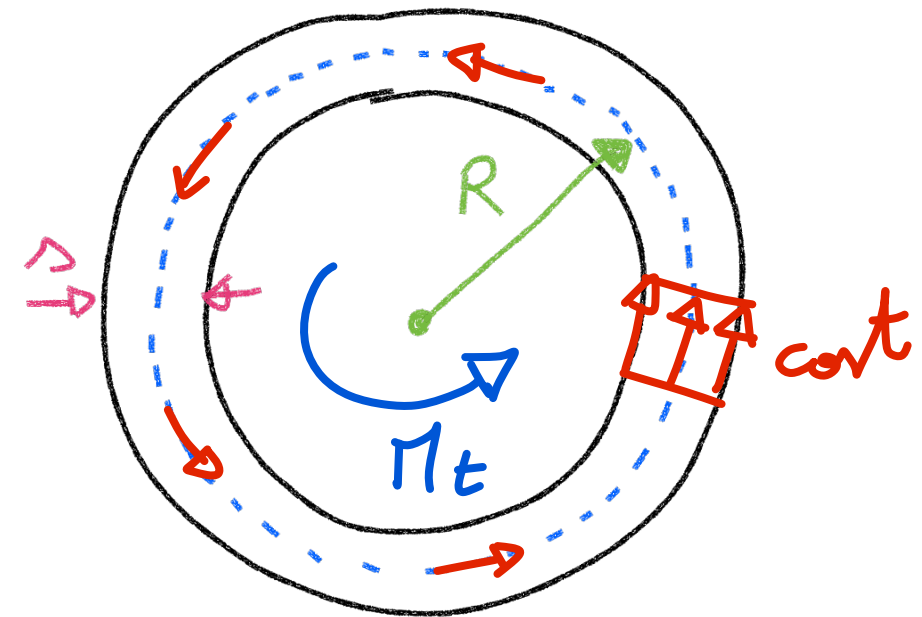
SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

$R \gg \Delta$

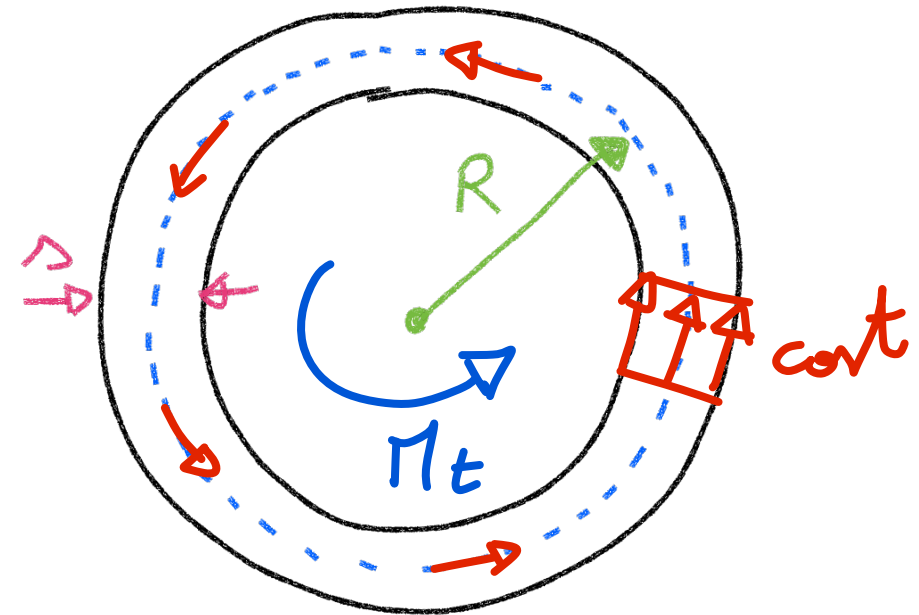


SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA :
(BREDT)

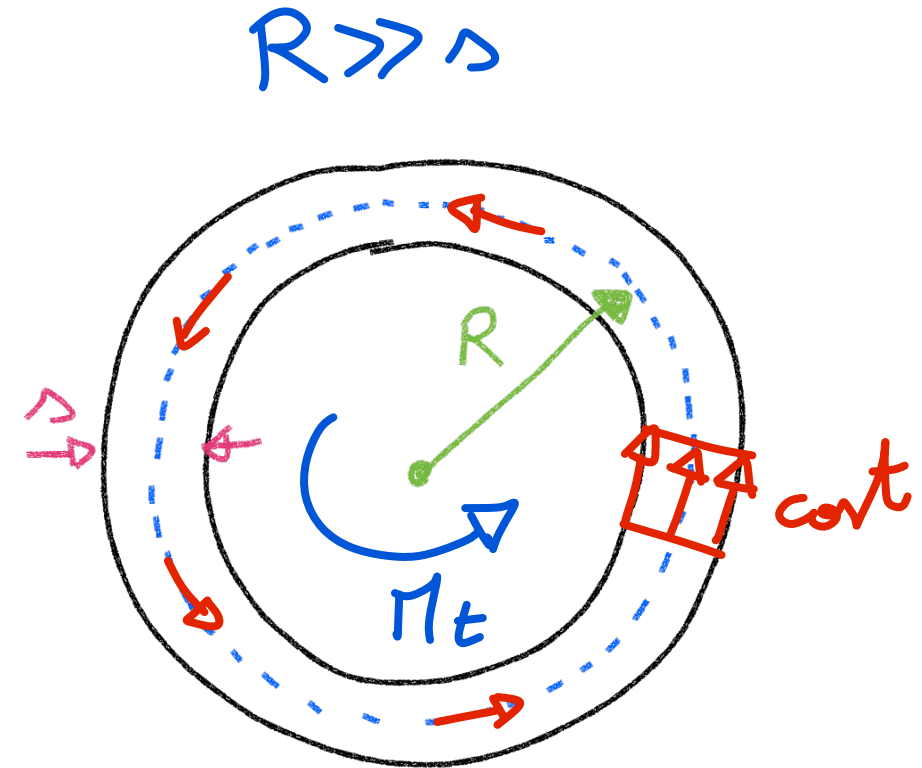
$R \gg \delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA (BREDT) : $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

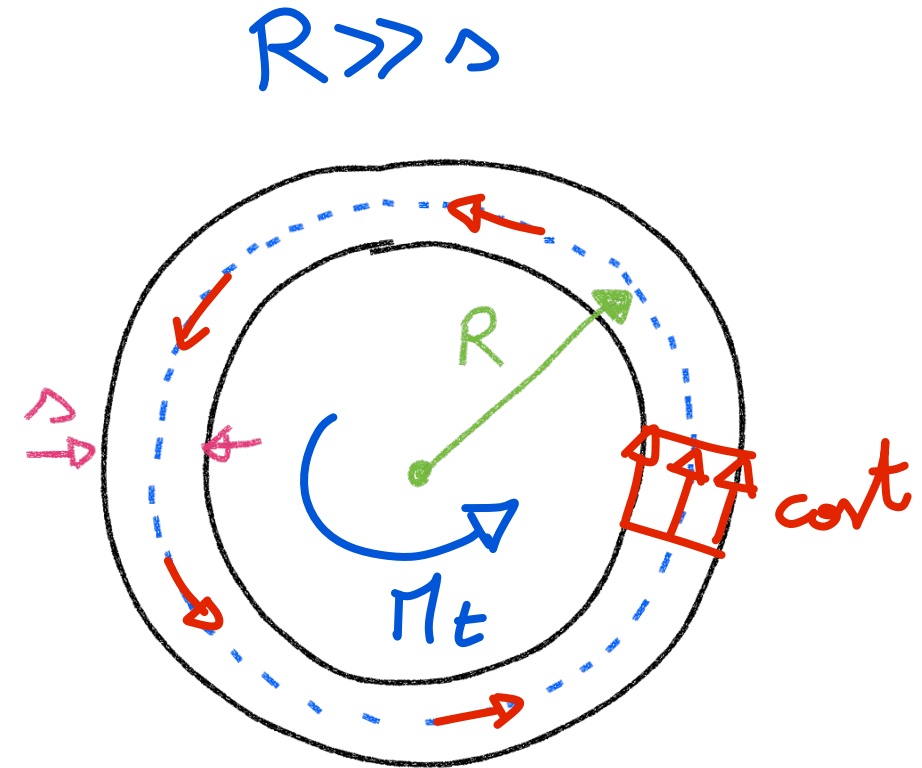


SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA (BREDT) : $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\alpha(t)}} = 2\pi R^3 \Delta \rightarrow \Theta = \frac{M_t}{2\pi R^3 \Delta G}$$



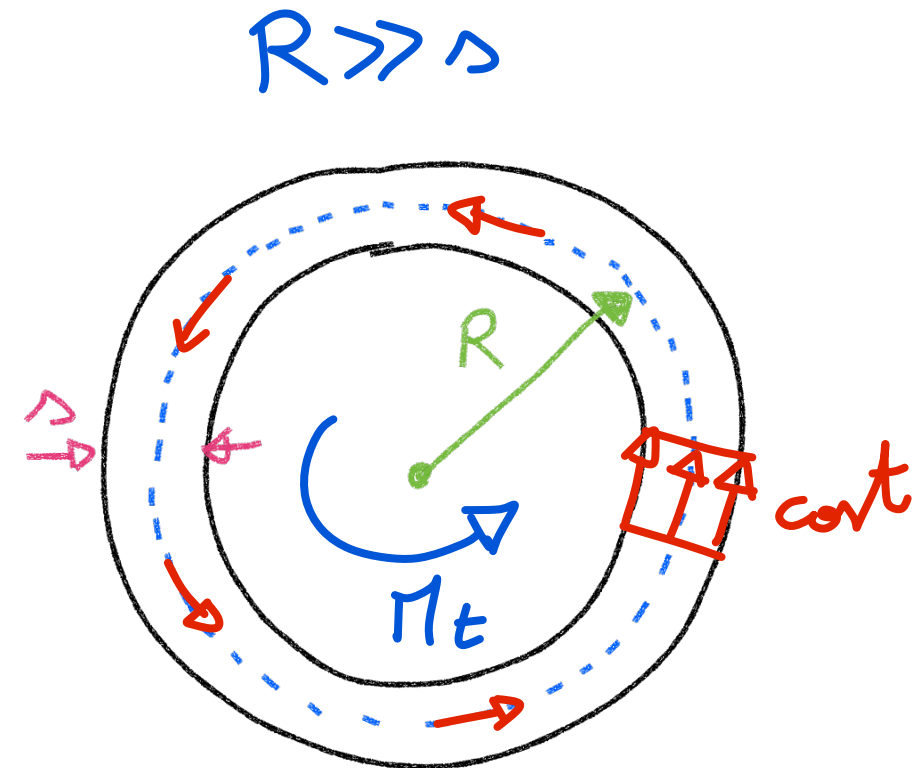
SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA (BREDT) : $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\alpha(t)}} = 2\pi R^3 \Delta \rightarrow \Theta = \frac{M_t}{2\pi R^3 \Delta G}$$

$\rightarrow Z \propto R^2 \Delta ; \Theta \propto R^3 \Delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

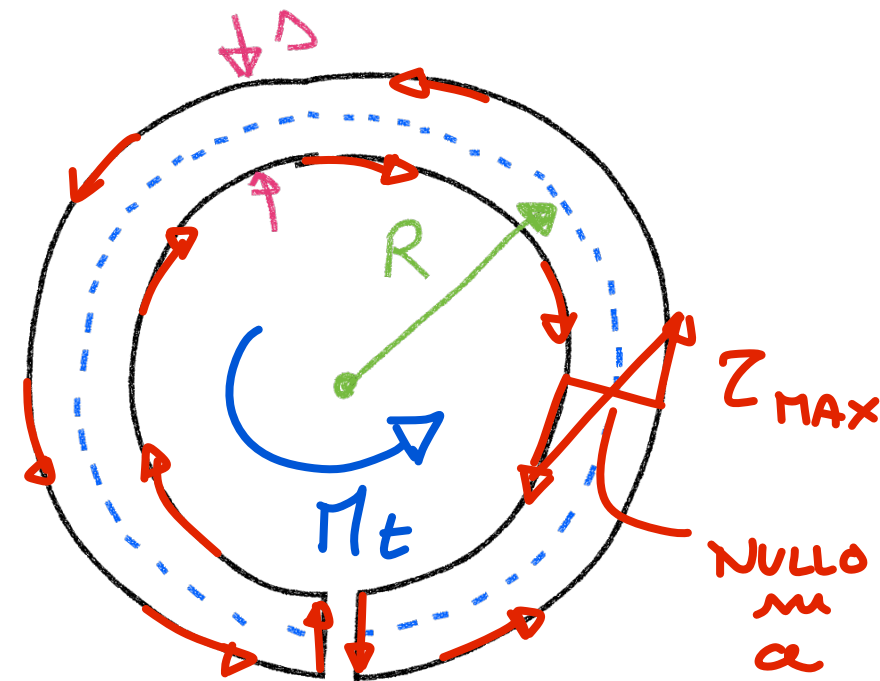
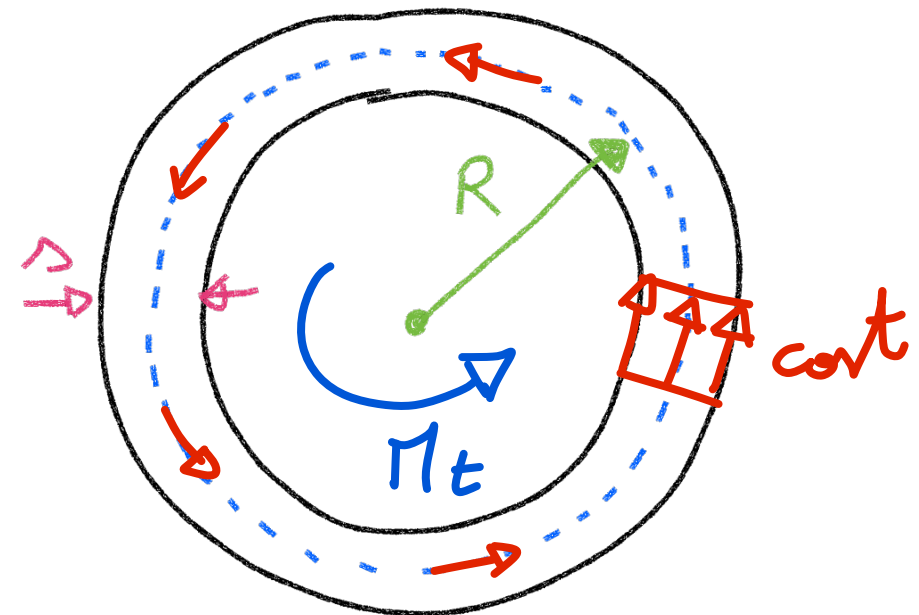
DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA (BRETT) : $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\alpha(t)}} = 2\pi R^3 \Delta \rightarrow \Theta = \frac{M_t}{2\pi R^3 \Delta G}$$

$\rightarrow Z \propto R^2 \Delta ; \Theta \propto R^3 \Delta$

$R \gg \Delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

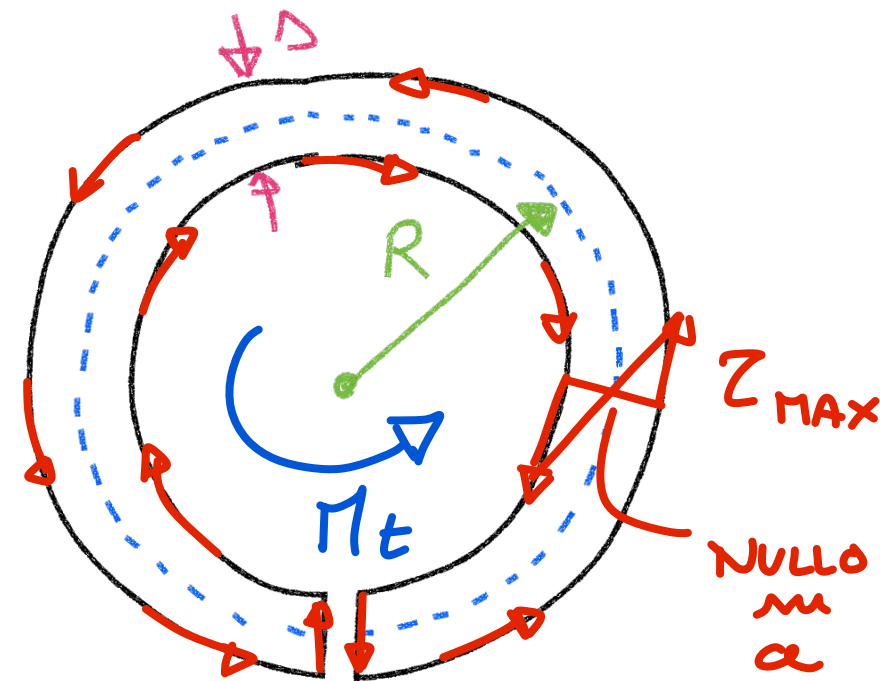
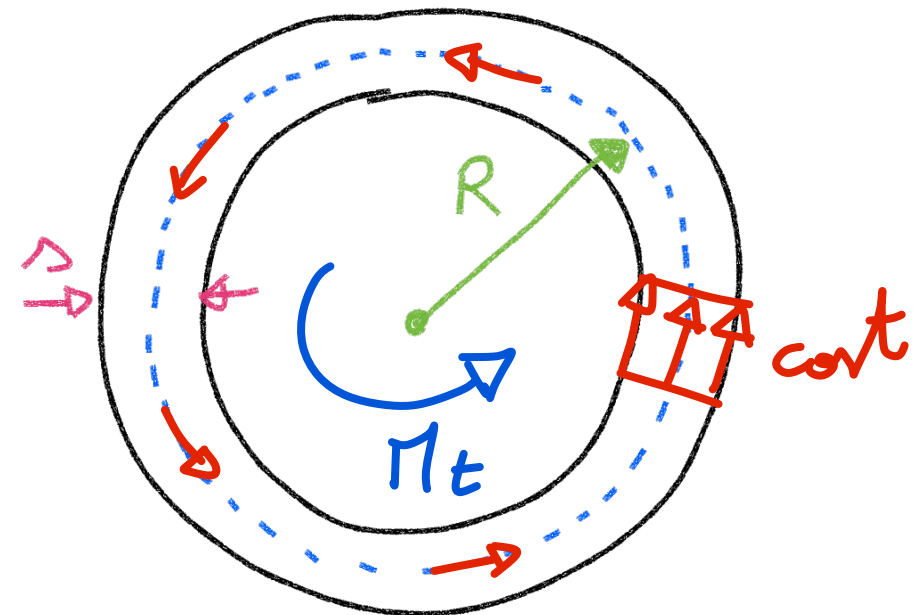
CHIUSA : $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$
(BREDT)

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\alpha(t)}} = 2\pi R^3 \Delta \rightarrow \Theta = \frac{M_t}{2\pi R^3 \Delta G}$$

$\rightarrow Z \propto R^2 \Delta ; \Theta \propto R^3 \Delta$

APERTA :
(RETT.)

$R \gg \Delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

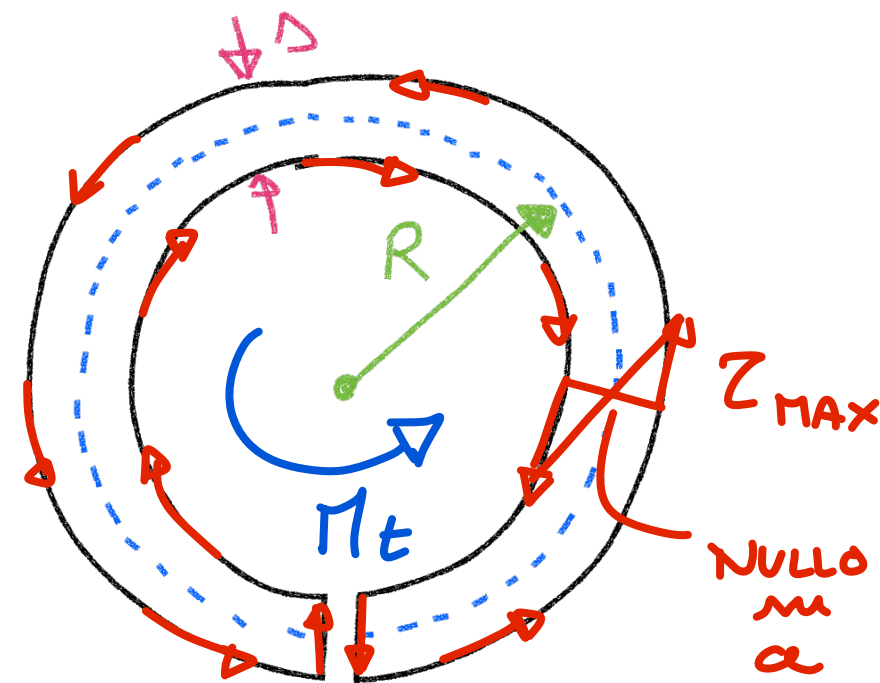
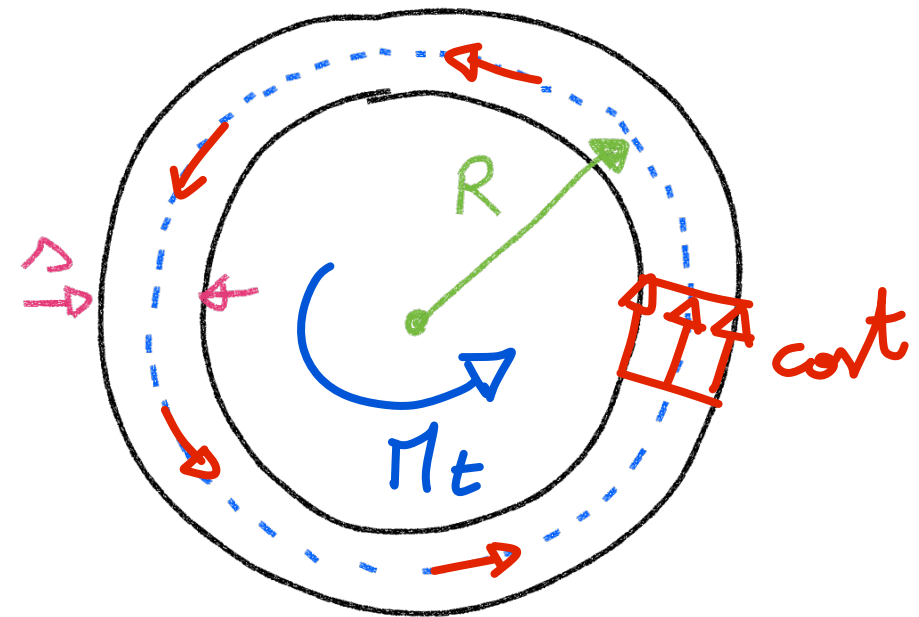
CHIUSA (BRETT) : $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\alpha(t)}} = 2\pi R^3 \Delta \rightarrow \Theta = \frac{M_t}{2\pi R^3 \Delta G}$$

$\rightarrow Z \propto R^2 \Delta ; \Theta \propto R^3 \Delta$

APERTA (RETT.) : $a = 2\pi R \rightarrow I_t = \frac{1}{3} a \Delta^3 = \frac{2}{3} \pi R \Delta^3$

$R \gg \Delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA (BREDT): $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

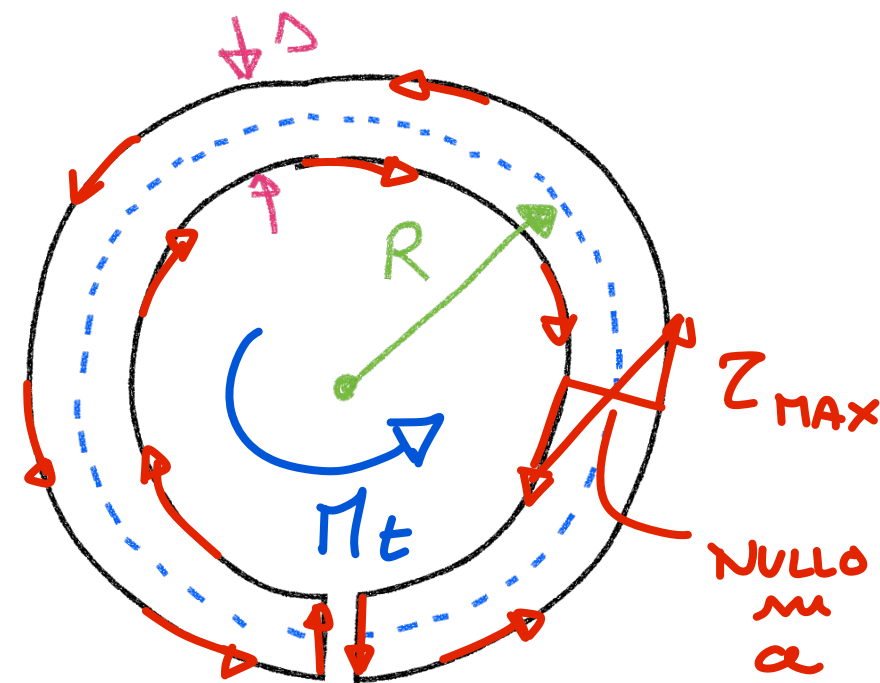
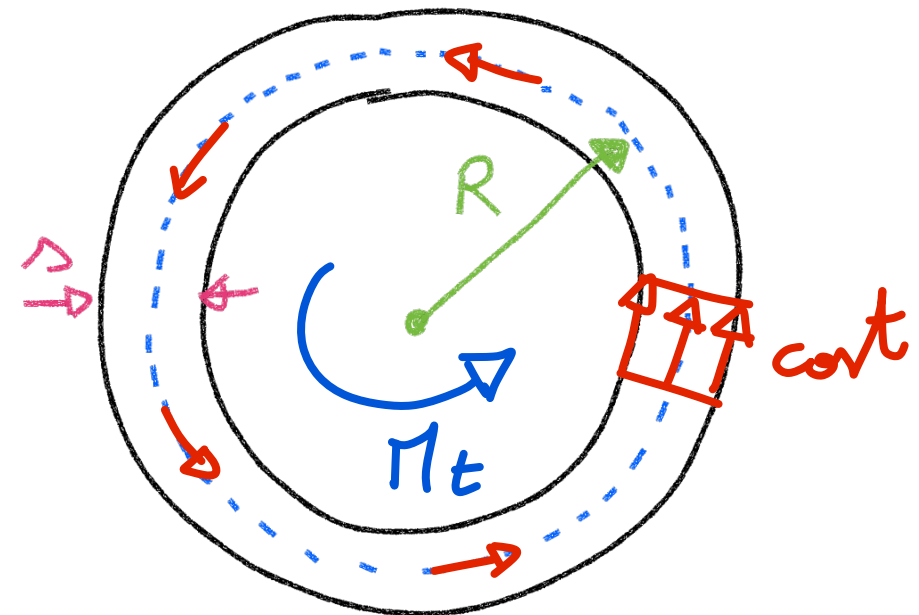
$$I_t = \frac{4\Omega^2}{\int_0^a \frac{dt}{\alpha(t)}} = 2\pi R^3 \Delta \rightarrow \Theta = \frac{M_t}{2\pi R^3 \Delta G}$$

$\rightarrow Z \propto R^2 \Delta; \Theta \propto R^3 \Delta$

APERTA (RETT.): $a = 2\pi R \rightarrow I_t = \frac{1}{3} a \Delta^3 = \frac{2}{3} \pi R \Delta^3$

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$R \gg \Delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA (BREDT): $R, \Omega = \pi R^2 \rightarrow Z = \frac{M_t}{2\Omega \Delta} = \frac{M_t}{2\pi R^2 \Delta}$

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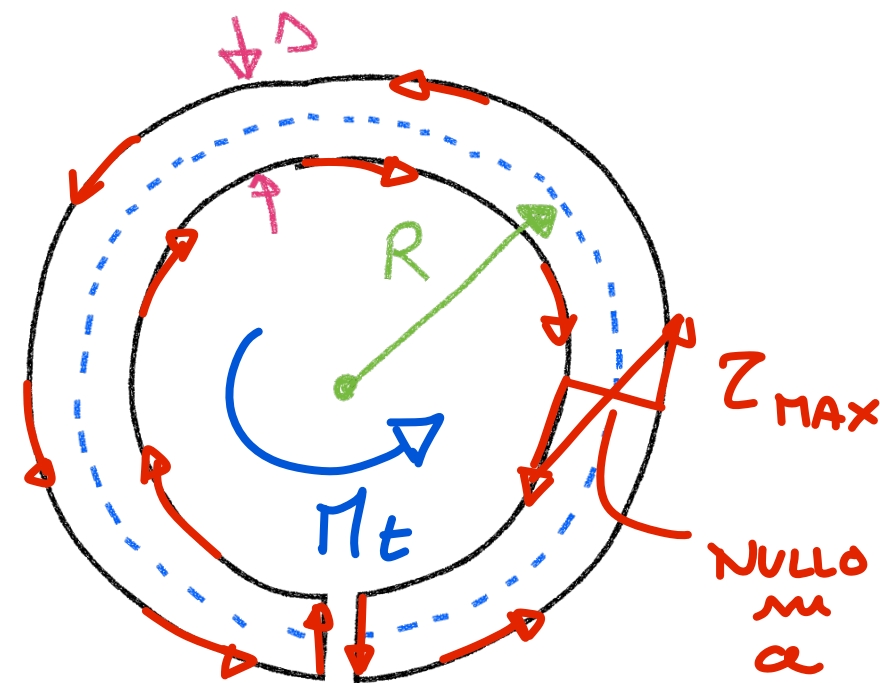
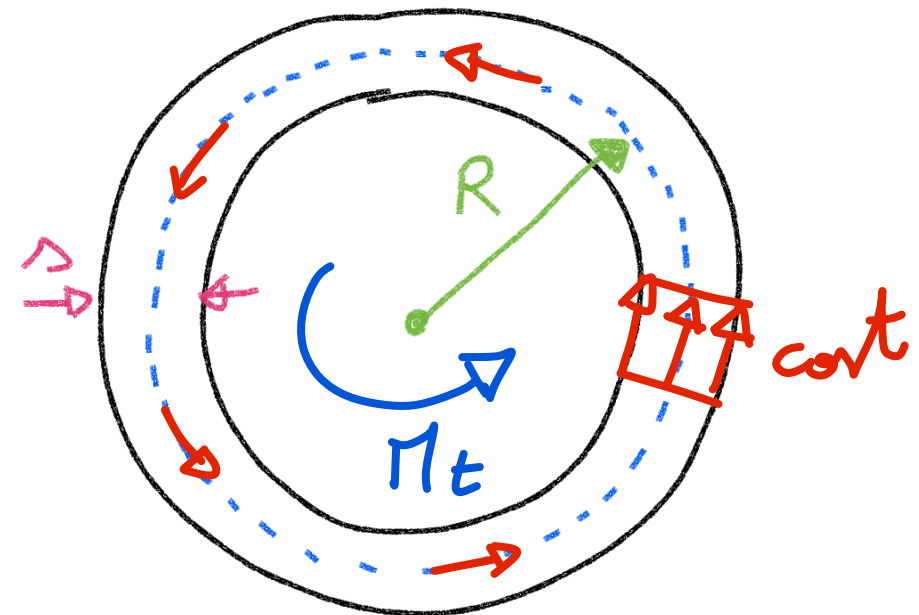
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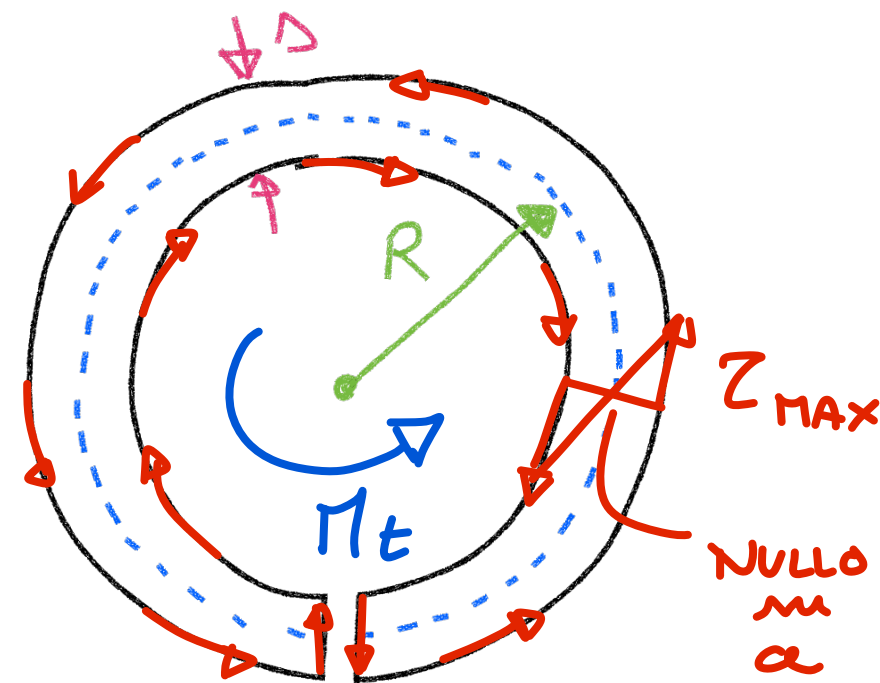
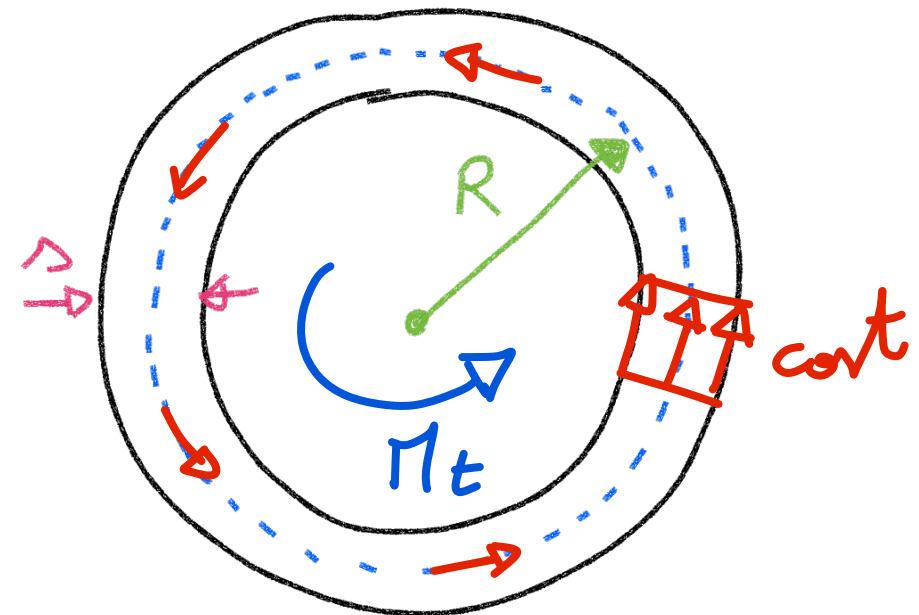
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CONFRONTO

$R \gg \Delta$



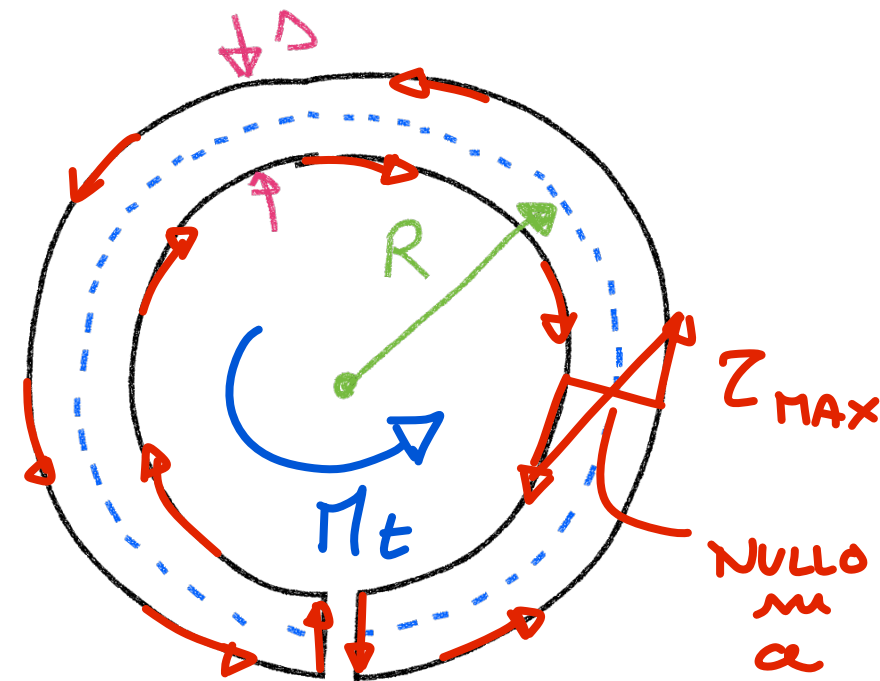
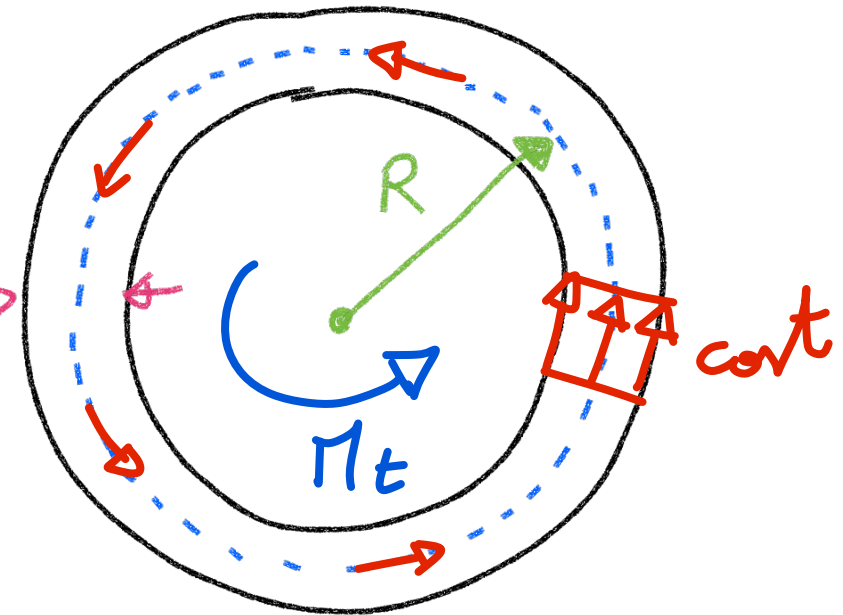
SEZIONI SOTTILI APERTE VS. CHIUSE

DIM. CHE SONO QUALIT. E QUANT. DIVERSI

CHIUSA
(BREDT) : $\tau \propto R^2 \Delta$; $\Theta \propto R^3 \Delta$

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$R \gg \Delta$



SEZIONI SOTTILI APERTE VS. CHIUSE

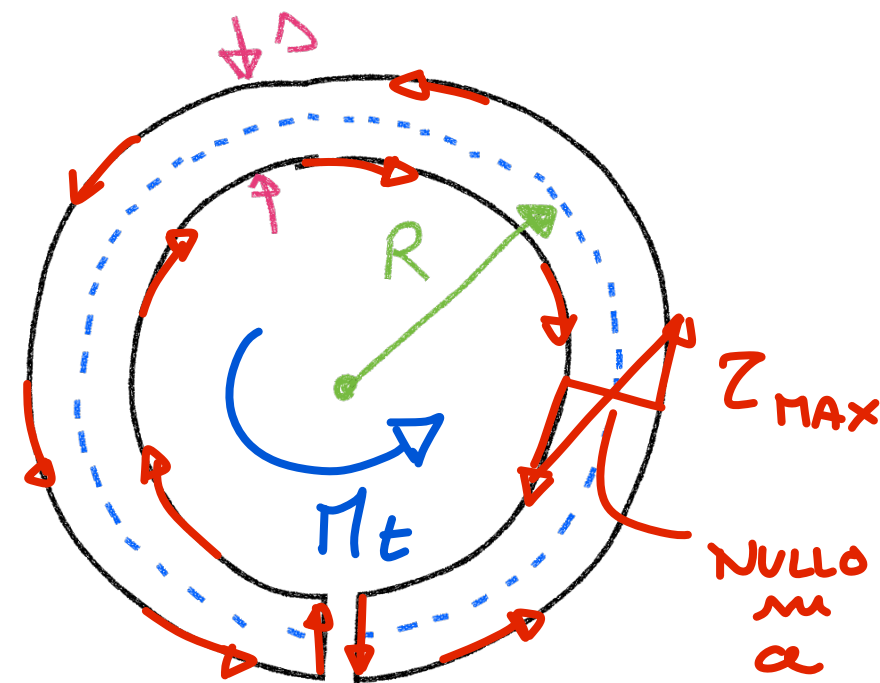
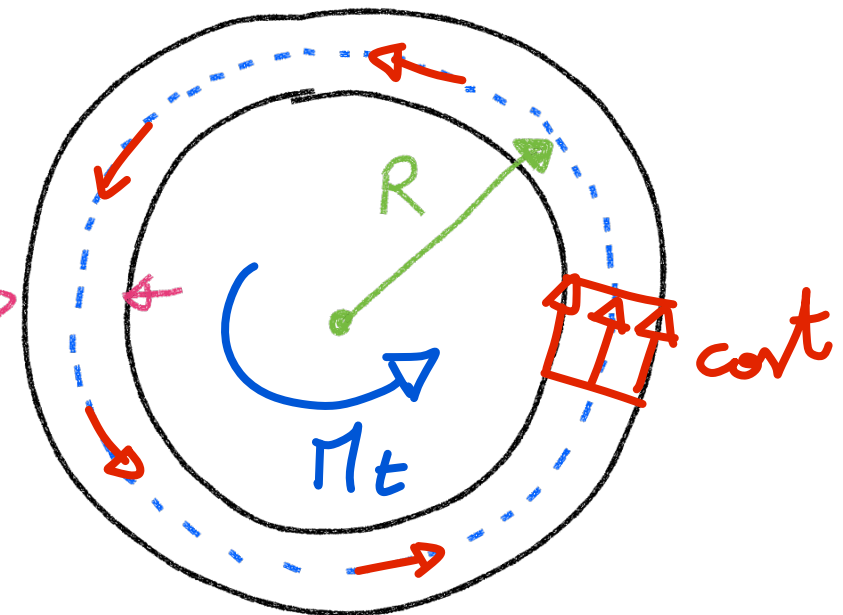
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$$\frac{\tau^{AP}}{\tau^{CH}} = \frac{3}{2} \frac{M_t}{\pi R \Delta^2} \cdot \frac{2\pi R^2 \Delta}{M_t} = 3 \frac{R}{\Delta}$$

$R \gg \Delta$



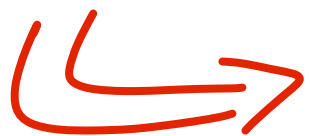
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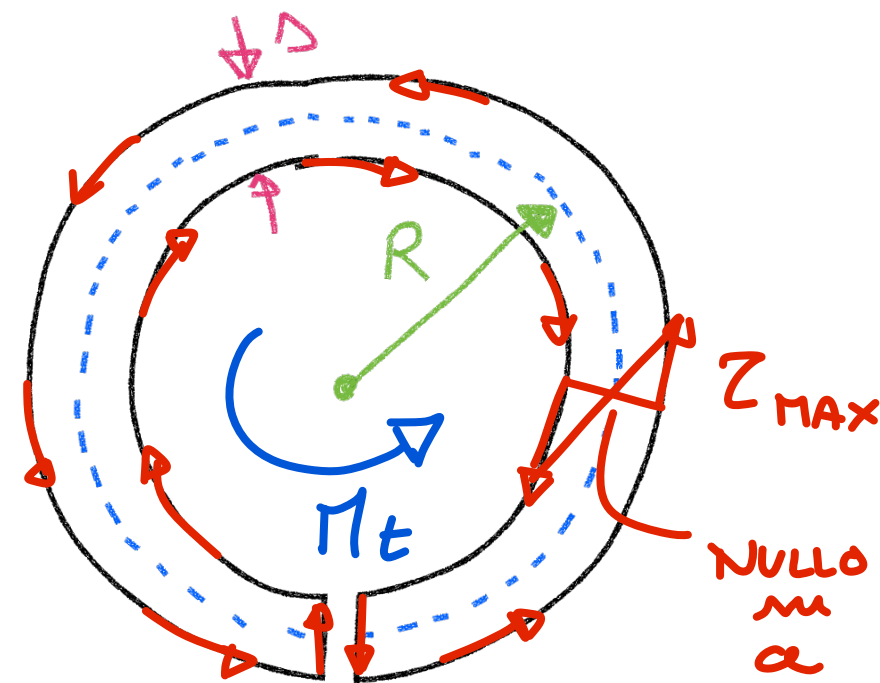
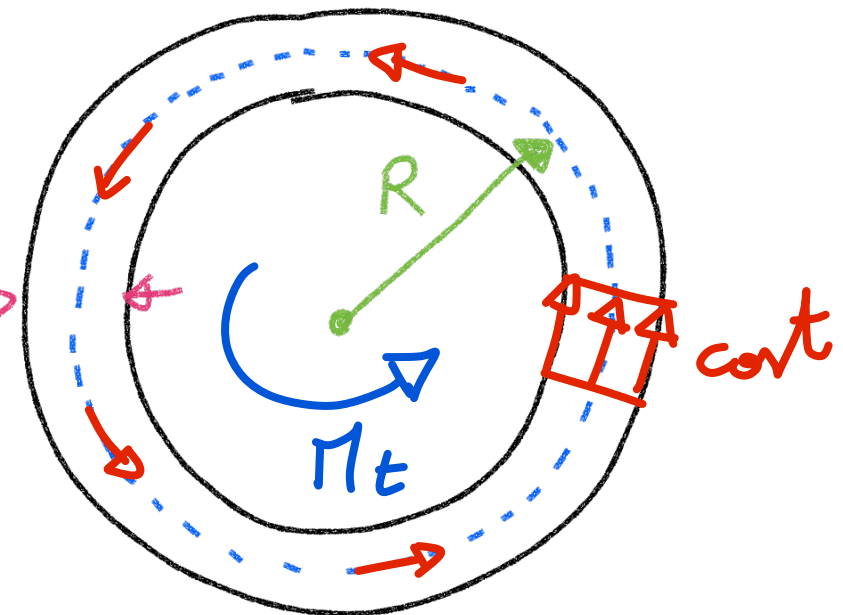
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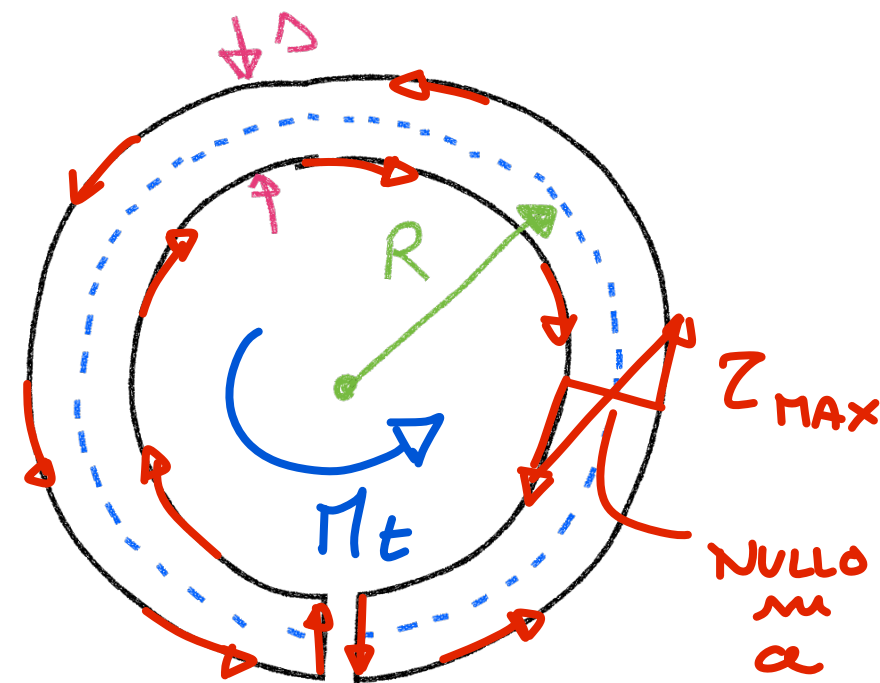
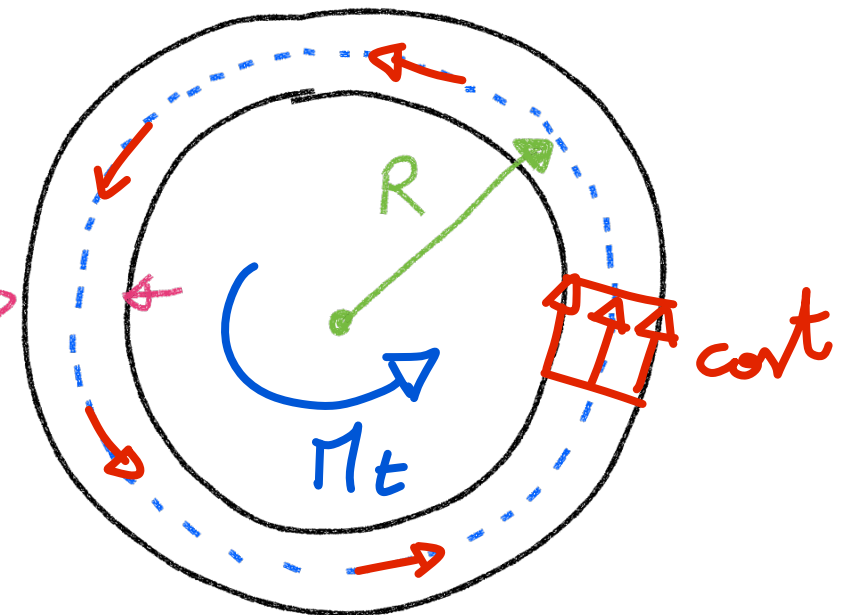
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$R \gg \Delta$



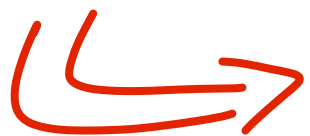
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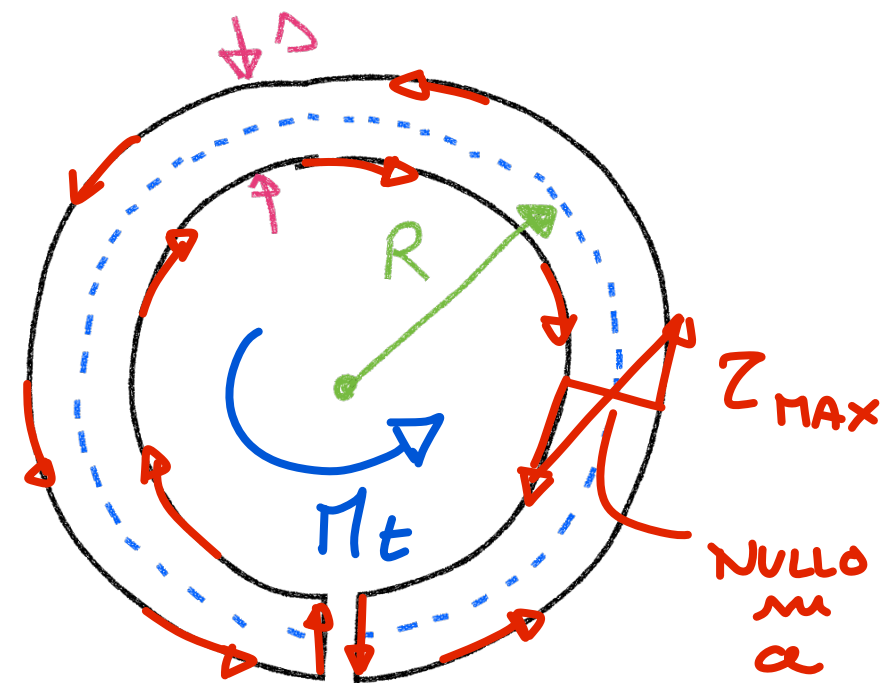
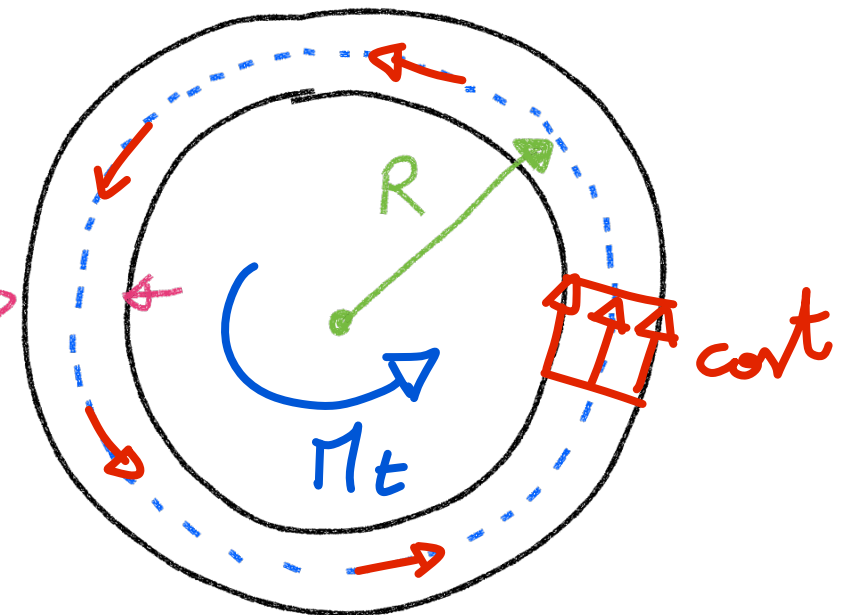
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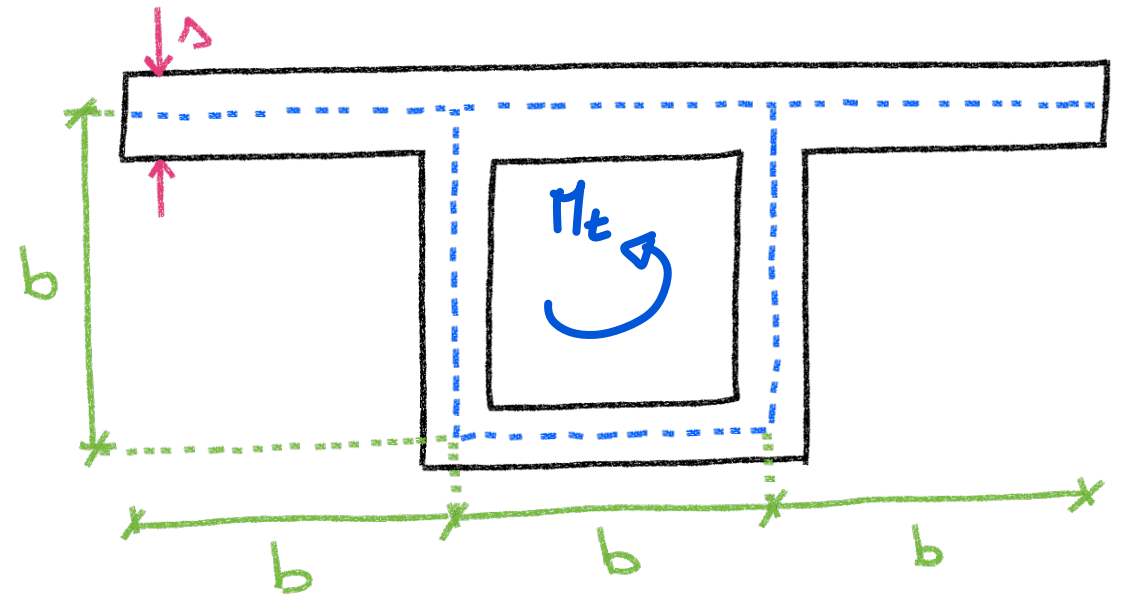


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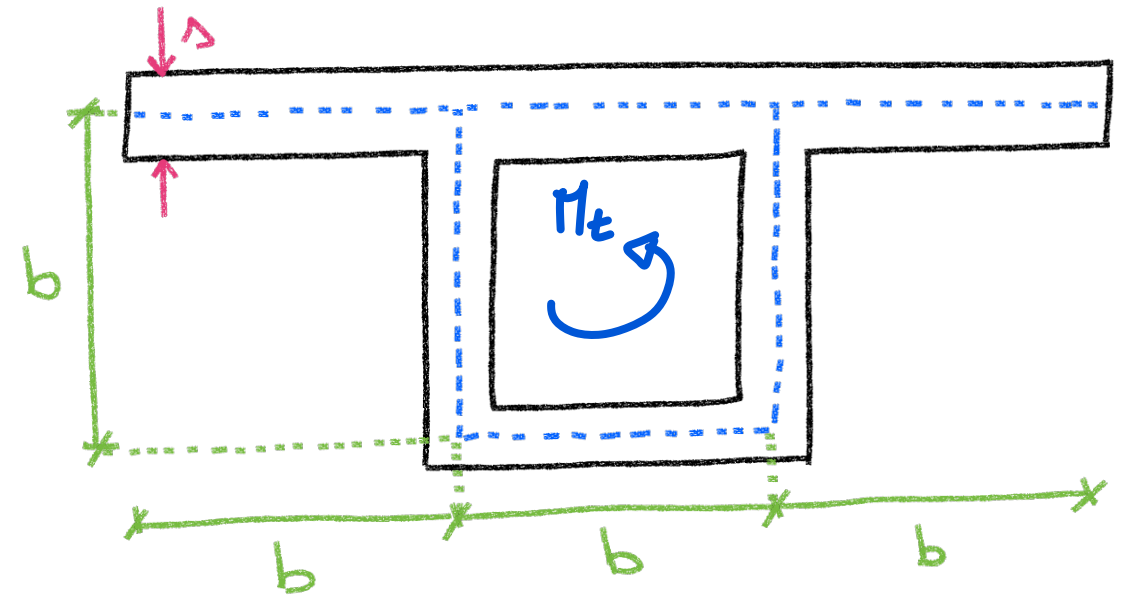
$R \gg \Delta$



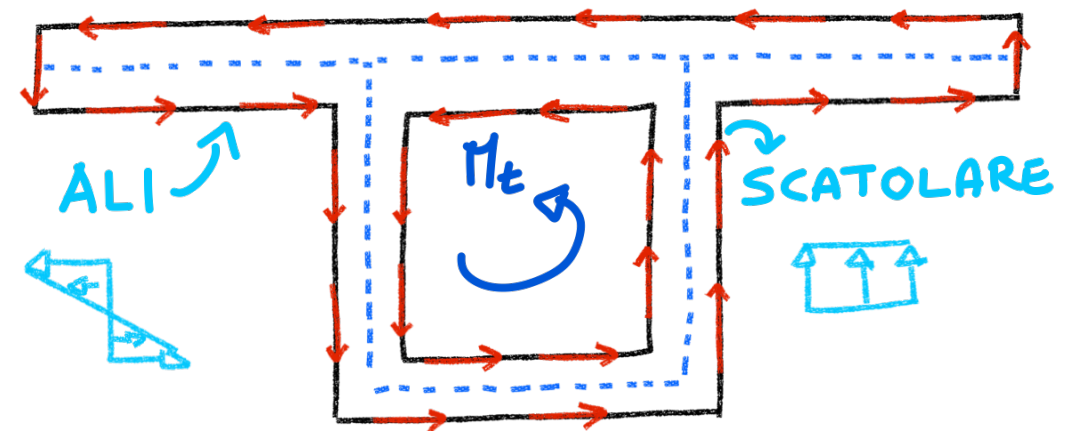
SEZIONI SOTTILI COMPOSITE



SEZIONI SOTTILI COMPOSTE



ANALOGIA IDRODINAMICA

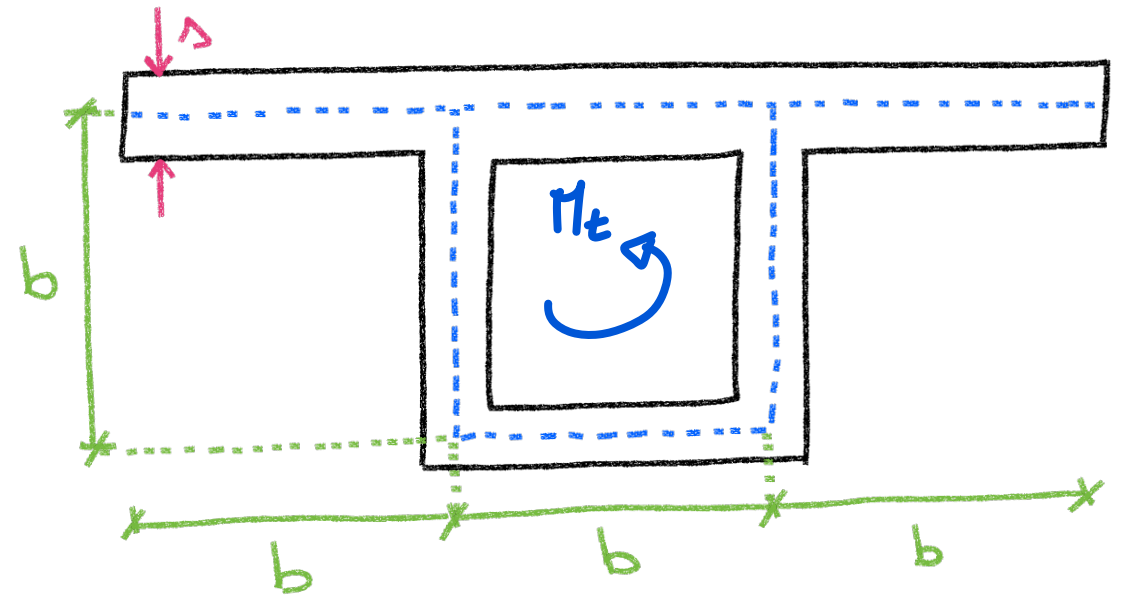


SEZIONI SOTTILI COMPOSTE

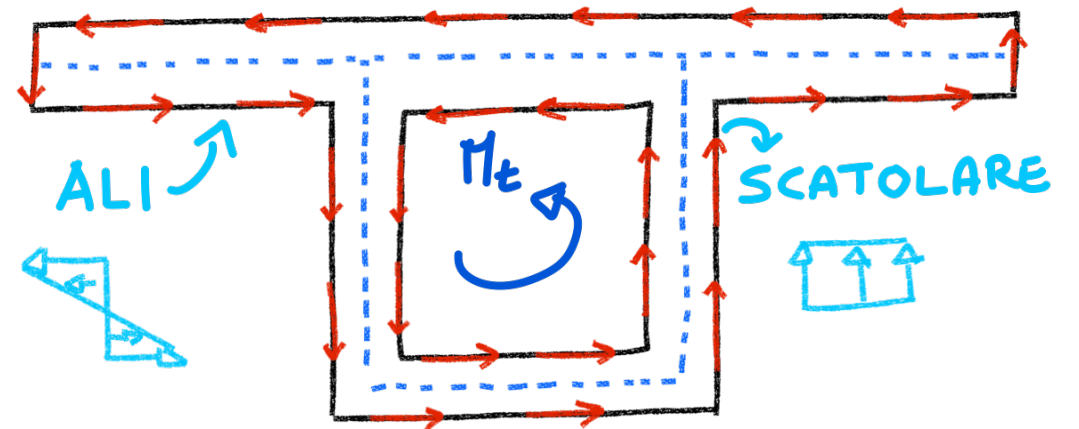
A_{LI} :

$$I_{t1} = I_{t2} = \frac{1}{3} b \Delta^3$$

$$z_{MAX}^1 = z_{MAX}^2 = \frac{M_{t1}}{I_{t1}} \Delta = \frac{M_{t2}}{I_{t2}} \Delta$$



ANALOGIA IDRODINAMICA



SEZIONI SOTTILI COMPOSTE

ALI:

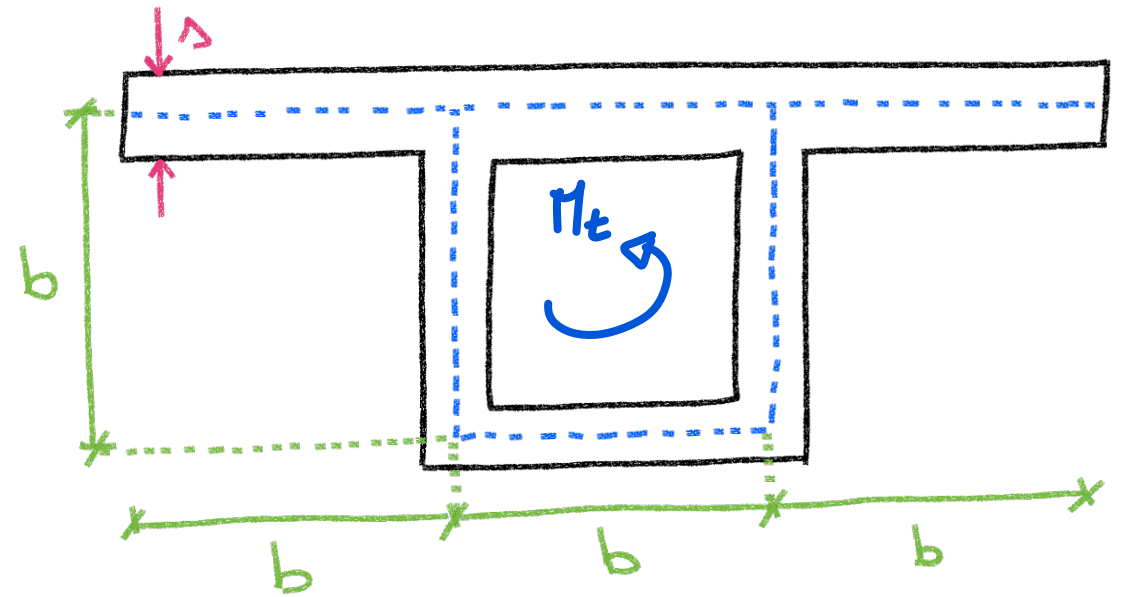
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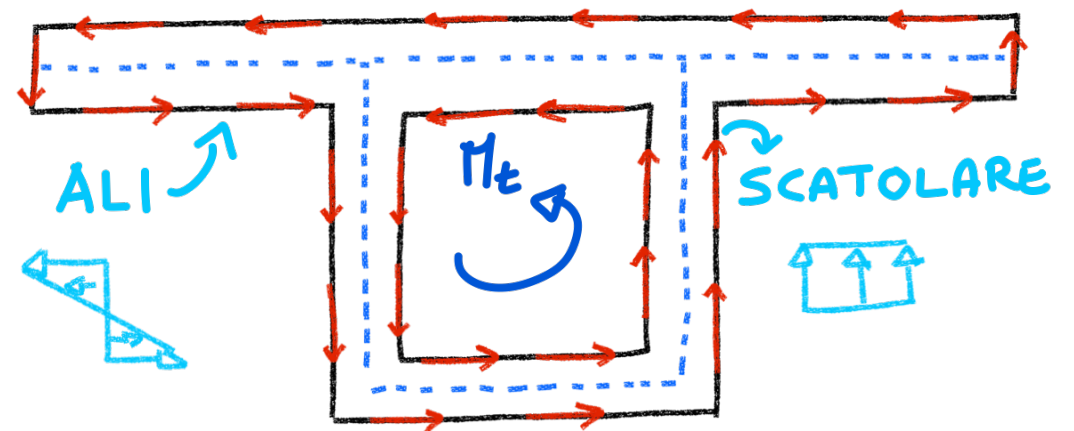
SCATOLARE:

$$I_{t3} = \frac{4 \Omega^2}{\oint \frac{ds}{\Delta}} = \frac{4 b^2}{\Delta} = b^3 \Delta$$

$$z_{MAX}^3 = z_3 = \frac{M_{t3}}{2 \Omega \Delta} = \frac{M_{t3}}{2 b^2 \Delta}$$



ANALOGIA IDRODINAMICA



SEZIONI SOTTILI COMPOSITE

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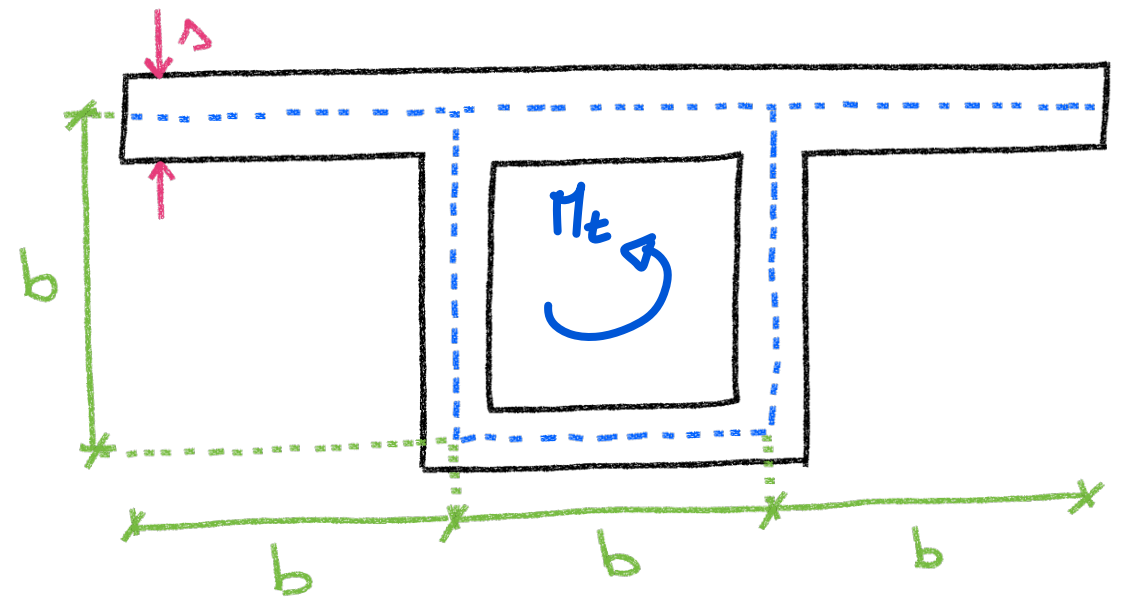
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SCATOLARE:

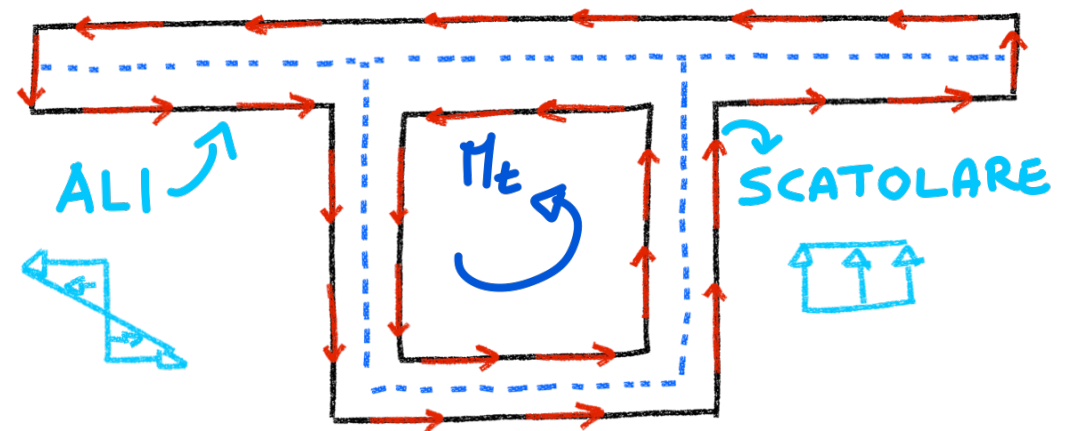
$$I_{t3} = \frac{4 \Omega^2}{\oint \frac{dt}{\Delta}} = \frac{4 b^2}{\frac{4b}{\Delta}} = b^3 \Delta$$

$$Z_{MAX}^3 = Z_3 = \frac{M_{t3}}{2 \Omega \Delta} = \frac{M_{t3}}{2 b^2 \Delta}$$

$$I_{t3} \gg I_{t1}, I_{t2}$$



ANALOGIA IDRODINAMICA



SEZIONI SOTTILI COMPOSTE

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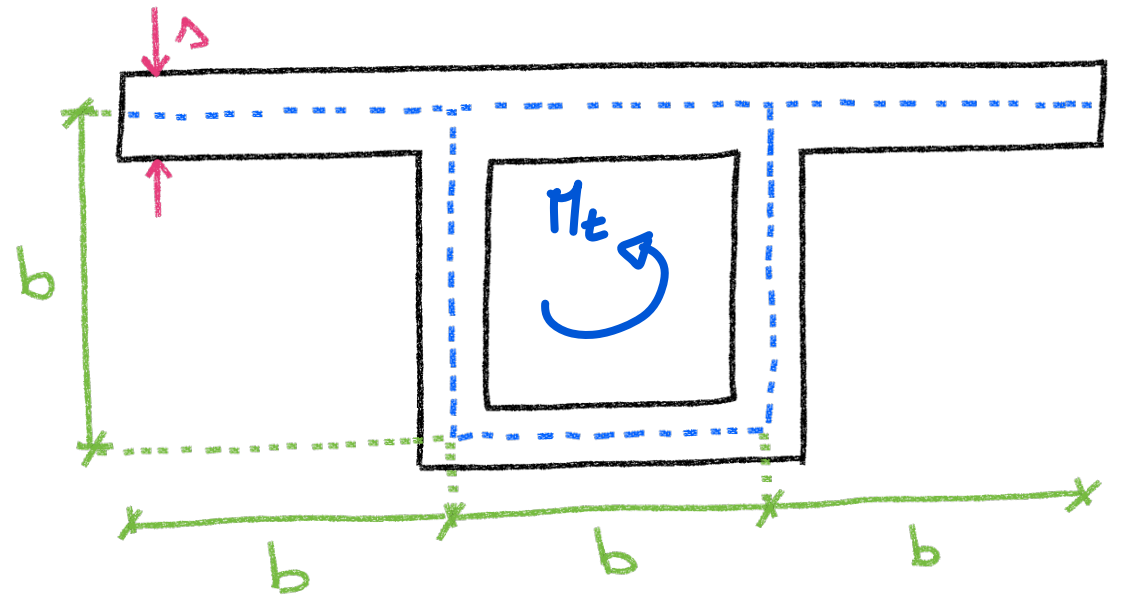
$$I_{t3} \gg I_{t1}, I_{t2}$$

INERZIA TORSIONALE:

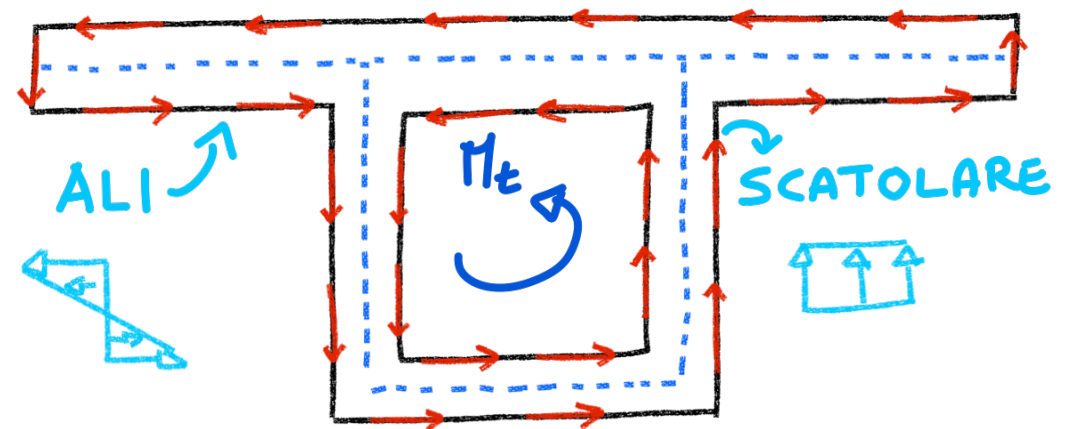
$$I_t = I_{t1} + I_{t2} + I_{t3} = \frac{2}{3} b \Delta^3 + b^3 \Delta$$

$$M_{t1} = M_{t2} = \frac{I_{t1}}{I_t} M_t$$

$$M_{t3} = \frac{I_{t3}}{I_t} M_t$$



ANALOGIA IDRODINAMICA



SEZIONI SOTTILI COMPOSTE

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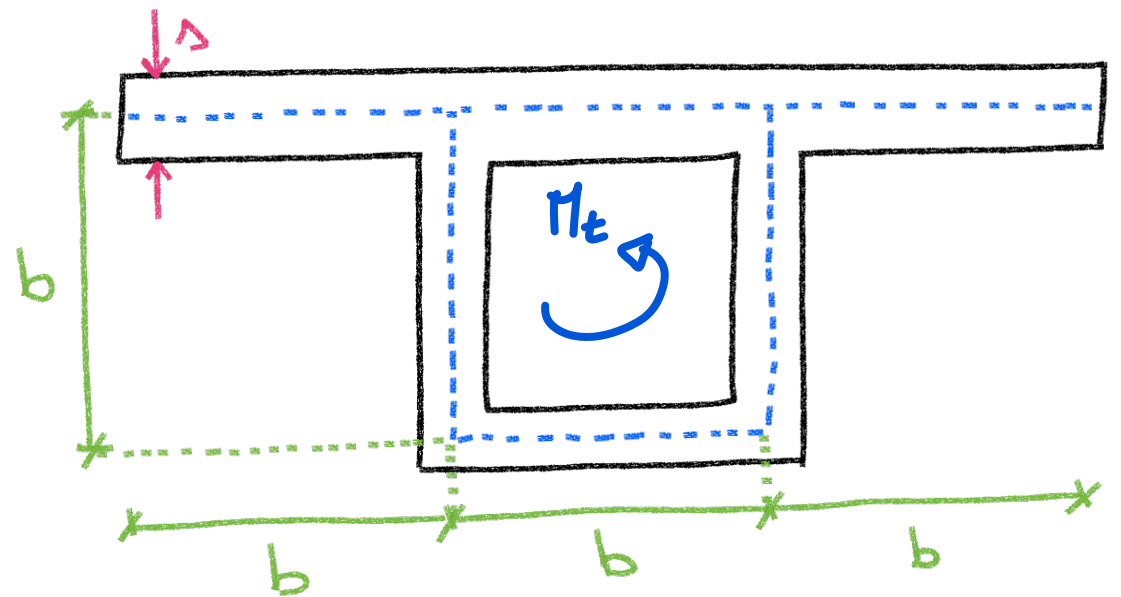
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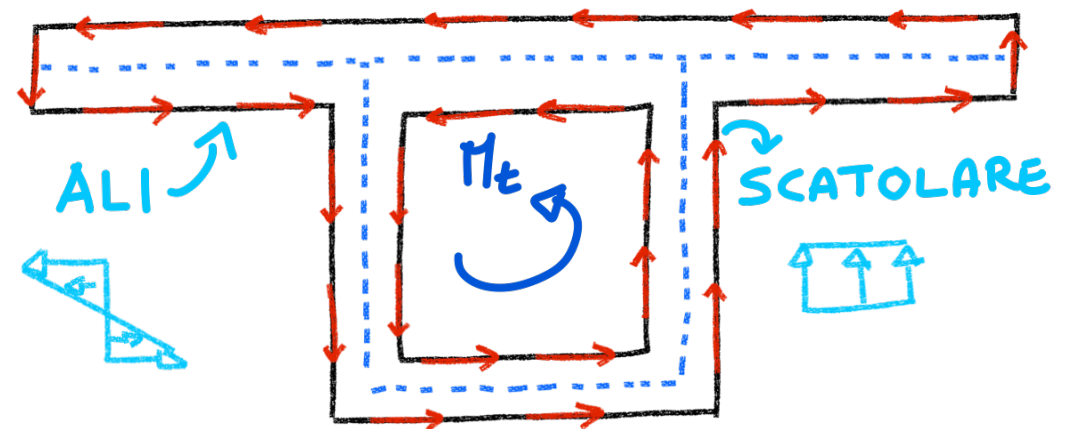
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ANALOGIA IDRODINAMICA

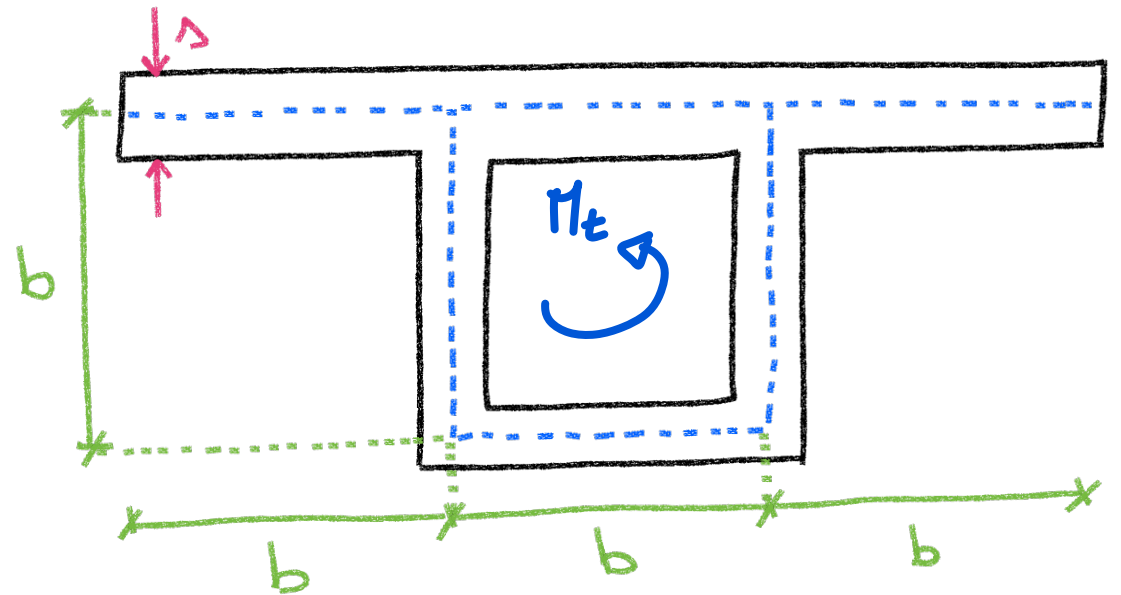


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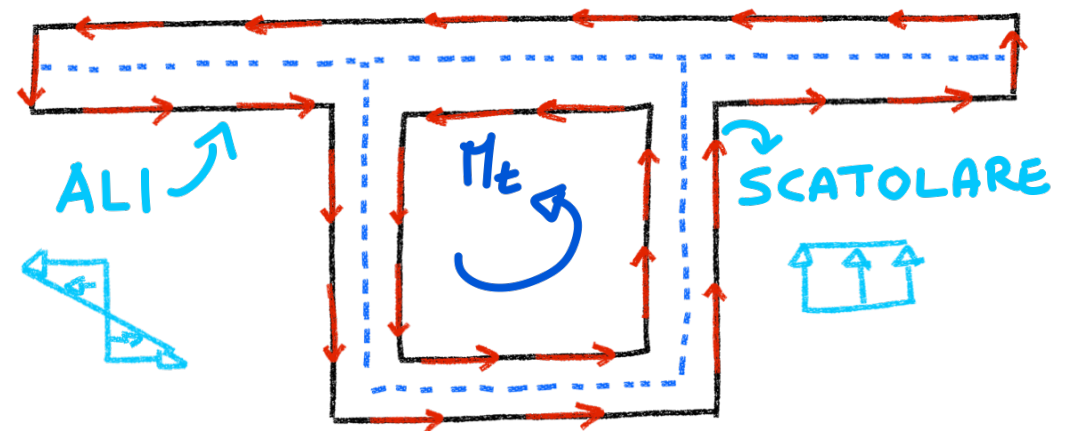
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ANALOGIA IDRODINAMICA



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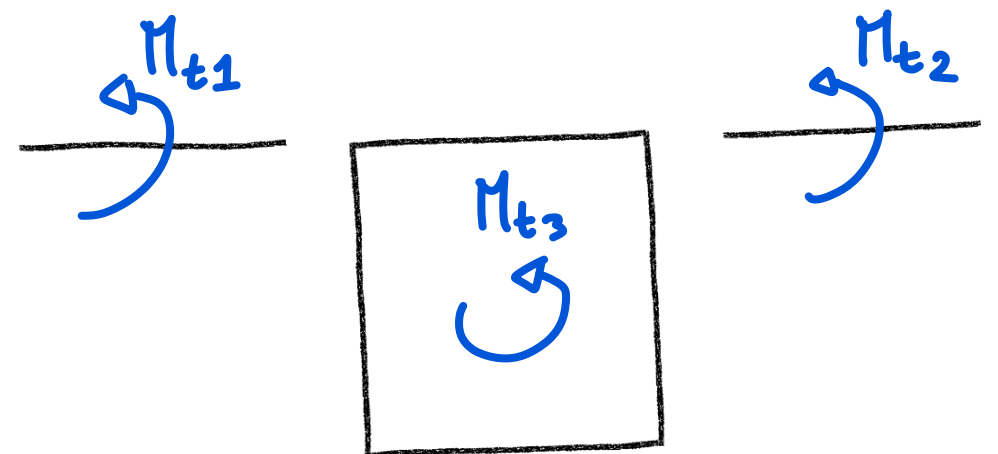
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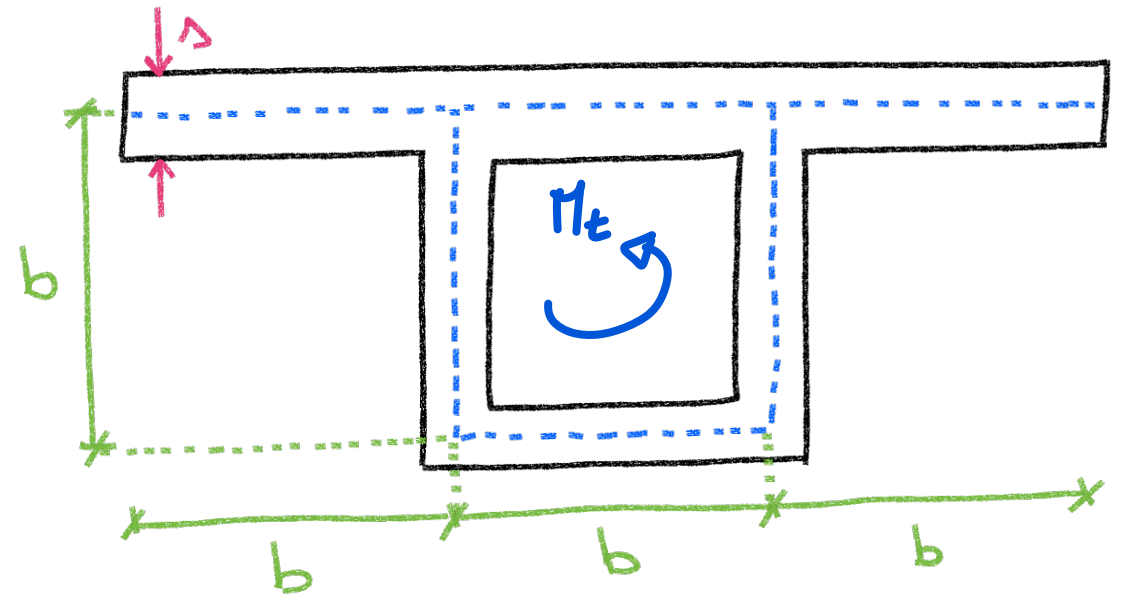
RIPARTIZIONE MOMENTO TORC.



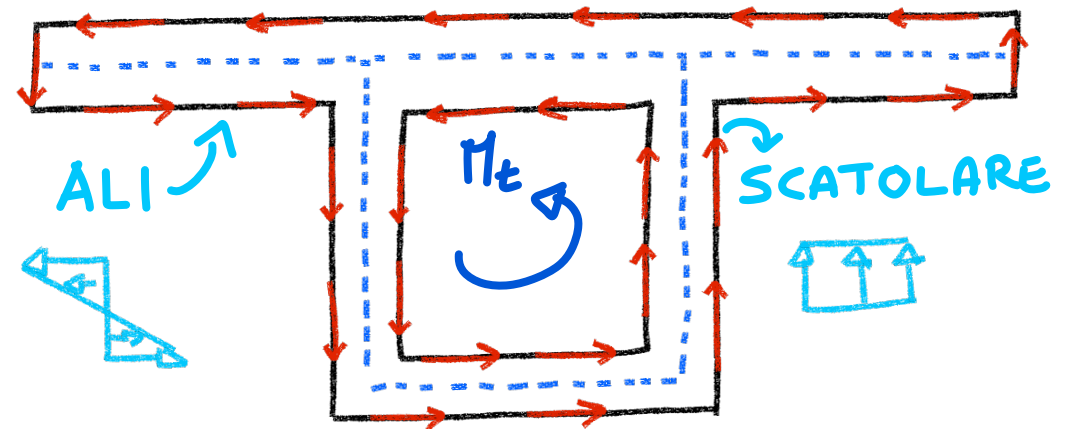
SEZIONI SOTTILI COMPOSTE

$$I_{t3} \gg I_{t1}, I_{t2}$$

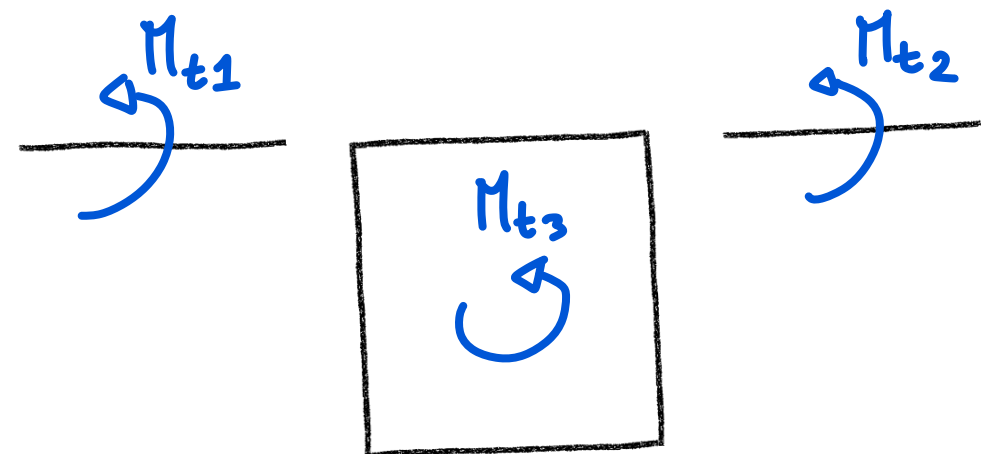
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ANALOGIA IDRODINAMICA



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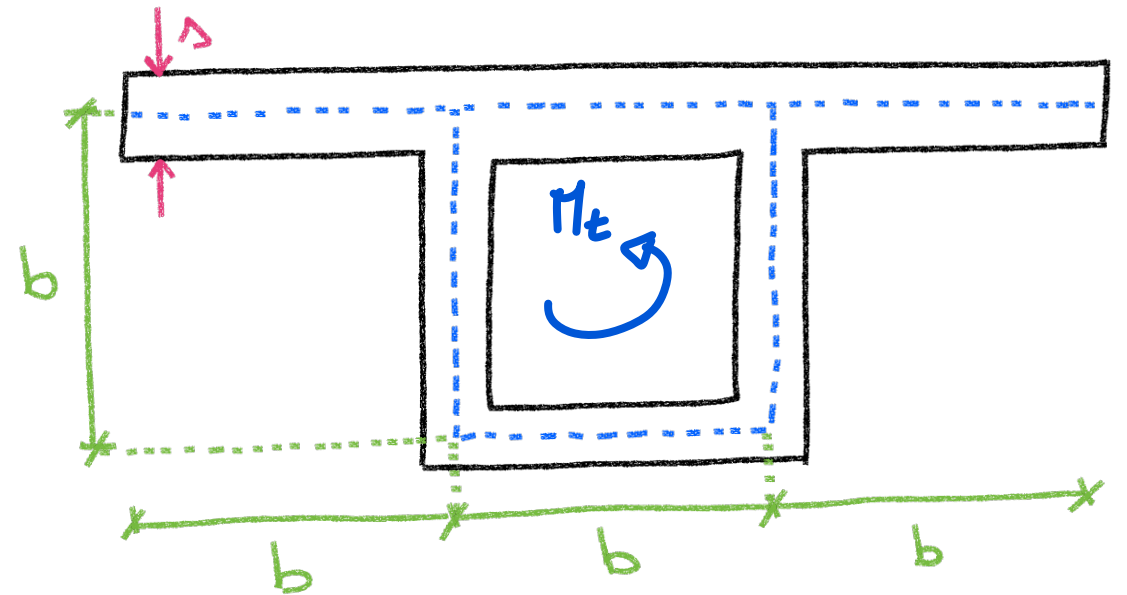
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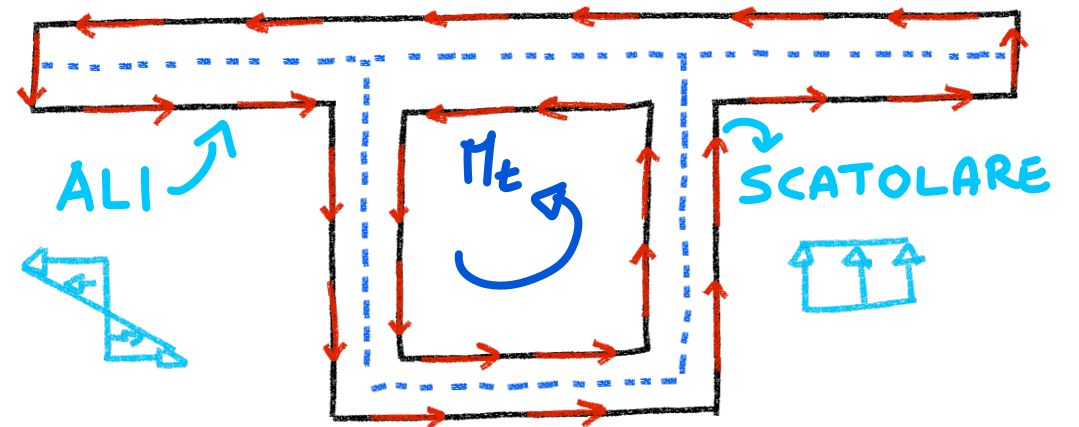
$$M_{t3} \gg M_{t1}, M_{t2}$$



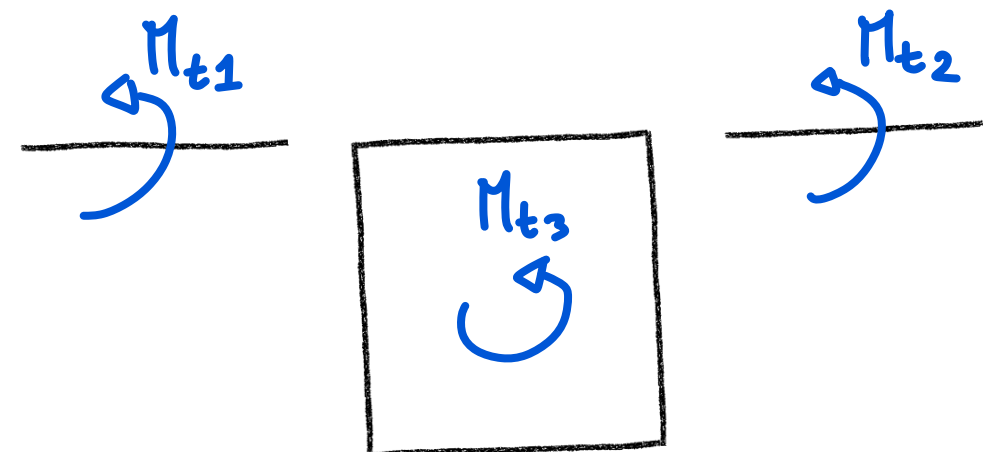
$$Z_{MAX}^1, Z_{MAX}^2 < Z_3$$



ANALOGIA IDRODINAMICA



RIPARTIZIONE MOMENTO TORC.



SEZIONI SOTTILI COMPOSTE

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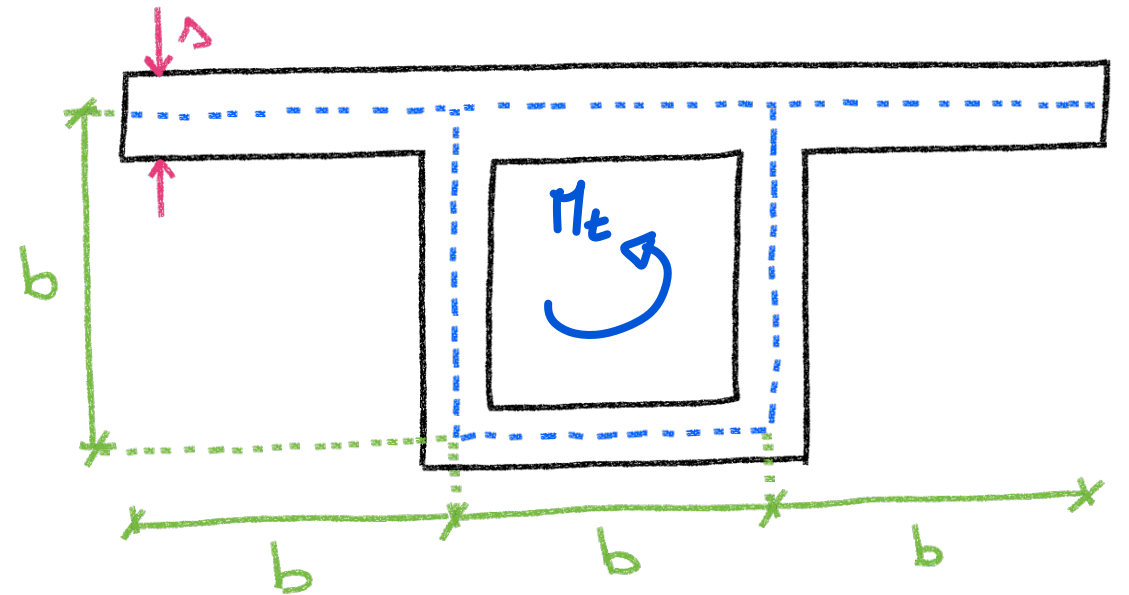
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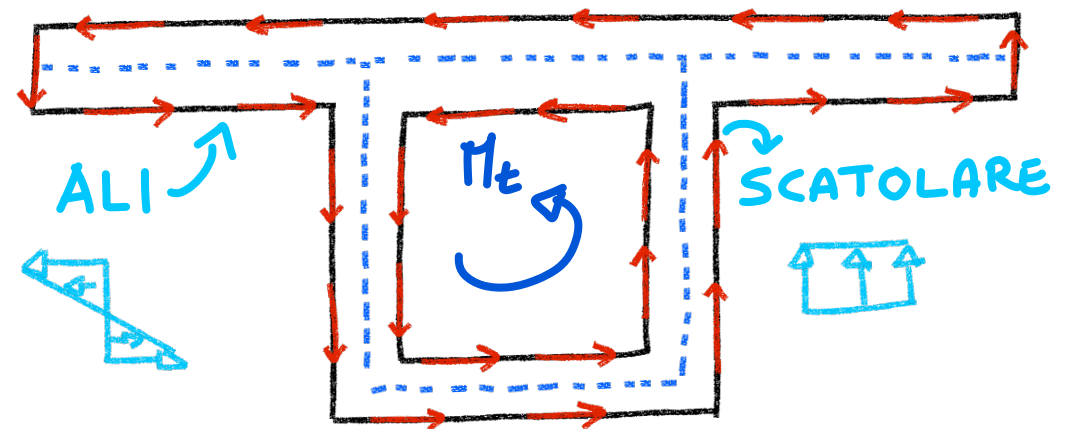
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NON <<

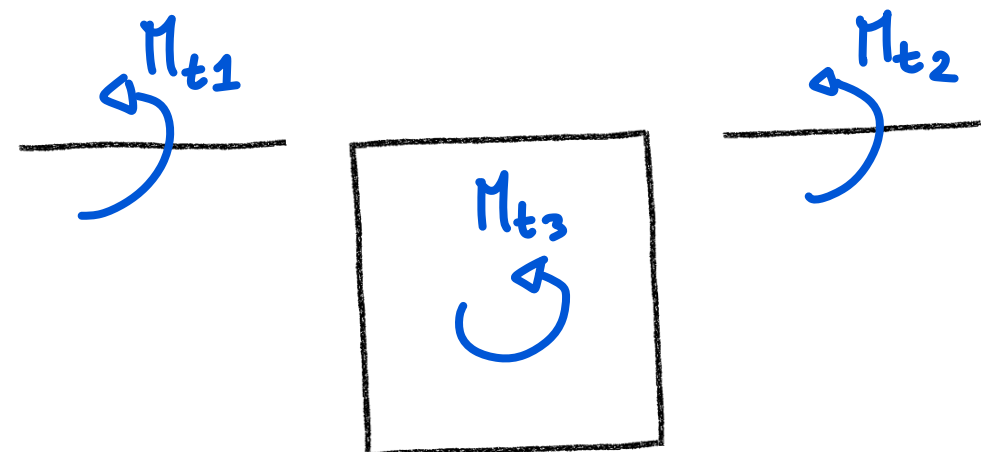
(*) ANCHE SE $I_{t3} \gg I_{t1}, I_{t2}$
LE Z SONO PARAGONABILI



ANALOGIA IDRODINAMICA



RIPARTIZIONE MOMENTO TORC.



TORSIONE: RIASSUNTO

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- $H_p: \mathcal{J}(z) = \Theta z$ con $[\Theta] = [L^{-1}]$

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↳ CIRCOLARE:
$$\begin{cases} \gamma(r) = \Theta r \\ \tau(r) = G \Theta r \end{cases}$$

EQUIV. STATICA:
$$\Theta = \frac{M_t}{G I_p}$$

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- NEUMANN: $\gamma(x, y) \Rightarrow \begin{cases} \frac{\partial^2 \gamma}{\partial x^2} + \frac{\partial^2 \gamma}{\partial y^2} = 0 \\ \frac{\partial \gamma}{\partial n} = \alpha y - \beta x \end{cases}$ con $\underline{z} \cdot \underline{n} = 0$ su Γ

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↳ INERZIA GENERALIZZ.
$$I_t = I_p + \int_A \left(\frac{\partial \gamma}{\partial y} x - \frac{\partial \gamma}{\partial x} y \right) dA \Rightarrow$$

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TORSIONE: RIASSUNTO

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- ANALOGIA IDRODINAMICA: $\underline{\tau} = [\tau_{zx} \quad \tau_{zy}] \Rightarrow \begin{cases} \tau_{zx} = G \Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G \Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases}$

TORSIONE: RIASSUNTO

• $H_p: \vartheta(z) = \Theta z$ con $[\Theta] = [L^{-1}]$

↳ CIRCOLARE: $\begin{cases} \vartheta(r) = \Theta r \\ \tau(r) = G \Theta r \end{cases}$ EQUIV. STATICA: $\Theta = \frac{M_t}{G I_p}$

• NEUMANN: $\gamma(x, y) \Rightarrow \begin{cases} \frac{\partial^2 \gamma}{\partial x^2} + \frac{\partial^2 \gamma}{\partial y^2} = 0 \\ \frac{\partial \gamma}{\partial n} = \alpha y - \beta x \end{cases}$ con $\underline{z} \cdot \underline{n} = 0$ su Γ

↳ INERZIA GENERALIZZ. $I_t = I_p + \int_A \left(\frac{\partial \gamma}{\partial y} x - \frac{\partial \gamma}{\partial x} y \right) dA \Rightarrow \boxed{\Theta = \frac{M_t}{G I_t}}$

• ANALOGIA IDRODINAMICA: $\underline{\tau} = [\tau_{zx} \ \tau_{zy}] \Rightarrow \begin{cases} \tau_{zx} = G \Theta \left(\frac{\partial \gamma}{\partial x} - y \right) \\ \tau_{zy} = G \Theta \left(\frac{\partial \gamma}{\partial y} + x \right) \end{cases}$

↳ $\text{div}(\underline{\tau}) = 0$ SOLENOIDALE (LINEE CHIUSE)

↳ $\text{rot}(\underline{\tau}) = 2 \Theta G \hat{k}$ FLUSSO COSTANTE

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- SEZIONE RETTANGOLARE SOTTILE (DISTRIBUZIONE LINEARE)

$$I_t = \frac{1}{3} a s^3 ; \quad z_{\max} = \frac{M_t}{I_t} s$$

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$$\frac{M_{ti}}{I_{ti}} = \frac{M_t}{I_t} ; \quad \tau_{\max,i} = \frac{M_t}{I_t} s_i$$

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- SEZIONE CAVE IN PARETE SOTTILE (DISTRIBUZIONE COST.)

$$q = \tau(t) s(t) = \text{cost}$$

$$\tau(t) = \frac{M_t}{2 \Omega s(t)}$$

$$I_t = \frac{4 \Omega^2}{\oint \frac{ds}{s(t)}}$$

$$\Rightarrow \Theta = \frac{M_t}{G I_t}$$